# PS65 / PS105

PRO P

MAINTENANCE DRAWINGS

**Document No: V210I-C1E-XG**

**Edition No.: G**

**MAKINO**
**MAKINO ASIA PTE LTD**

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 10/1/2023 | AVINASH |
| ② | | CHECKED | 10/1/2023 | JAVAHAR |
| ① | | DRAWN | 10/1/2023 | BALA |

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| CIRCUIT DRAWING | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | A01 |

# HISTORY OF MODIFICATION

| EDITION | DATE | DRAWING UPDATE RELEASE NO. | CONTENT | DRAWING NO | REMARKS | PREPARED BY |
|---|---|---|---|---|---|---|
| A | 20-12-2015 | ---- | FIRST RELEASE | V210I-C1E-XA | PROTOTYPE MC | NASEEF |
| B | 20-02-2016 | ---- | NEW JB CONN DESIGN, TSC, DISPLAY PCB | V210I-C1E-XB | PILOT MC | NASEEF |
| C | 12-04-2016 | MV21-0184 | OPTION CIRCUITS ADDED, LAYOUTS UPDATED | V210I-C1E-XC | PRODUCTION MC | NASEEF |
| D | 31-08-2016 | MV21-0289 | OPTION CIRCUITS ADDED,<br>CONNECTOR NAME, PIN NUMBER AND LAYOUT UPDATED,<br>LAYOUT DRAWINGS AND BOM UPDATED. | V210I-C1E-XD | -- | NASEEF |
| E | 24-08-2017 | MV21-0415 | UPDATE CROSS REFERENCE DIRECTION ARROWS,<br>UPDATE PFM CIRCUIT TO ADAPT CE CIRCUIT,<br>WORKWASH GUN NOTES ADDED,<br>AIRCON FEEDBACK SIGNAL REMOVED,<br>VISION BTS AND SPINDLE ACTIVE CARE CIRCUITS ARE ADDED,<br>RCH1 CONNECTION UPDATED,<br>ATV 320 CIRCUIT ADDED (ATC ARM MOTOR INVERTER),<br>LUCC CIRCUIT UPDATED. | V210I-C1E-XE | -- | NASEEF |
| F | 13-06-2019 | MV21-0635 | TSC 7MPA, OIL AIR FLOW SENSOR AND RFID OPTIONS<br>ADDED (PAGE NO. C36,C37,C114 & C115. | V210I-C1E-XF | -- | JAVAHAR |
| G | 10-01-2023 | MV21-1086 | OPTION-SPD#50 ADDED<br>PAGE NO-C55.1-SPINDLE FOOT SWITCH CIRCUIT ADDED.<br>PAGE NO-C58.1-TLS TOOL LOCK/UNLOCK CIRCUIT ADDED.<br>PAGE NO-C59.1&C59.2-ATC SHUTTER CIRCUIT ADDED.<br>PAGE NO-C59.3-ATC FOOT SWITCH CIRCUIT ADDED. | V210I-C1E-XG | -- | BALA |

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|---|---|---|---|---|
| HISTORY OF MODIFICATION | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | A02 |

DRAWING CROSS REFERENCE:

1. FORMAT:

C01/C05

- C01 — PAGE NO.
- C05 — ZONE NO.

2. ZONE NO. REFERENCE

C 05

- C — HORIZONTAL (DWG SHEET)
- 05 — VERTICAL (DWG SHEET)

3. PAGE NO. REFERENCE

C 01

- C — GROUP NO. SEE TABLE-1 FOR PAGE GROUP REFERENCE
- 01 — PAGE NO. UNDER PARTICULAR GROUP

TABLE 1: PAGE GROUP REFERENCE

| Sheet | Description |
|---|---|
| A | COVER, MODIFICATION HISTORY, LEGEND |
| B | TABLE OF CONTENTS |
| C | SCHEMATIC DRAWING |
| D | CUSTOM OPTIONS SCHEMATIC DRAWING |
| E | DI/DO ADDRESS TABLE |
| F | ASSY DRAWINGS |

DRAWING NO. REFERENCE:

**V210i - C1E - XA**

- V — MACHINE SERIES
- 2 — CONTROLLER TYPE
- 1 — CIRCUIT DRAWING
- E — 1 = STD M/C; 9 = CUSTOM CKT
- N — ENGLISH
- X — REVISION X= STD SPEC., Y= CE SPEC.

| REVISIONS | | | DATE | NAME |
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| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| LEGEND | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | A03 |

# Table of Contents


| PAGE NO. | PAGE DESCRIPTION |
|---|---|
| A01 | CIRCUIT DRAWING |
| A02 | HISTORY OF MODIFICATION |
| A03 | LEGEND |
| B01-B07 | TABLE OF CONTENTS |
| C01 | MAIN INCOMING POWER |
| C02 | TB0 CONNECTIONS |
| C03 | CIRCUIT PROTECTOR CP11 |
| C04 | CIRCUIT PROTECTOR CP12 |
| C05 | CIRCUIT PROTECTOR CP13/CP14/CP21 |
| C06 | CIRCUIT PROTECTOR CP22 (OPT) |
| C07 | CB OVERLOAD CONTROL |
| C08 | CONTROL VOLTAGE DISTRIBUTION (24VDC) |
| C09 | CONTROL VOLTAGE DISTRIBUTION (C24HB) |
| C10 | CONTROL VOLTAGE DISTRIBUTION (E24H) |
| C11 | PCB POWER INPUT |
| C12 | POWER OUT |
| C13 | POWER SUPPLY MODULE |
| C14 | SPINDLE AMPLIFIER MODULE |
| C15 | SERVO AMPLIFIER CONNECTION |
| C16 | X, Y & Z AXIS SERVO AMPLIFIER MODULE |
| C17 | ATC AXIS SERVO AMPLIFIER MODULE |
| C18 | 4th & 5th AXIS SERVO AMP MODULE (OPT) |
| C19 | FEEDBACK SCALE (OPT) |


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| ① | | DRAWN | 13/06/2019 | |

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| TITLE | MACHINE TYPE | | DRAWING No. | GROUP |
|---|---|---|---|---|
| TSC 7.0MPa CONTROL (OPT) | NC TYPE | | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | B01 |

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| PAGE NO. | PAGE DESCRIPTION |
|---|---|
| C20 | Z AXIS MOTOR BRAKE |
| C21 | CONTROLLER UNIT |
| C22 | CONTROLLER UNIT & PUNCH PANEL |
| C23 | I/O LINK CONNECTION |
| C24 | I/O LINK CONNECTION |
| C25 | NC POWER ON CONTROL |
| C26 | EMERGENCY STOP CIRCUIT |
| C27 | REFERENCE AND OVERTRAVEL (OPT) |
| C28 | SCREW CONVEYOR |
| C29 | NOZZLE COOLANT & WW GUN PUMP |
| C30 | BASE / FIXTURE COOLANT (OPT) |
| C31 | SHOWER COOLANT (OPT) |
| C32 | TSC 1.5MPa & 3MPa PWR (OPT) |
| C33 | TSC 1.5MPa & 3MPa CONTROL (OPT) |
| C34 | TSC 1.5MPa CONTROL (OPT) |
| C35 | TSC 3.0MPa CONTROL (OPT) |
| C36 | TSC 7.0MPa CONTROL (OPT) |
| C37 | TSC 7.0MPa CONTROL (OPT) |
| C38 | CYCLONE FILTER PRESSURE SW (OPT) |
| C39 | SECONDARY TRANSFER PUMP (OPT) |
| C40 | LUCC DRUM (OPT) |
| C41 | LIFT UP CHIP CYR PWR (OPT) |
| C42 | LIFT UP CHIP CYR CONTROL (OPT) |


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|---|---|---|---|---|
| TSC 7.0MPa CONTROL (OPT) | NC TYPE | | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | B02 |

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| PAGE NO. | PAGE DESCRIPTION |
|---|---|
| C43 | OIL SKIMMER (OPT) |
| C44 | MIST COLLECTOR (OPT) |
| C45 | COOLANT LEVEL SENSOR - MAIN TANK |
| C46 | COOLANT LEVEL SENSOR - SEC TANK (OPT) |
| C47 | AIR BLOW & TSA (OPT) |
| C48 | MASTER SOLENOID |
| C49 | POWER PACK (TOOL CL/UCL) |
| C50 | OIL AIR UNIT (OPT) |
| C51 | COOLANT CHILLER (OPT) |
| C52 | SPINDLE SWITCHING CONTROL |
| C53 | SPINDLE TEMPERATURE CONTROLLER |
| C54 | TOOL CLAMP/UNCLAMP SOL |
| C55 | TOOL CLAMP/UNCLAMP LIMIT SW |
| C56 | ATC ARM MOTOR (ATV312) |
| C56-1 | ATC ARM MOTOR (ATV320) |
| C57 | ATC ARM ENCODER |
| C58 | ATC POT CONTROL |
| C59 | ATC SHUTTER |
| C60 | ATC PANEL |
| C61 | AWM & ATLM INTERFACE (OPT) |
| C62 | ATLM SOLENOIDS (OPT) |
| C63 | AWM MARPOSS (OPT) |
| C64 | AWM RENISHAW (OPT) |

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| PAGE NO. | PAGE DESCRIPTION |
|---|---|
| C65 | AWM BLUM (OPT) |
| C66 | ATLM TP MARPOSS (OPT) |
| C67 | ATLM TP RENISHAW (OPT) |
| C68 | ATLM TP BLUM (OPT) |
| C69 | ATLM LASER RENISHAW (OPT) |
| C70 | ATLM LASER BLUM (OPT) |
| C71 | USER M-CODE (OPT) |
| C72 | S/G DOOR LOCK W INTERLOCK |
| C73 | ATC DOOR INTERLOCK |
| C74 | ATC DOOR LOCK W INTERLOCK (OPT) |
| C75 | OPERATOR SIDE(S/G) AUTO DOOR (OPT) |
| C76 | LIGHT CURTAIN (OPT) |
| C77 | 4TH AXIS CONTROL (PNEUMATIC) |
| C78 | 4TH AXIS CONTROL (HYDRAULIC) |
| C79 | 5TH AXIS CONTROL (PNEUMATIC) |
| C80 | 5TH AXIS CONTROL (HYDRAULIC) |
| C81 | RWH POWER PACK (OPT) |
| C82 | MACHINE LIGHT |
| C83 | AUTO GREASE |
| C84 | MTC FAN |
| C85 | MTC AIR CONDITIONER (OPT) |
| C86 | AIR DRYER (OPT) |
| C87 | TWO HAND PUSH BUTTON (OPT) |


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| TSC 7.0MPa CONTROL (OPT) | NC TYPE | | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | B04 |

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|---|---|
| C88 | STRIKER SWITCH (OPT) |
| C89 | AIR SEAT CHECK (1-8 SENSOR) (OPT) |
| C90 | HYDRAULIC FIXTURE (OPT) |
| C91 | HYDRAULIC FIXTURE SOL/PS (OPT) |
| C92 | HYDRAULIC PANEL (OPT) |
| C93 | ROBOT INTERFACE (OPT) |
| C94 | ROBOT INTERFACE (OPT) |
| C95 | EXTERNAL HYDRAULIC PUMP I/F(OPT) |
| C96 | MAIN RELAY PCB - OPERATOR PANEL PCB INTERFACE |
| C97 | 24V FOR OPE PCB & I/O MODULE |
| C98 | OPERATOR PANEL EMG CIRCUIT |
| C99 | OPERATOR PANEL SWITCHES |
| C100 | OPERATOR PANEL SWITCHES |
| C101 | CUTTING/JOG OVERRIDE CONTROL |
| C102 | RAPID/HANDLE OVERRIDE CONTROL |
| C103 | RAPID OVERRIDE CONTROL |
| C104 | MANUAL PULSE GENERATOR |
| C105 | MANUAL PULSE GENERATOR (PORTABLE) |
| C106 | SIGNAL / PATROL LIGHT |
| C107 | OPERATOR PANEL SOFT KEYS 1 |
| C108 | OPERATOR PANEL SOFT KEYS 2 |
| C109 | AXIS SELECTION |
| C110 | INDICATION LED |

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| C112 | VISION BTS (OPT) |
| C113 | SPINDLE ACTIVE CARE (OPT) |
| C114 | OIL AIR FLOW SENSOR |
| C115 | RFID (OPTION) |
| E01 | DI/DO ADDRESS TABLE - IO MODULE 1 |
| E02 | DI/DO ADDRESS TABLE - IO MODULE 1 |
| E03 | DI/DO ADDRESS TABLE - IO MODULE 2 |
| E04 | DI/DO ADDRESS TABLE - IO MODULE 2 |
| E05 | DI/DO ADDRESS TABLE - IO MODULE 3 |
| E06 | DI/DO ADDRESS TABLE - IO MODULE 3 |
| E07 | DI/DO ADDRESS TABLE - IO MODULE 4 |
| E08 | DI/DO ADDRESS TABLE - IO MODULE 4 |
| E09 | DI/DO ADDRESS TABLE - IO MODULE 5 |
| E10 | DI/DO ADDRESS TABLE - IO MODULE 5 |
| E11 | DI/DO ADDRESS TABLE - IO MODULE 6 |
| E12 | DI/DO ADDRESS TABLE - IO MODULE 6 |
| E13 | DI/DO ADDRESS TABLE - IO MODULE 7 |
| E14 | DI/DO ADDRESS TABLE - IO MODULE 7 |
| E15 | DI/DO ADDRESS TABLE - IO MODULE 8 |
| E16 | DI/DO ADDRESS TABLE - IO MODULE 8 |
| F01 | MTC LAYOUT DETAILS |
| F02 | MTC LAYOUT DETAILS |

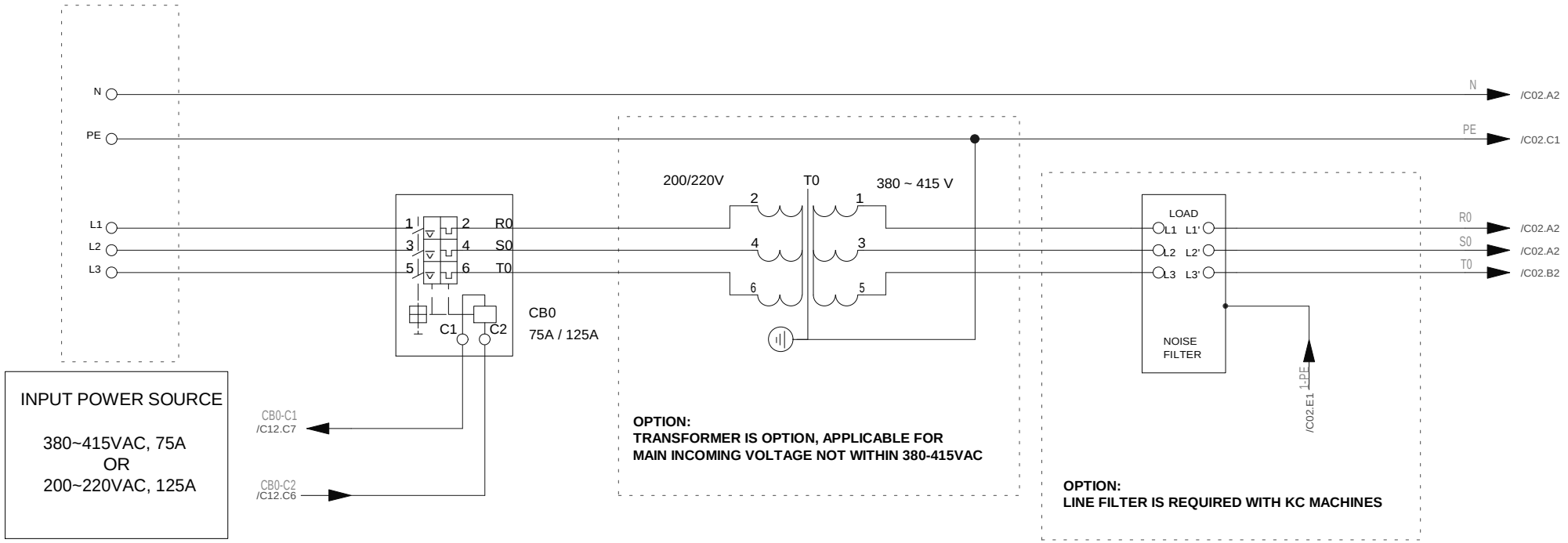
# Table of Contents


| PAGE NO. | PAGE DESCRIPTION |
|---|---|
| F03 | MTC COMPONENT DETAILS |
| F04 | MTC COMPONENT DETAILS |
| F05 | CONNECTOR PIN DETAILS |
| F06 | CONNECTOR PIN DETAILS |
| F07 | TB10 / TB20 / TB30 LAYOUT |
| F08 | TB40 SPINDLE TERMINAL BOX |
| F09 | TB50 COOLANT TERMINAL BOX |
| F10 | TB60 FLUID PANEL TERMINAL BOX |
| F11 | TB70 ATC TERMINAL BOX |
| F12 | TB90 SEC TANK TERMINAL BOX |
| F13 | SWING PANEL -TB100/TB110 LAYOUT |
| F14 | RELAY PCB LAYOUT & PARTS DETAILS |
| F15 | OPTION PCB LAYOUT & PARTS DETAILS |
| F16 | FUSE PCB LAYOUT & PARTS DETAILS |
| F17 | OPERATOR PCB LAYOUT DETAILS |
| F18 | OPERATOR PCB PARTS DETAILS |
| F19 | DISPLAY PCB LAYOUT & PARTS DETAILS |
| F20 | OPERATORS PANEL LAYOUT, PARTS |
| F21 | PORTABLE / FIXTURE MPG PLATE |
| | |
| | |
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|---|---|---|---|---|
| TSC 7.0MPa CONTROL (OPT) | NC TYPE | | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | B07 |

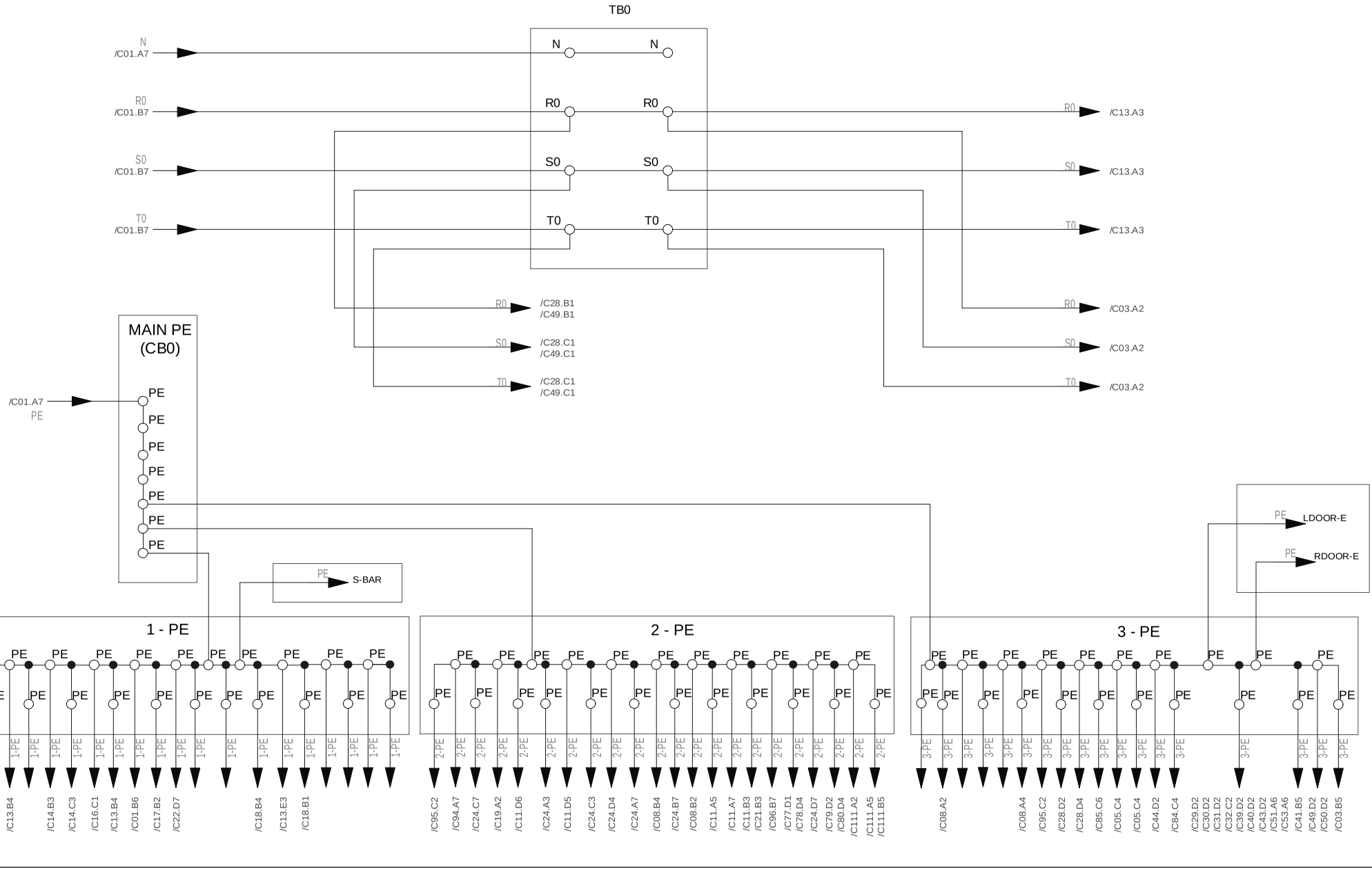



| 3 PHASE AC SOURCE | CB0 SHUNT TRIP CONTROL | MAIN CIRCUIT BREAKER CB0 | MAIN TRANSFORMER | LINE FILTER |
|---|---|---|---|---|

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| MAIN INCOMING POWER | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C01 |

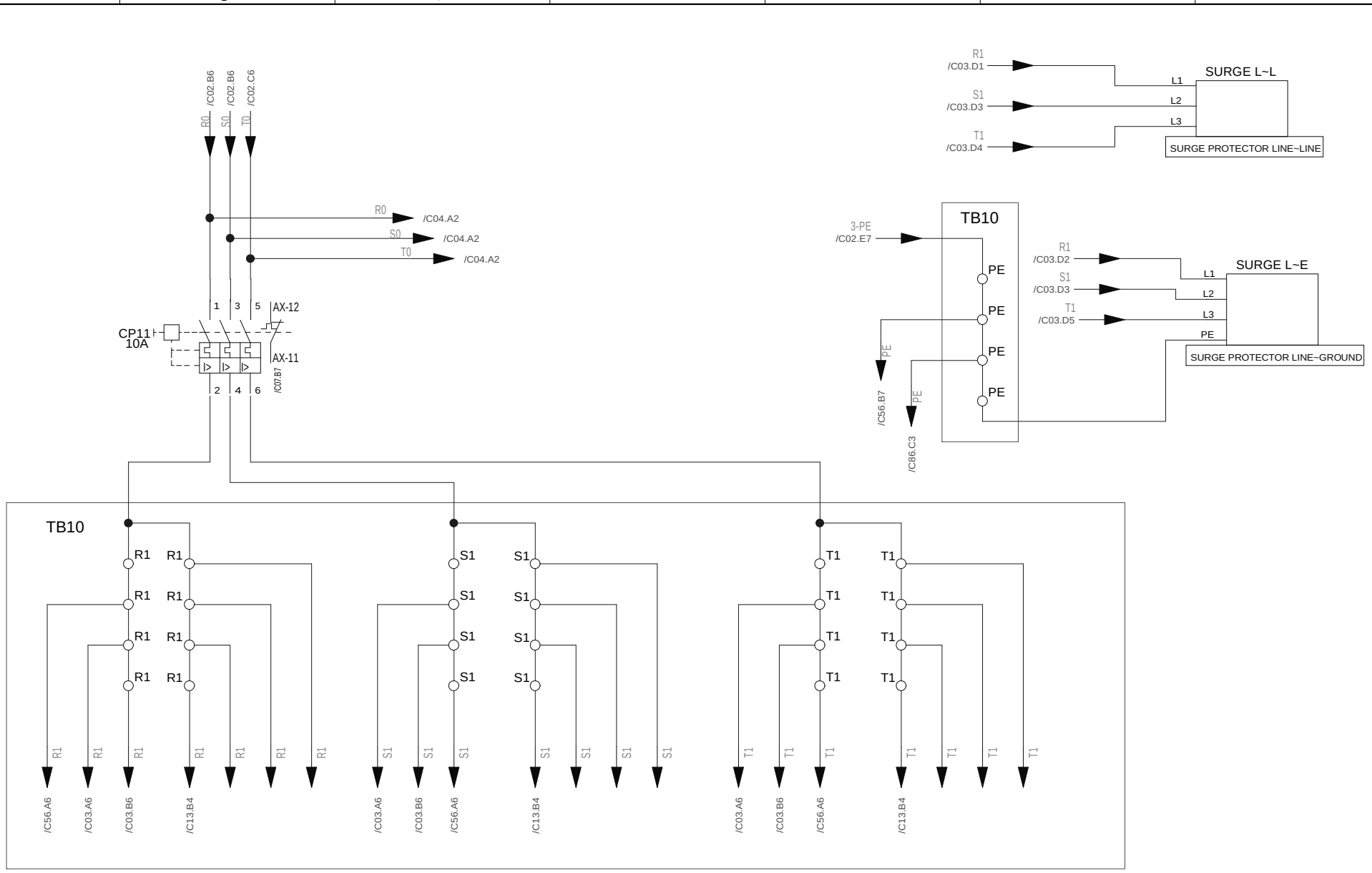



THREE PHASE AC POWER DISTRIBUTION

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| TB0 CONNECTIONS | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C02 |



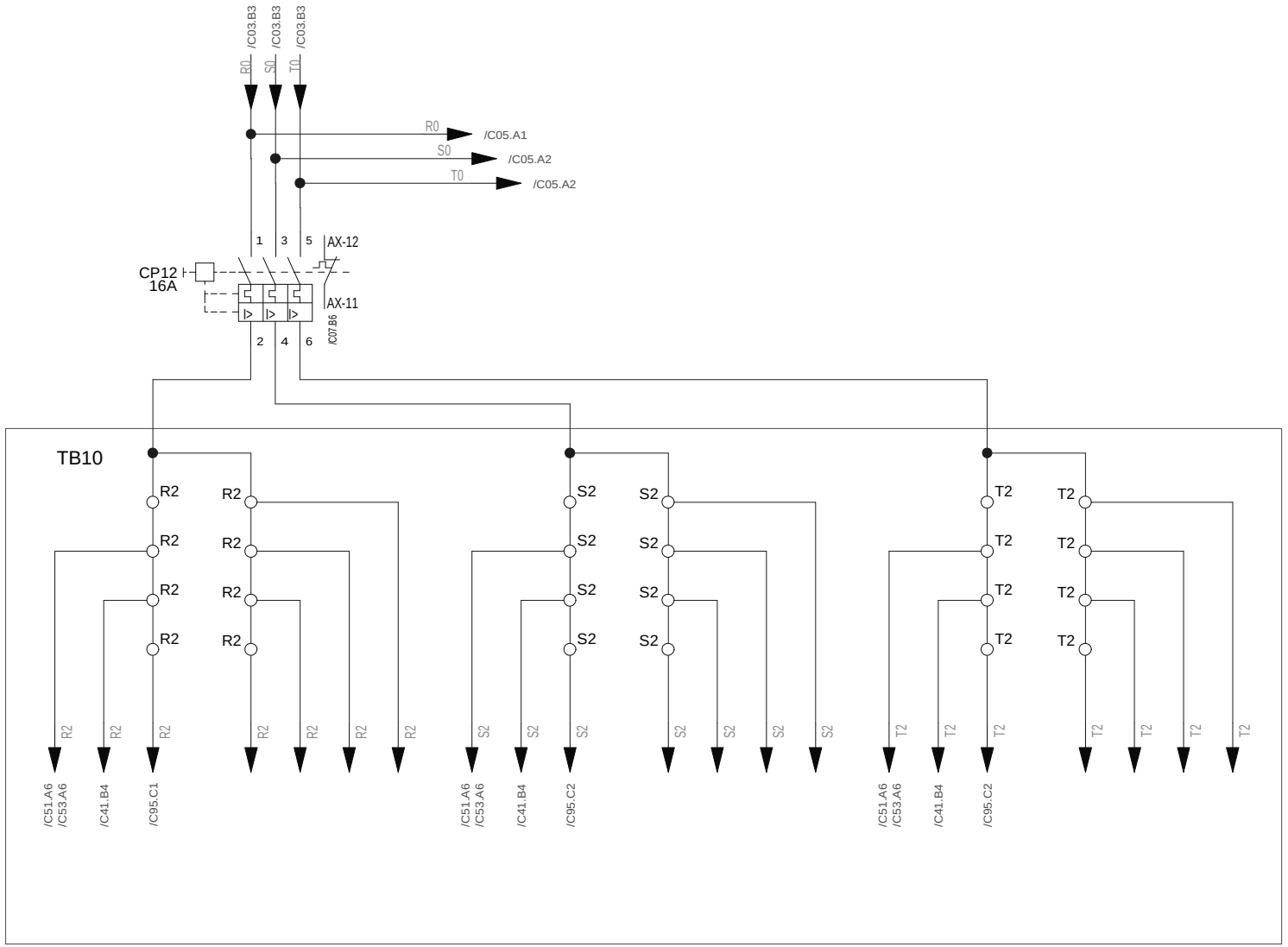

THREE PHASE AC POWER DISTRIBUTION

SURGE SUPPRESSOR

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
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| | ① | DRAWN | 13/06/2019 | JAVAHAR |

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| CIRCUIT PROTECTOR CP11 | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C03 |

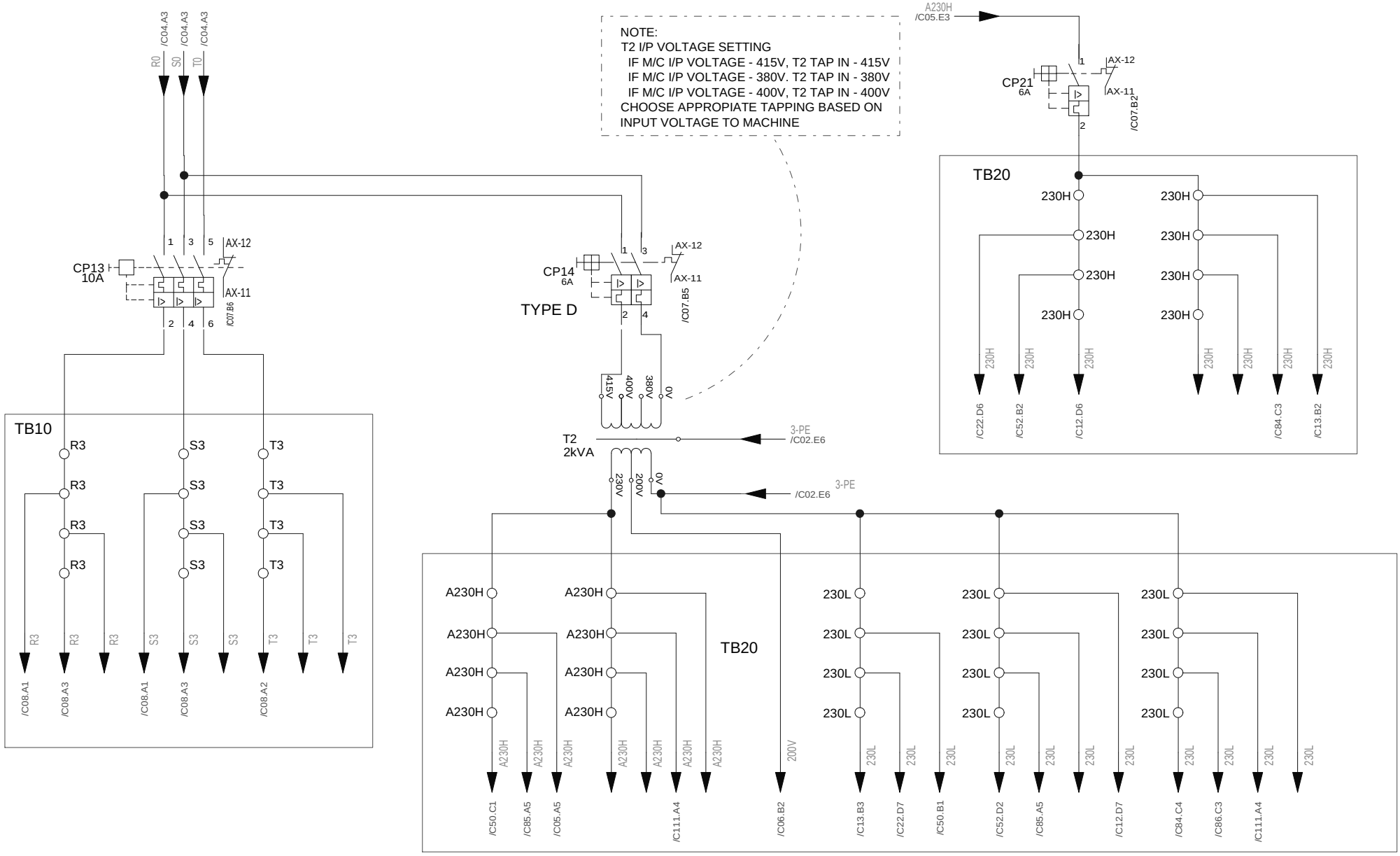



THREE PHASE AC POWER DISTRIBUTION

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|---|---|---|---|---|
| | | | | = |
| CIRCUIT PROTECTOR CP12 | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C04 |



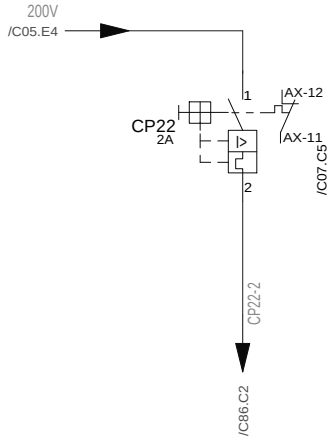

THREE PHASE AC POWER DISTRIBUTION

SINGLE PHASE AC POWER DISTRIBUTION

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| CIRCUIT PROTECTOR CP13/CP14/CP21 | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | C05 |

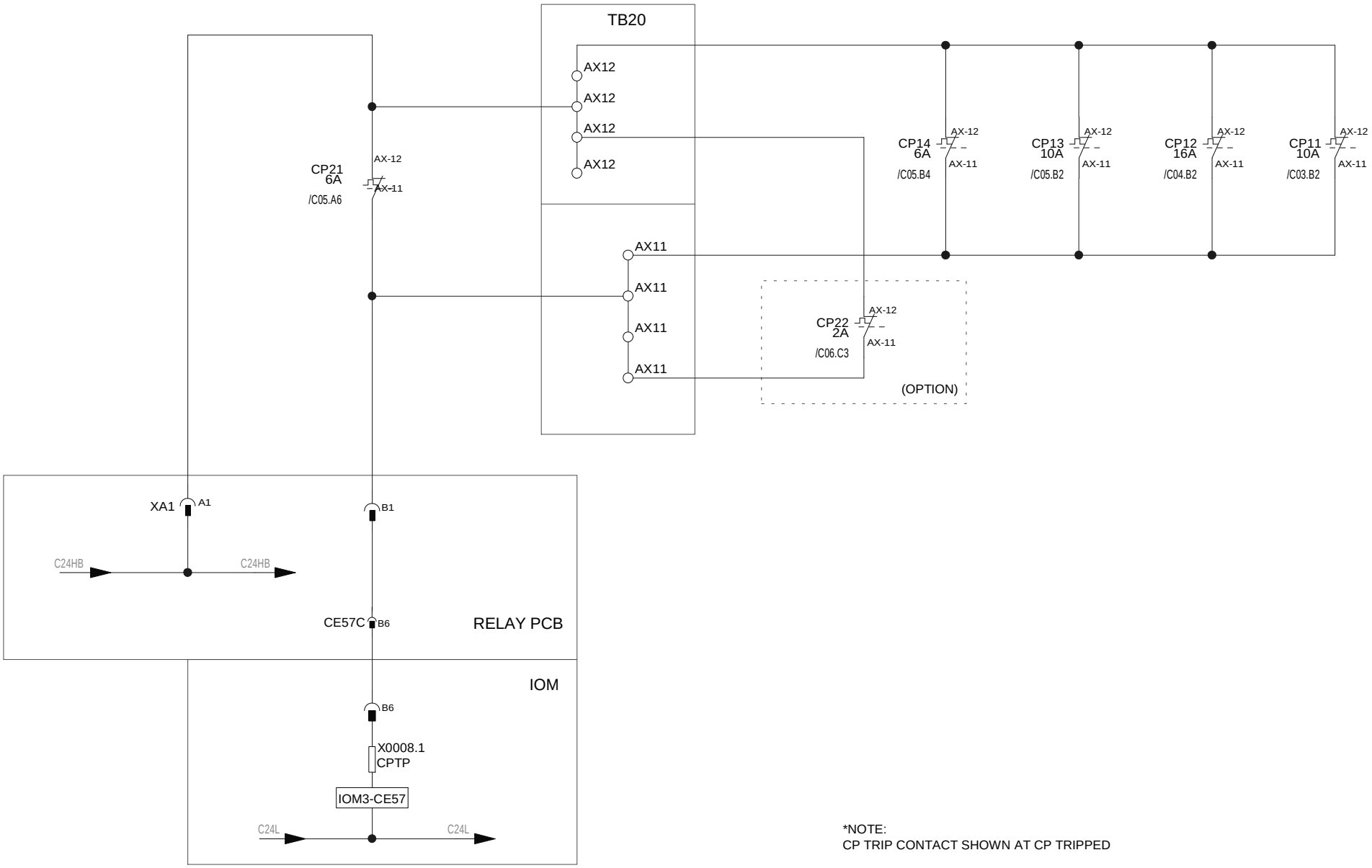



SINGLE PHASE AC POWER DISTRIBUTION

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|---|---|---|---|---|
| | | | | = |
| CIRCUIT PROTECTOR CP22 (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C06 |



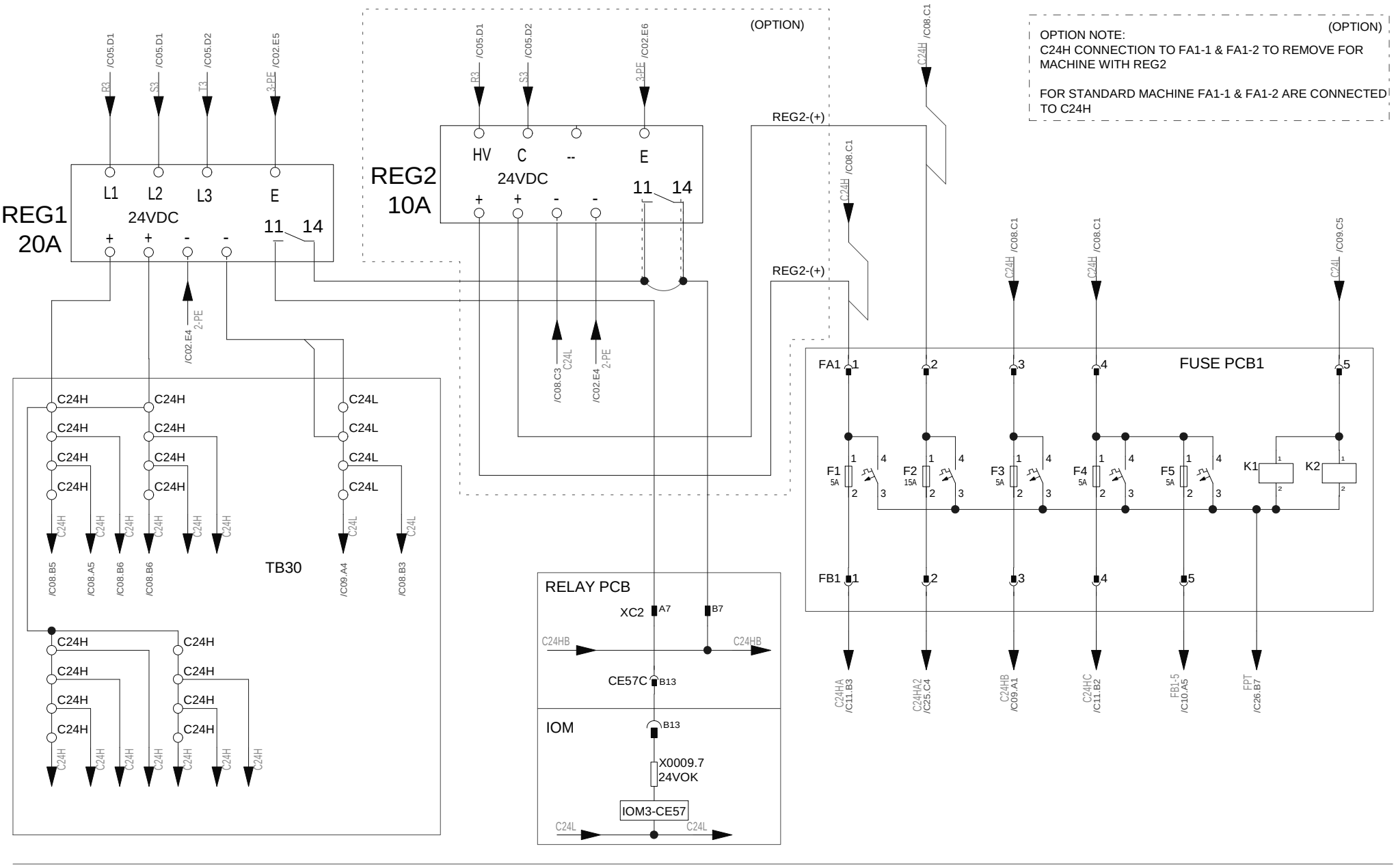

*NOTE:
CP TRIP CONTACT SHOWN AT CP TRIPPED

CP TRIP

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| CB OVERLOAD CONTROL | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | C07 |

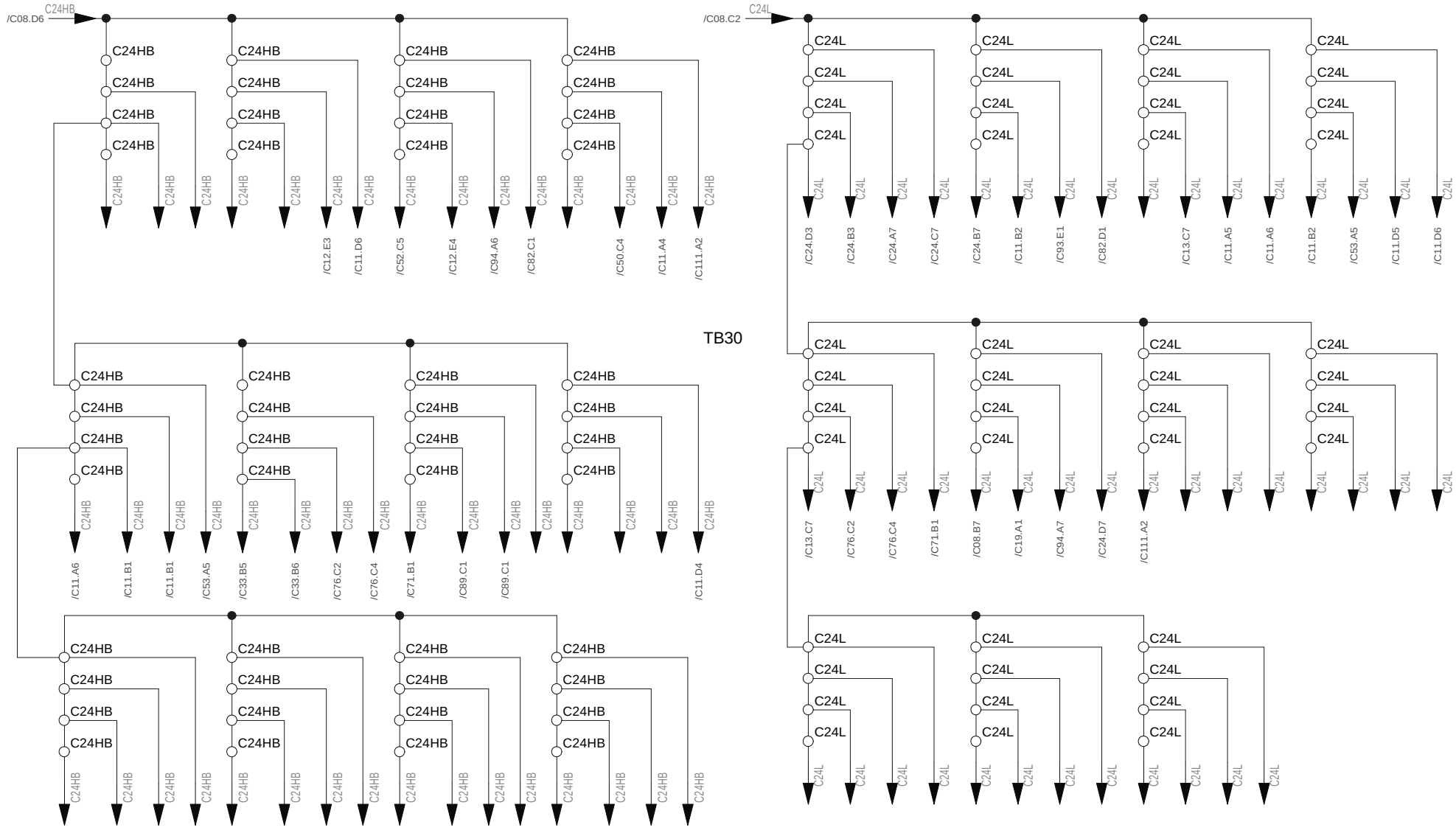



DC REG AND 24DC SOURCE (C24H) | DC REG STATUS | FUSE PCB

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| CONTROL VOLTAGE DISTRIBUTION (24VDC) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C08 |

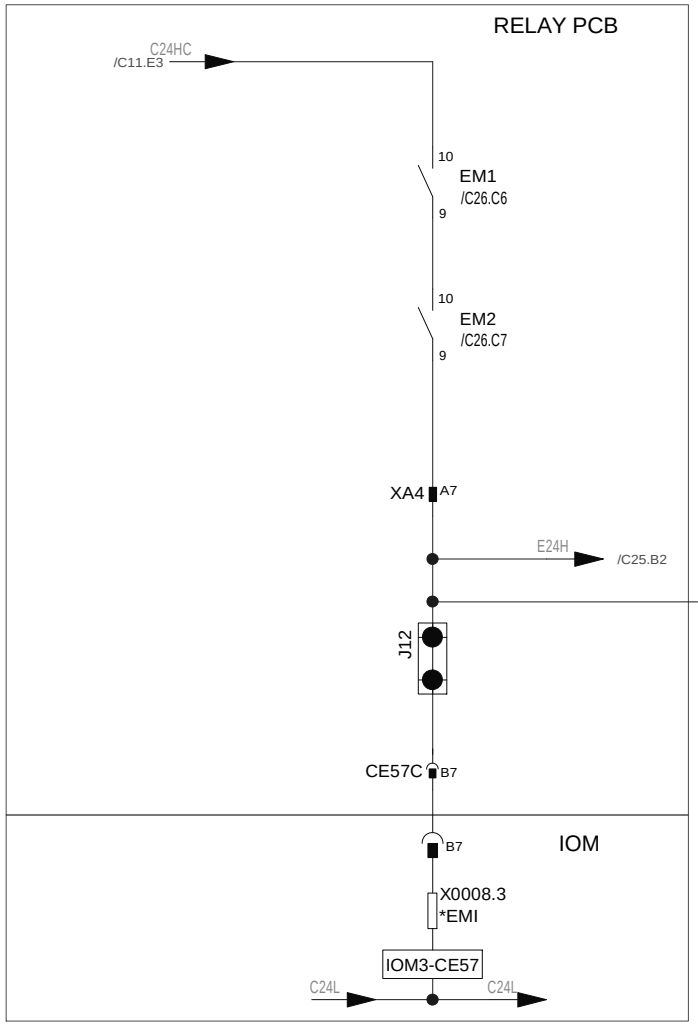



24DC SOURCE (C24HB)

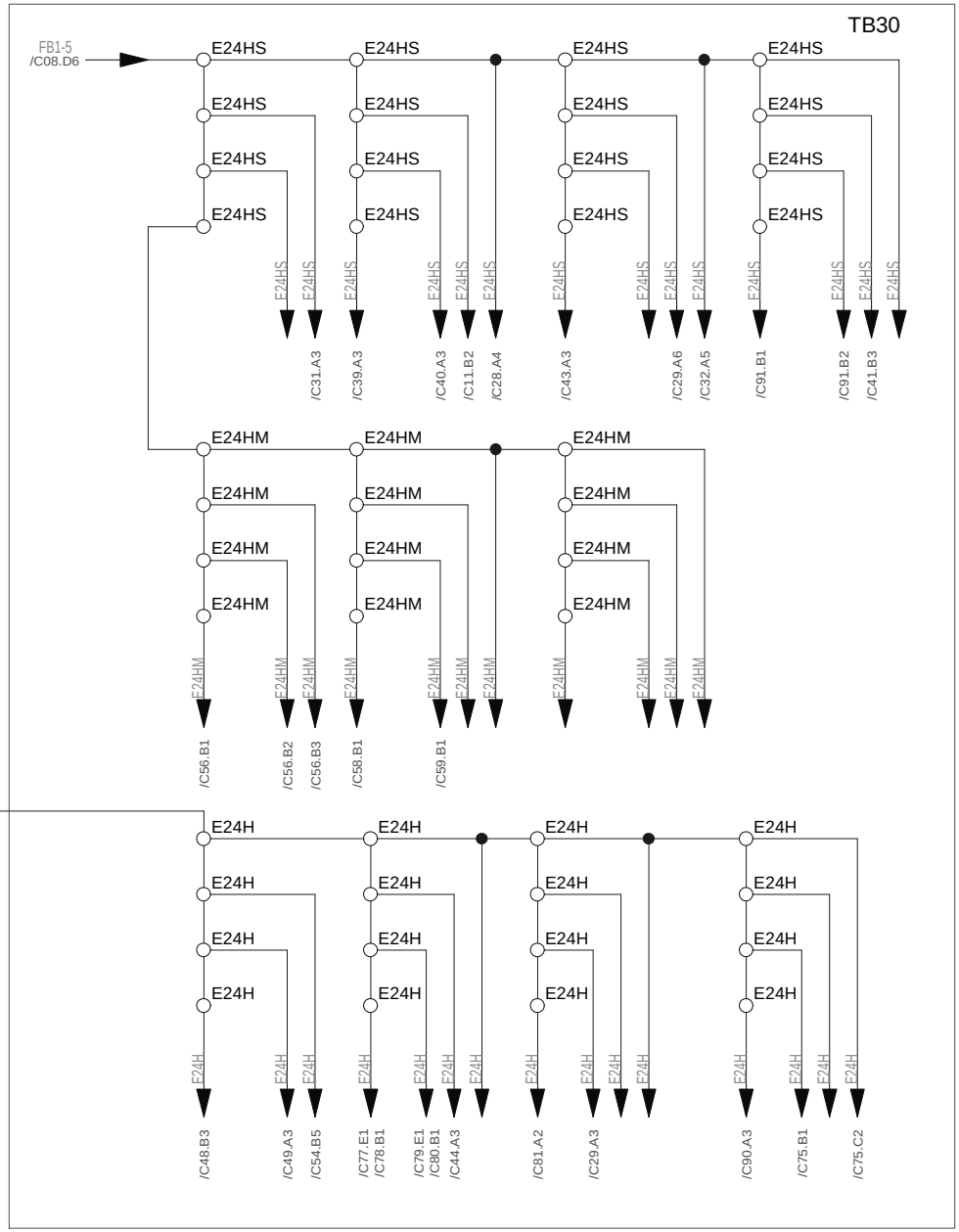
| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| | ④ | | | |
| | ③ | APPROVED | 13/06/2019 | AVINASH |
| | ② | CHECKED | 13/06/2019 | AVINASH |
| | ① | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| CONTROL VOLTAGE DISTRIBUTION (C24HB) | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C09 |




EMG STOP SW

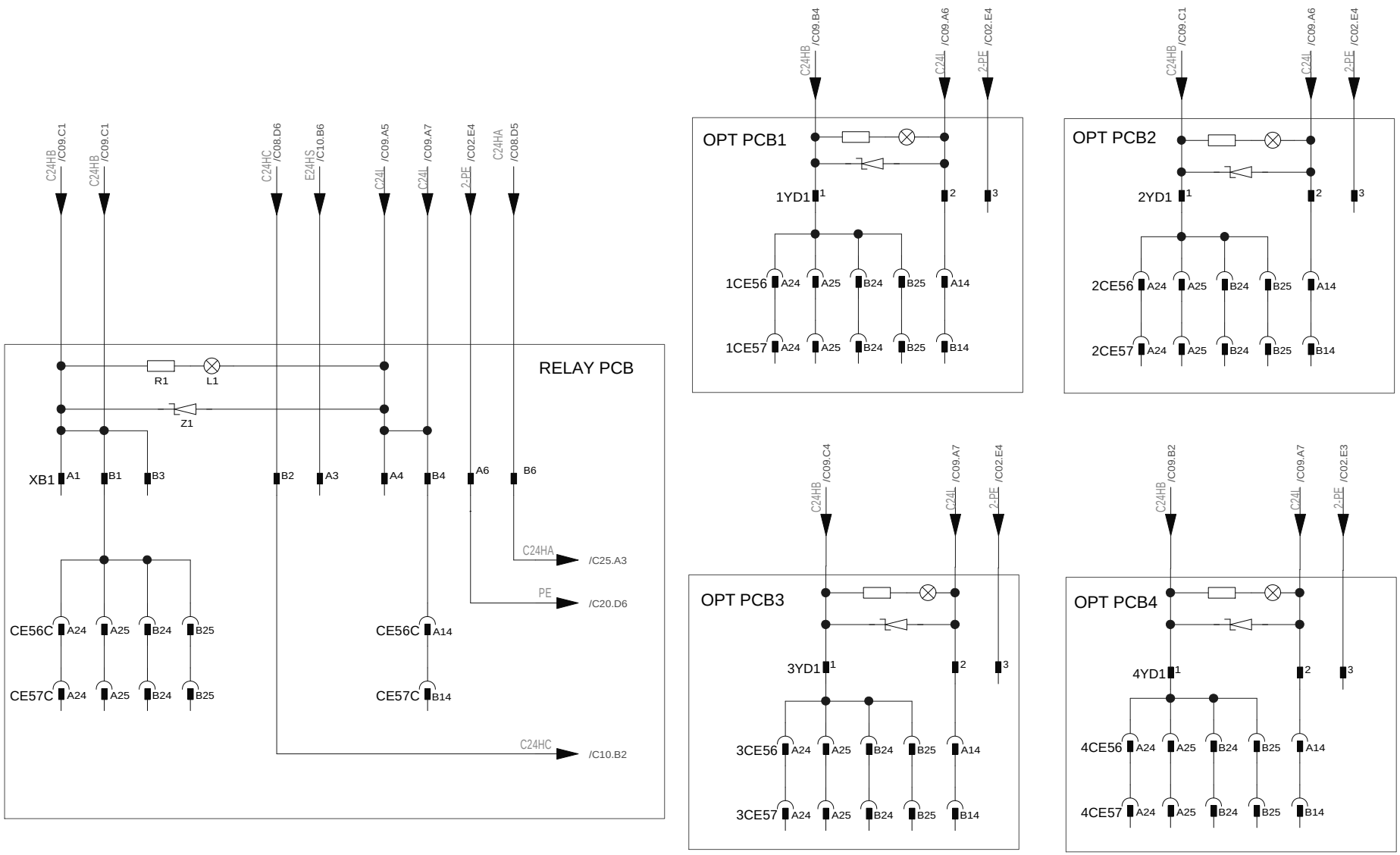



E24H/E24HS/E24HM DISTRIBUTION

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

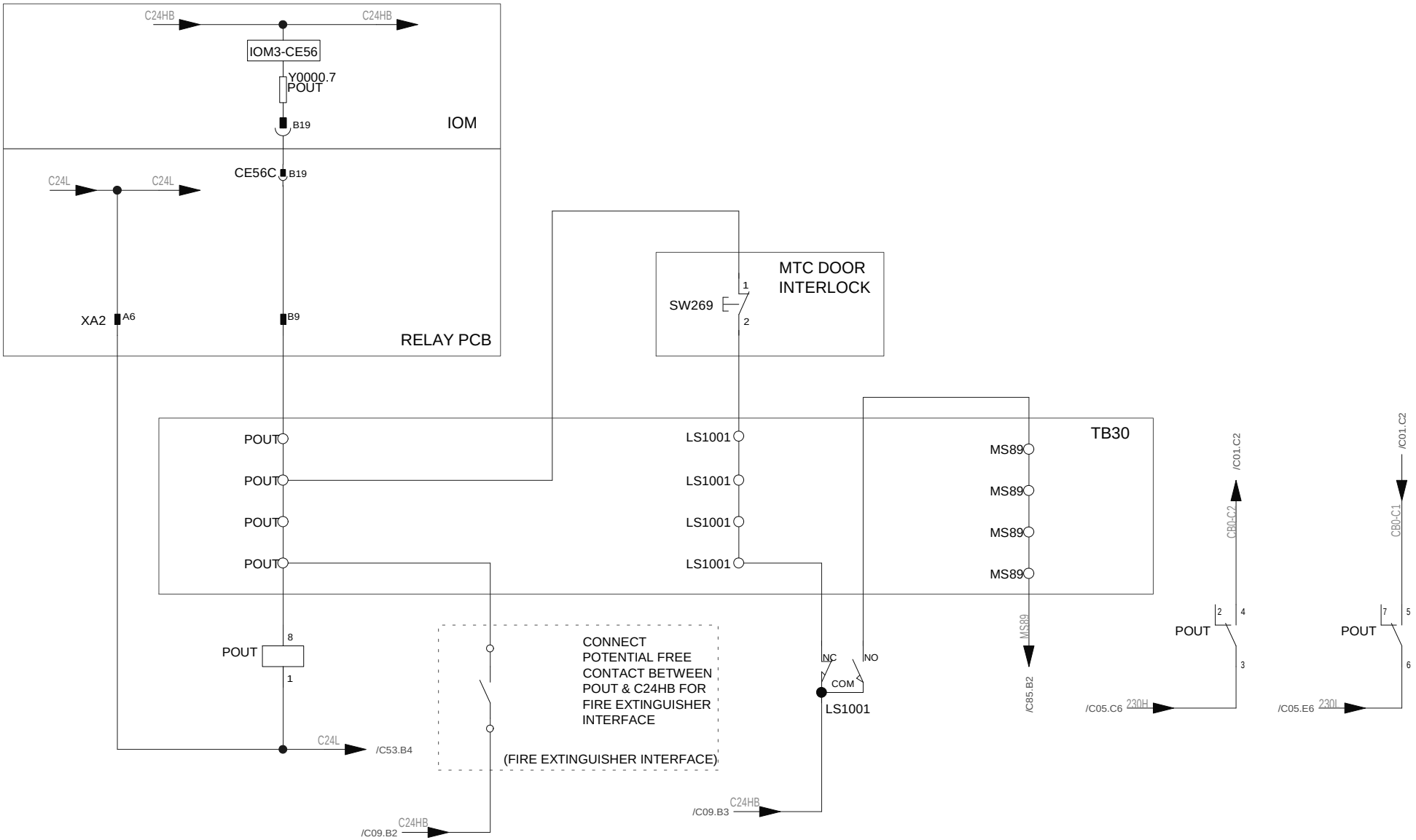
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| CONTROL VOLTAGE DISTRIBUTION (E24H) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C10 |




| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| PCB POWER INPUT | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C11 |

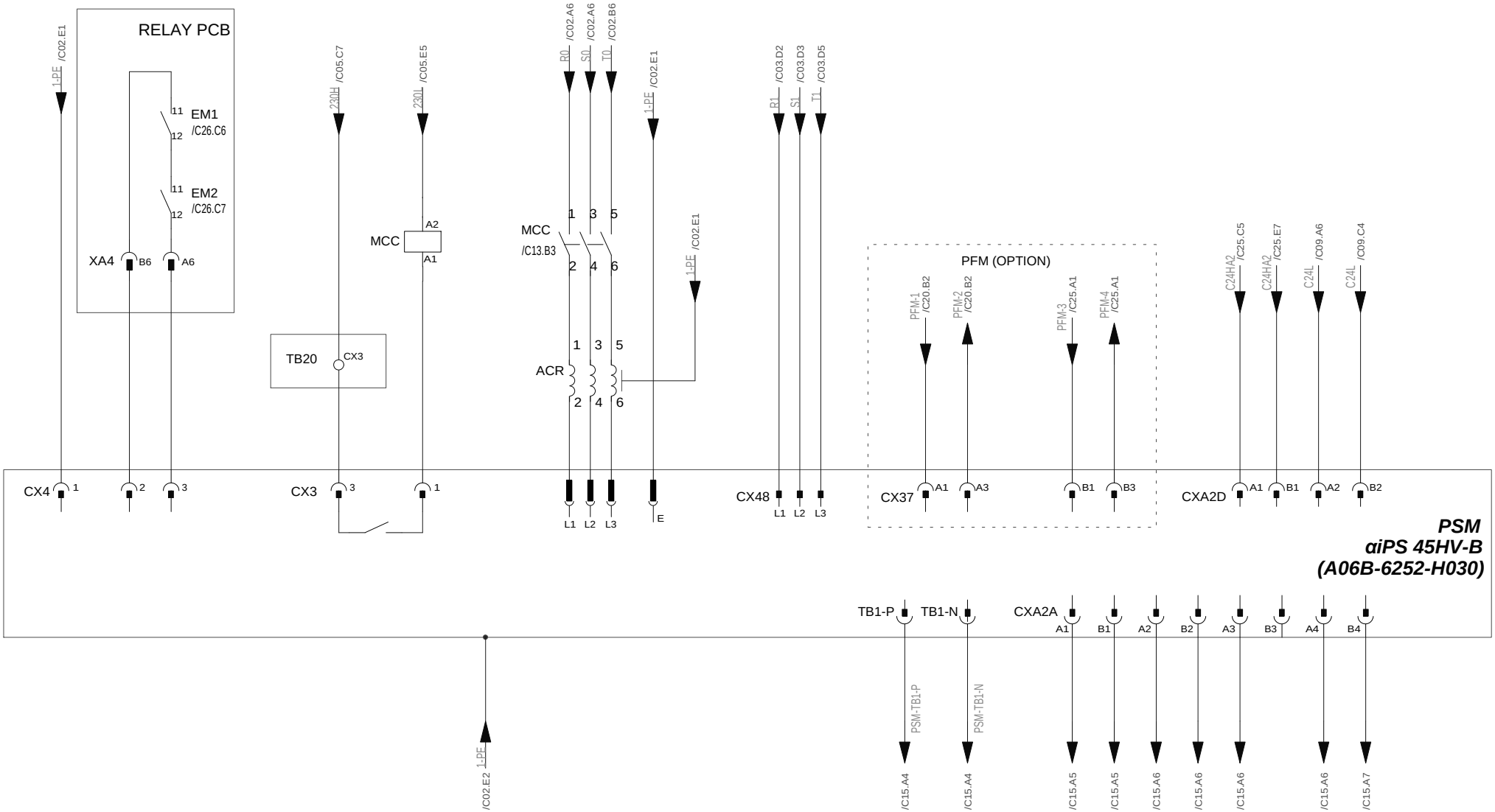



| AUTO POWER OFF | FIRE EXTINGUISHER | MTC CABINET DOOR INTERLOCK & LIMIT SWITCH | AIR CON ON CONTROL | CB0 TRIP |
|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| POWER OUT | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C12 |

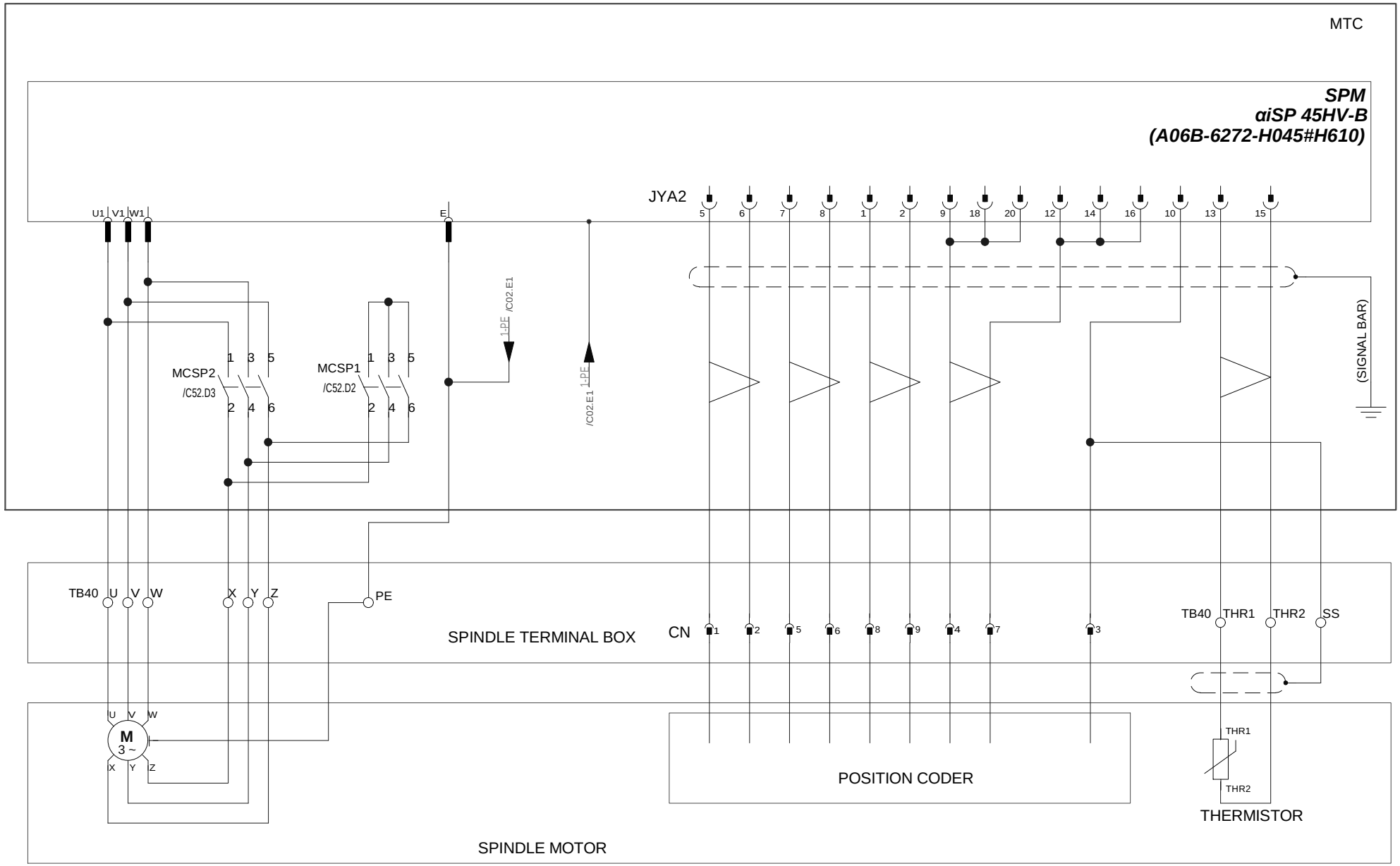



| EMERGENCY INTERLOCK | DRIVE CONTROL | POWER SUPPLY | MONITORING | DC BUS | INTERFACE SIGNALS |
|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| POWER SUPPLY MODULE | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C13 |

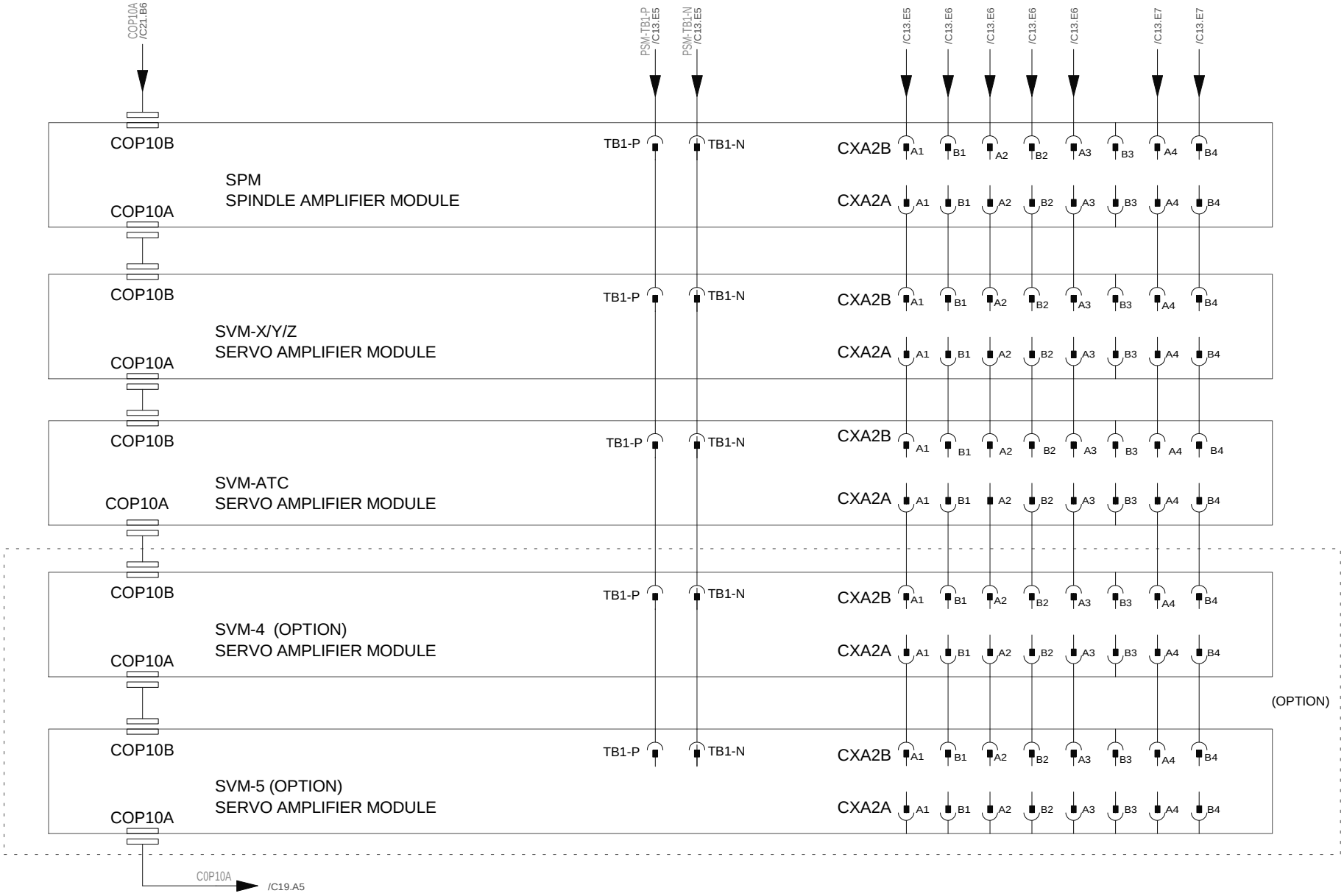



| SPINDLE MOTOR POWER | SPINDLE MOTOR POSITION SIGNAL | OVERHEAT CONTROL |
|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| SPINDLE AMPLIFIER MODULE | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C14 |




SERVO AMPLIFIER CONNECTIONS

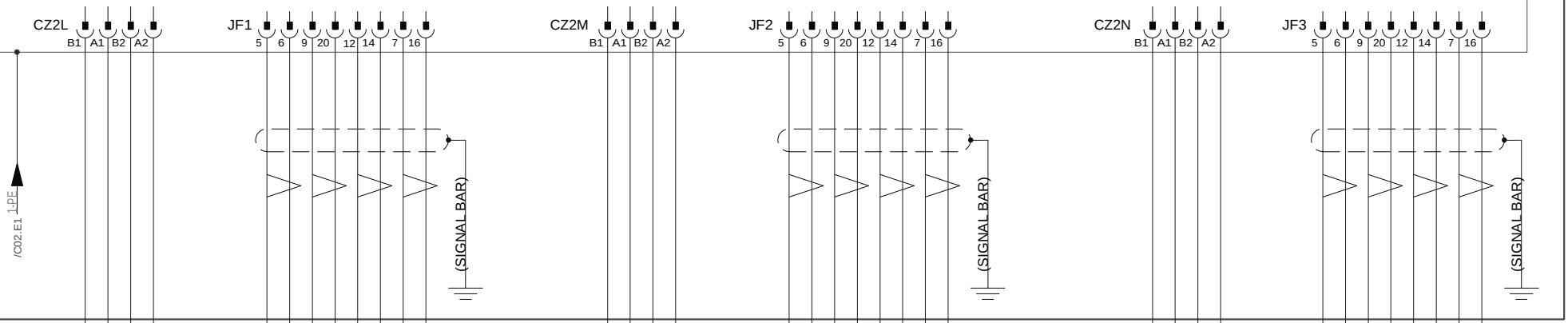
| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| | ④ | | | |
| | ③ | APPROVED | 13/06/2019 | AVINASH |
| | ② | CHECKED | 13/06/2019 | AVINASH |
| | ① | DRAWN | 13/06/2019 | JAVAHAR |

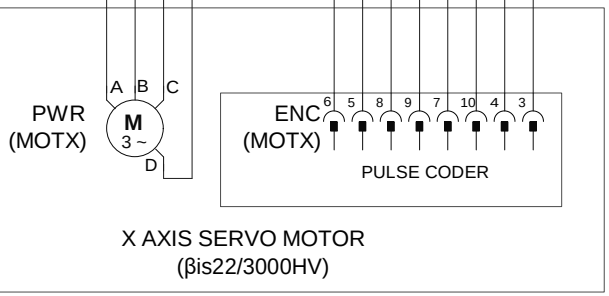
MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| SERVO AMPLIFIER CONNECTION | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C15 |

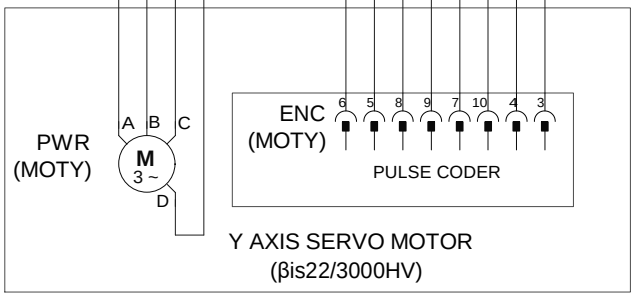
MTC

***SVM3 X/Y/Z***
***αiSV 40/40/40HV-B***
***(A06B-6290-H328)***

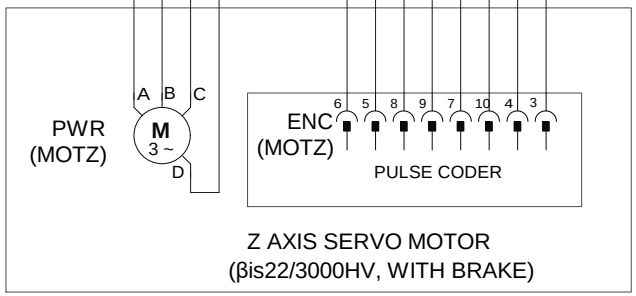



X AXIS SERVO MOTOR
(βis22/3000HV)




Y AXIS SERVO MOTOR
(βis22/3000HV)



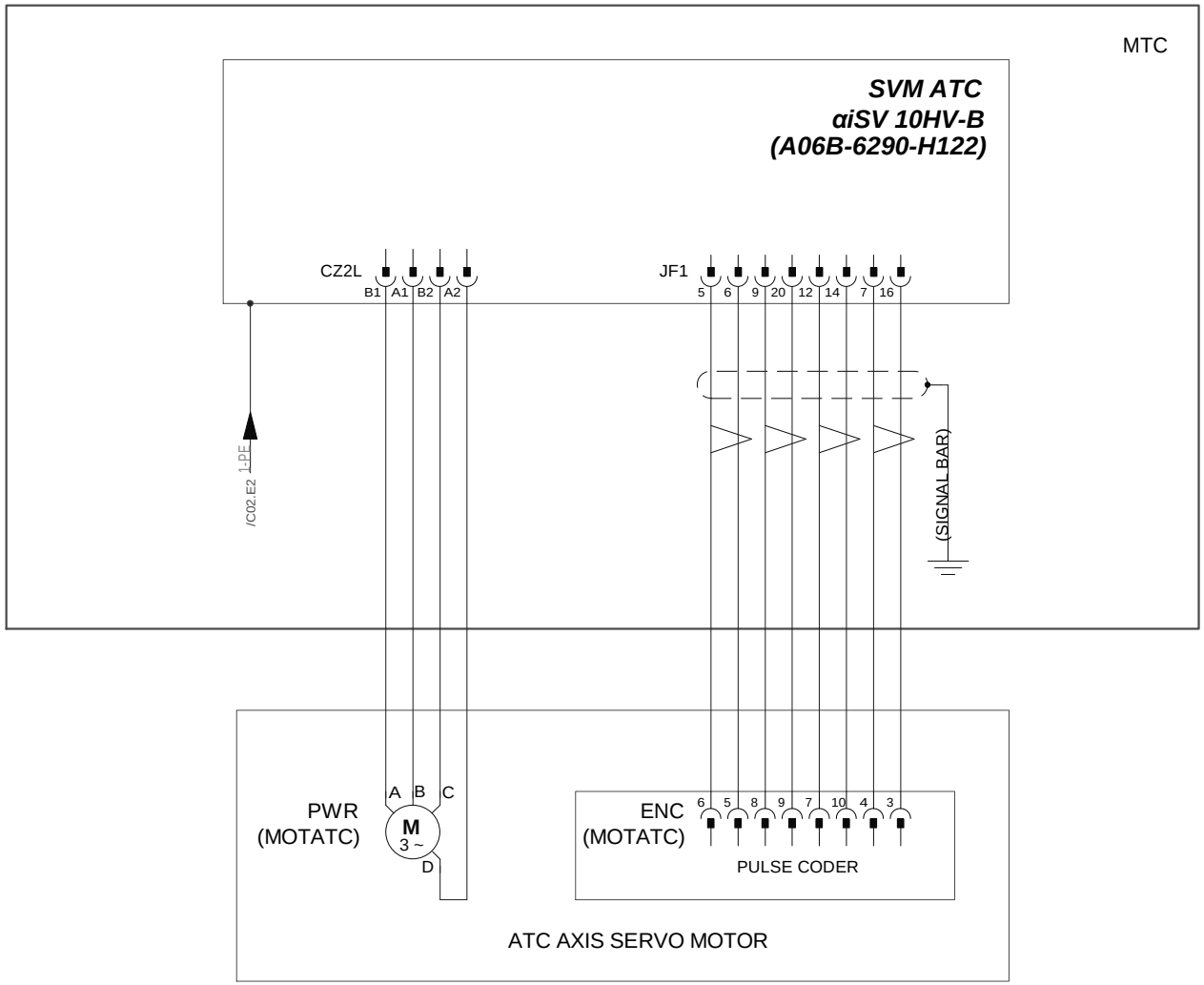

Z AXIS SERVO MOTOR
(βis22/3000HV, WITH BRAKE)

| X - AXIS POWER / SIGNAL | Y - AXIS POWER / SIGNAL | Z - AXIS POWER / SIGNAL |
|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| X, Y & Z AXIS SERVO AMPLIFIER MODULE | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C16 |

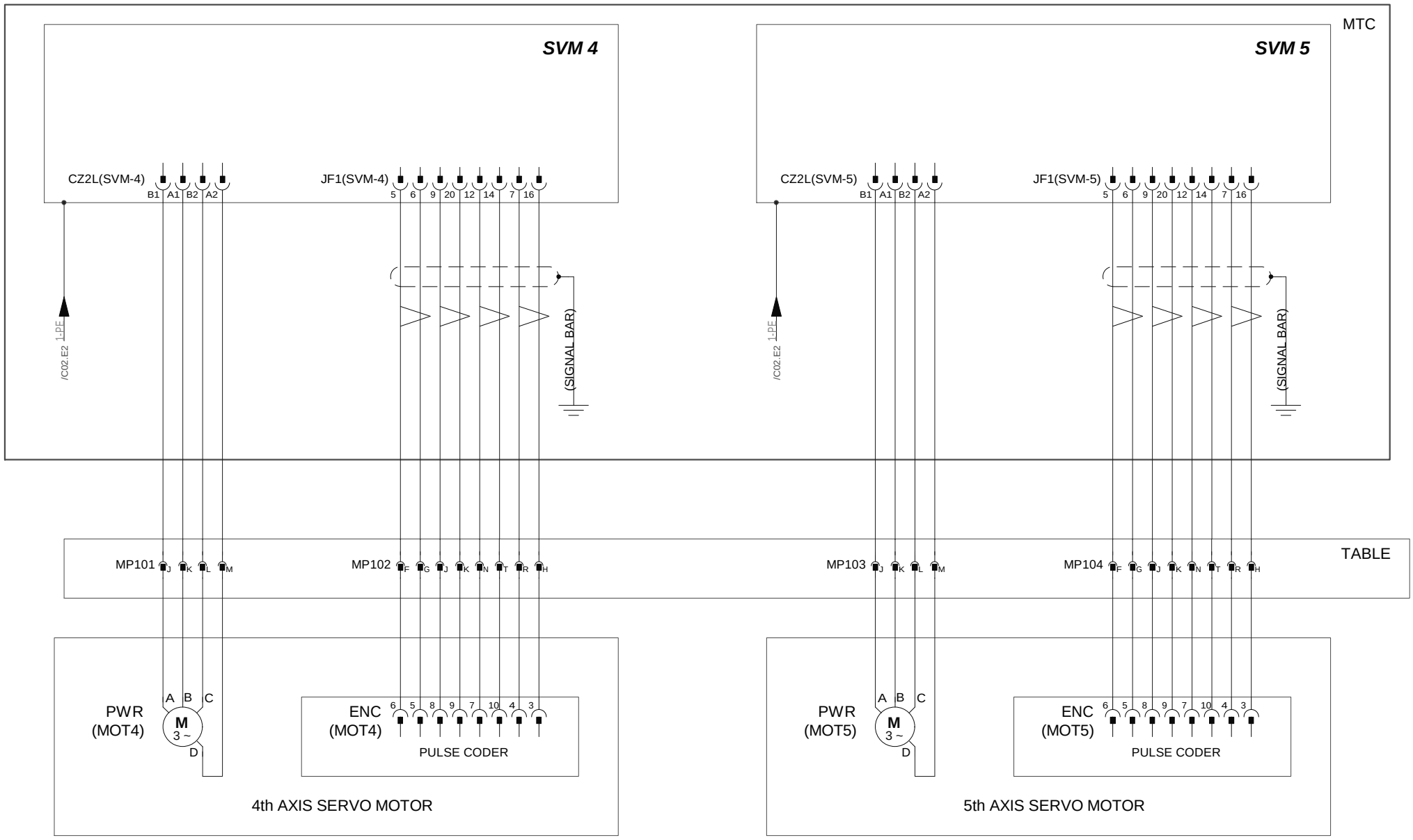



ATC - AXIS POWER / SIGNAL

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| ATC AXIS SERVO AMPLIFIER MODULE | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C17 |



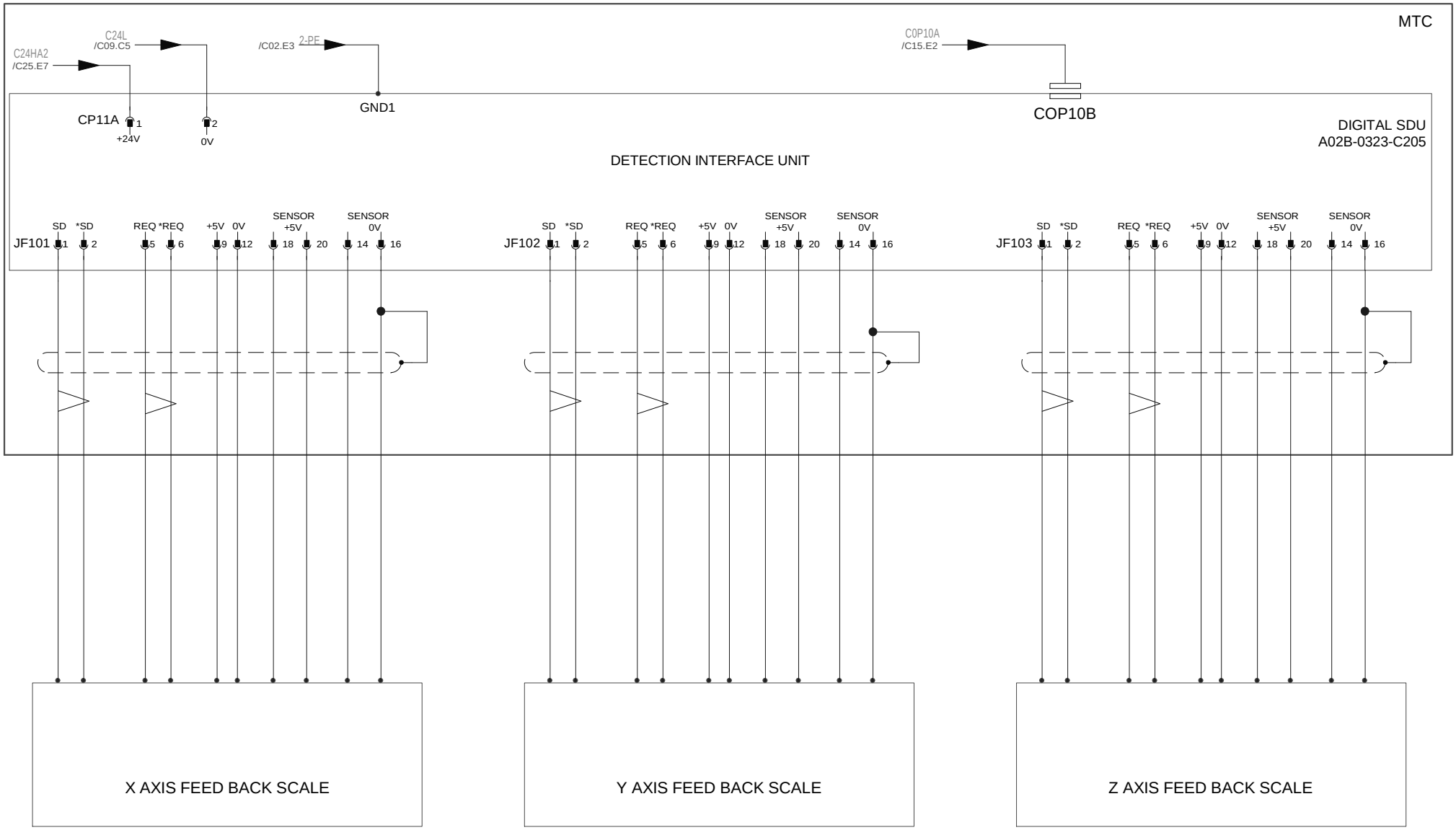

4th - AXIS POWER / SIGNAL

5th - AXIS POWER / SIGNAL

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| 4th & 5th AXIS SERVO AMP MODULE (OPT) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C18 |

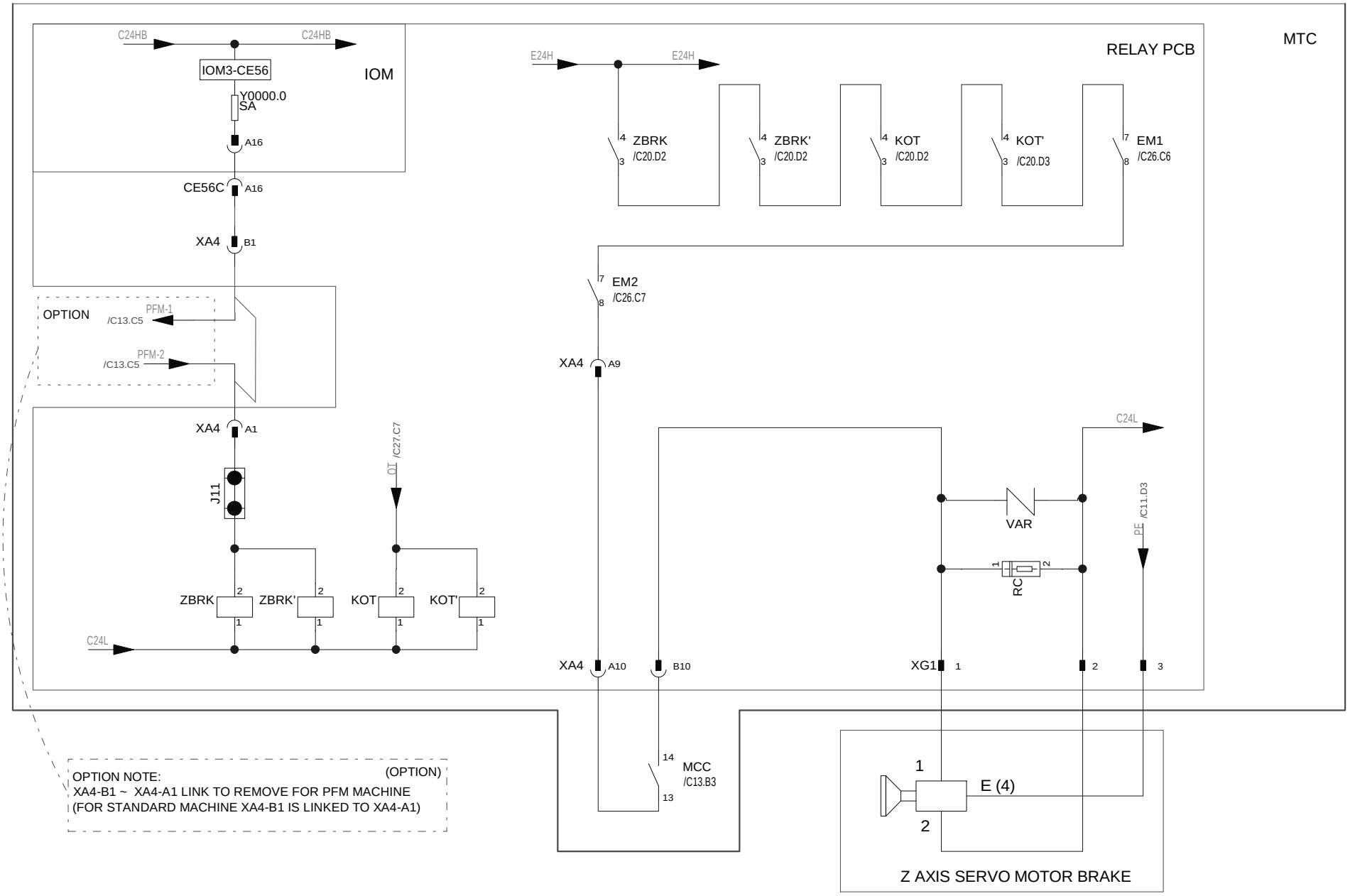



FEED BACK SCALE CONNECTION

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| FEEDBACK SCALE (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C19 |

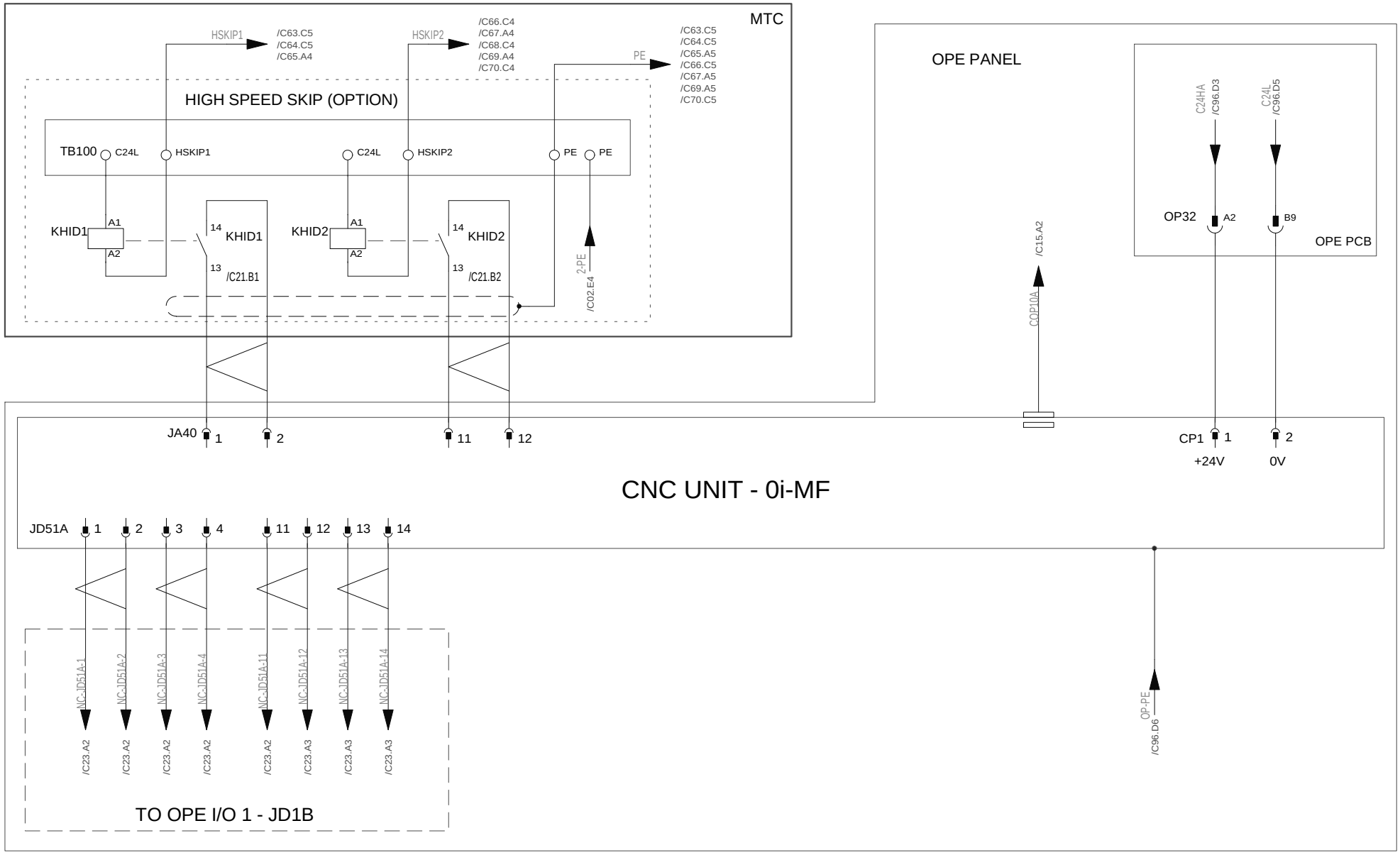



| SERVO READY | Z AXIS MOTOR BRAKE CONTROL | Z AXIS MOTOR BRAKE |
|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| Z AXIS MOTOR BRAKE | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C20 |

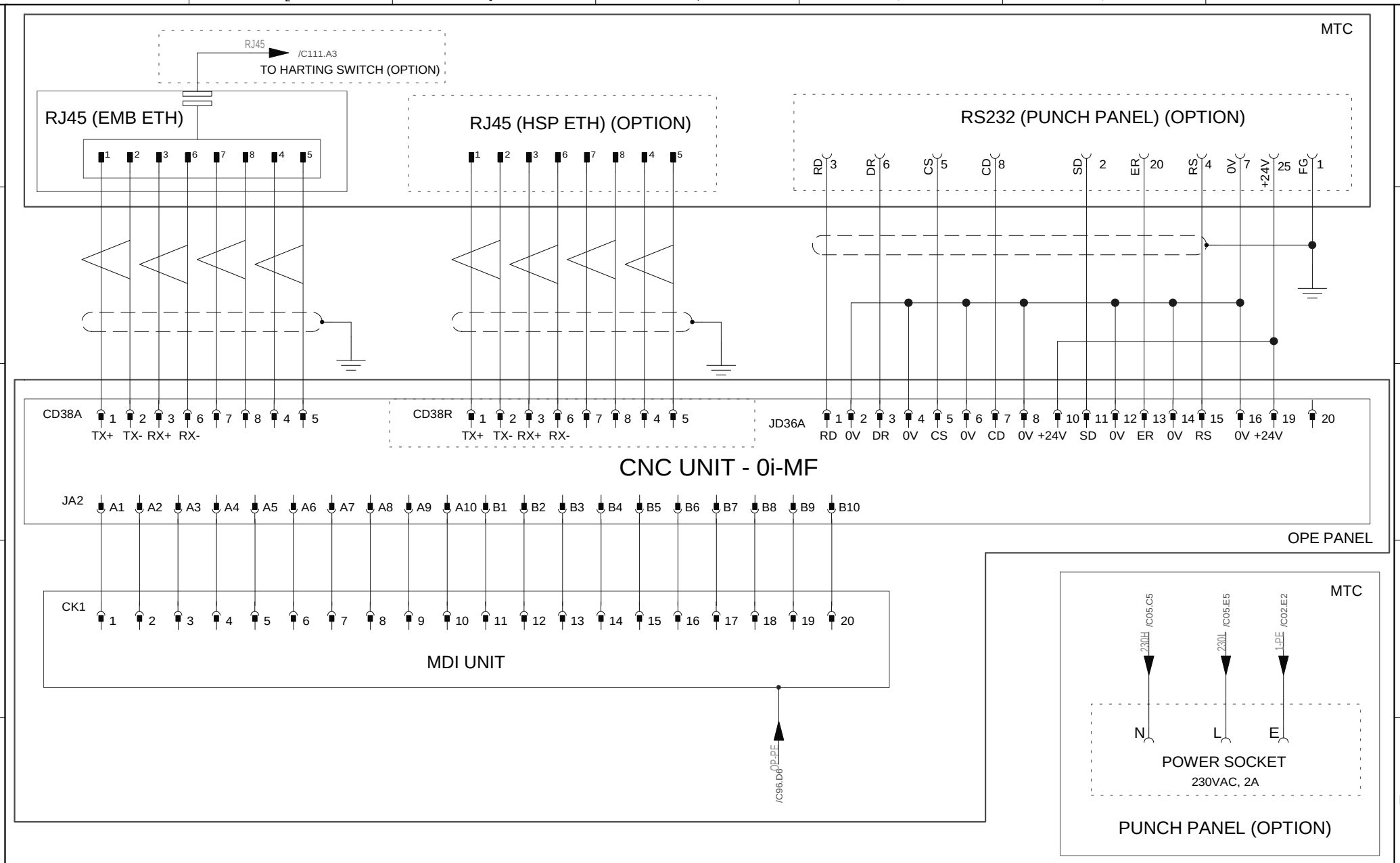



| I/O LINK CONNECTION | HSKIP | | POWER |
|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| CONTROLLER UNIT | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C21 |

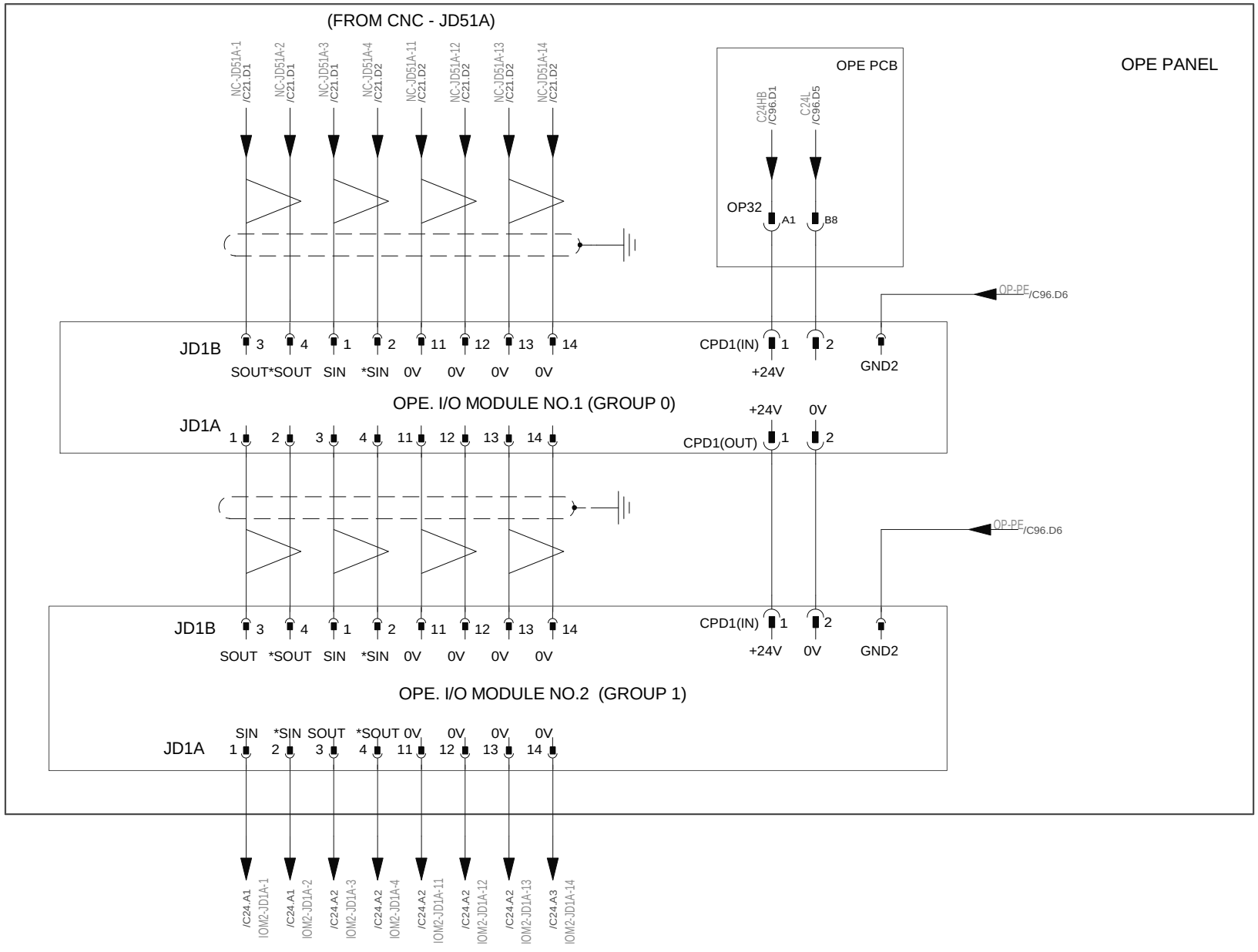



CNC UNIT CONNECTIONS

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| | ④ | | | |
| | ③ | APPROVED | 13/06/2019 | AVINASH |
| | ② | CHECKED | 13/06/2019 | AVINASH |
| | ① | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| CONTROLLER UNIT & PUNCH PANEL | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C22 |

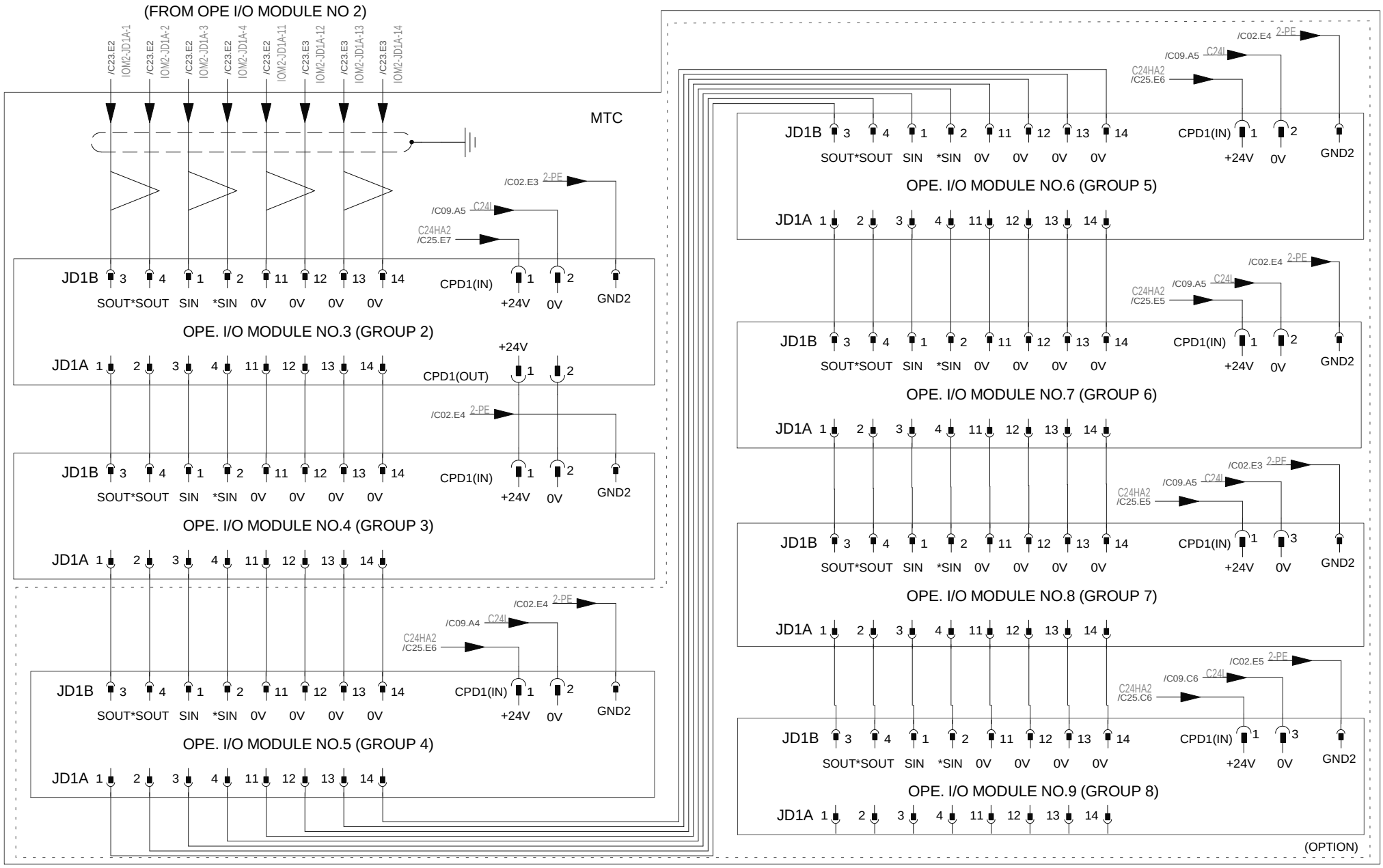



I/O LINK CONNECTION | I/O POWER

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| | ④ | | | |
| | ③ | APPROVED | 13/06/2019 | AVINASH |
| | ② | CHECKED | 13/06/2019 | AVINASH |
| | ① | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| I/O LINK CONNECTION | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C23 |

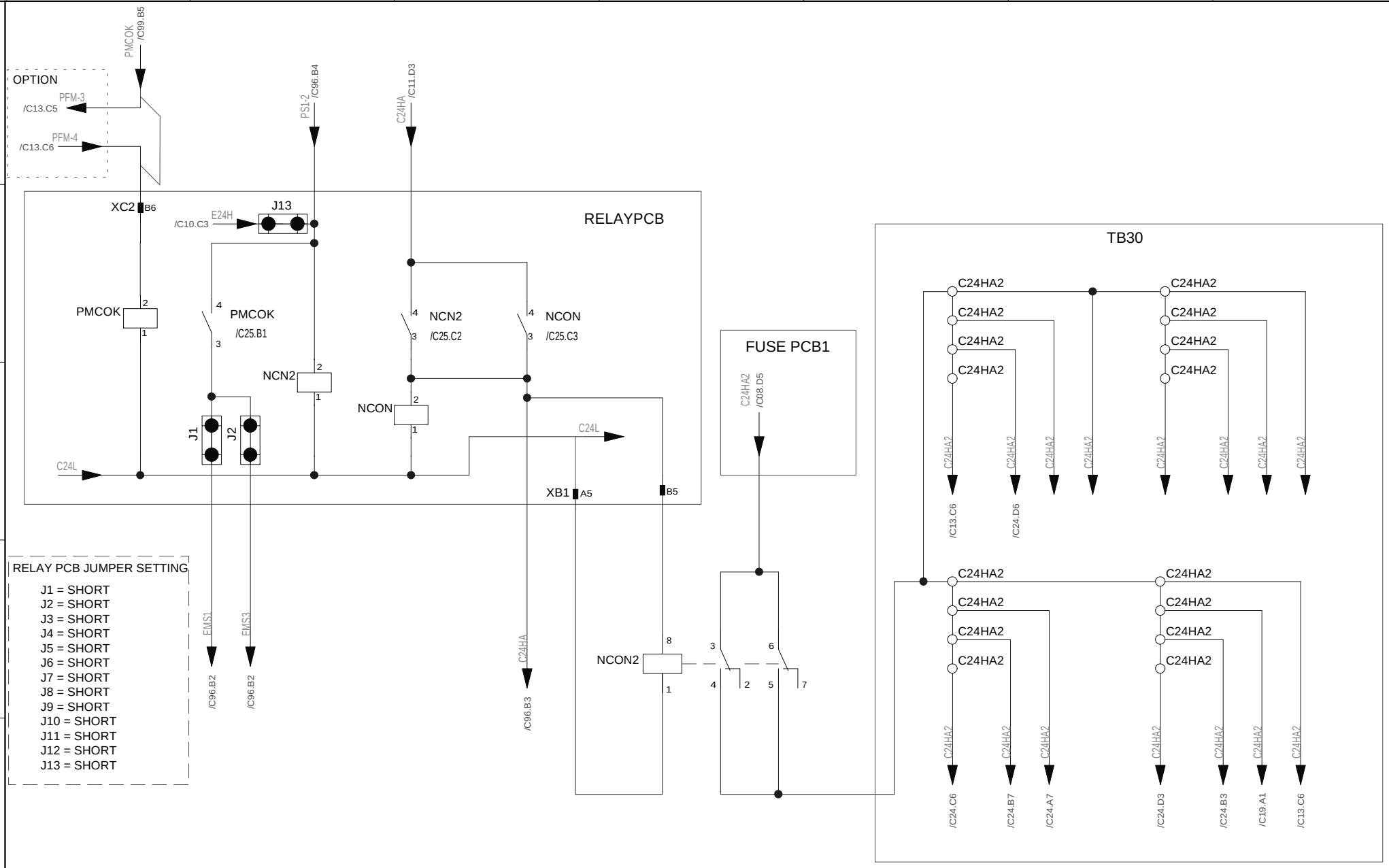



| I/O LINK CONNECTION | I/O POWER | I/O LINK CONNECTION | I/O POWER |
|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| | ④ | | | |
| | ③ | APPROVED | 13/06/2019 | AVINASH |
| | ② | CHECKED | 13/06/2019 | AVINASH |
| | ① | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| I/O LINK CONNECTION | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C24 |

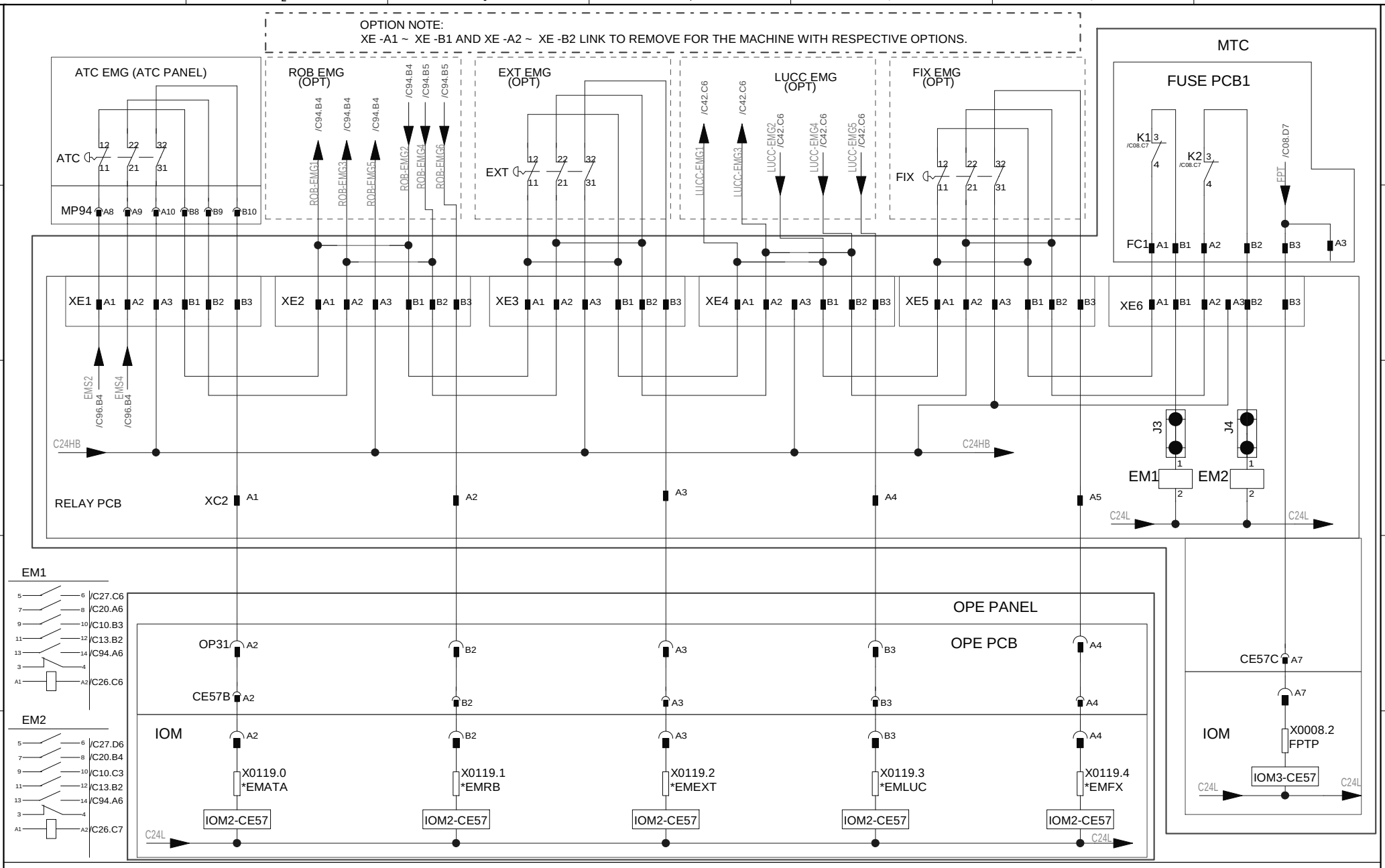



NC ON/OFF CONTROL | NC POWER | C24HA2

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP = |
|---|---|---|---|---|
| NC POWER ON CONTROL | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C25 |



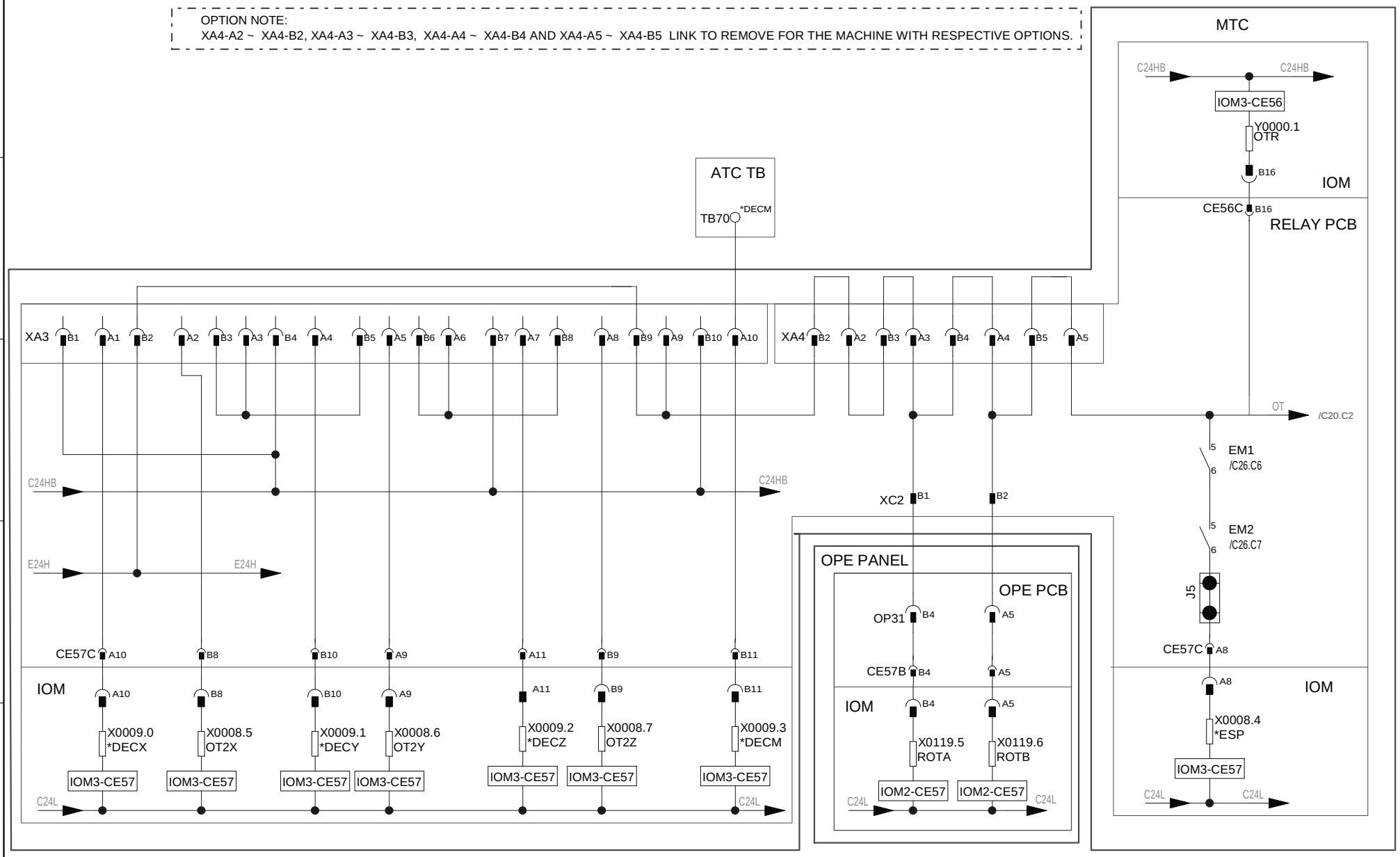

EMERGENCY STOP CONTROL

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| EMERGENCY STOP CIRCUIT | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C26 |

OPTION NOTE:
XA4-A2 ~ XA4-B2, XA4-A3 ~ XA4-B3, XA4-A4 ~ XA4-B4 AND XA4-A5 ~ XA4-B5 LINK TO REMOVE FOR THE MACHINE WITH RESPECTIVE OPTIONS.

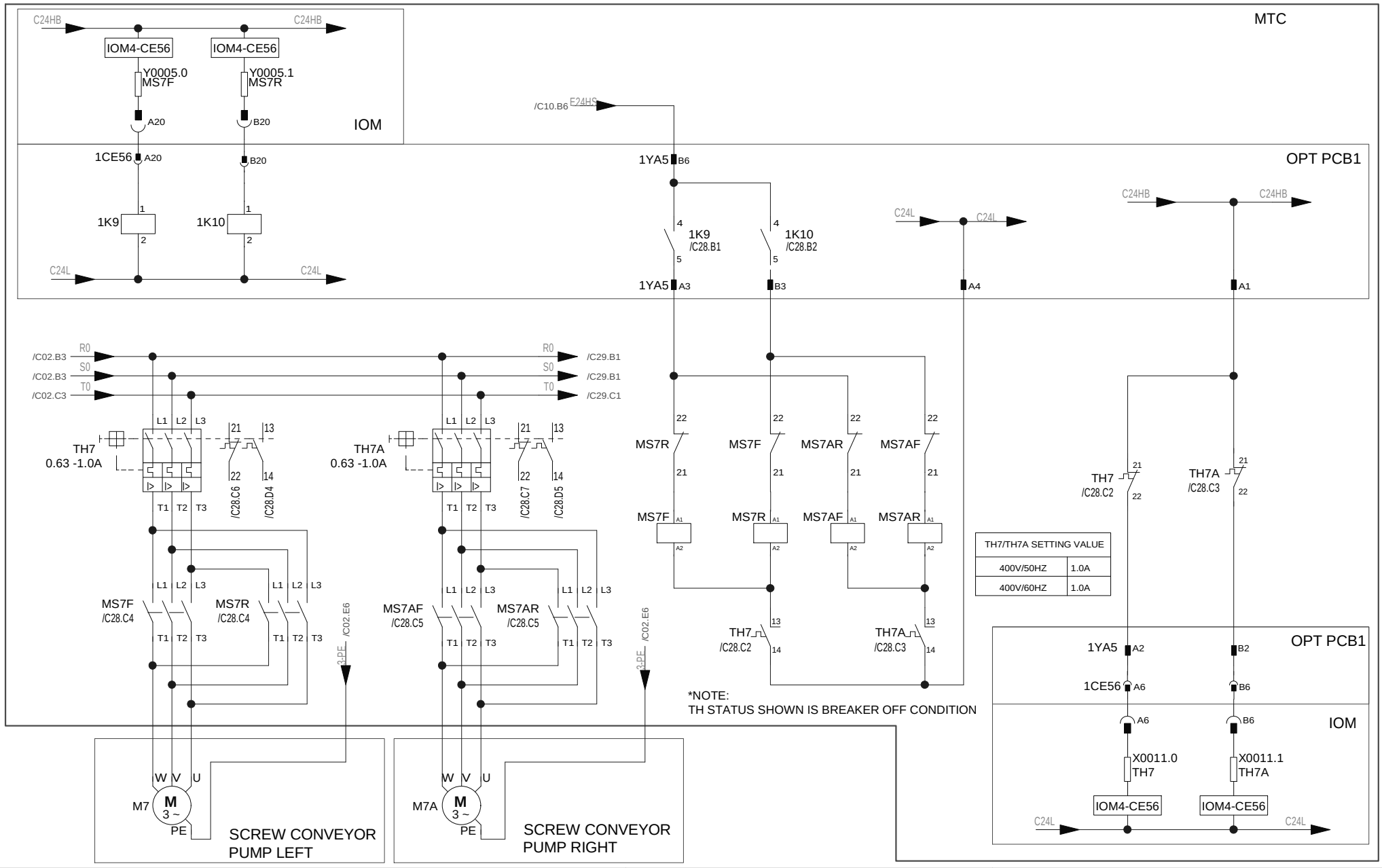



| REFERENCE SLOW DOWN -X | OVERTRAVEL X AXIS | REFERENCE SLOW DOWN -Y | OVERTRAVEL Y AXIS | REFERENCE SLOW DOWN -Z | OVERTRAVEL Z AXIS | REFERENCE SLOW DOWN - MAGAZINE | | EMERGENCY STOP | OT |
|---|---|---|---|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| REFERENCE AND OVERTRAVEL (OPT) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C27 |

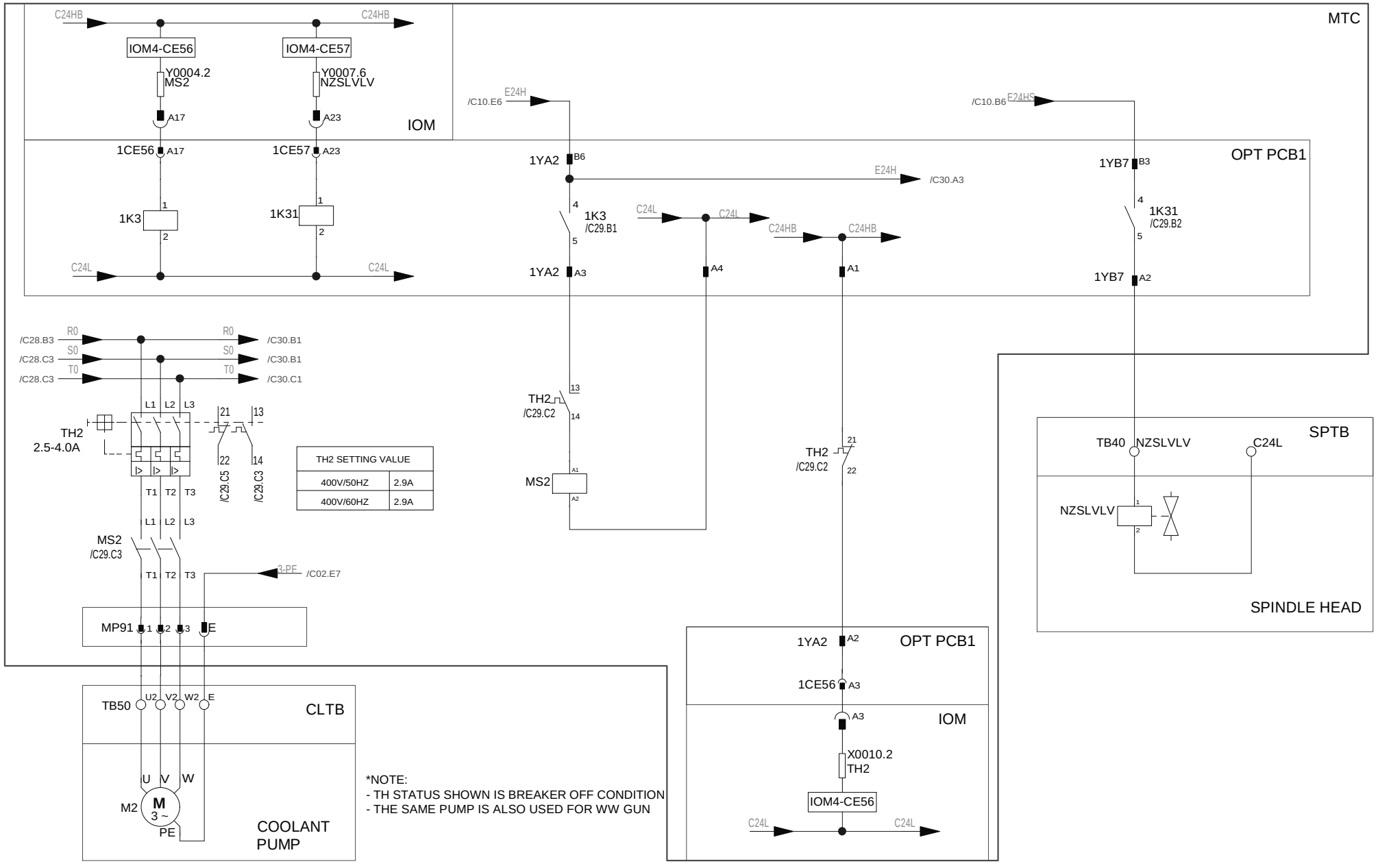



SCREW CONVEYOR FORWARD AND REVERSE | FORWARD AND REVERSE CONTROL | SCREW CONVEYOR THERMAL ALARM

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| | ④ | | | |
| | ③ | APPROVED | 13/06/2019 | AVINASH |
| | ② | CHECKED | 13/06/2019 | AVINASH |
| | ① | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| SCREW CONVEYOR | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C28 |

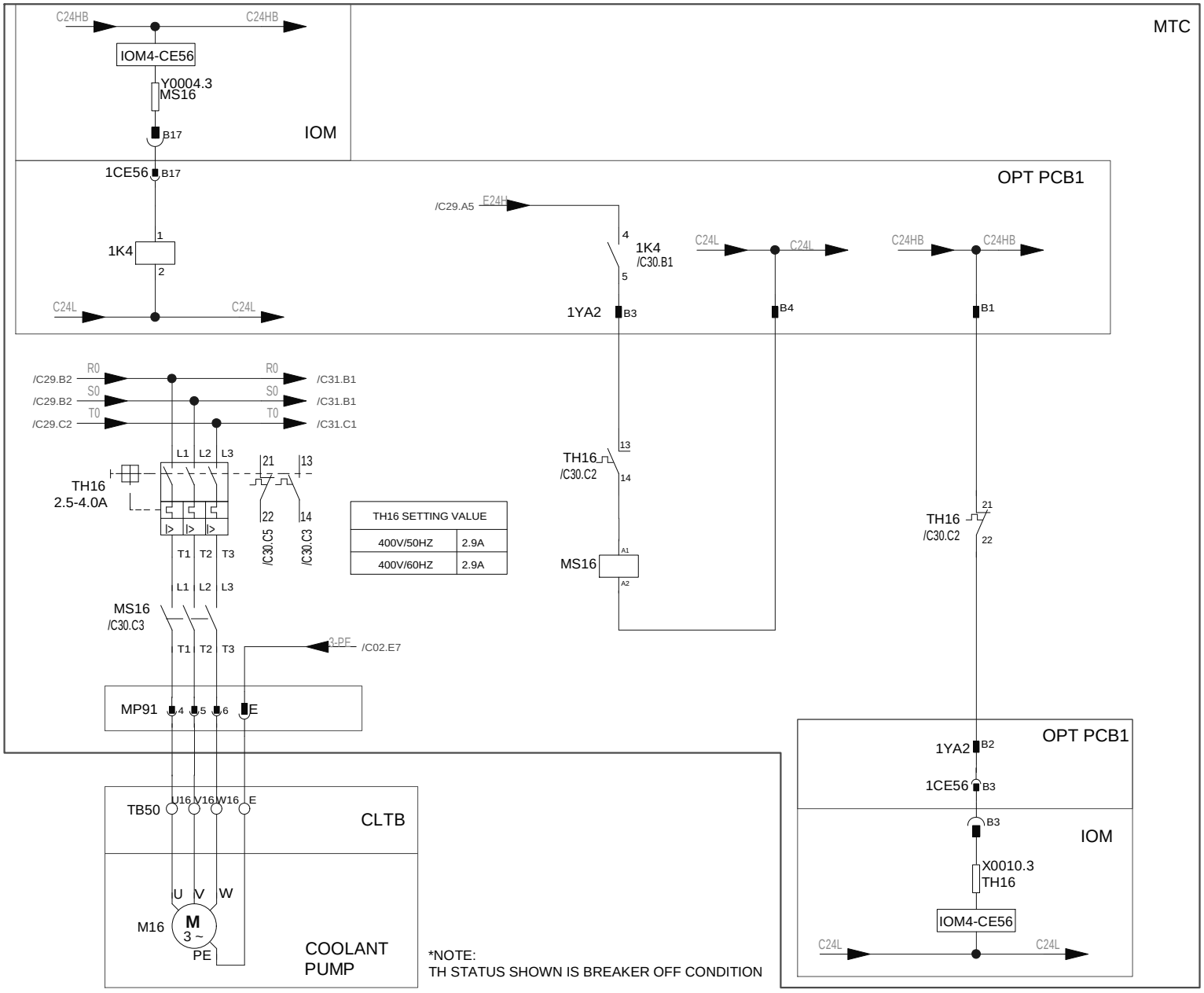



NOZZLE COOLANT PUMP ON | | NOZZLE COOLANT THERMAL ALARM | NOZZLE SOL VALVE

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| NOZZLE COOLANT & WW GUN PUMP | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C29 |



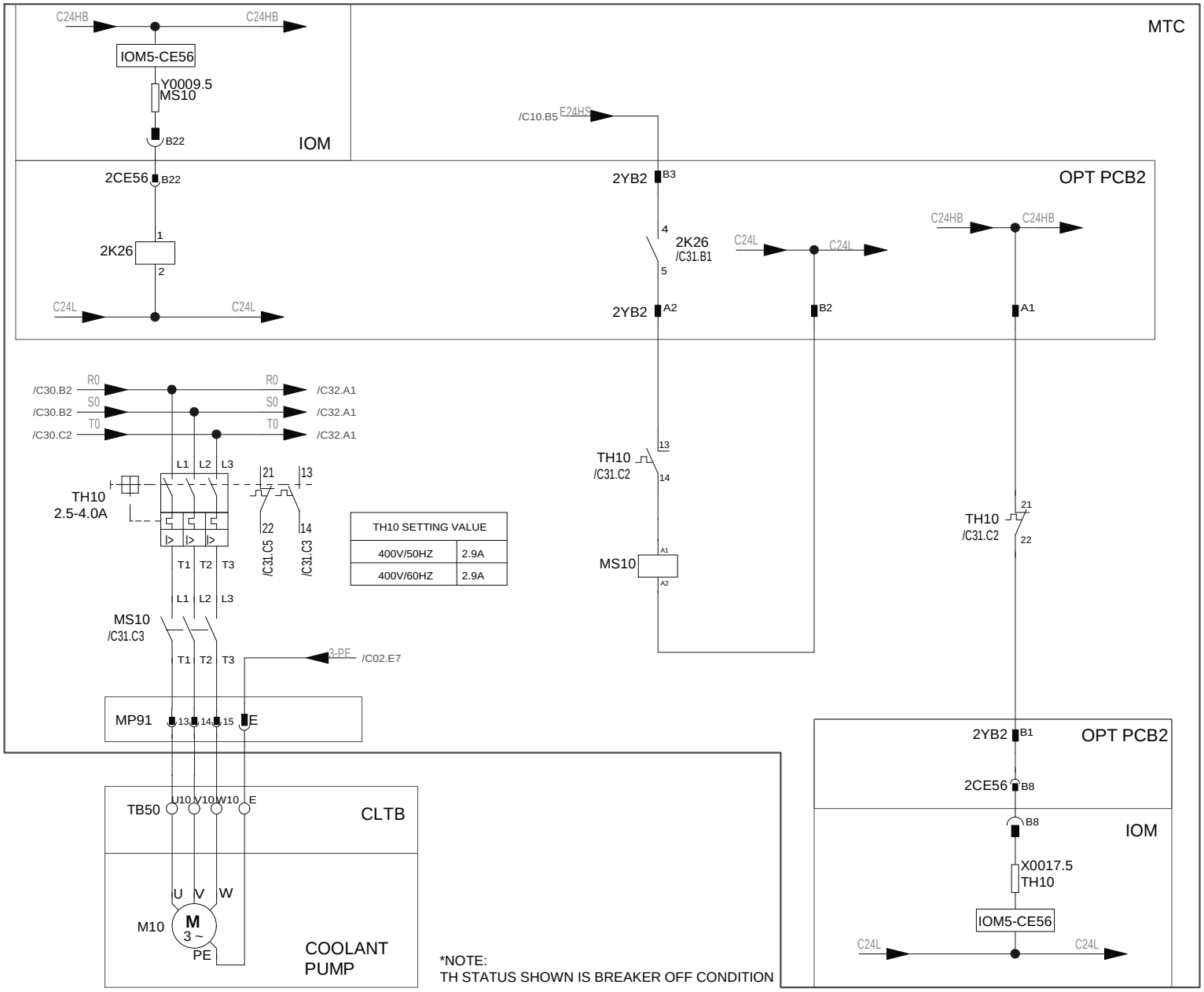

BASE COOLANT PUMP ON

BASE COOLANT THERMAL ALARM

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| | ④ | | | |
| | ③ | APPROVED | 13/06/2019 | AVINASH |
| | ② | CHECKED | 13/06/2019 | AVINASH |
| | ① | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| BASE / FIXTURE COOLANT (OPT) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C30 |



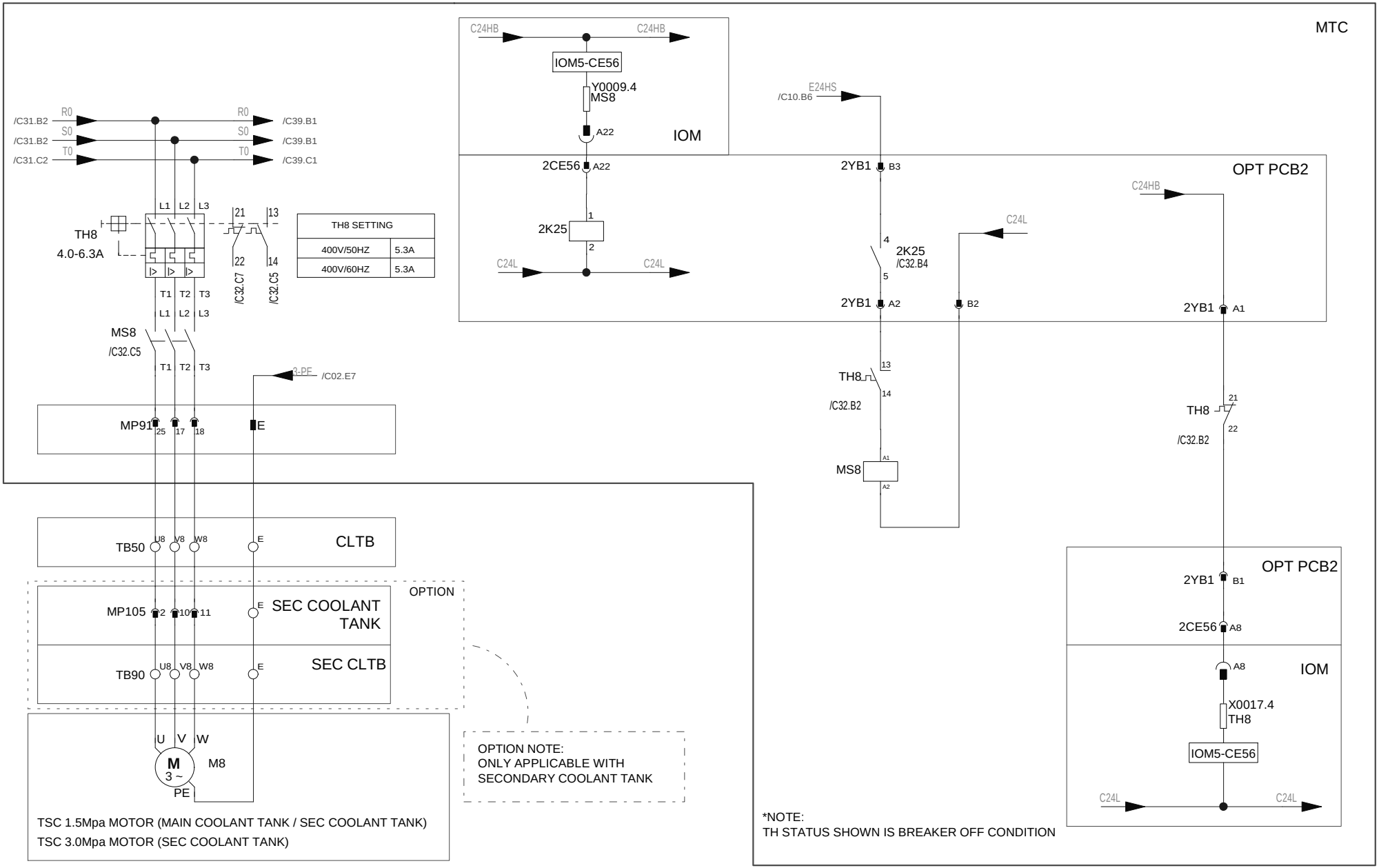

SHOWER COOLANT PUMP ON

SHOWER COOLANT THERMAL ALARM

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| | ④ | | | |
| | ③ | APPROVED | 13/06/2019 | AVINASH |
| | ② | CHECKED | 13/06/2019 | AVINASH |
| | ① | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| SHOWER COOLANT (OPT) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C31 |

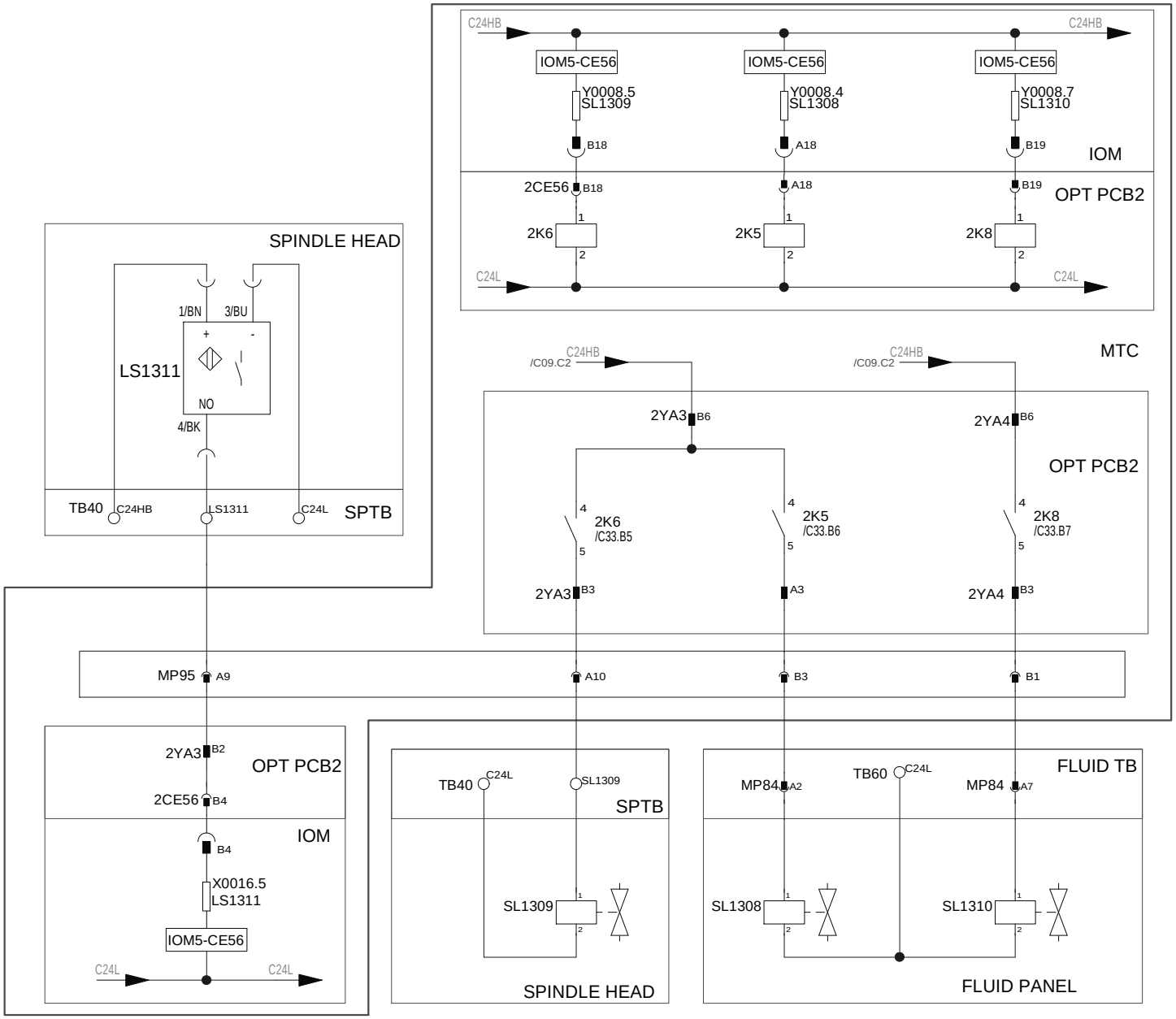



| TSC PUMP | TSC PUMP ON | TSC THERMAL ALARM |
|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| TSC 1.5MPa & 3MPa PWR (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | C32 |

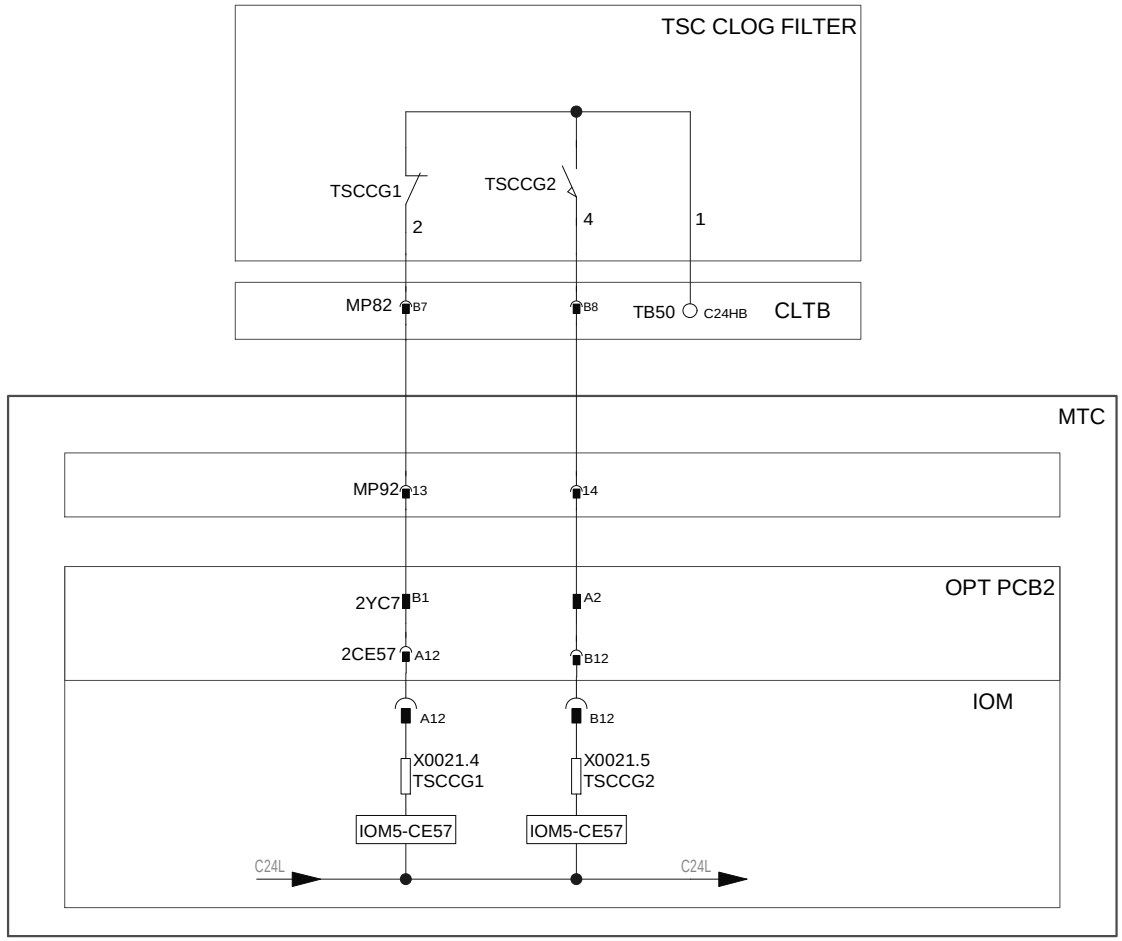



TSC LEAKAGE DETECT | TSC UNION DRAIN | TSC UNION BACKUP AIR | TSC SUCTION

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| | ④ | | | |
| | ③ | APPROVED | 13/06/2019 | AVINASH |
| | ② | CHECKED | 13/06/2019 | AVINASH |
| | ① | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| TSC 1.5MPa & 3MPa CONTROL (OPT) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C33 |



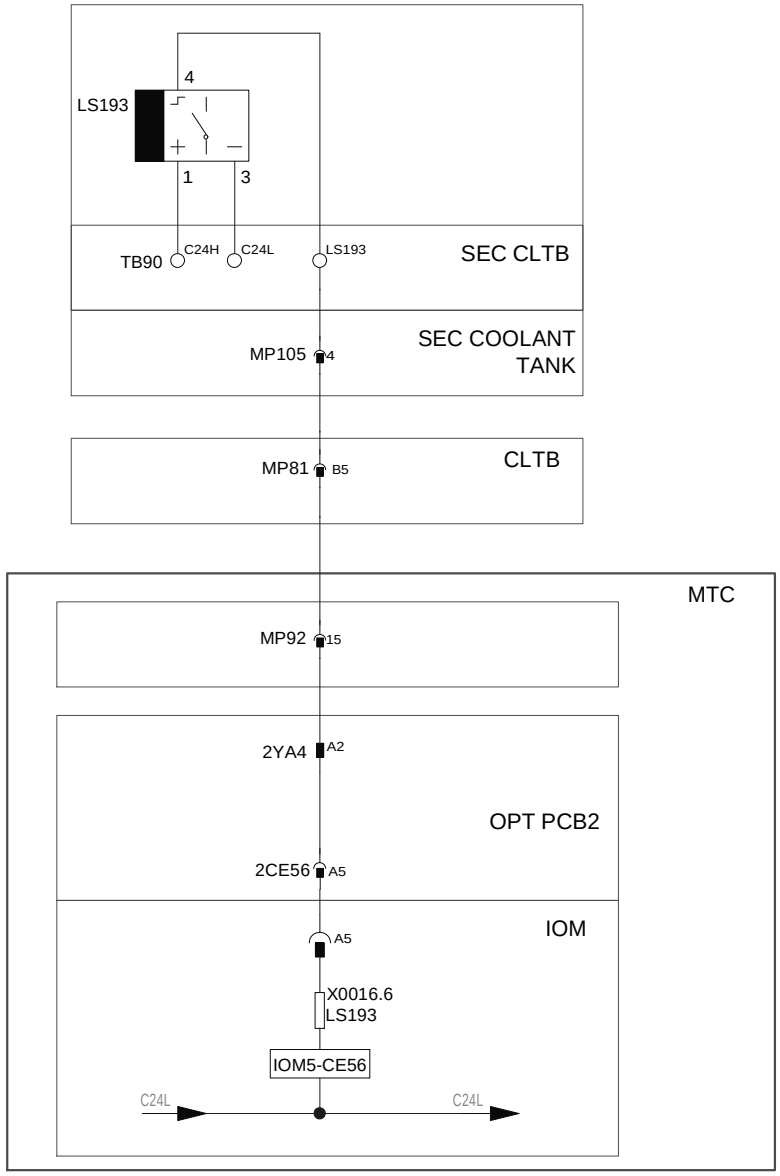

*NOTE:
ONLY APPLICABLE WITH 1.5MPa IN MAIN COOLANT TANK

100% CLOG - ALARM | 75% CLOG - WARNING

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| TSC 1.5MPa CONTROL (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C34 |



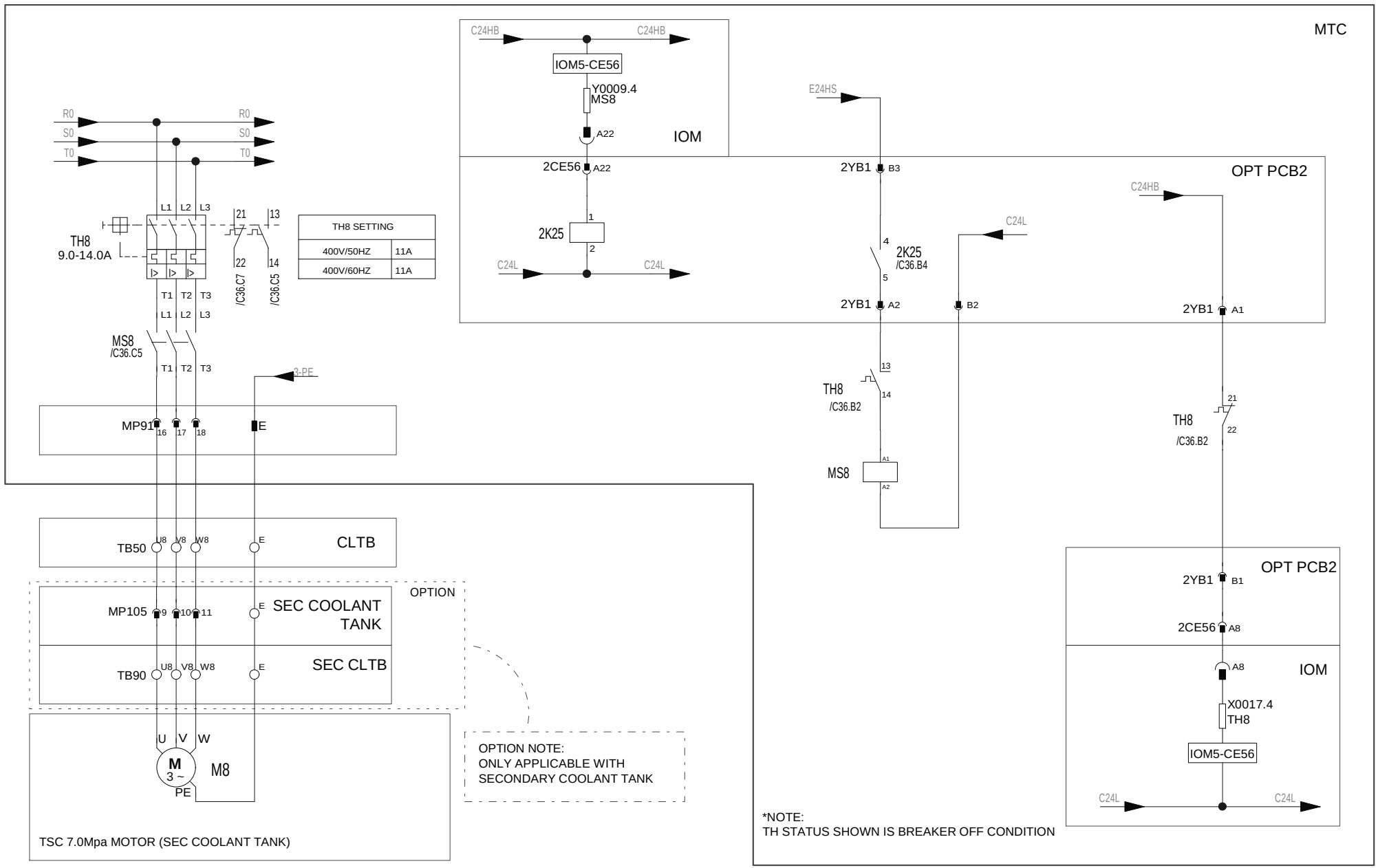

*NOTE:
ONLY APPLICABLE WITH 3.0MPa

TSC FLOW SWITCH

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| TSC 3.0MPa CONTROL (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C35 |

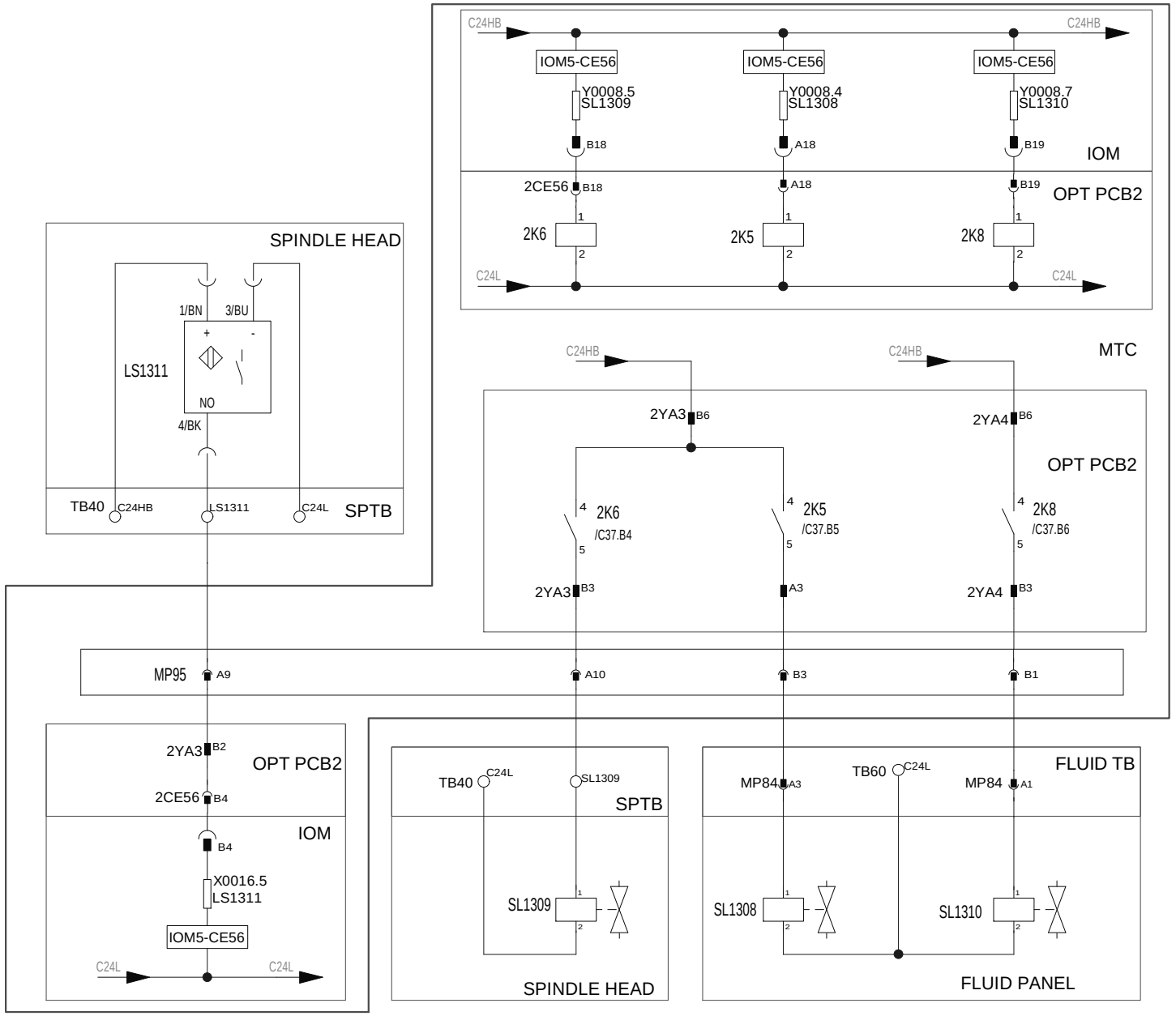



| TSC PUMP | TSC PUMP ON | TSC THERMAL ALARM |
|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| TSC 7.0MPa CONTROL (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | C36 |

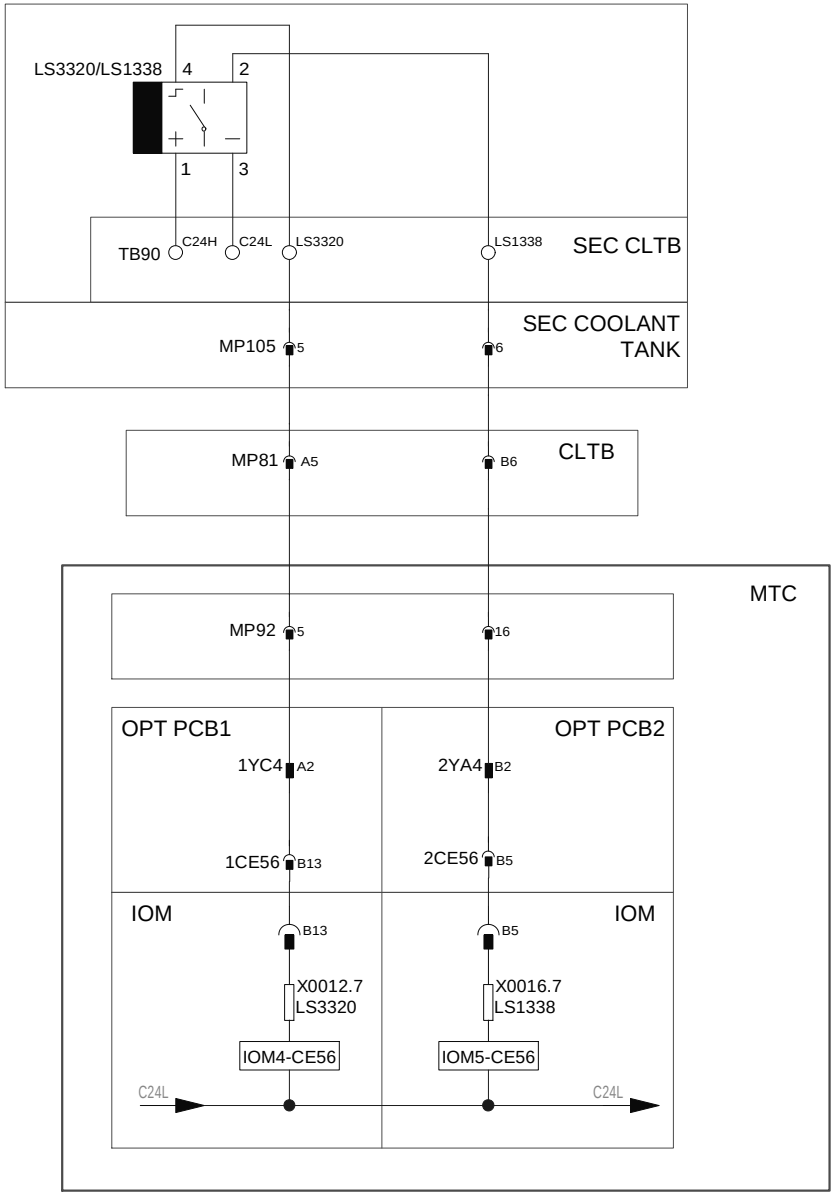



| TSC LEAKAGE DETECT | TSC UNION DRAIN | TSC UNION BACKUP AIR | TSC SUCTION |
|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| TSC 7.0MPa CONTROL (OPT) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C37 |

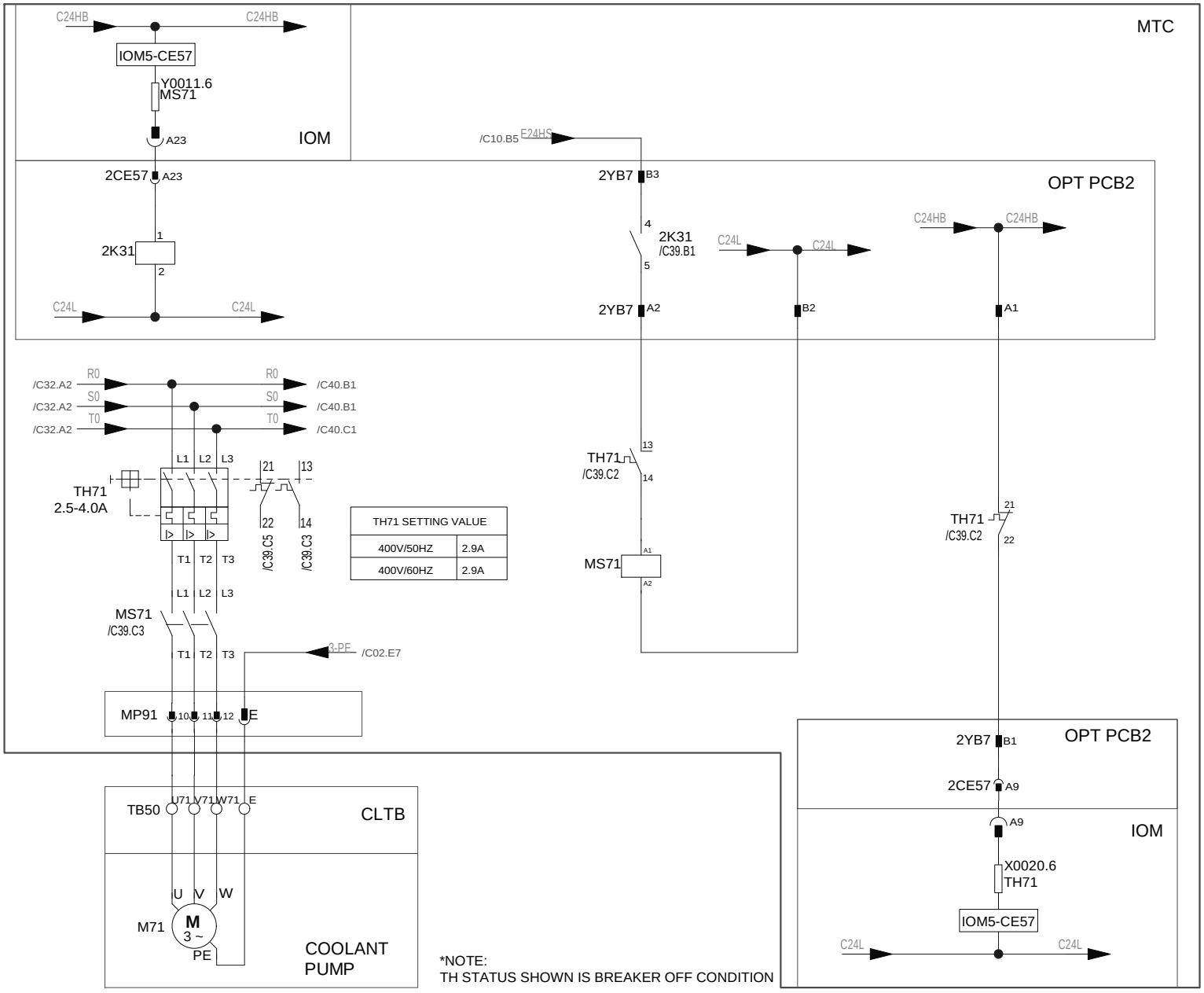



| CYCLONE FILTER (TSC) PRESSURE SWITCH: | LOW | HIGH |
|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| CYCLONE FILTER PRESSURE SW (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C38 |



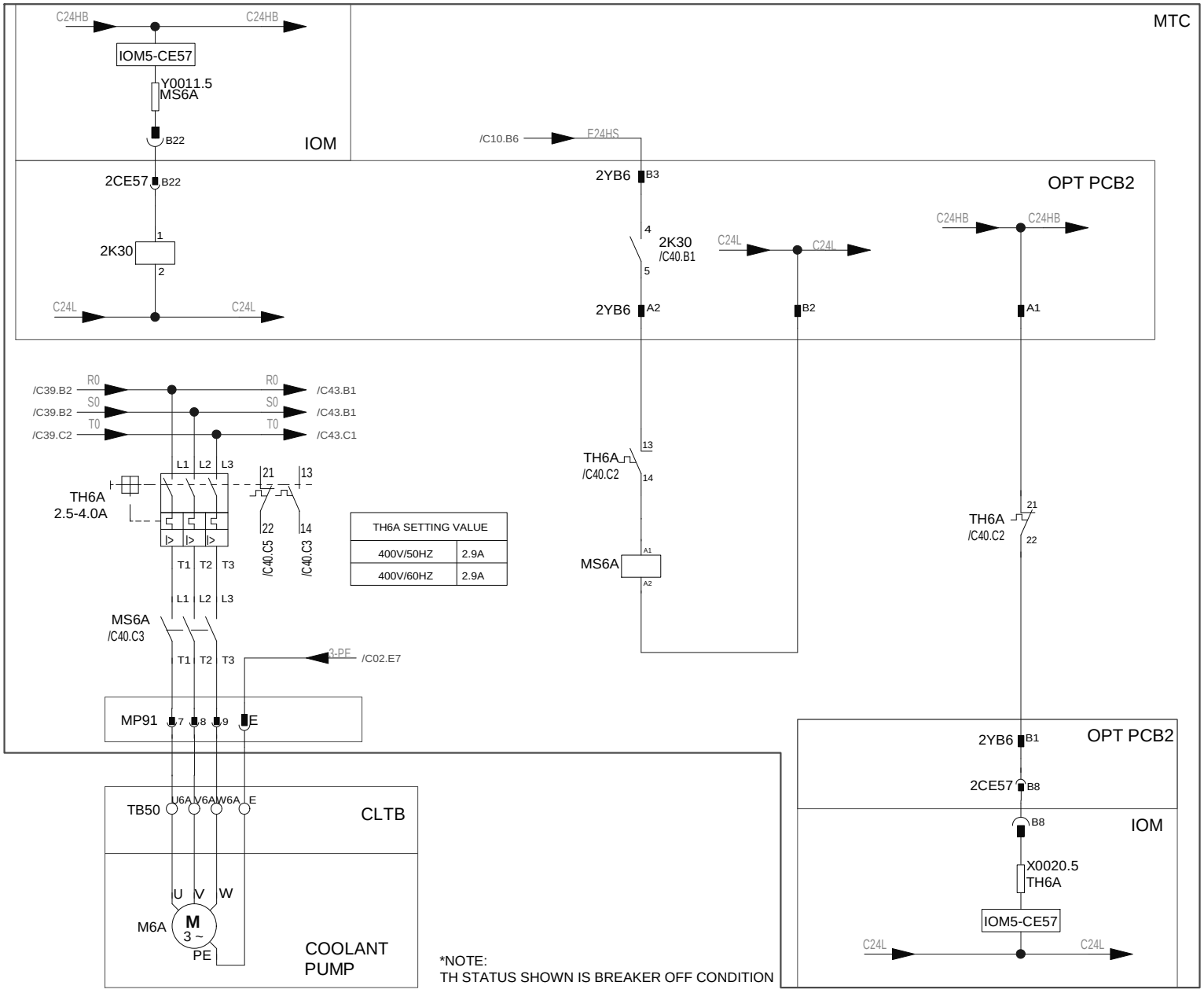

SECONDARY TRANSFER PUMP ON

SECONDARY TRANSFER PUMP THERMAL ALARM

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| | ④ | | | |
| | ③ | APPROVED | 13/06/2019 | AVINASH |
| | ② | CHECKED | 13/06/2019 | AVINASH |
| | ① | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| SECONDARY TRANSFER PUMP (OPT) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C39 |



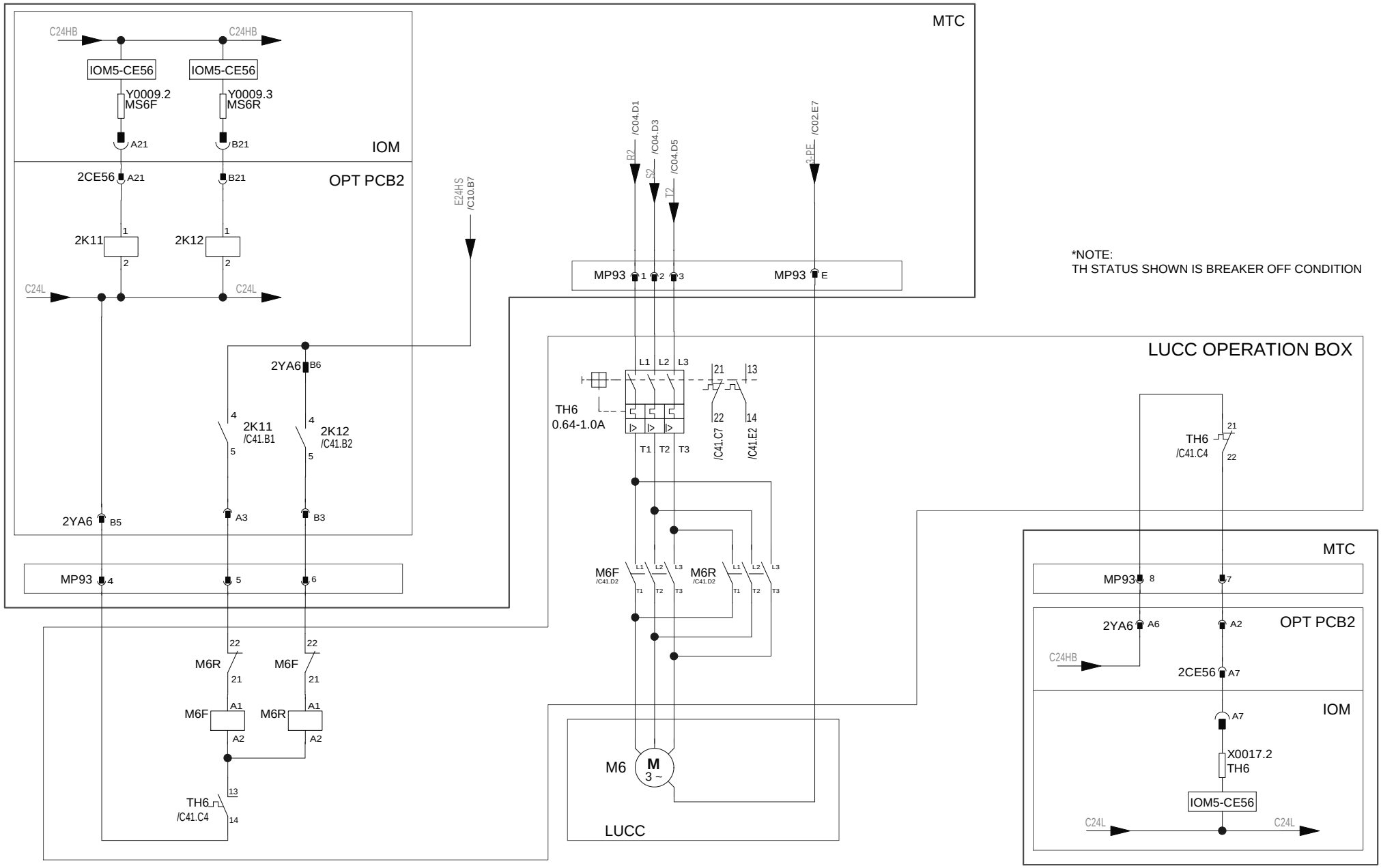

LUCC DRUM PUMP ON

LUCC DRUM THERMAL ALARM

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| | ④ | | | |
| | ③ | APPROVED | 13/06/2019 | AVINASH |
| | ② | CHECKED | 13/06/2019 | AVINASH |
| | ① | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| LUCC DRUM (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C40 |

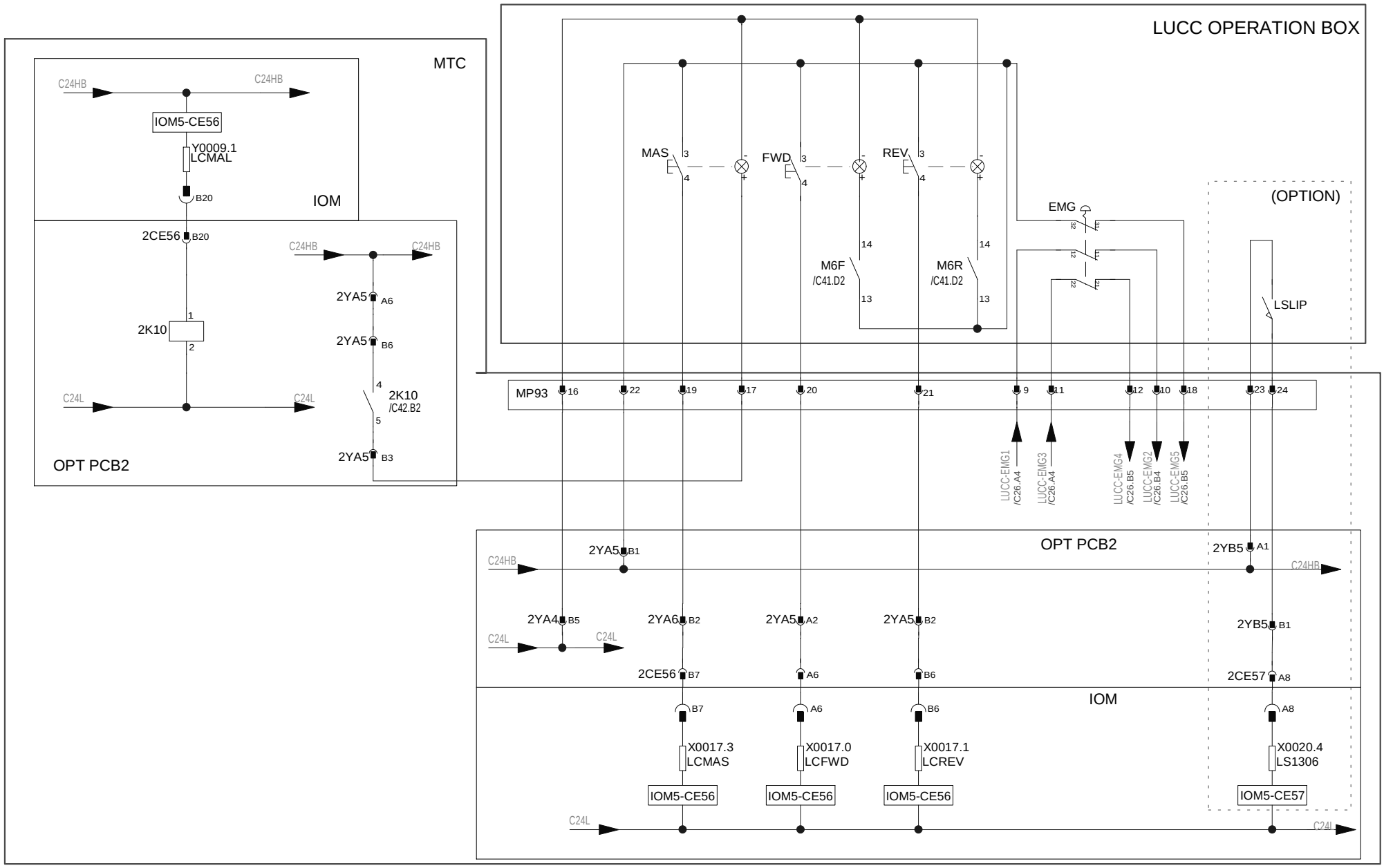



| LUCC FORWARD CONTROL | LUCC REVERSE CONTROL | LUCC MOTOR POWER | LUCC THERMAL ALARM |
|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| LIFT UP CHIP CYR PWR (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | C41 |

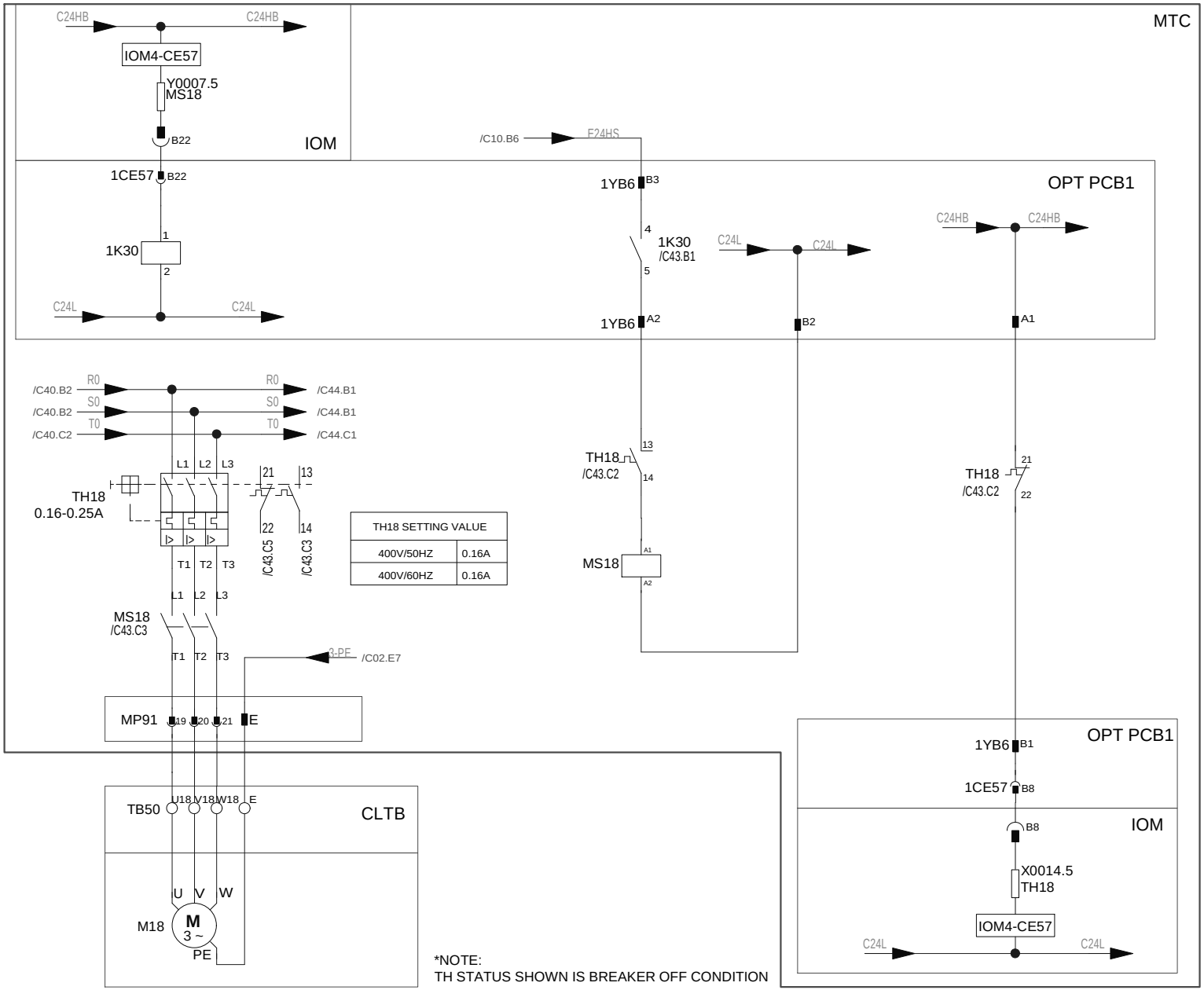



| MANUAL PB | MANUAL LAMP | LUCC FORWARD | LUCC REVERSE | LUCC EMERGENCY CONTROL | LSLIP RESET |
|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| LIFT UP CHIP CYR CONTROL (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C42 |



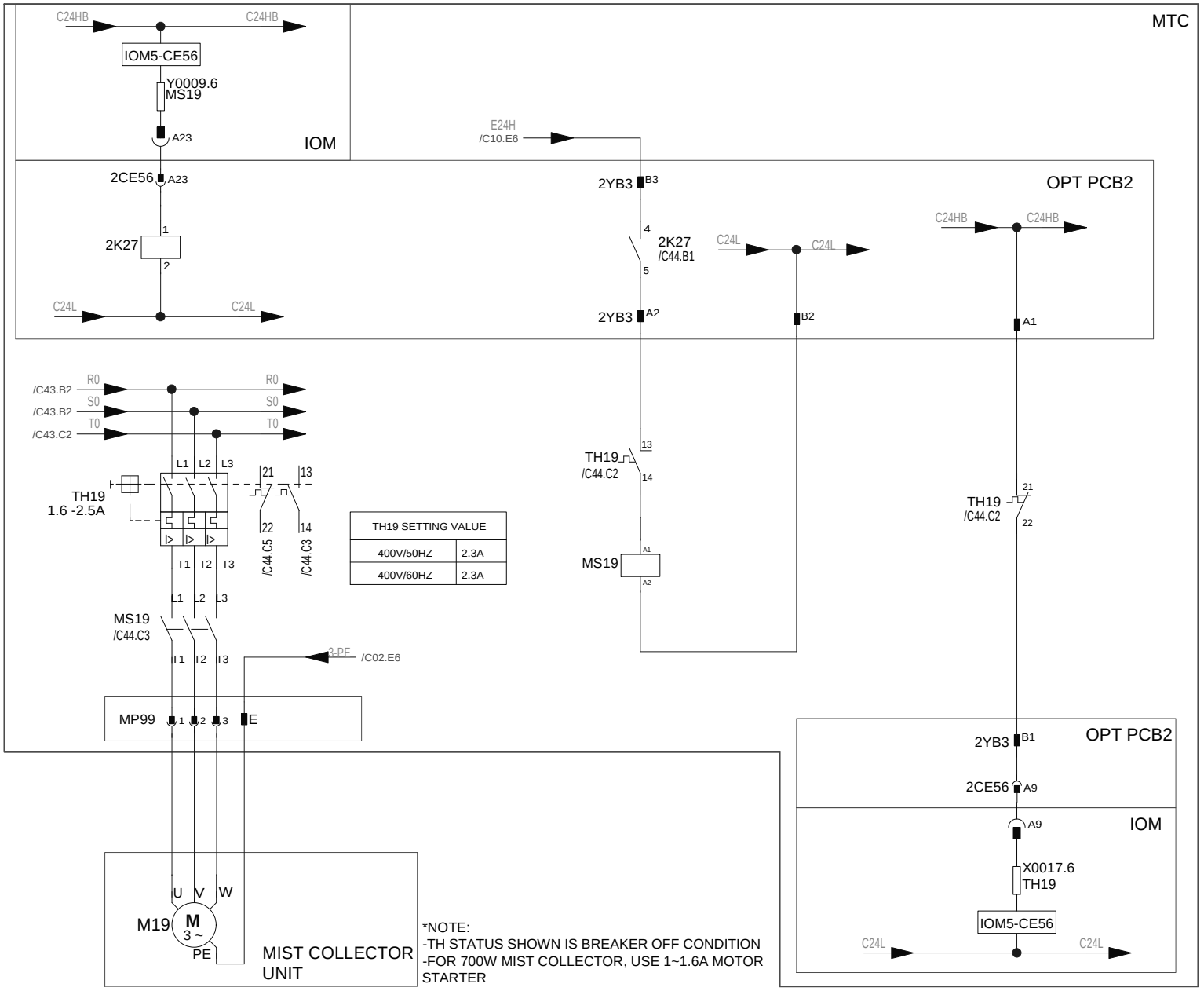

OIL SKIMMER PUMP ON

OIL SKIMMER THERMAL ALARM

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| OIL SKIMMER (OPT) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C43 |



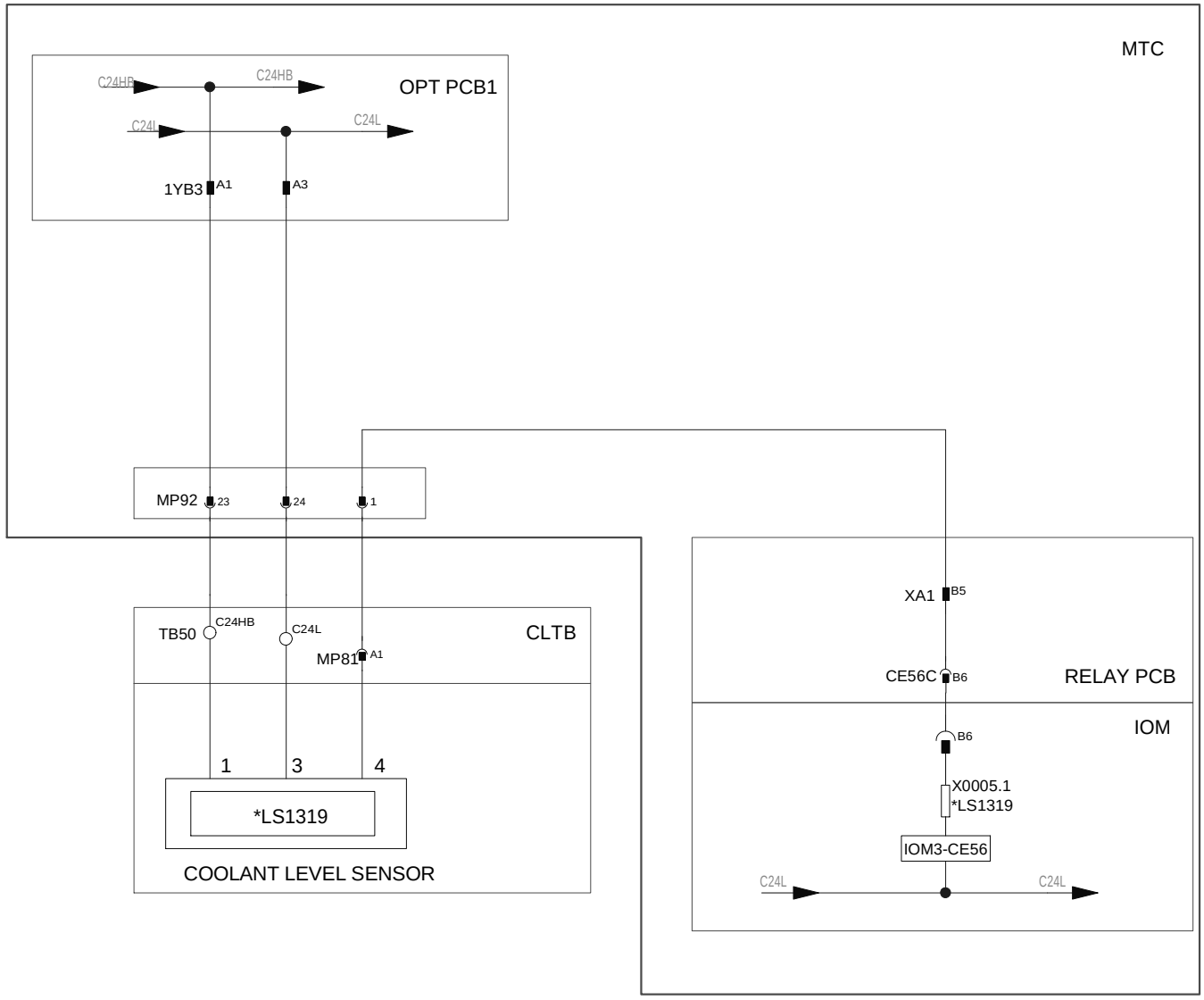

MIST COLLECTOR PUMP ON

MIST COLLECTOR THERMAL ALARM

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| | ④ | | | |
| | ③ | APPROVED | 13/06/2019 | AVINASH |
| | ② | CHECKED | 13/06/2019 | AVINASH |
| | ① | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| MIST COLLECTOR (OPT) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C44 |



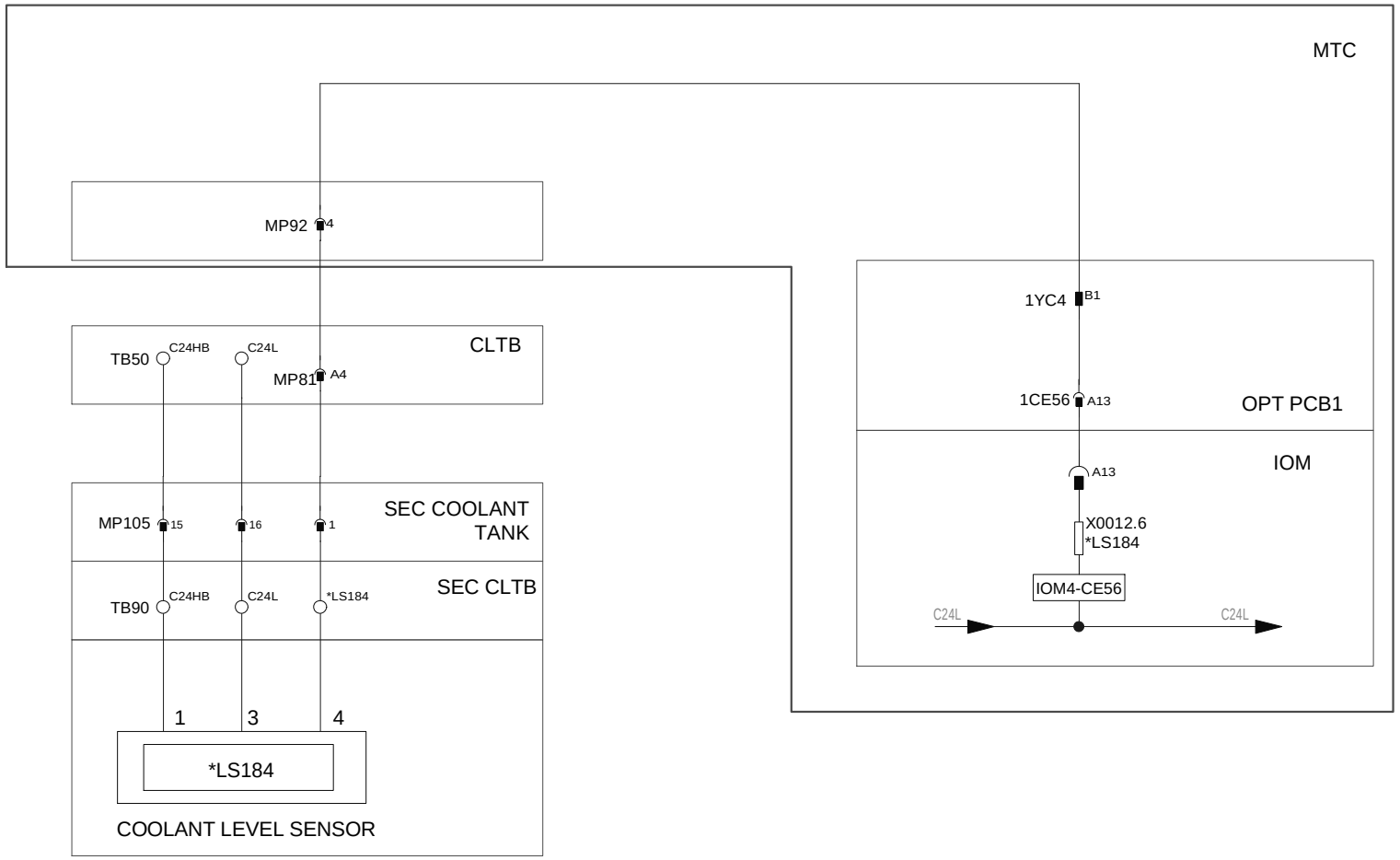

COOLANT LOW LEVEL SENSOR

MAIN COOLANT TANK LOW LEVEL

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| COOLANT LEVEL SENSOR - MAIN TANK | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C45 |



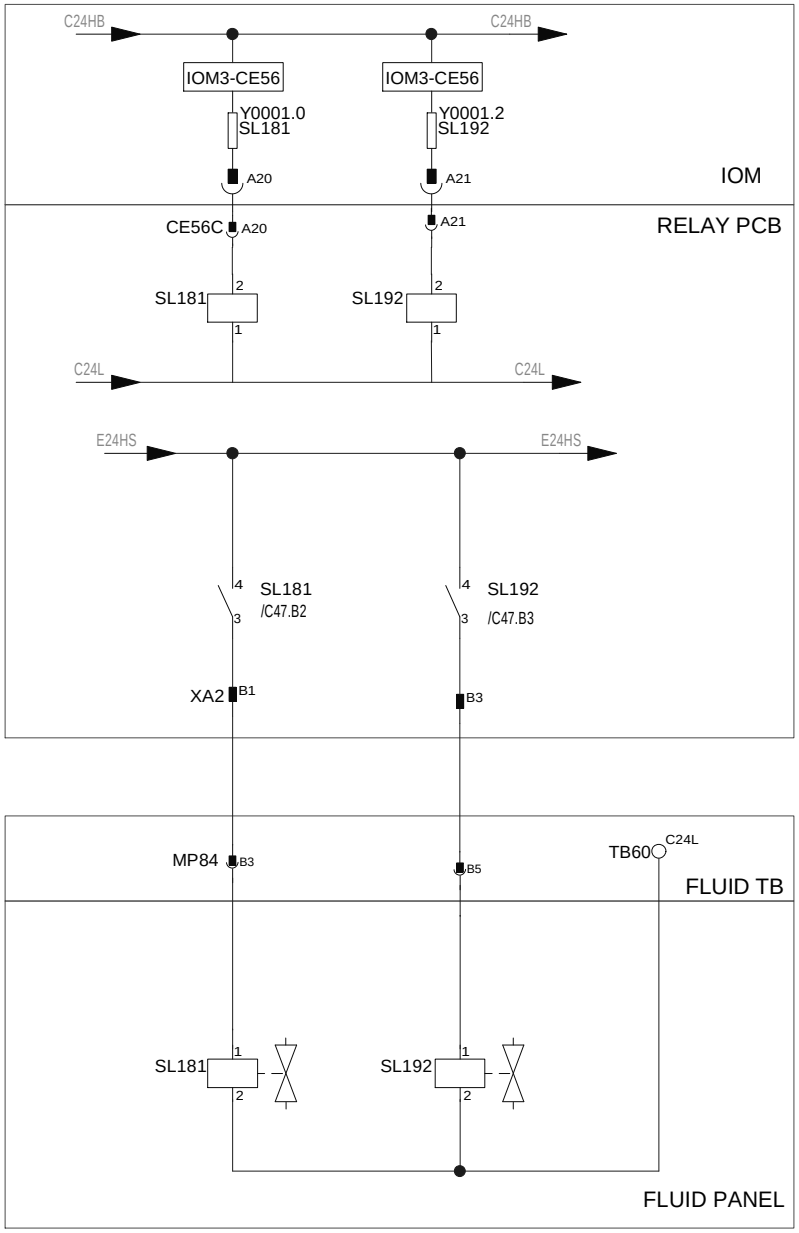

COOLANT LOW LEVEL SENSOR

SEC COOLANT TANK LOW LEVEL DETECT

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| COOLANT LEVEL SENSOR - SEC TANK (OPT) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C46 |

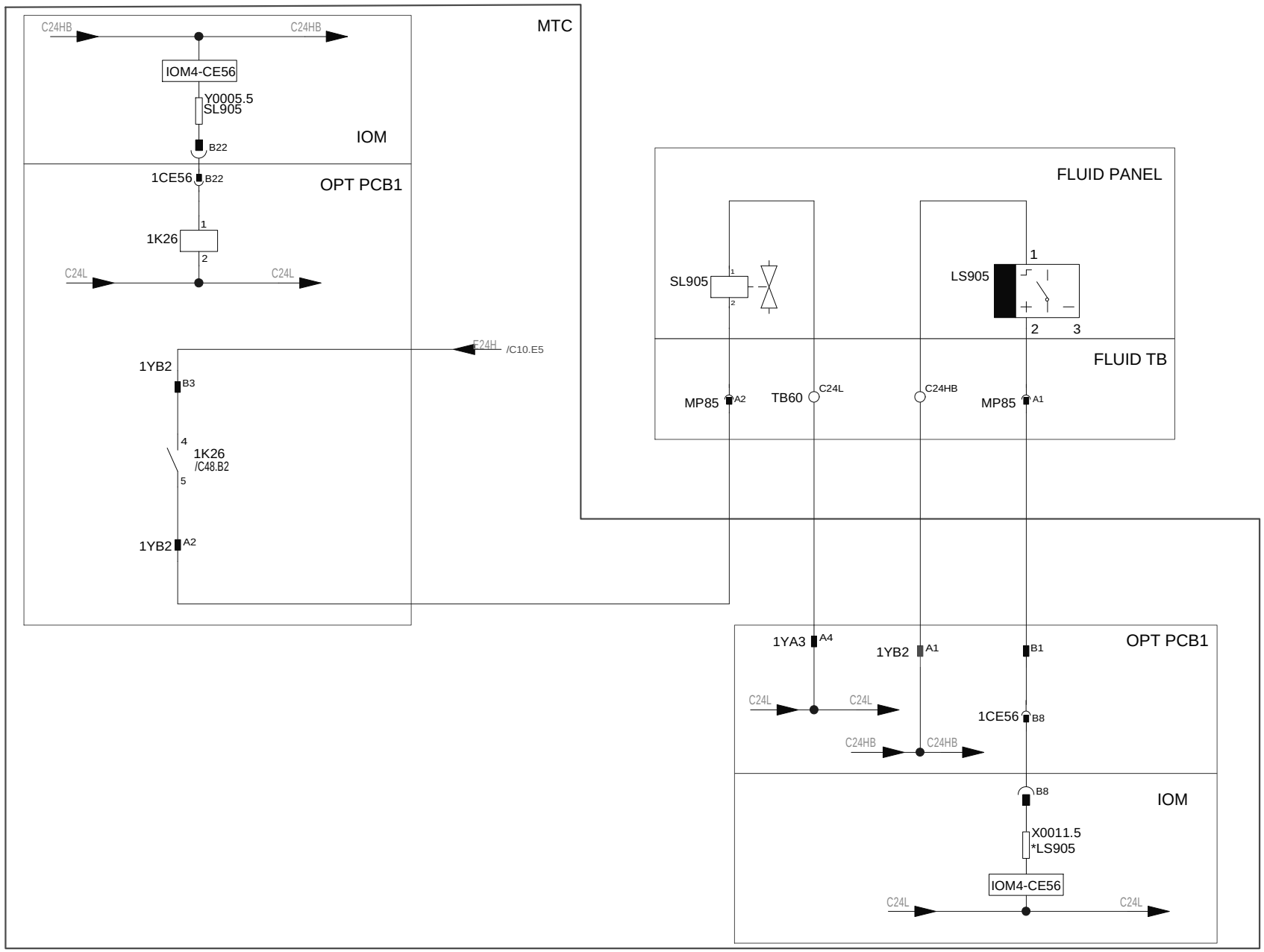



| | |
|---|---|
| AIR BLOW | THROUGH SPINDLE AIR (OPT) |

| REVISIONS | | | | |
|---|---|---|---|---|
| ④ | | | DATE | NAME |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| AIR BLOW & TSA (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C47 |



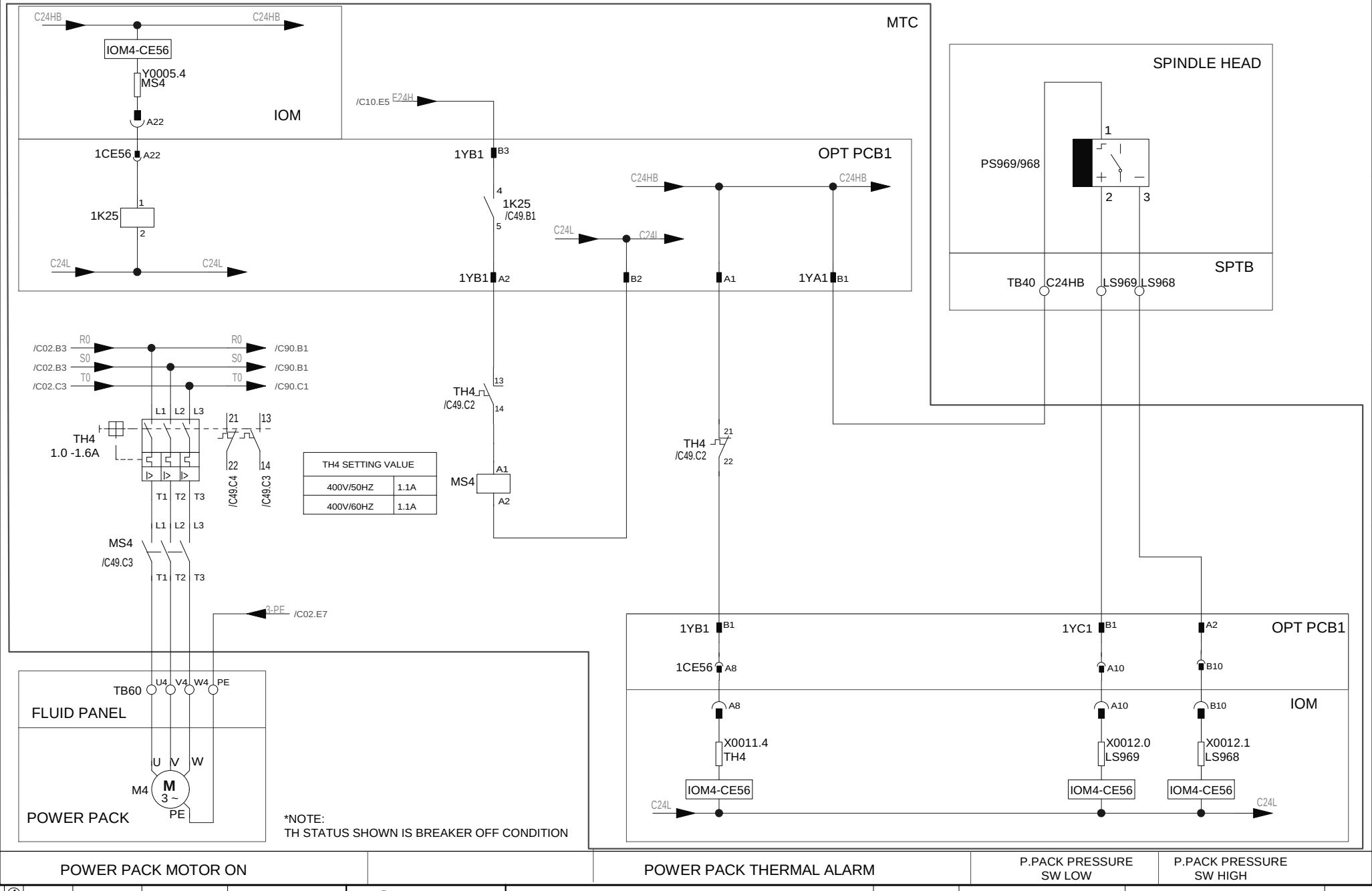

MASTER SOL VALVE

AIR PRESSURE SWITCH

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| MASTER SOLENOID | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C48 |

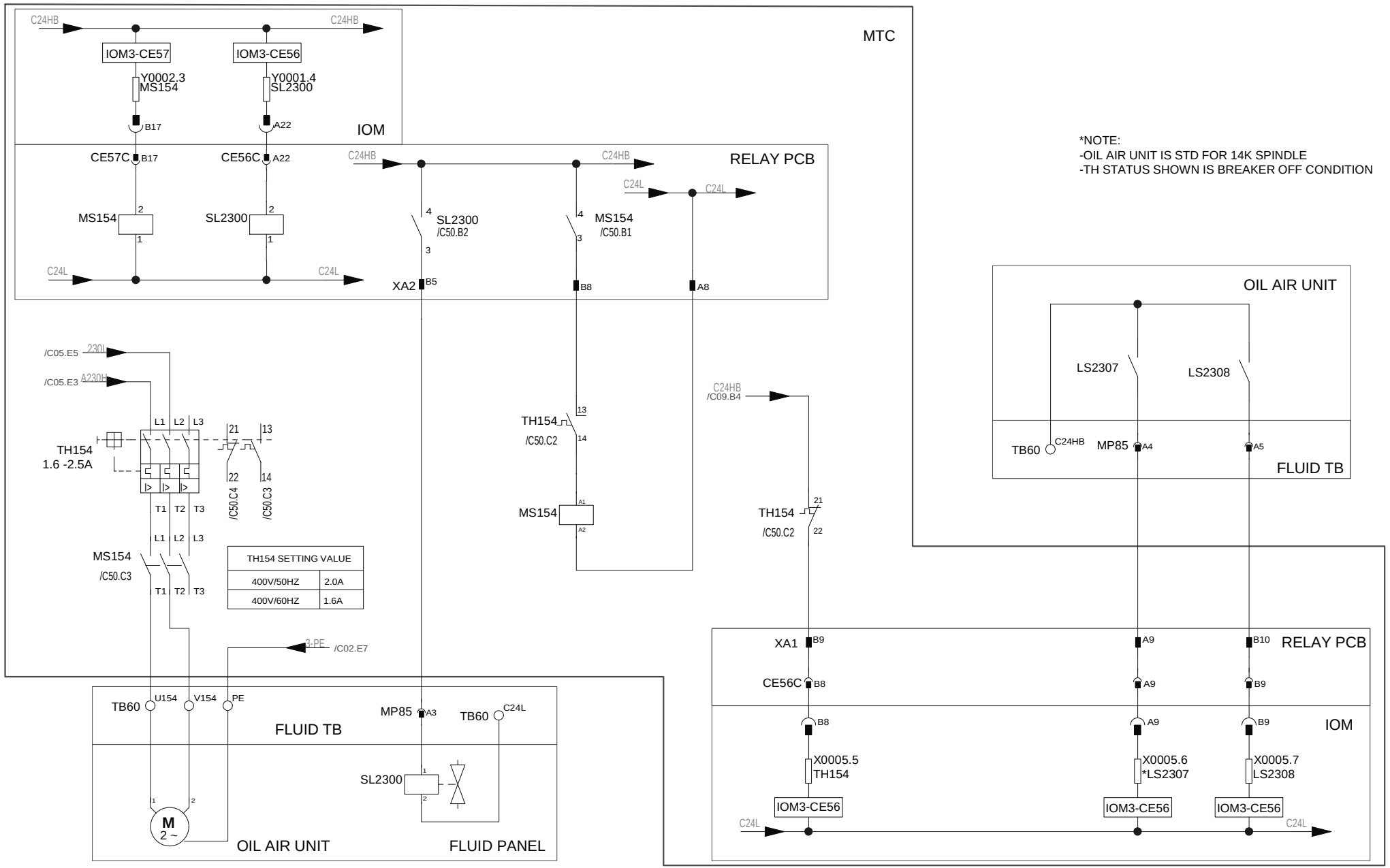



| POWER PACK MOTOR ON | | POWER PACK THERMAL ALARM | P.PACK PRESSURE SW LOW | P.PACK PRESSURE SW HIGH |
|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| POWER PACK (TOOL CL/UCL) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | C49 |

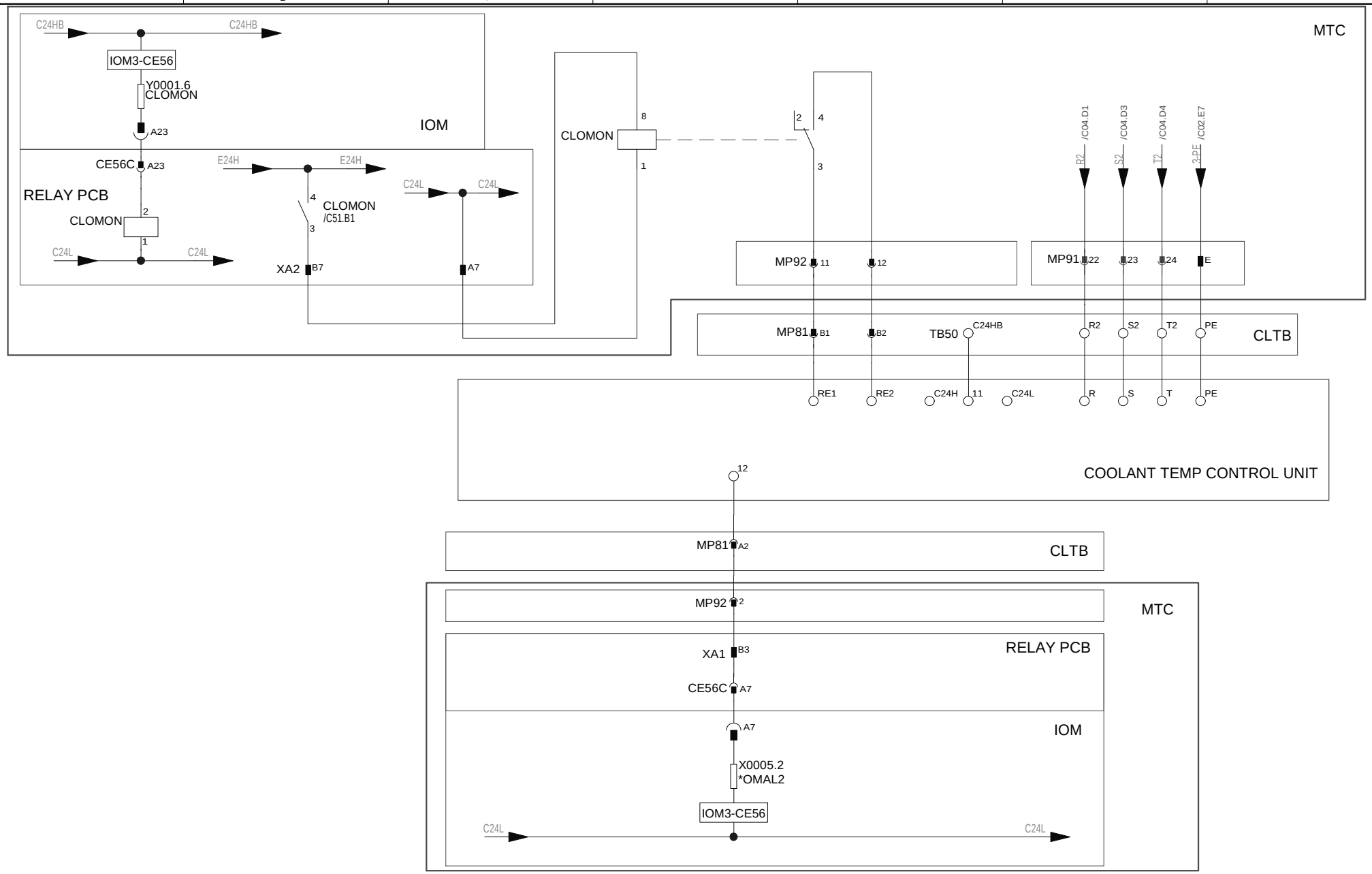



| OIL AIR PUMP ON | SP LUB AIR | | OIL AIR THERMAL ALARM | LOW LEVEL | PRESSURE DETECTION |
|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| OIL AIR UNIT (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | C50 |



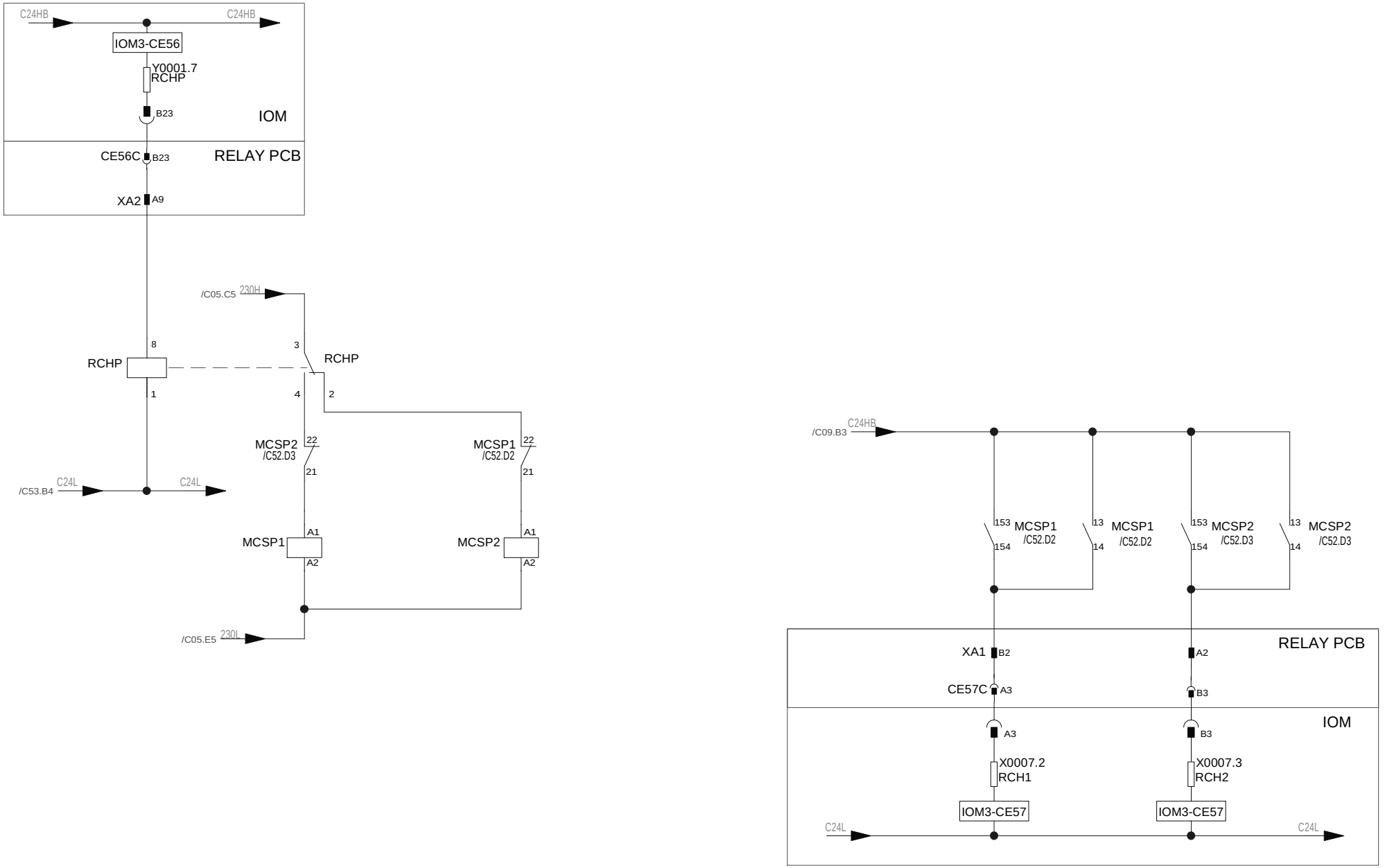

COOLANT TEMP CONTROL ON

COOLANT TEMP CONTROL UNIT ALARM

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| COOLANT CHILLER (OPT) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C51 |

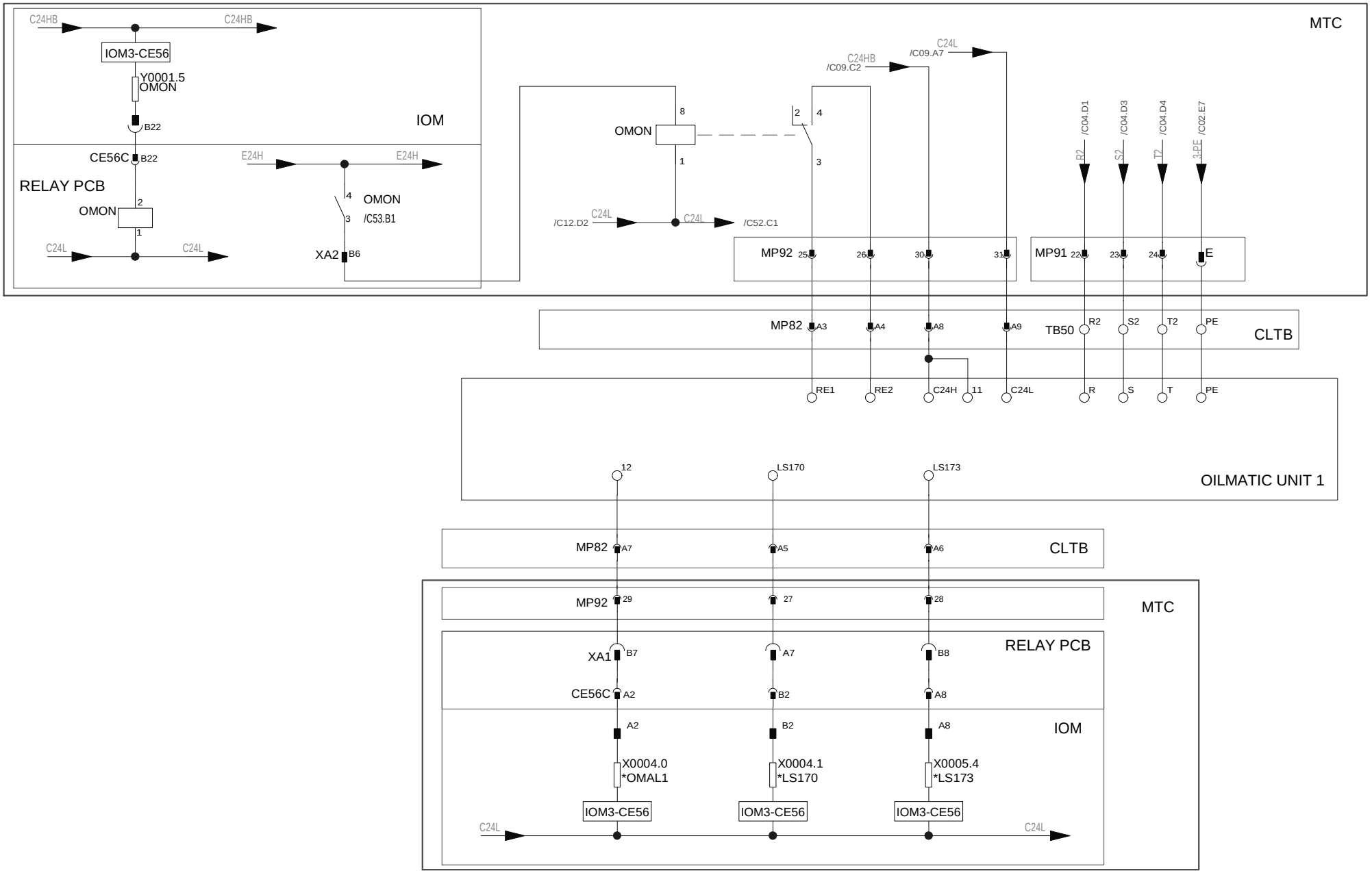



| SPINDLE SPEED CONTROL | LOW SPEED SELECTION | HIGH SPEED SELECTION | SPINDLE LOW SPEED CONFIRM | HIGH SPEED CONFIRM |
|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| SPINDLE SWITCHING CONTROL | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C52 |

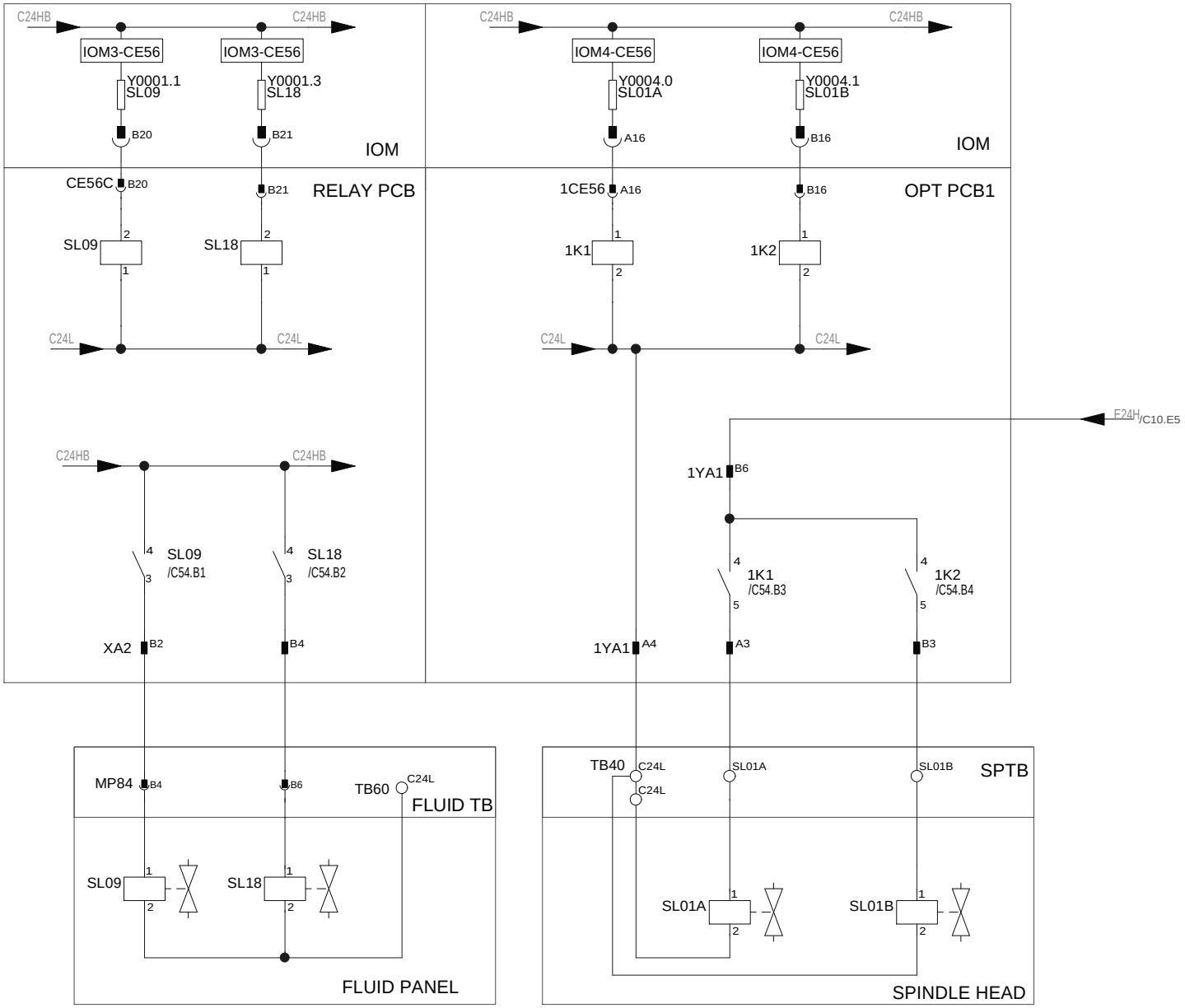



| SPINDLE TEMP CONTROL ON | | SPINDLE OILMATIC UNIT ALARM | FLOW SWITCH | PRESSURE LIMIT SWITCH |
|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| SPINDLE TEMPERATURE CONTROLLER | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C53 |

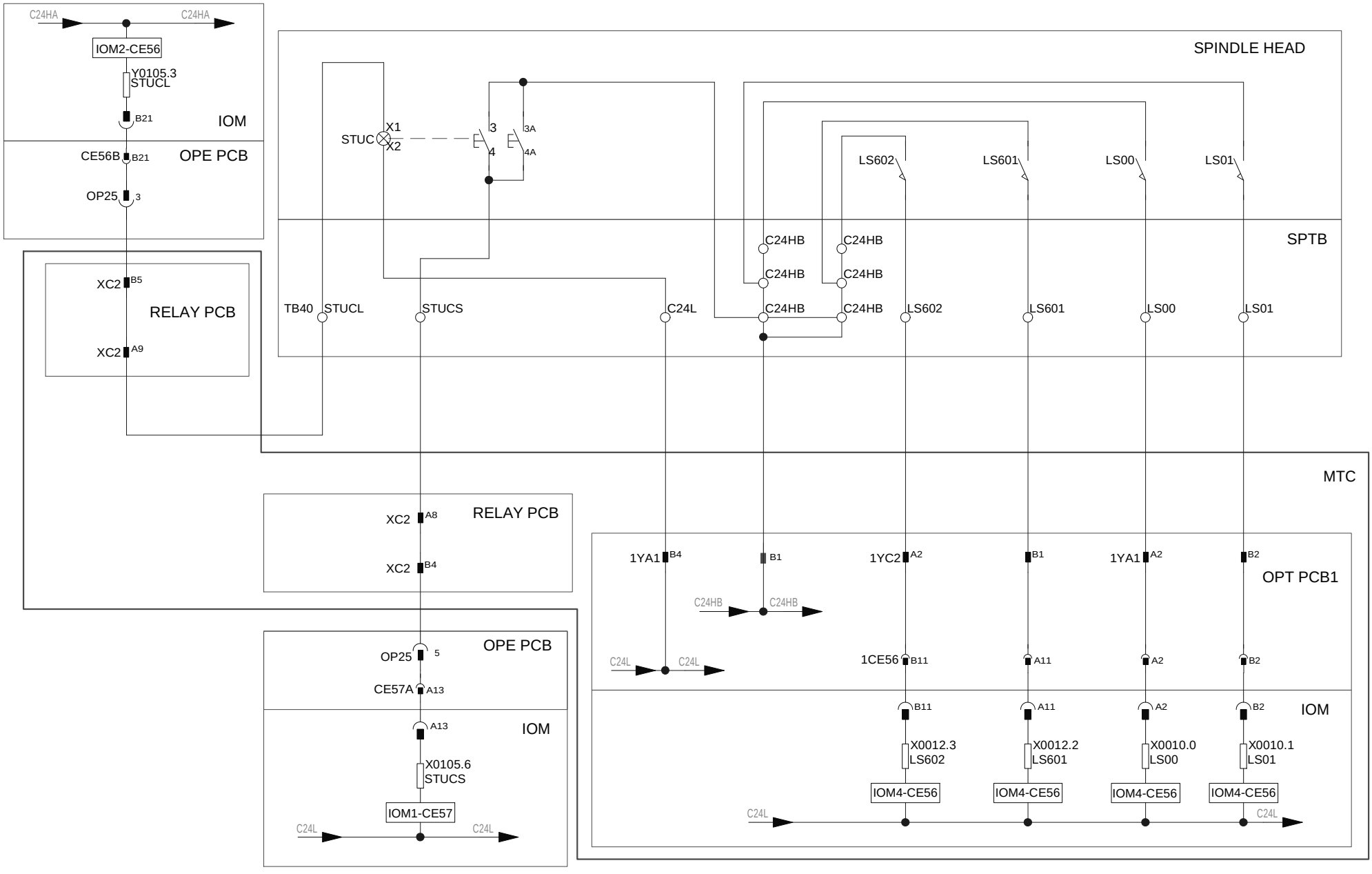



| TAPER CLEANING AIR 1 | SPINDLE AIR SEAL | SPINDLE TOOL CLAMP | SPINDLE TOOL UNCL |
|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| TOOL CLAMP/UNCLAMP SOL | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | C54 |



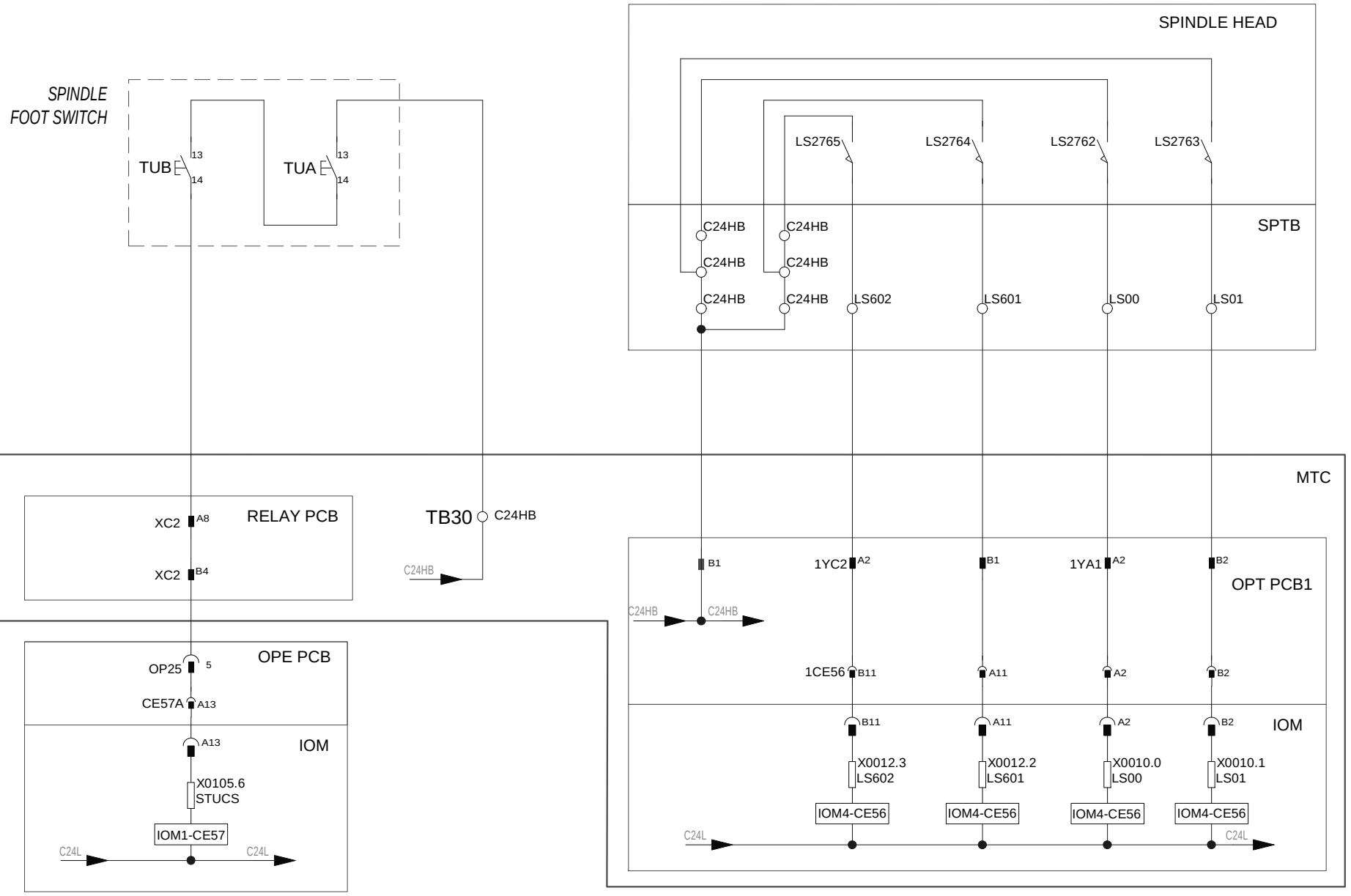

| SPINDLE TOOL UNCL SWITCH LAMP | SPINDLE TOOL UNCL SWITCH | SPINDLE TOOL CONFIRM | TOOL SEATING | TOOL CLAMP | TOOL UNCL |
|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| TOOL CLAMP/UNCLAMP LIMIT SW | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C55 |

*THIS PAGE ONLY FOR OPTION-SPD#50*

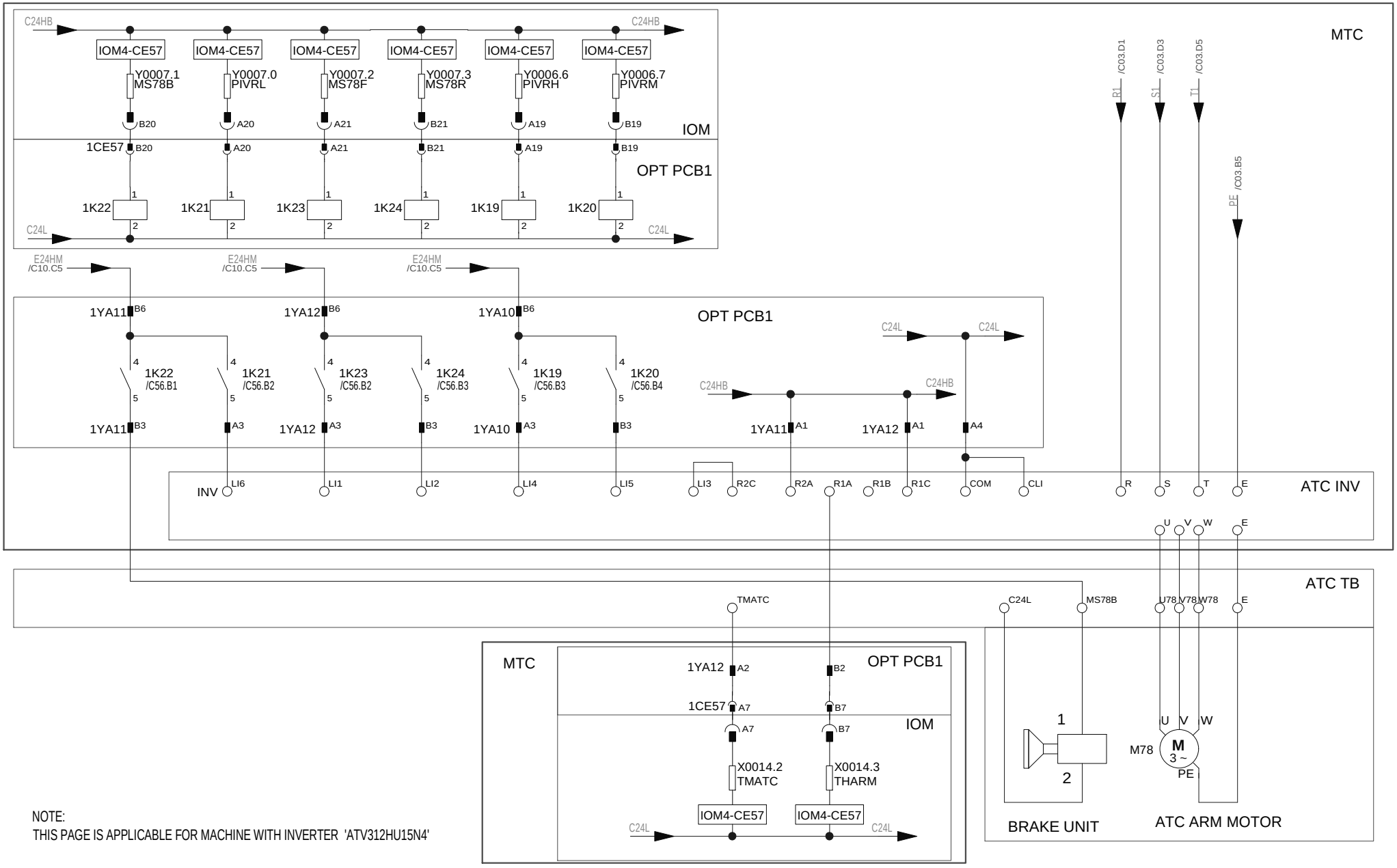



| SPINDLE FOOT SWITCH | SPINDLE TOOL CONFIRM | TOOL SEATING | TOOL CLAMP | TOOL UNCL |
|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 10/1/2023 | AVINASH |
| ② | | CHECKED | 10/1/2023 | JAVAHAR |
| ① | | DRAWN | 10/1/2023 | BALA |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| TOOL CLAMP/UNCLAMP LIMIT SW (OPTION SPD #50) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C55.1 |



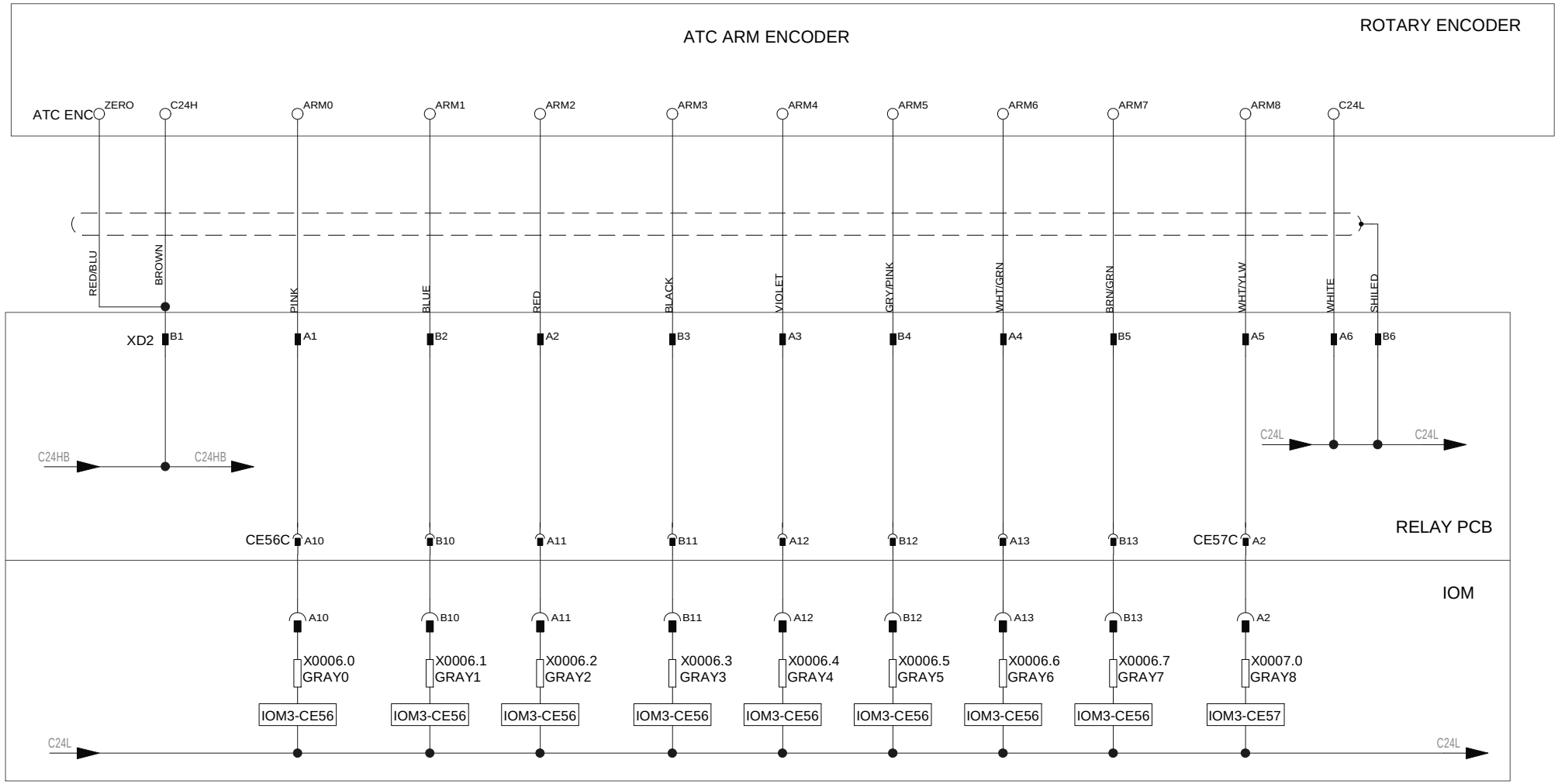

NOTE:
THIS PAGE IS APPLICABLE FOR MACHINE WITH INVERTER 'ATV312HU15N4'

| ATC MOTOR BRAKE | SPEED 3(LOW) | FORWARD | REVERSE | SPEED 1,2 (HIGH&MID) | MOTOR OVERHEAT | INVERTER ALARM |
|---|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

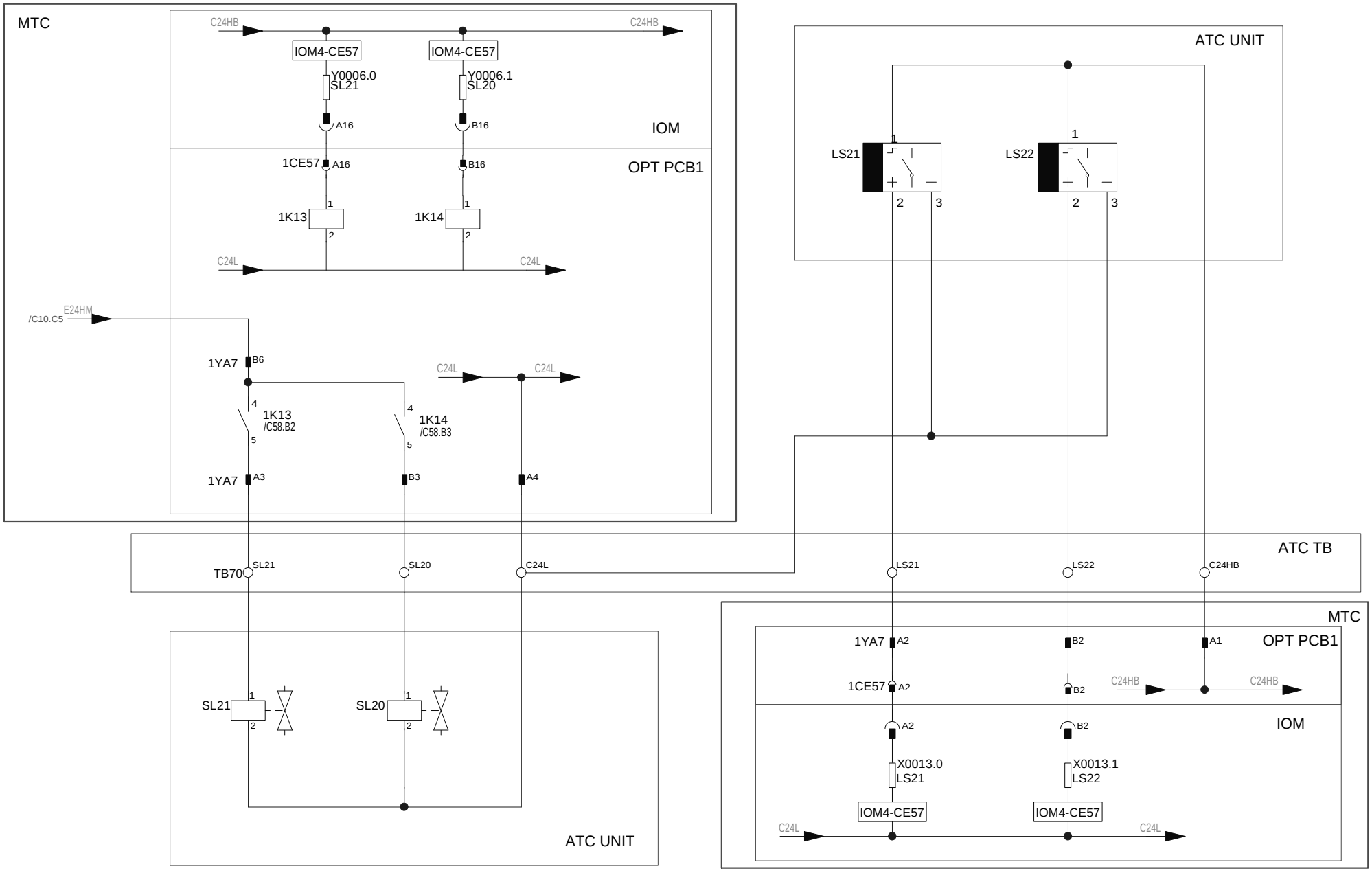
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| ATC ARM MOTOR (ATV312) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C56 |




| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| ATC ARM ENCODER | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C57 |



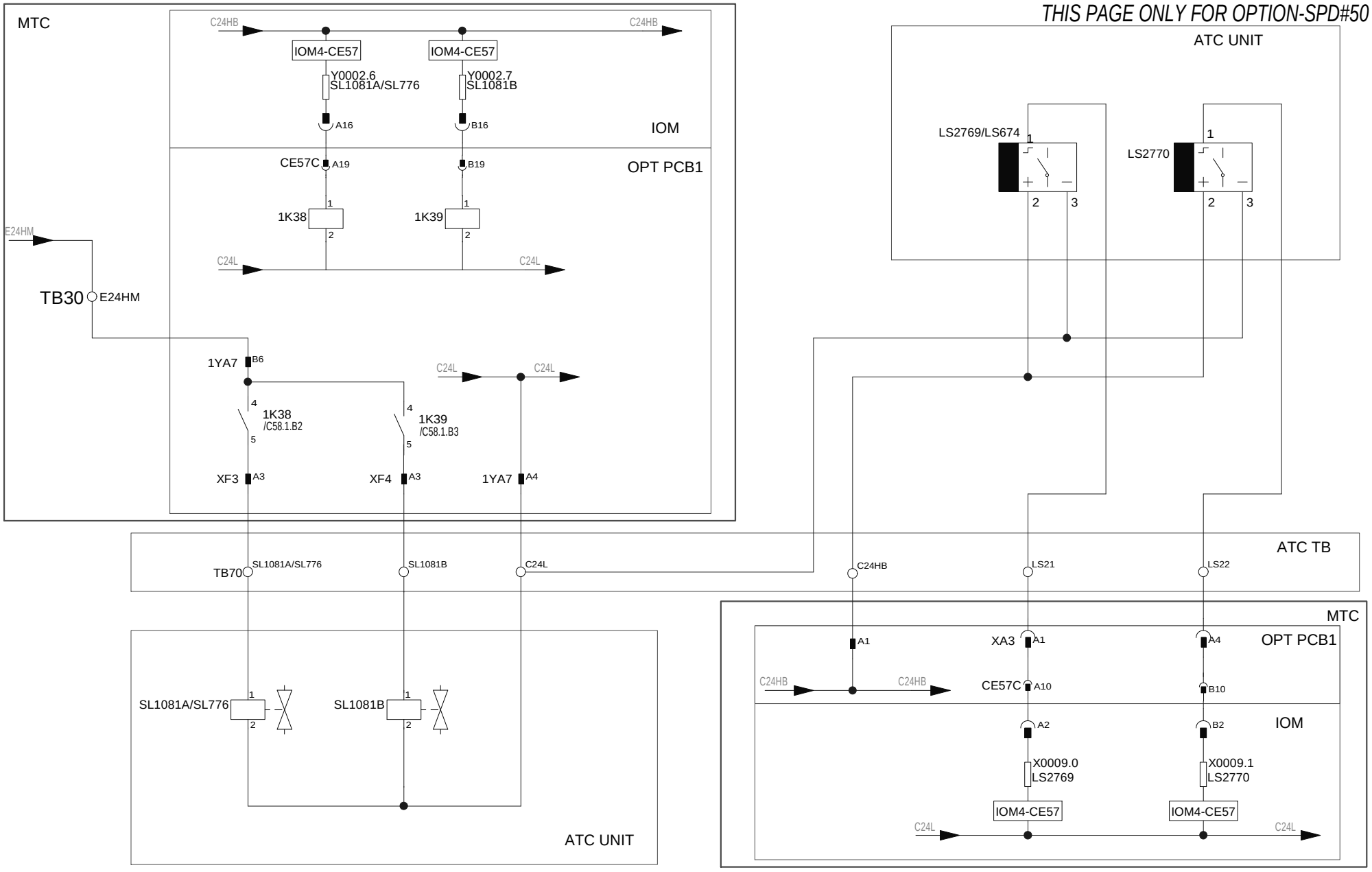

POT VERTICAL SOL | POT HORIZONTAL SOL | POT HORIZONTAL LS | POT VERTICAL LS

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| ATC POT CONTROL | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | C58 |

*THIS PAGE ONLY FOR OPTION-SPD#50*

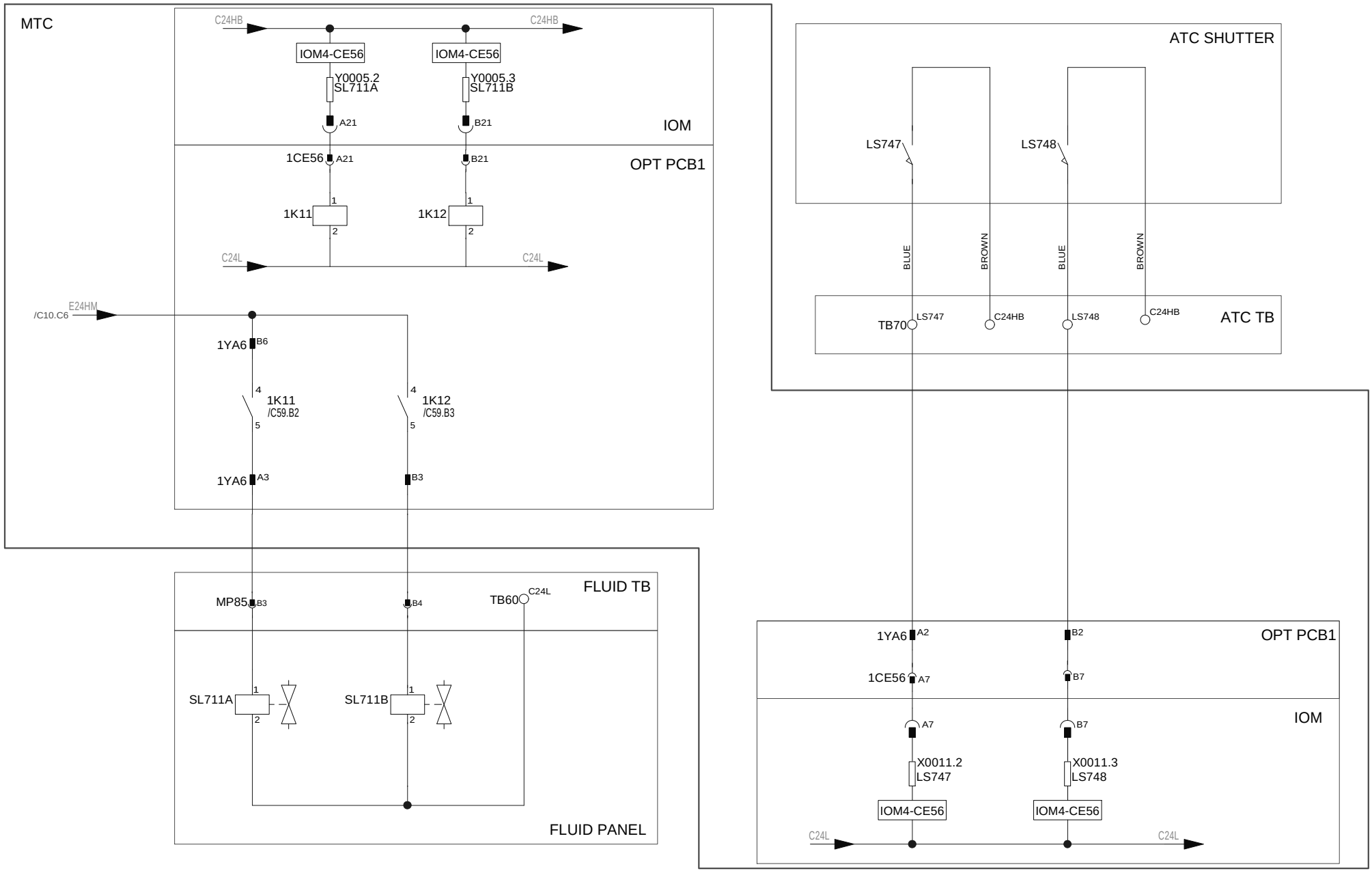



| TLS TOOL LOCK | TLS TOOL UNLOCK | TLS TOOL LOCK REED SWITCH | TLS TOOL UNLOCK REED SWITCH |
|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 10/1/2023 | AVINASH |
| ② | | CHECKED | 10/1/2023 | JAVAHAR |
| ① | | DRAWN | 10/1/2023 | BALA |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| ATC POT CONTROL (OPTION SPD#50) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C58.1 |



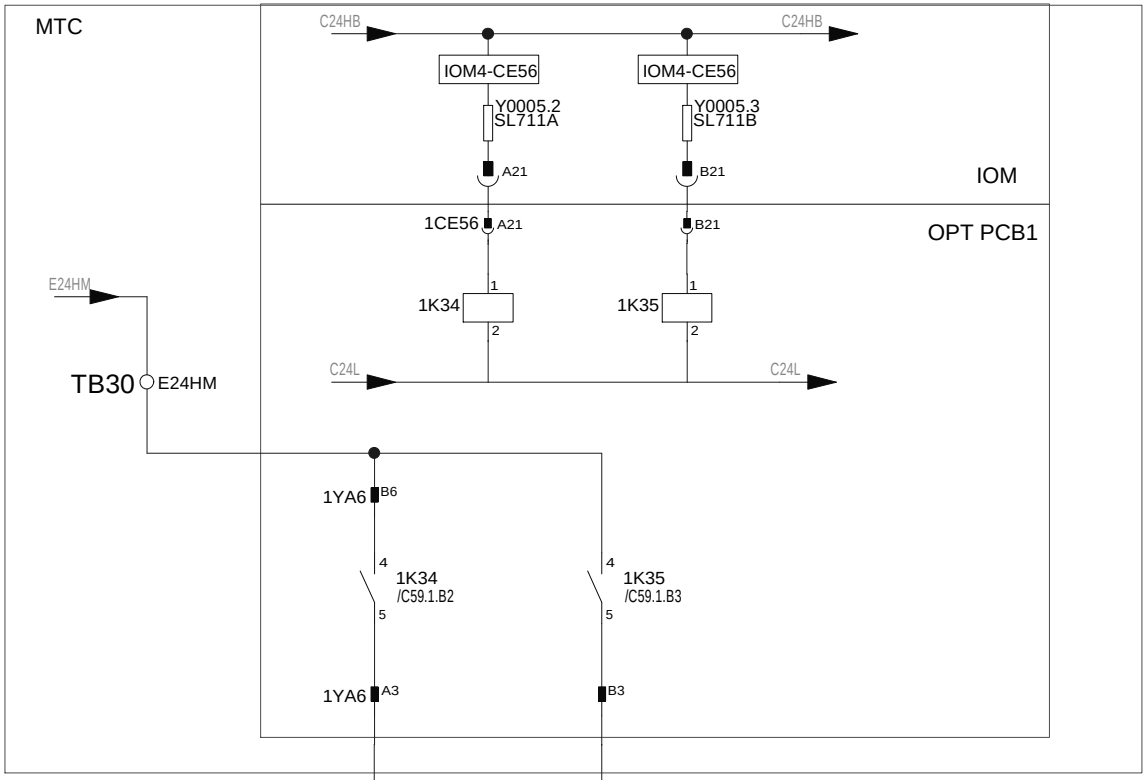

| ATC SHUTTER CLOSE SOL | ATC SHUTTER OPEN SOL | ATC SHUTTER CLOSE LS | ATC SHUTTER OPEN LS |
|---|---|---|---|

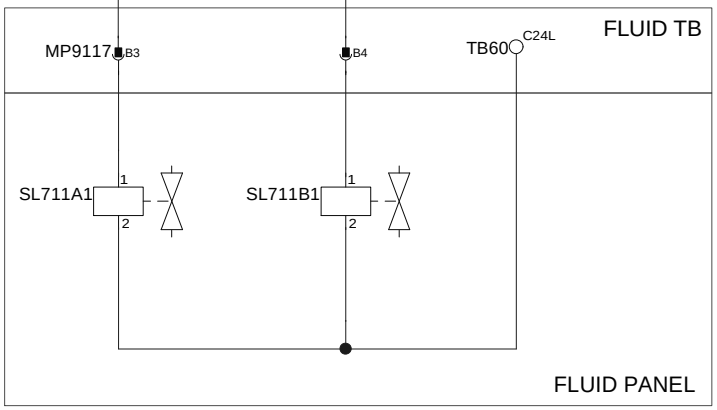
| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| ATC SHUTTER | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | C59 |

*THIS PAGE ONLY FOR OPTION-SPD#50*



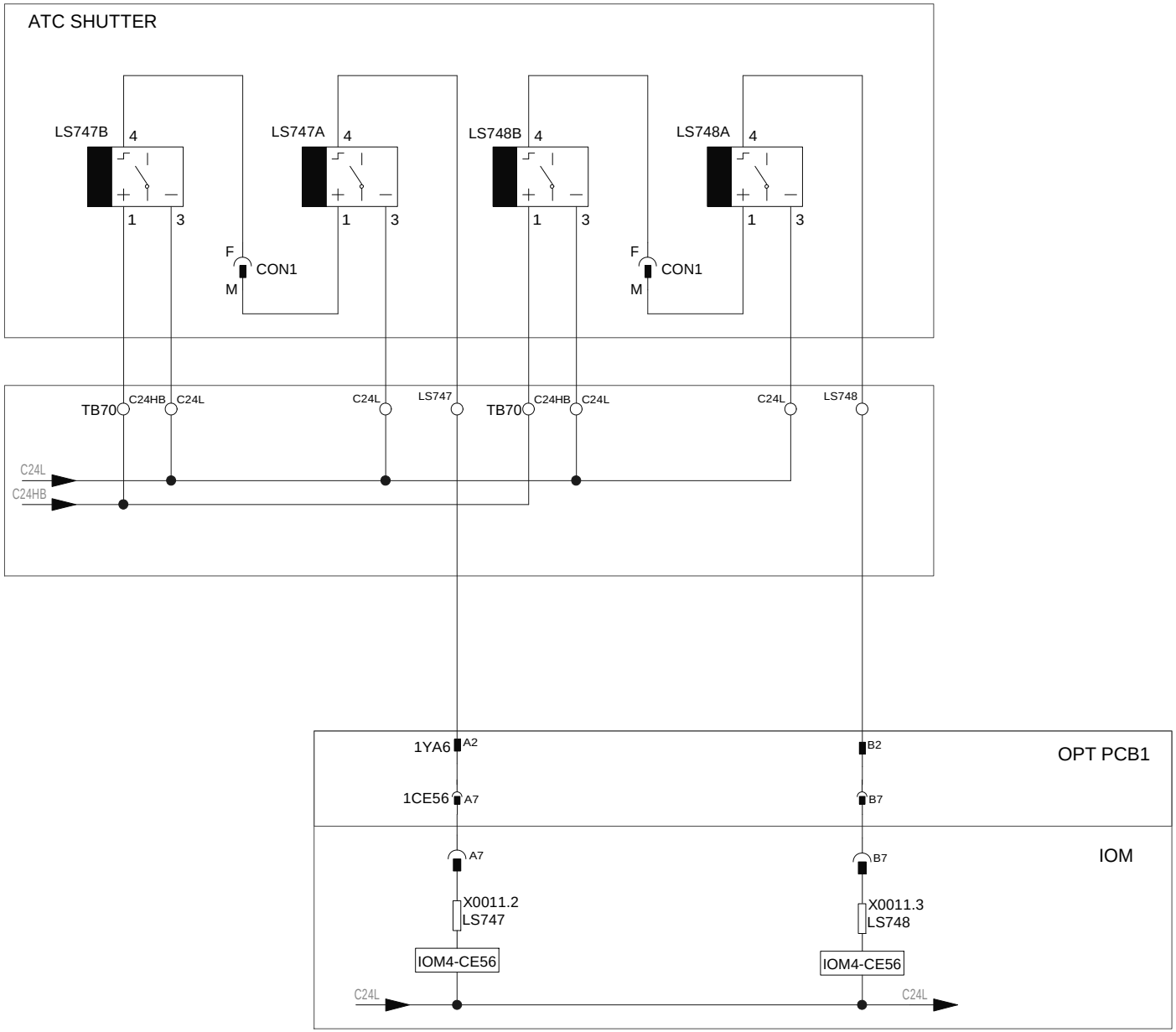

ATC SHUTTER CLOSE SOL | ATC SHUTTER OPEN SOL

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 10/1/2023 | AVINASH |
| ② | | CHECKED | 10/1/2023 | JAVAHAR |
| ① | | DRAWN | 10/1/2023 | BALA |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| ATC SHUTTER (OPTION SPD#50) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C59.1 |

*THIS PAGE ONLY FOR OPTION-SPD#50*




ATC SHUTTER CLOSE LS

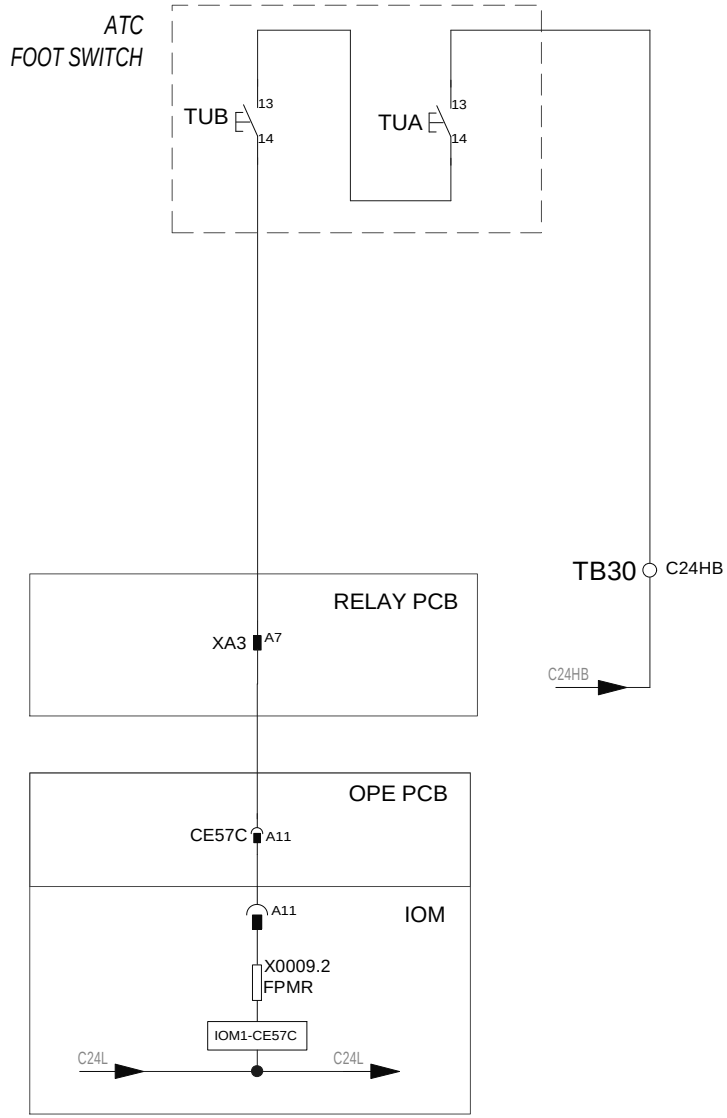
ATC SHUTTER OPEN LS

| REVISIONS | | | | |
|---|---|---|---|---|
| ④ | | | DATE | NAME |
| ③ | | APPROVED | 10/1/2023 | AVINASH |
| ② | | CHECKED | 10/1/2023 | JAVAHAR |
| ① | | DRAWN | 10/1/2023 | BALA |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| ATC SHUTTER (OPTION SPD#50) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C59.2 |

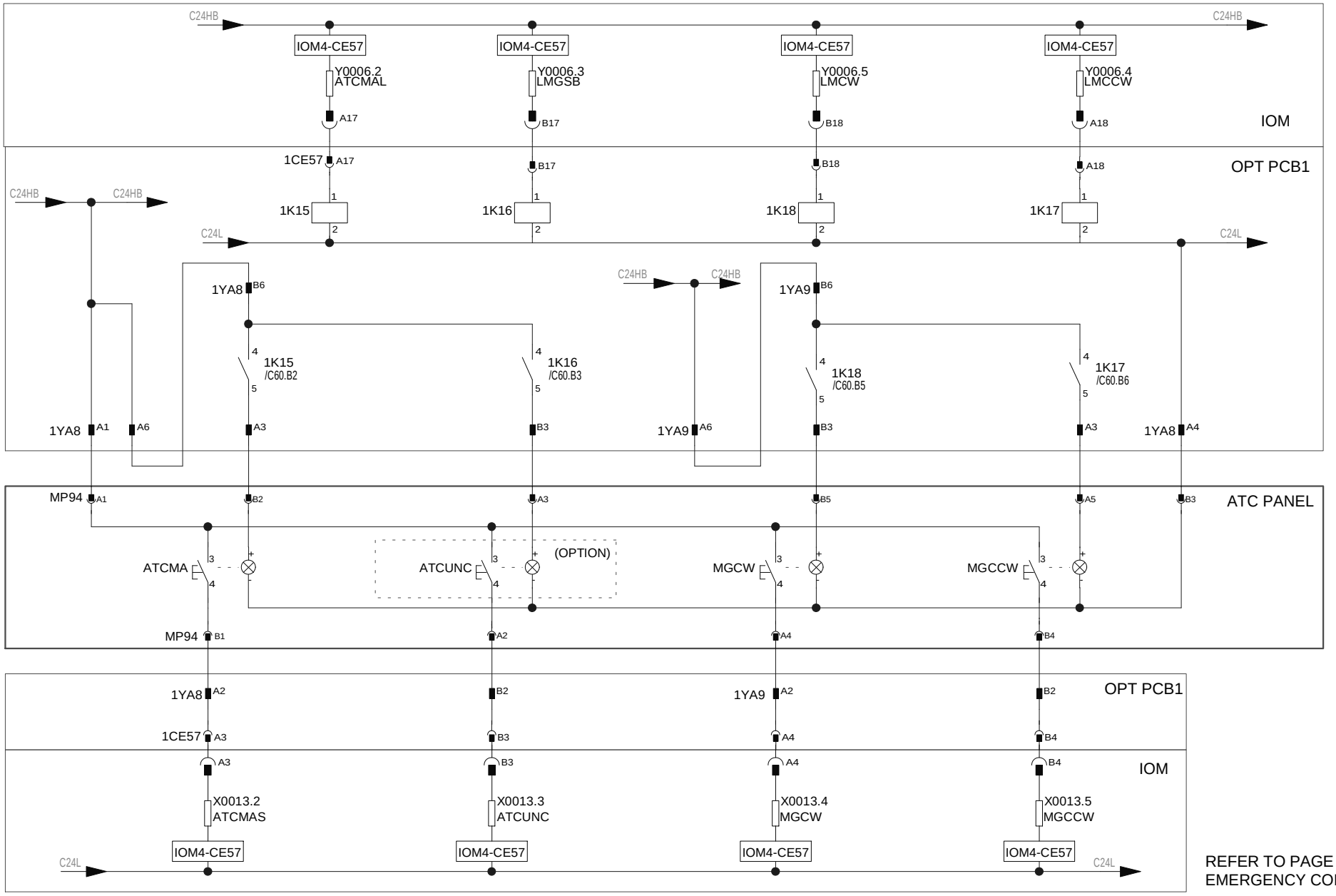
*THIS PAGE ONLY FOR OPTION-SPD#50*




| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 10/1/2023 | AVINASH |
| ② | | CHECKED | 10/1/2023 | JAVAHAR |
| ① | | DRAWN | 10/1/2023 | BALA |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| ATC SHUTTER (OPTION SPD#50) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C59.3 |

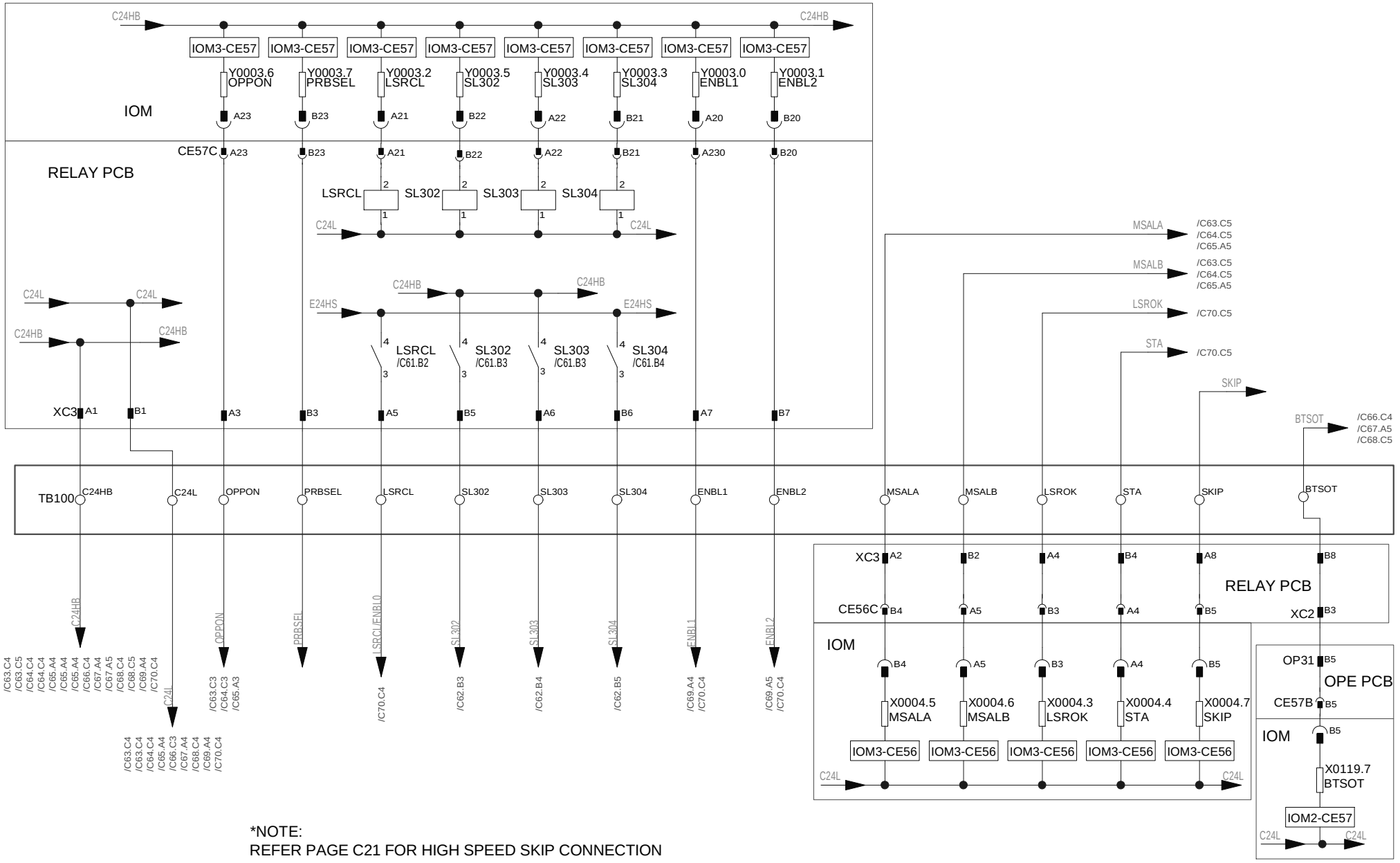



| ATC MANUAL INTERRUPT | ATC DOOR UNCLOK (OPT) | ATC MAGAZINE CCW | ATC MAGAZINE CW |
|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| ATC PANEL | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C60 |



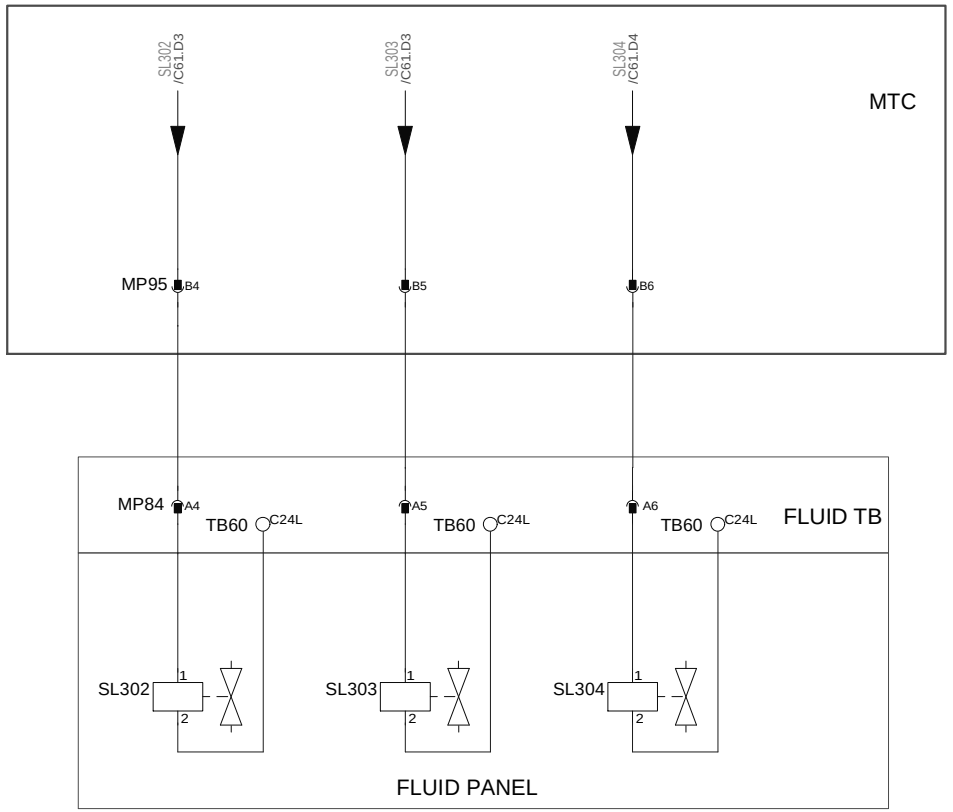

*NOTE:
REFER PAGE C21 FOR HIGH SPEED SKIP CONNECTION

| | MEASURING PROBE ON | PROBE SELECT | LASER CLEARANCE | TOOL CLEAN AIR | PURGE AIR | LASER SHUTTER OPEN | ENABLE 1 | ENABLE 2 | MEAS SYSTEM RECEP ALARM | MEAS SYSTEM BATTERY ALARM | LASER OK | STATIC | SKIP | BTS OVERTRAVEL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| AWM & ATLM INTERFACE (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C61 |

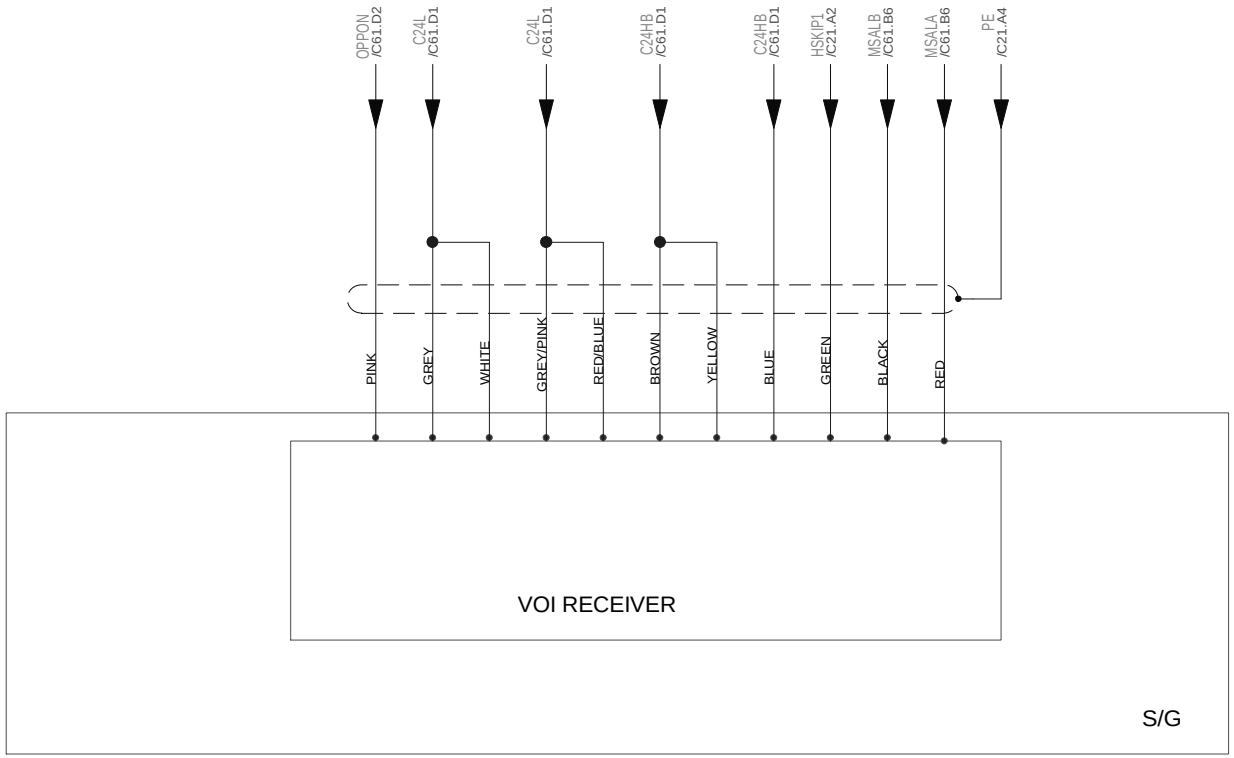



| TOOL CLEANING AIR | PURGE AIR | LASER SHUTTER OPEN |
|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

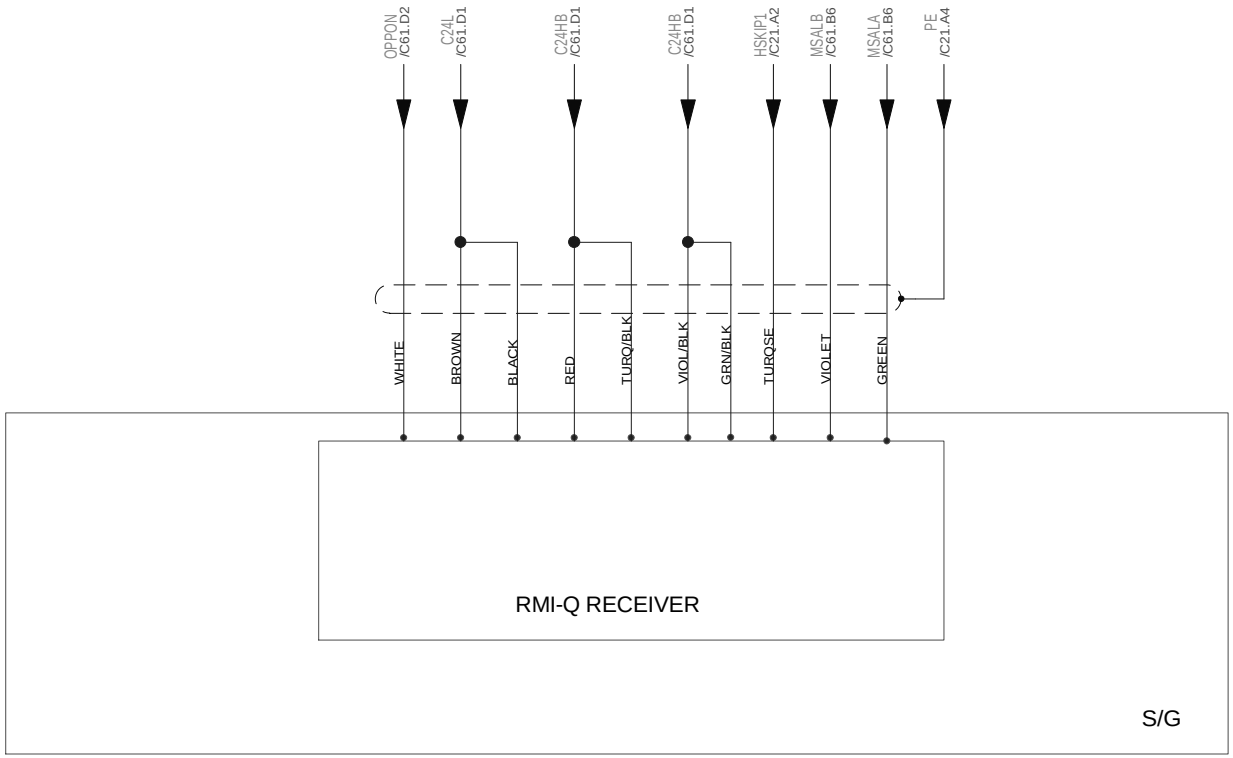
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| ATLM SOLENOIDS (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C62 |




| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

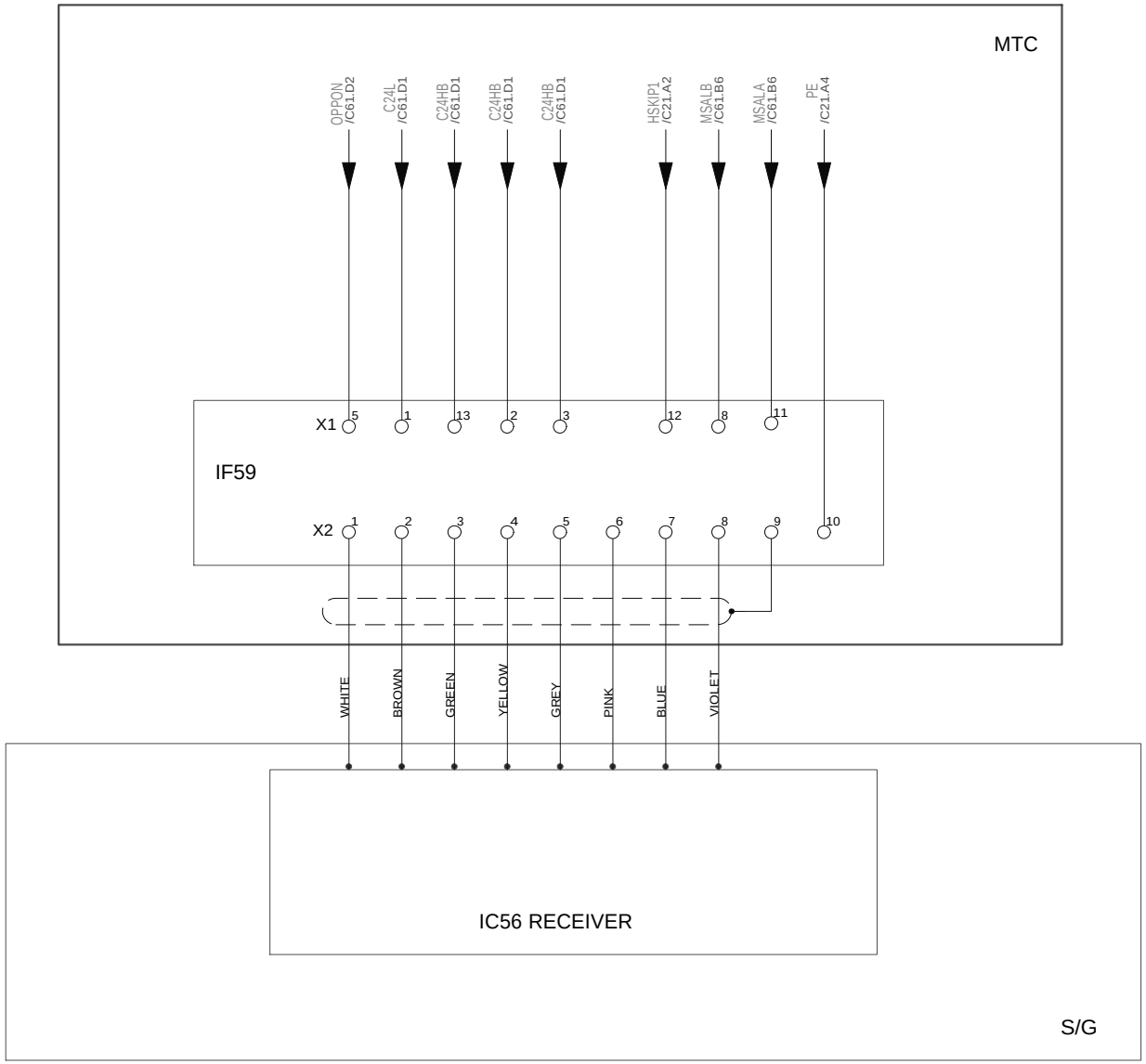
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| AWM MARPOSS (OPT) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C63 |




| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

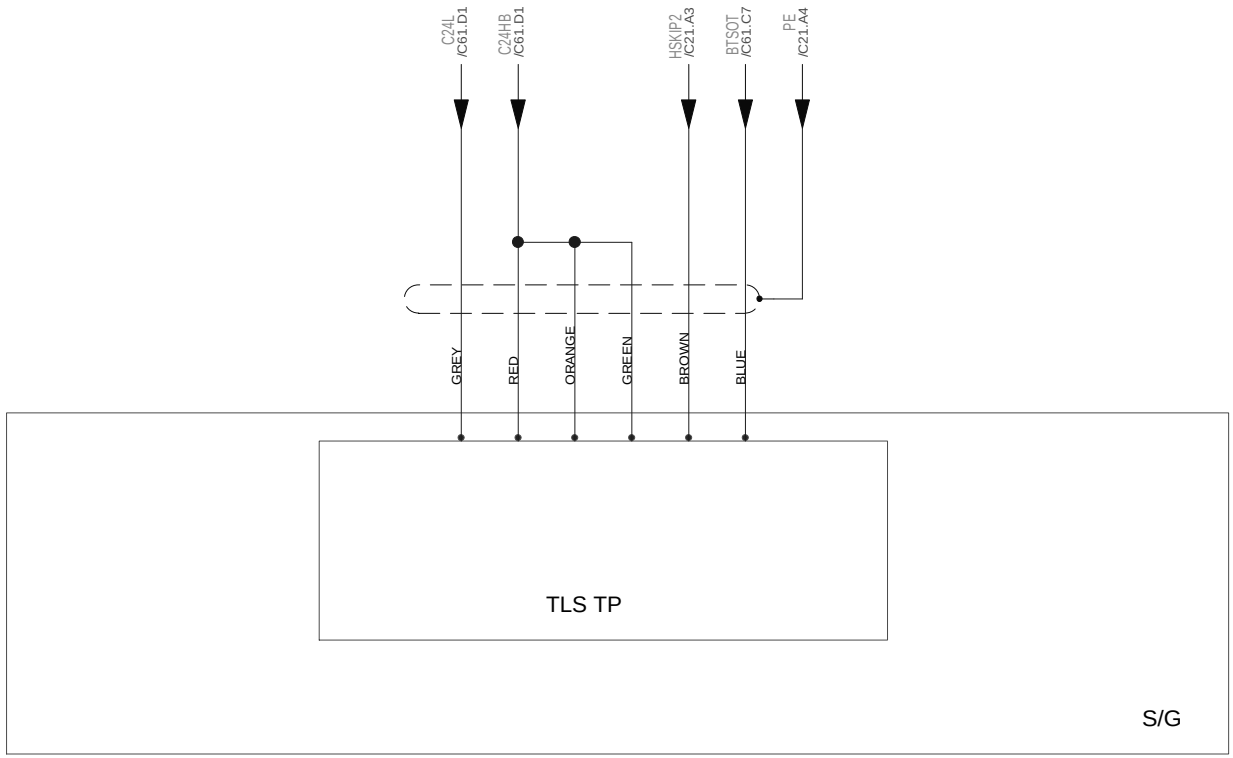
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| AWM RENISHAW (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C64 |




| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

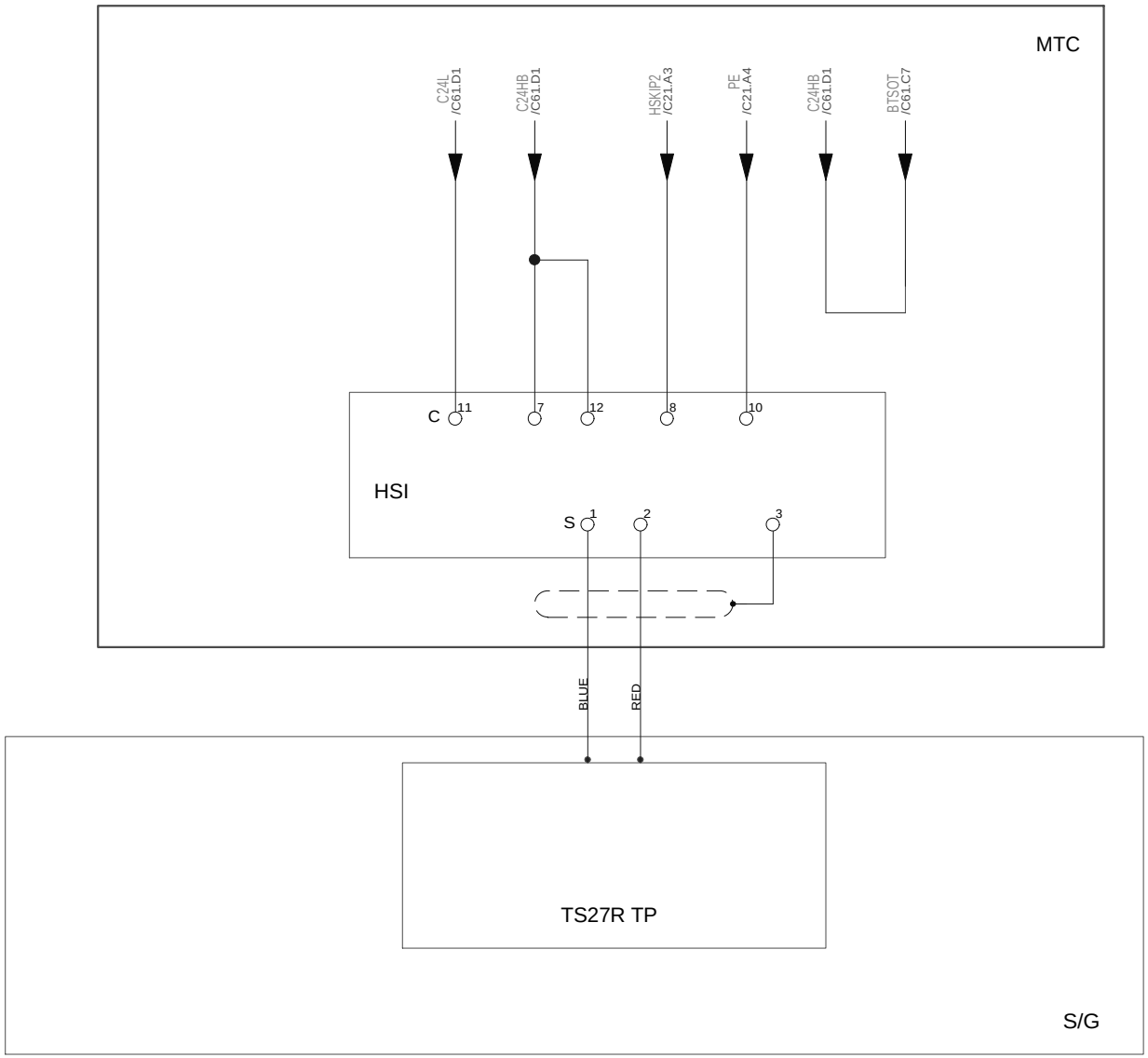
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| AWM BLUM (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C65 |




| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

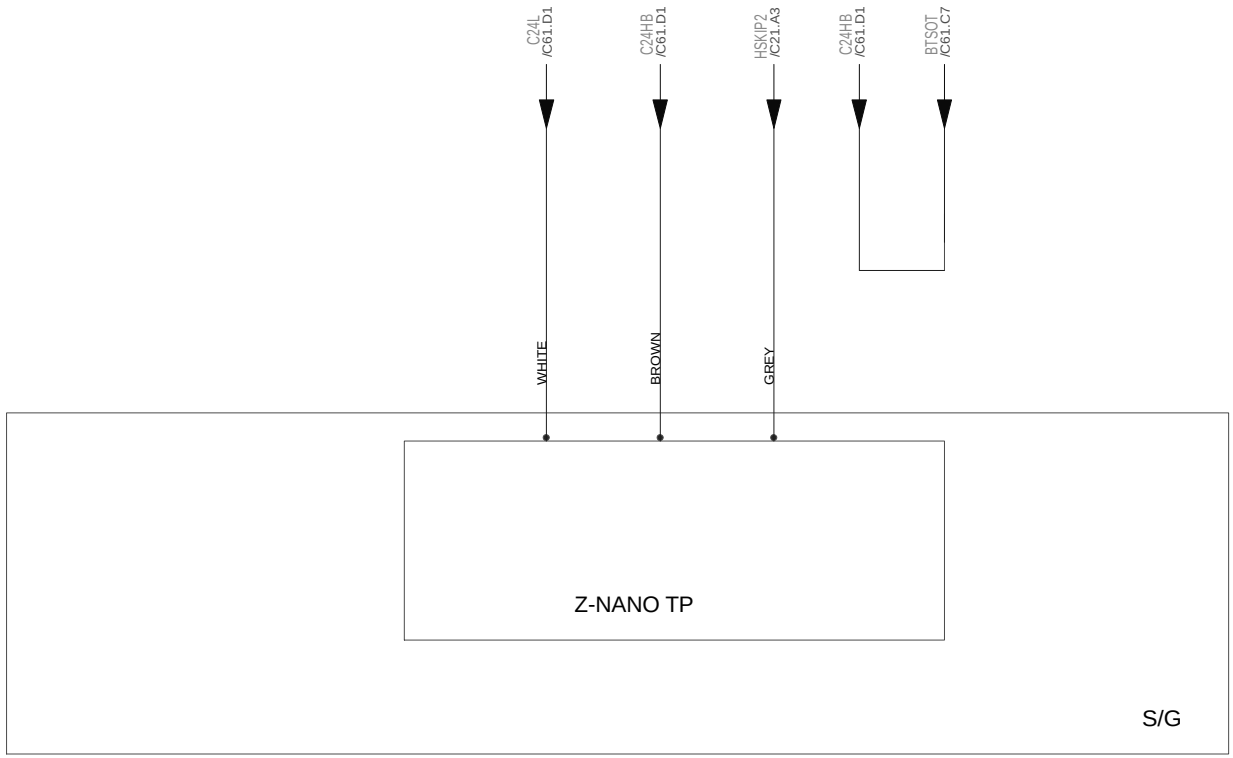
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| ATLM TP MARPOSS (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C66 |




| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

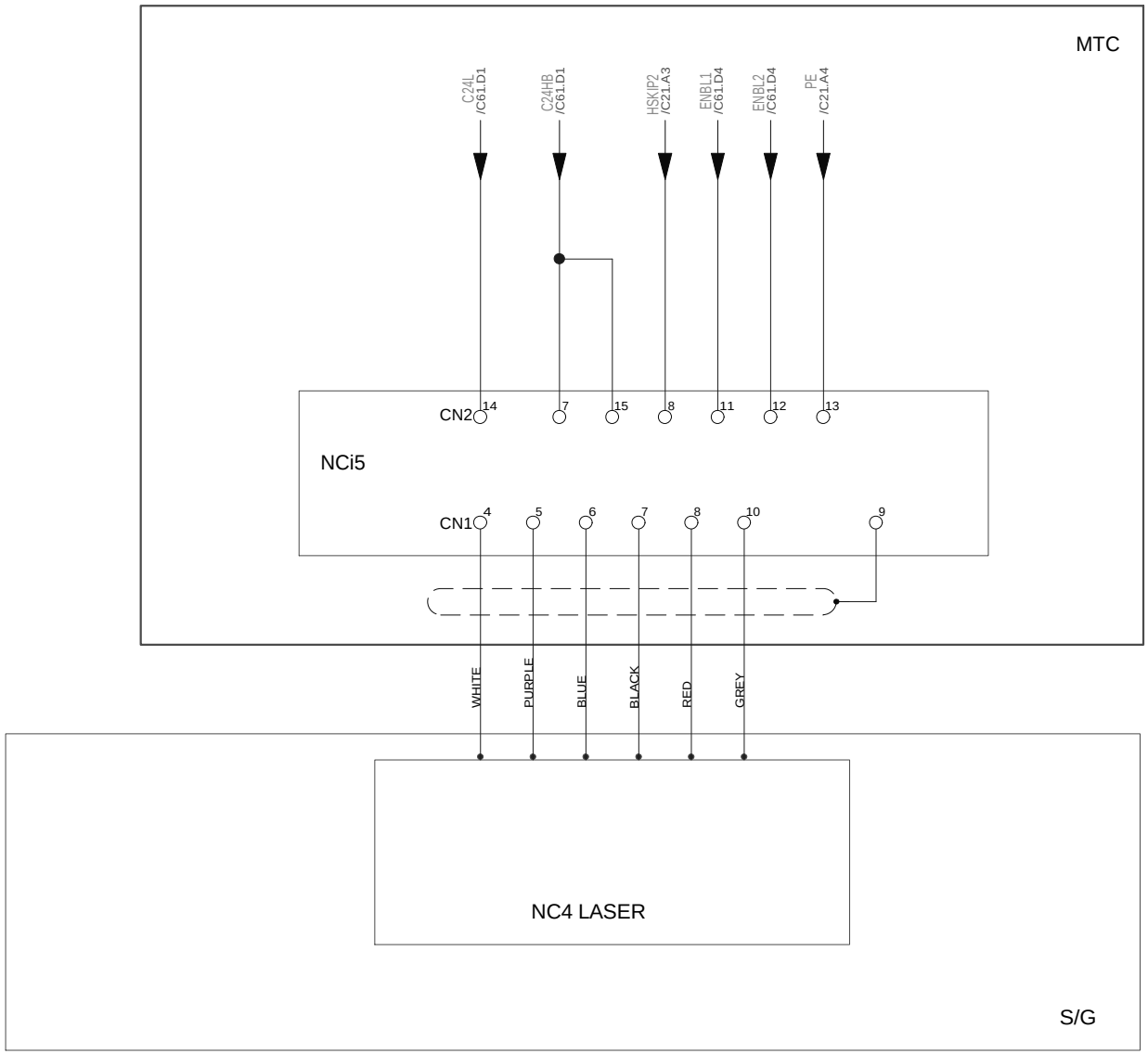
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| ATLM TP RENISHAW (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C67 |




| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

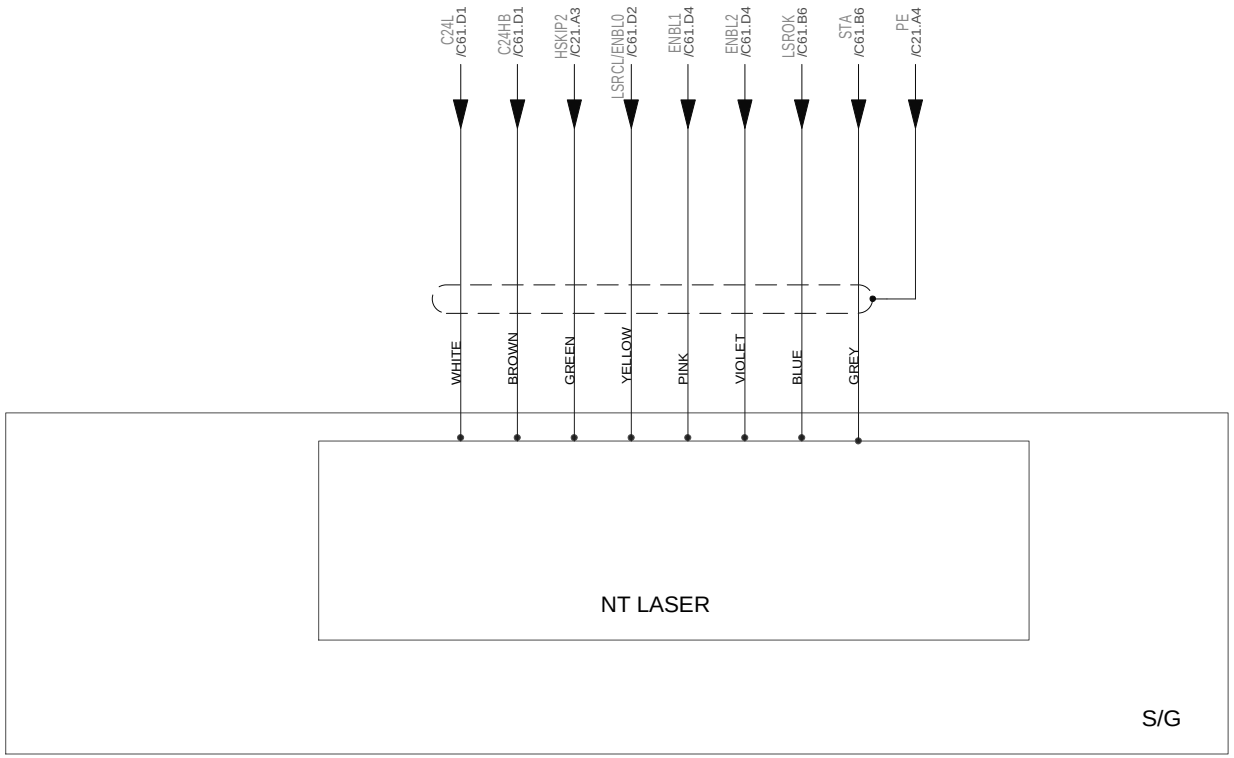
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| ATLM TP BLUM (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C68 |




| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

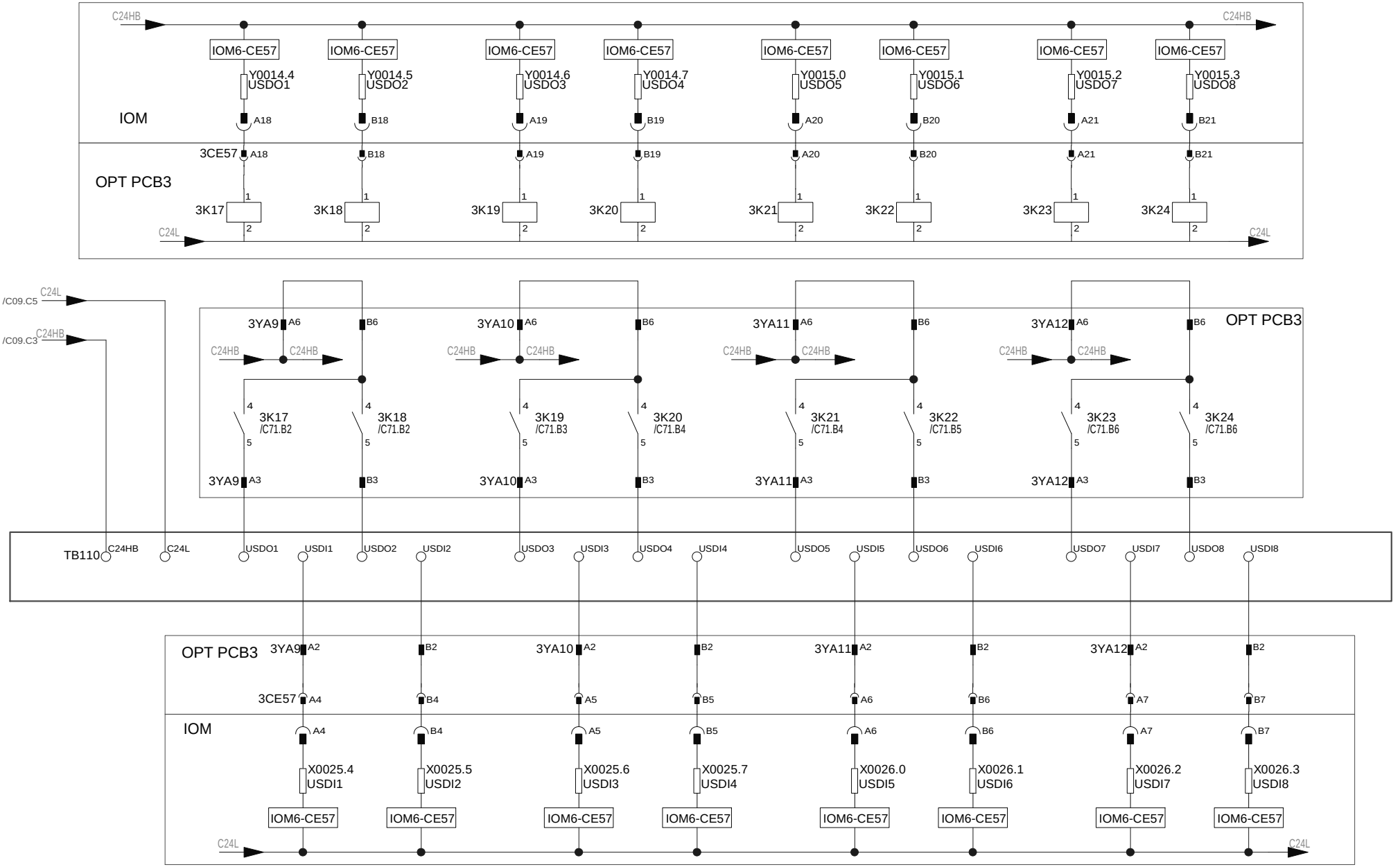
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| ATLM LASER RENISHAW (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C69 |




| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| ATLM LASER BLUM (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | C70 |

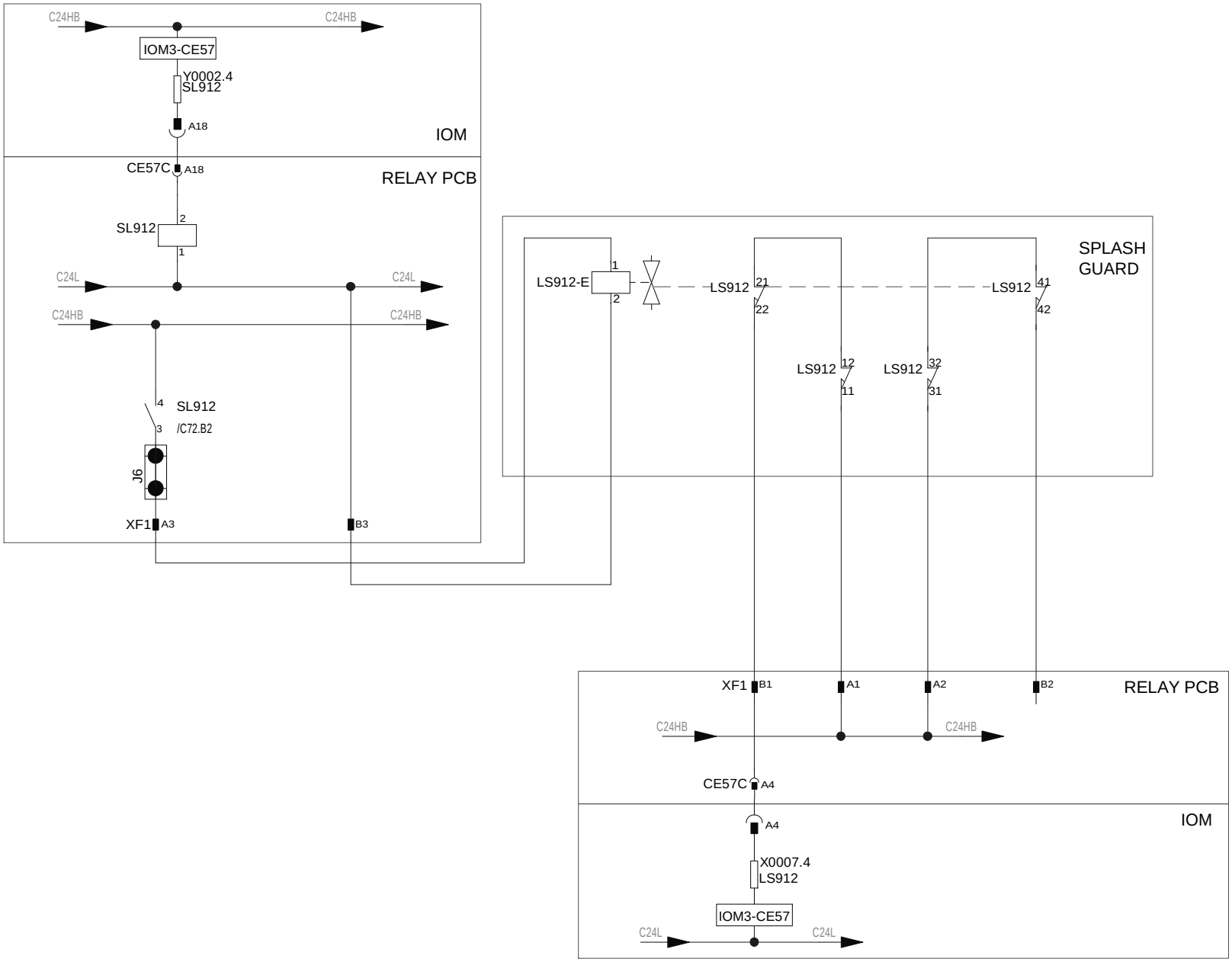



| USER DO-DI 1&2 | USER DO-DI 3&4 | USER DO-DI 5&6 | USER DO-DI 7&8 |
|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| USER M-CODE (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | C71 |



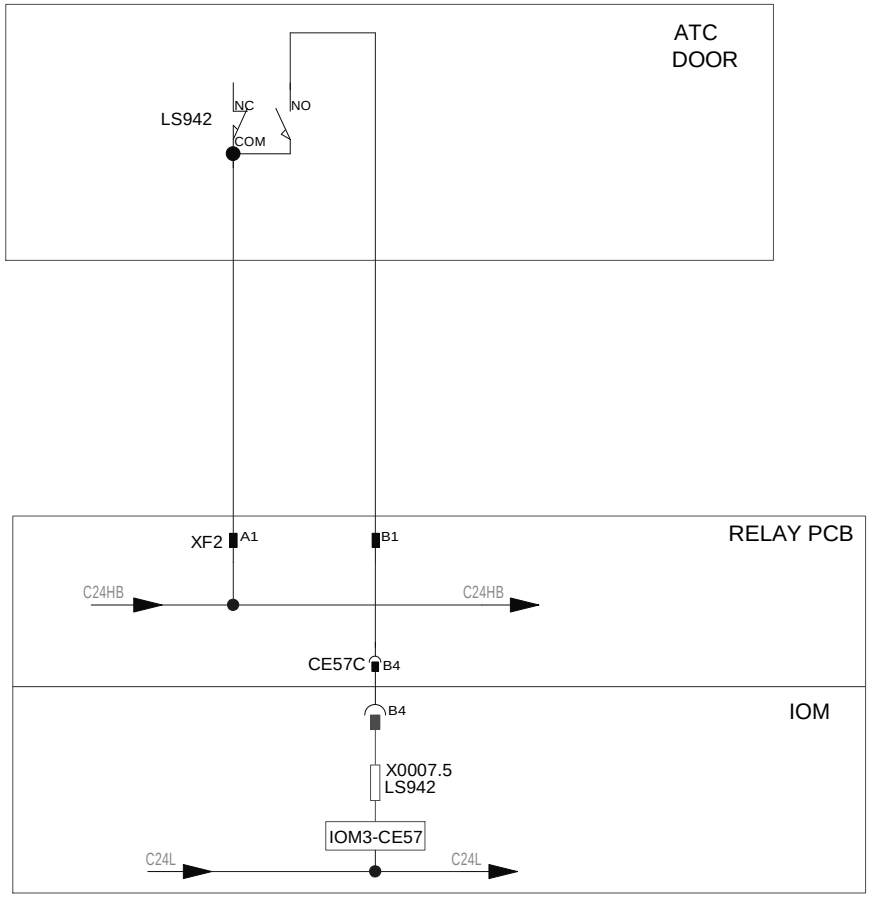

SG DOOR OPEN

SG DOOR INTERLOCK LS

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| S/G DOOR LOCK W INTERLOCK | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C72 |

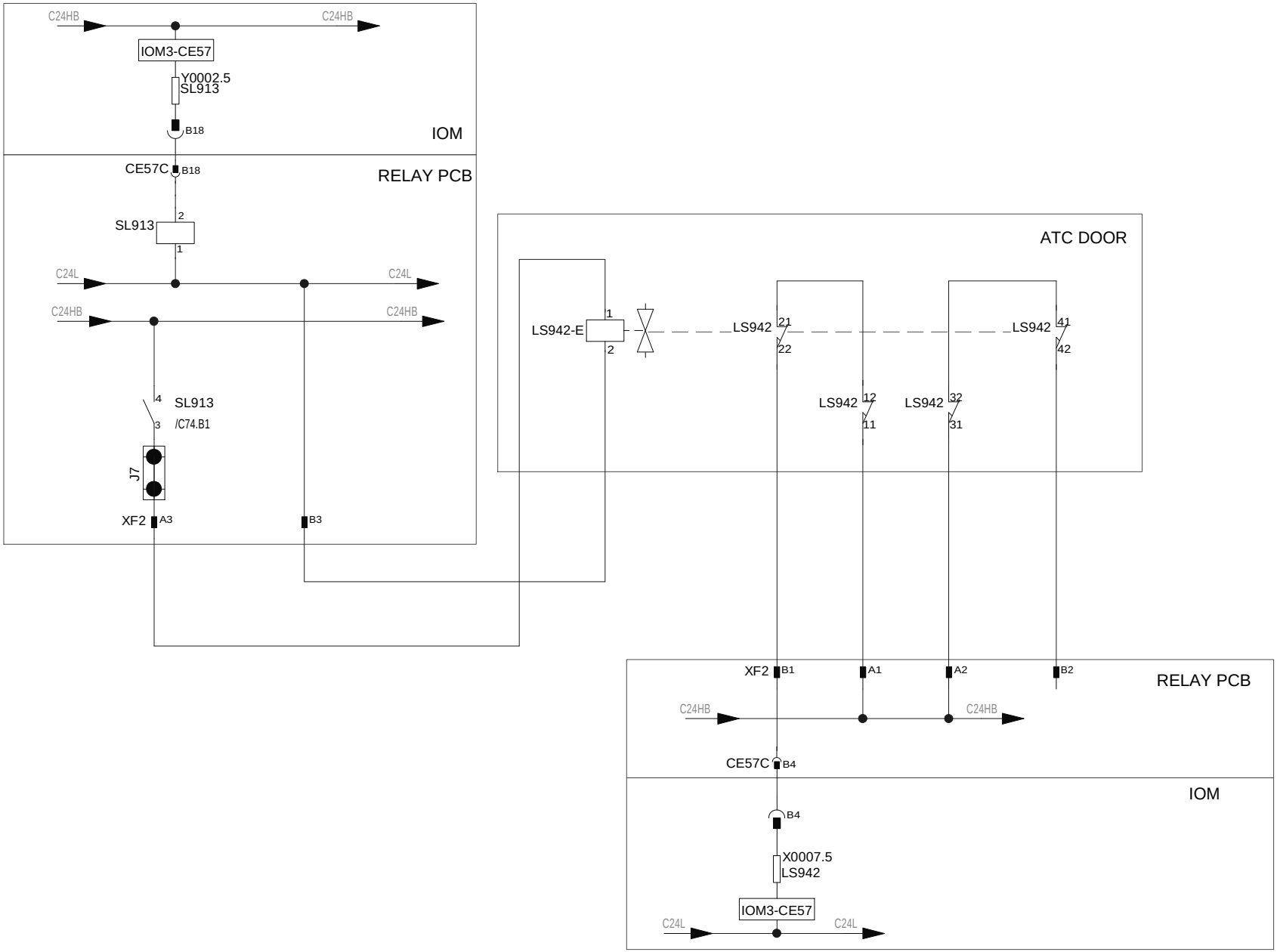



MAGAZINE DOOR CLOSE LS

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| ATC DOOR INTERLOCK | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C73 |



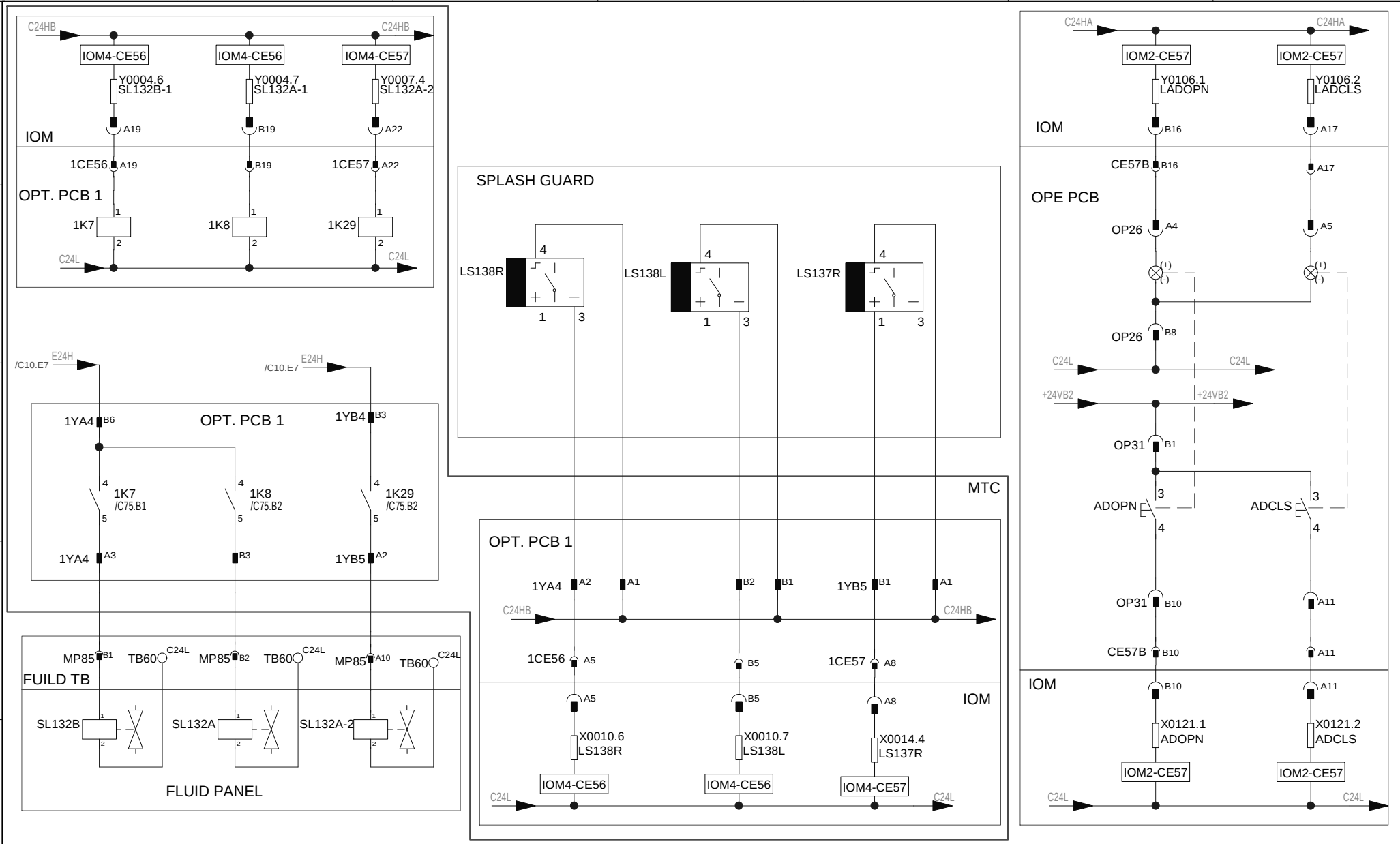

ATC DOOR OPEN

ATC DOOR INTERLOCK LS

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| ATC DOOR LOCK W INTERLOCK (OPT) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C74 |

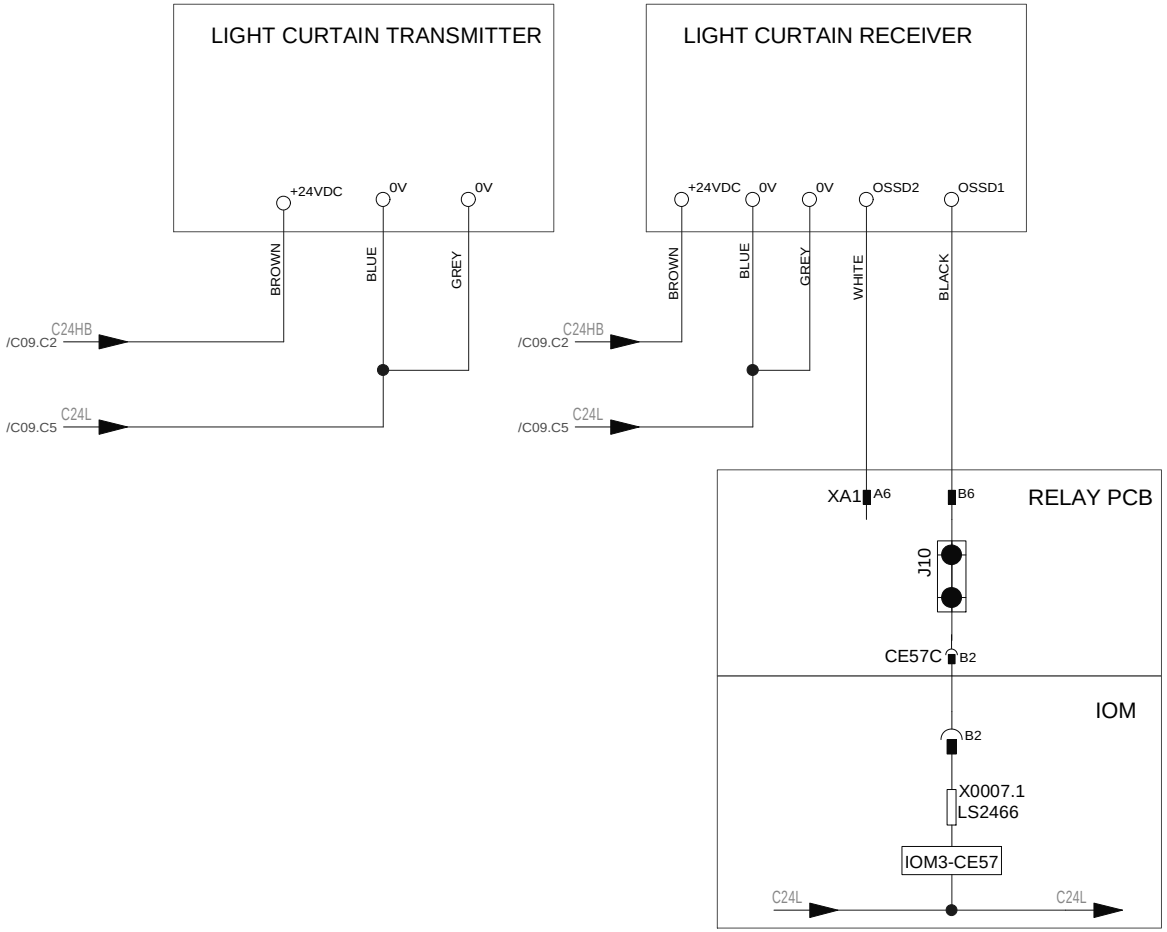



| AUTO DOOR OPEN | AUTO DOOR CLOSE | AUTO DOOR CLOSE SLOW DOWN | AUTO DOOR OPEN RIGHT | AUTO DOOR OPEN LEFT | AUTO DOOR CLOSE SLOW DOWN | DOOR OPEN PB | DOOR CLOSE PB |
|---|---|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| OPERATOR SIDE(S/G) AUTO DOOR (OPT) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C75 |

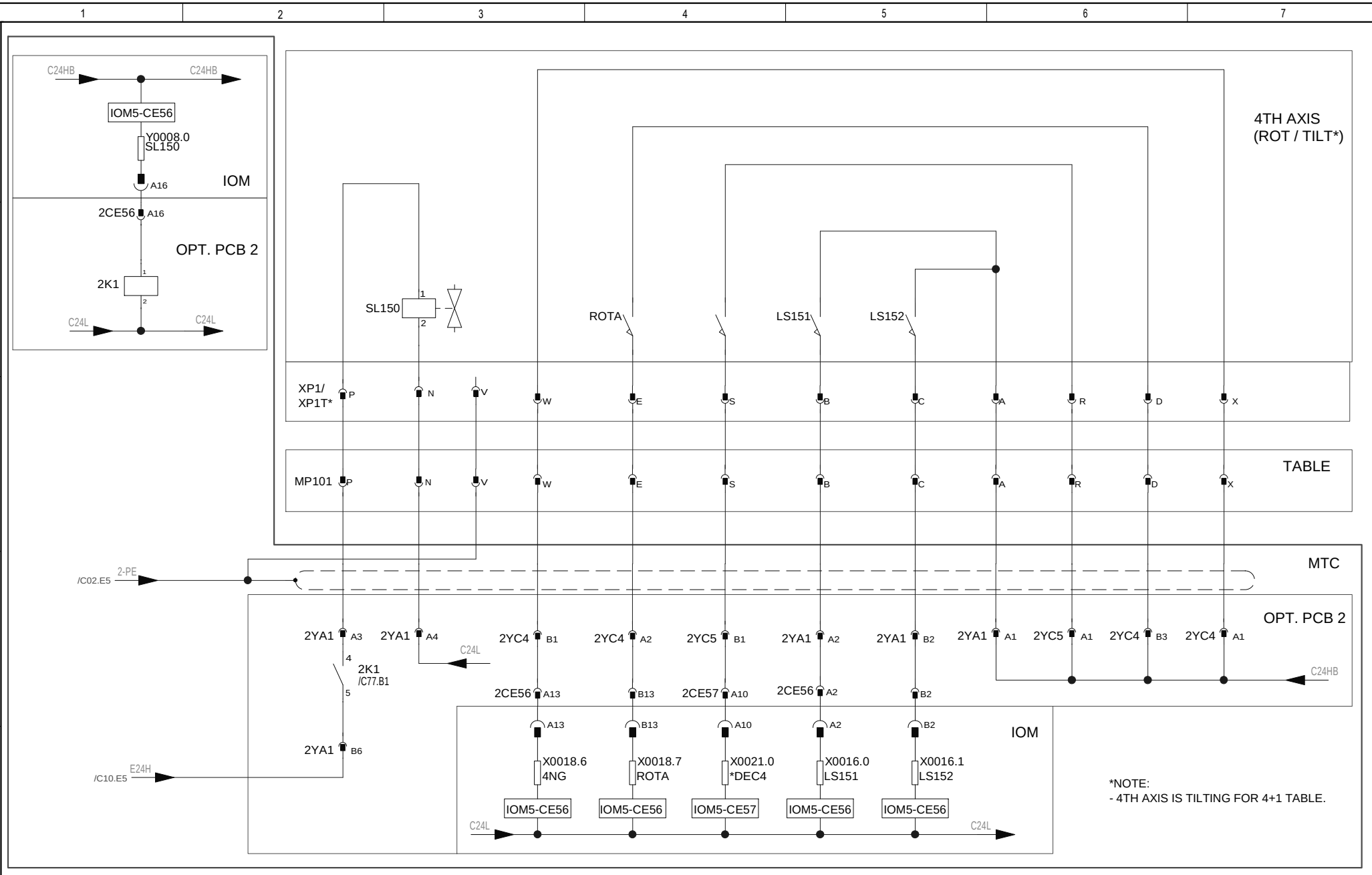



AREA SENSOR

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| LIGHT CURTAIN (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C76 |

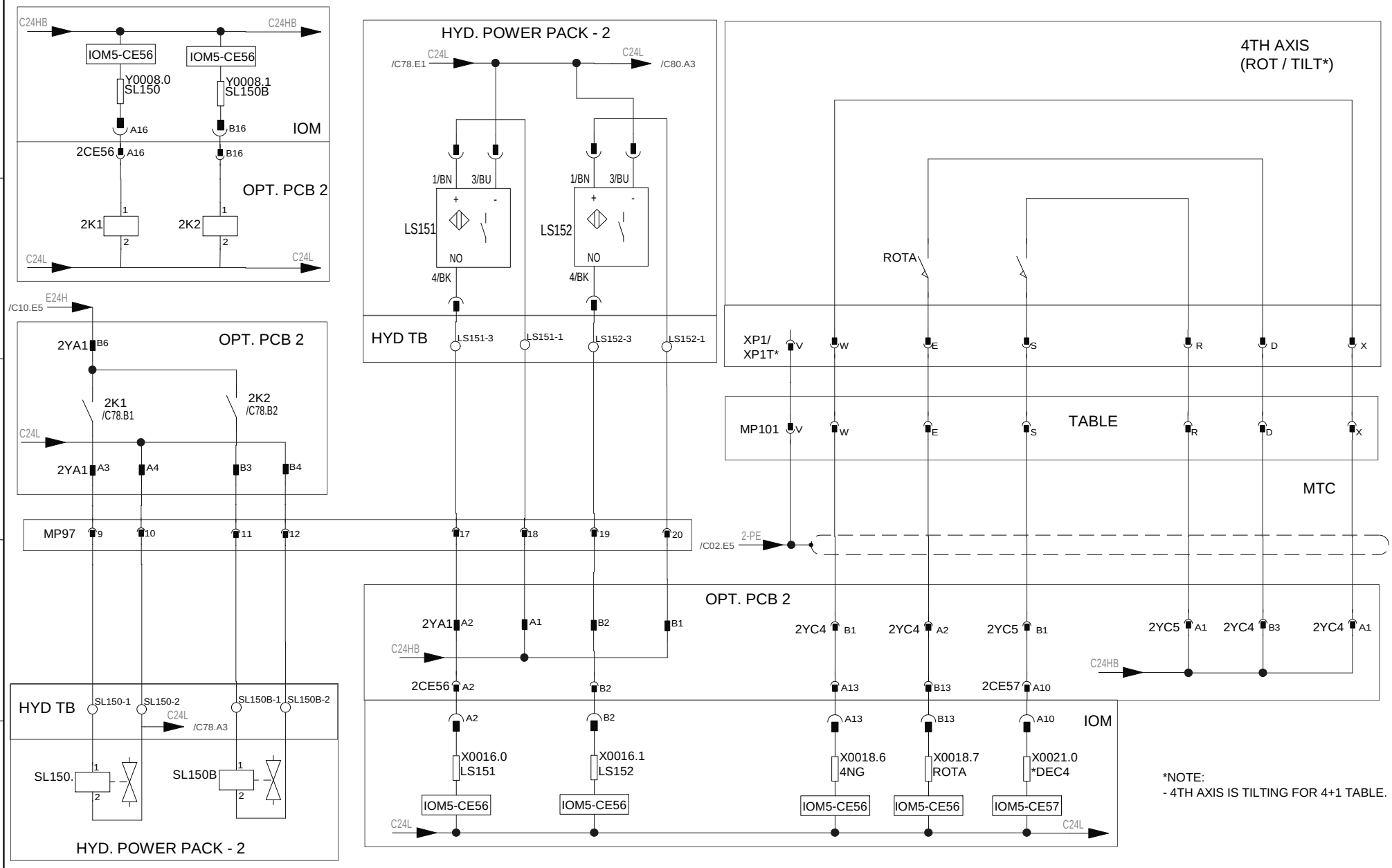



| 4TH AXIS UNCLAMP | 4TH AXIS NEGLECT | 4TH AXIS +OT ALARM | 4TH AXIS DECELERATION | 4TH AXIS CLAMP | 4TH AXIS UNCLAMP |
|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| 4TH AXIS CONTROL (PNEUMATIC) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C77 |

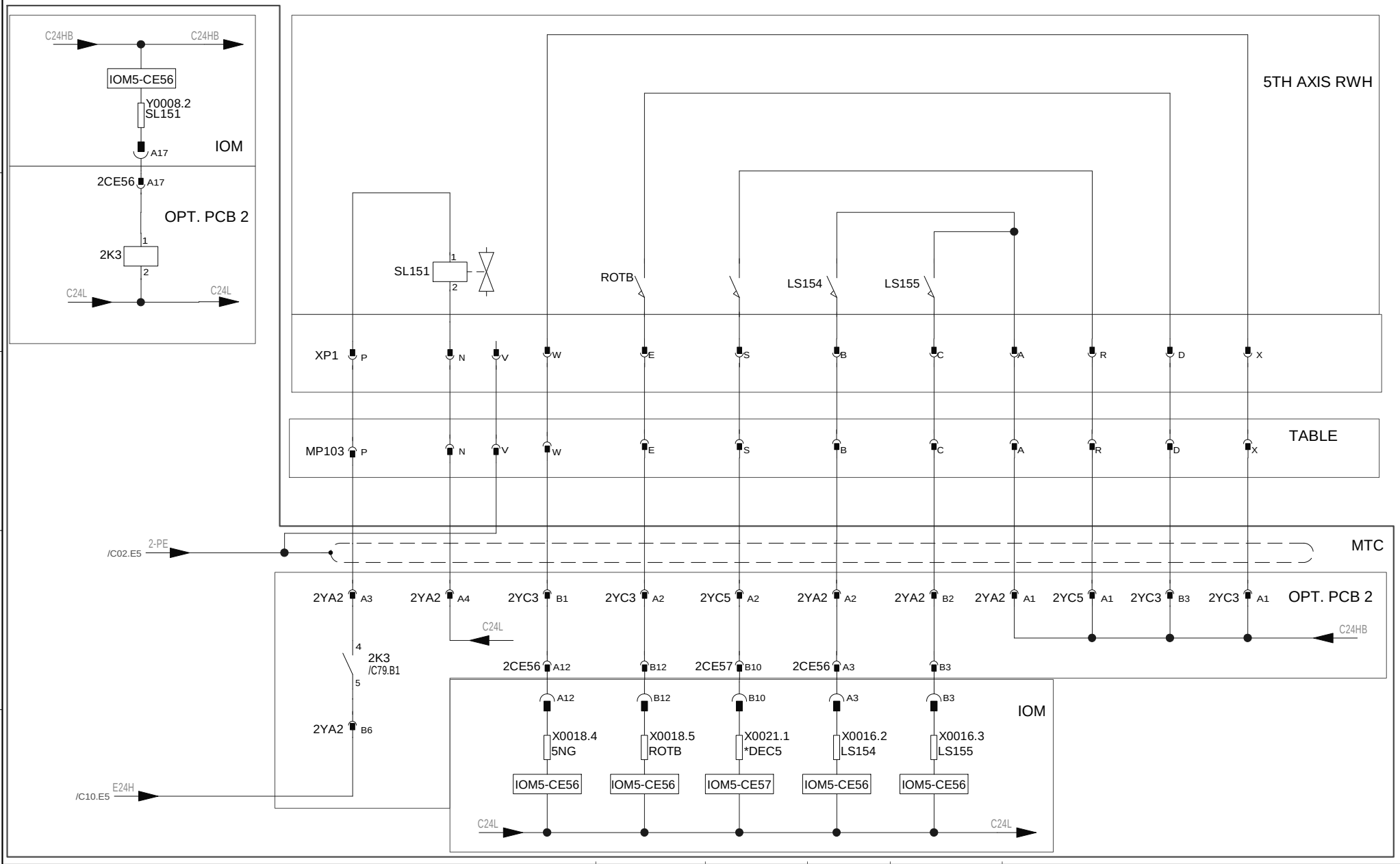



| 4TH AXIS UNCLAMP/ CLAMP | PRES SWITCH CLAMP (4th AXIS) | PRES SWITCH UNCLAMP (4th AXIS) | 4TH AXIS NEGLECT | 4TH AXIS +OT ALARM | 4TH AXIS DECELERATION |
|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| 4TH AXIS CONTROL (HYDRAULIC) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C78 |

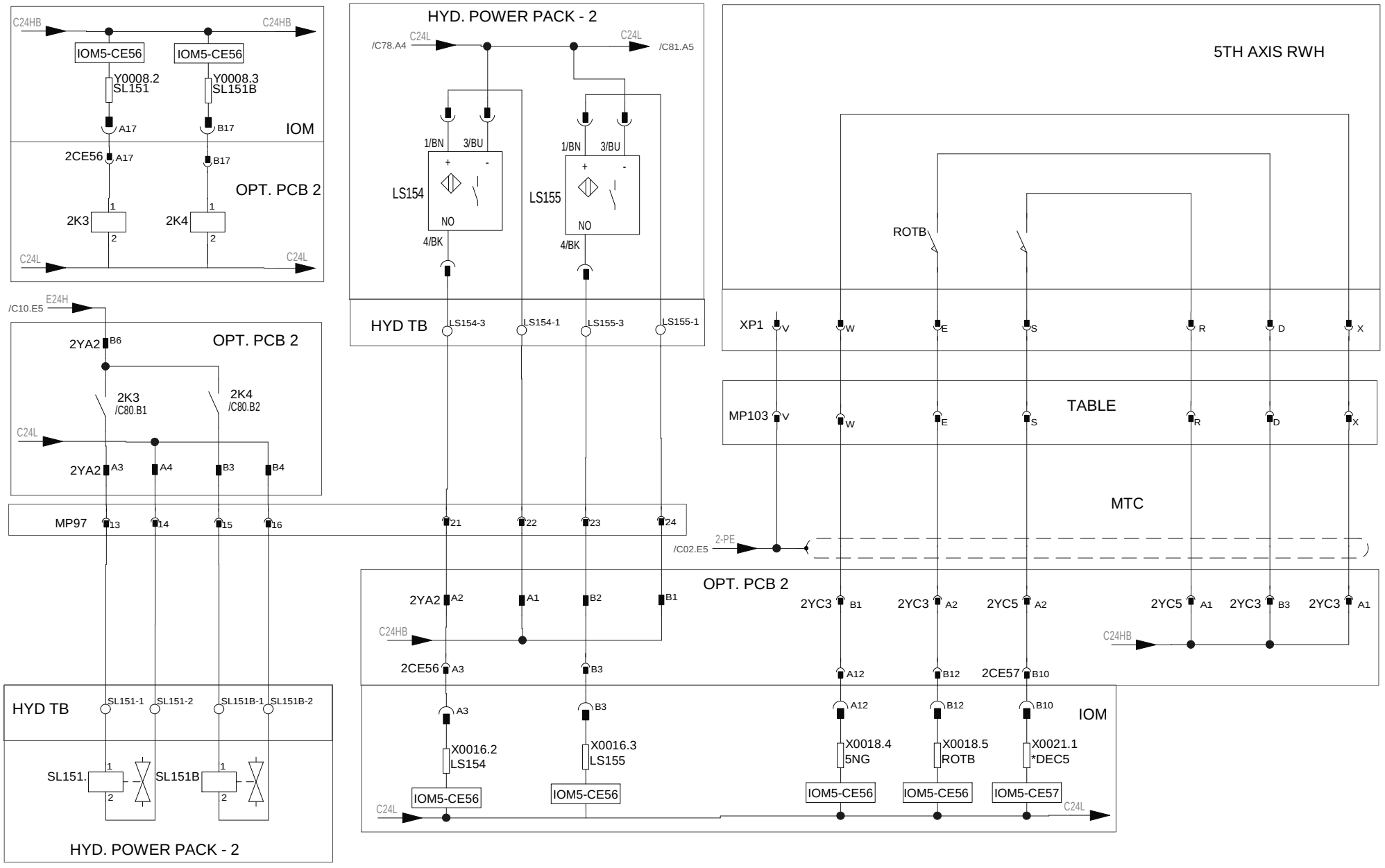



| 5TH AXIS UNCLAMP | 5TH AXIS NEGLECT | 5TH AXIS +OT ALARM | 5TH AXIS DECELERATION | 5TH AXIS CLAMP | 5TH AXIS UNCLAMP |
|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| 5TH AXIS CONTROL (PNEUMATIC) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C79 |

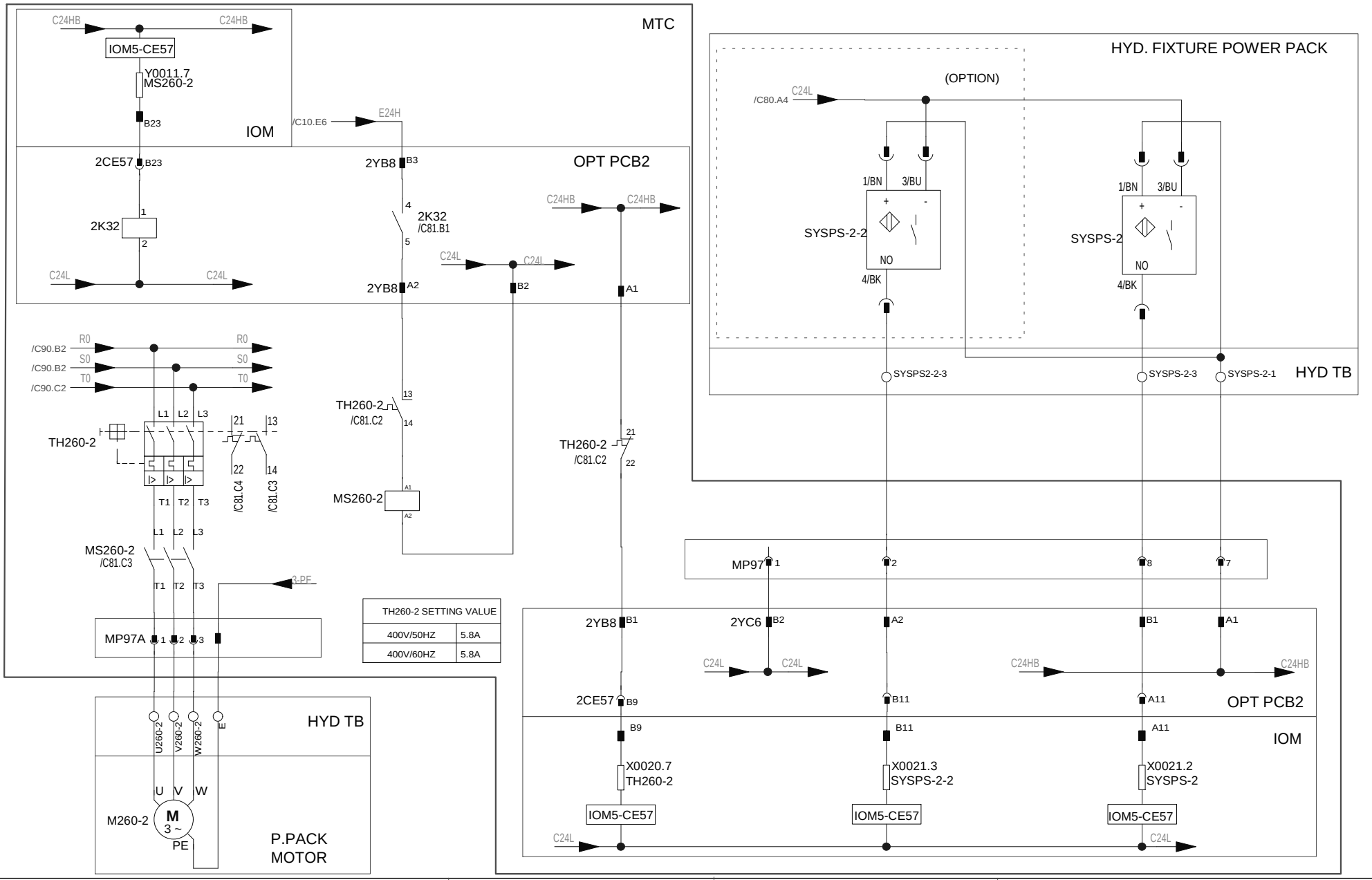



| 5TH AXIS UNCLAMP/ CLAMP | PRES SWITCH CLAMP (5th AXIS) | PRES SWITCH UNCLAMP (5th AXIS) | 5TH AXIS NEGLECT | 5TH AXIS +OT ALARM | 5TH AXIS DECELERATION |
|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| 5TH AXIS CONTROL (HYDRAULIC) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | C80 |

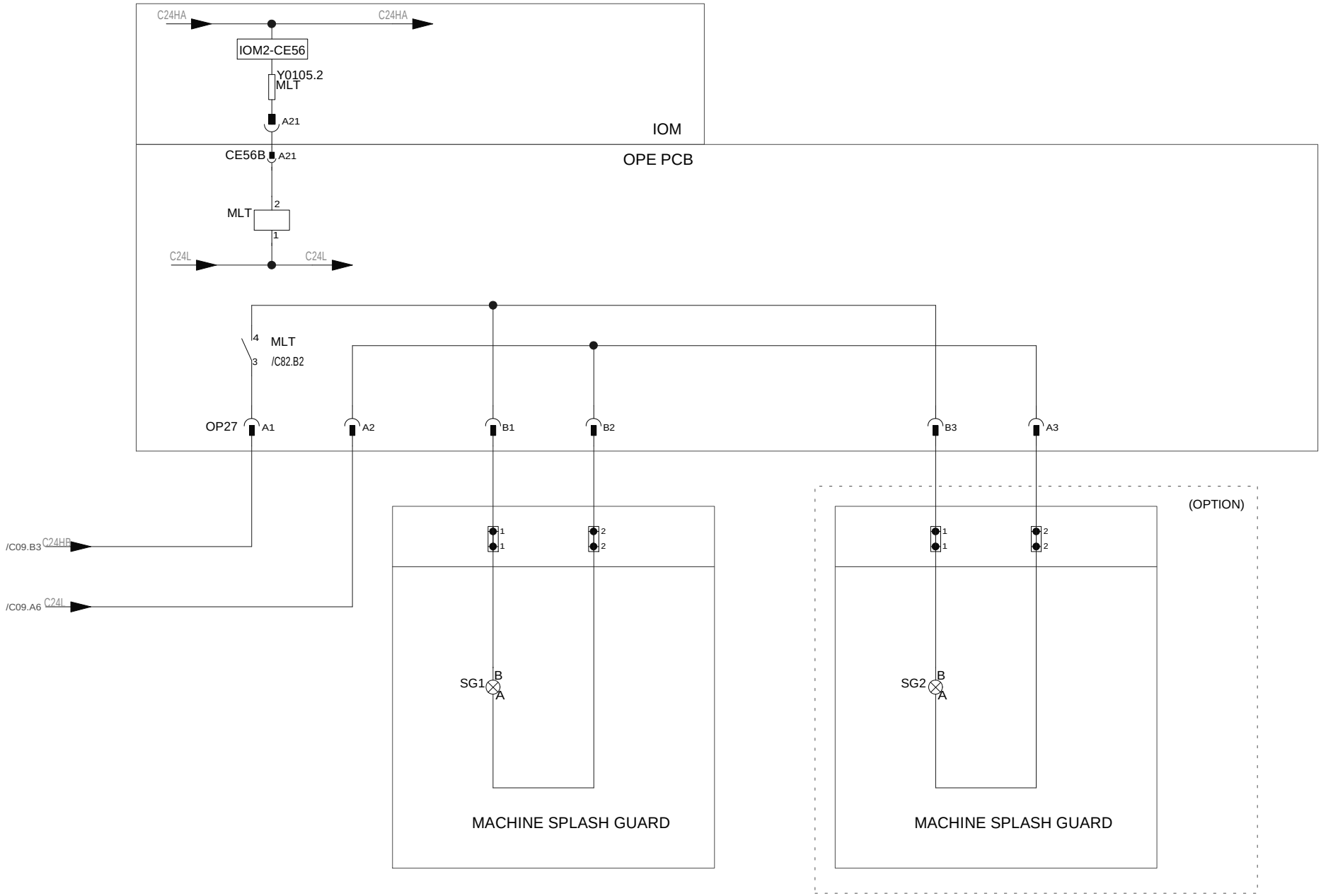



| HYDRAULIC PP ON | HYDRAULIC PP THERMAL | HYDRAULIC PP SYTEM PRESSURE SW (LOW) | HYDRAULIC PP SYTEM PRESSURE SW (HIGH) |
|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| RWH POWER PACK (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C81 |

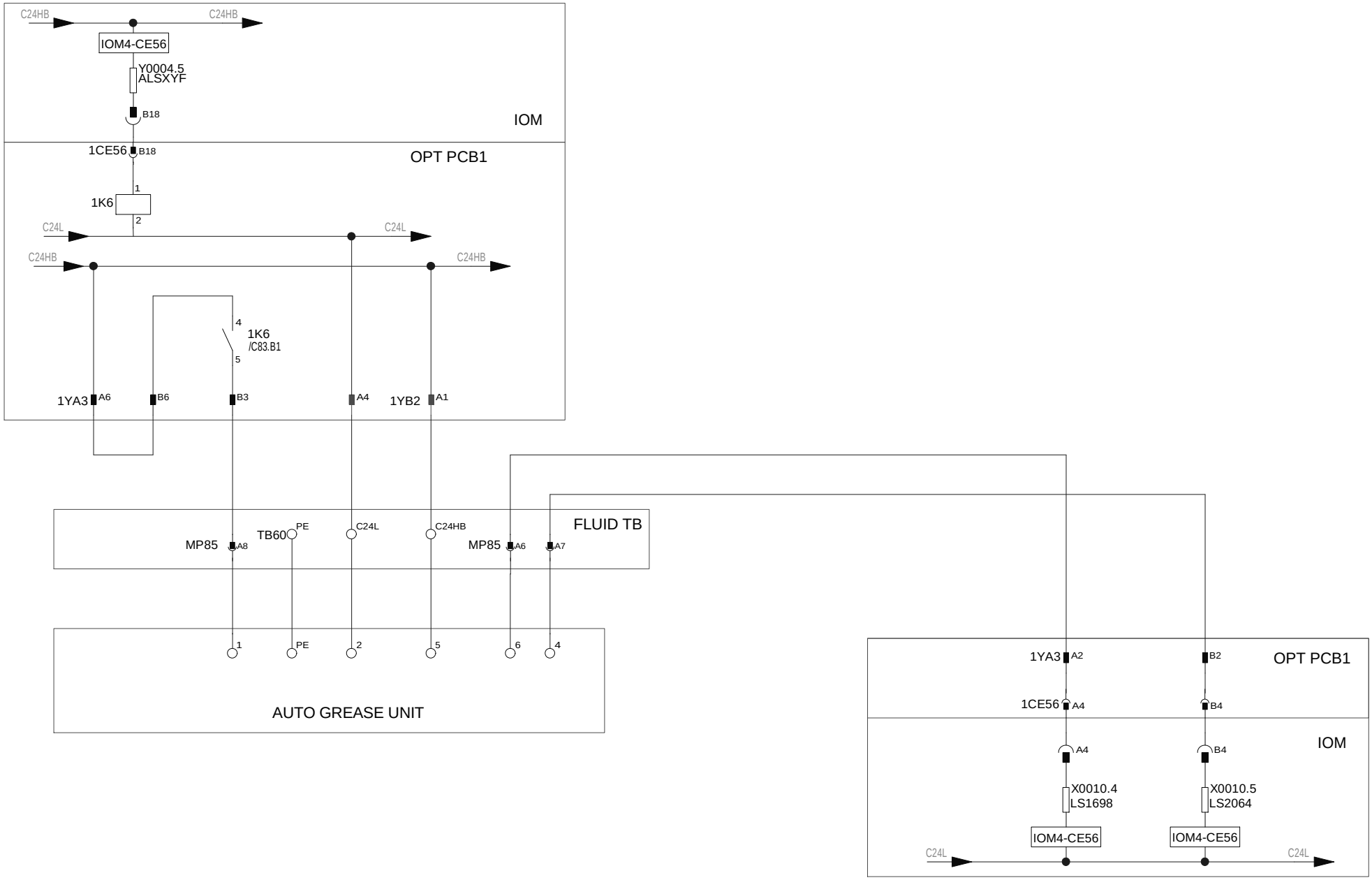



MACHINE LIGHT ON

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| MACHINE LIGHT | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C82 |




AUTO GREASE INJECTOR CW

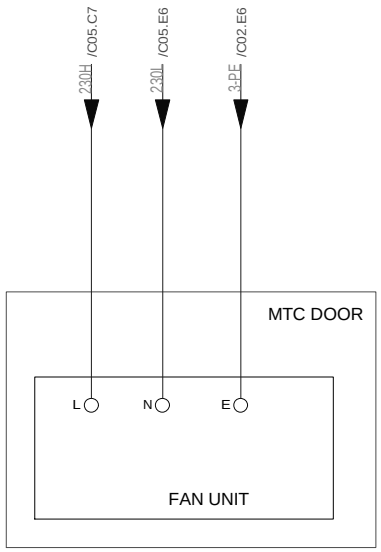
AUTO GREASE INJECTOR PRESSURE UP

SUPPLY UNIT LEVEL SW

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

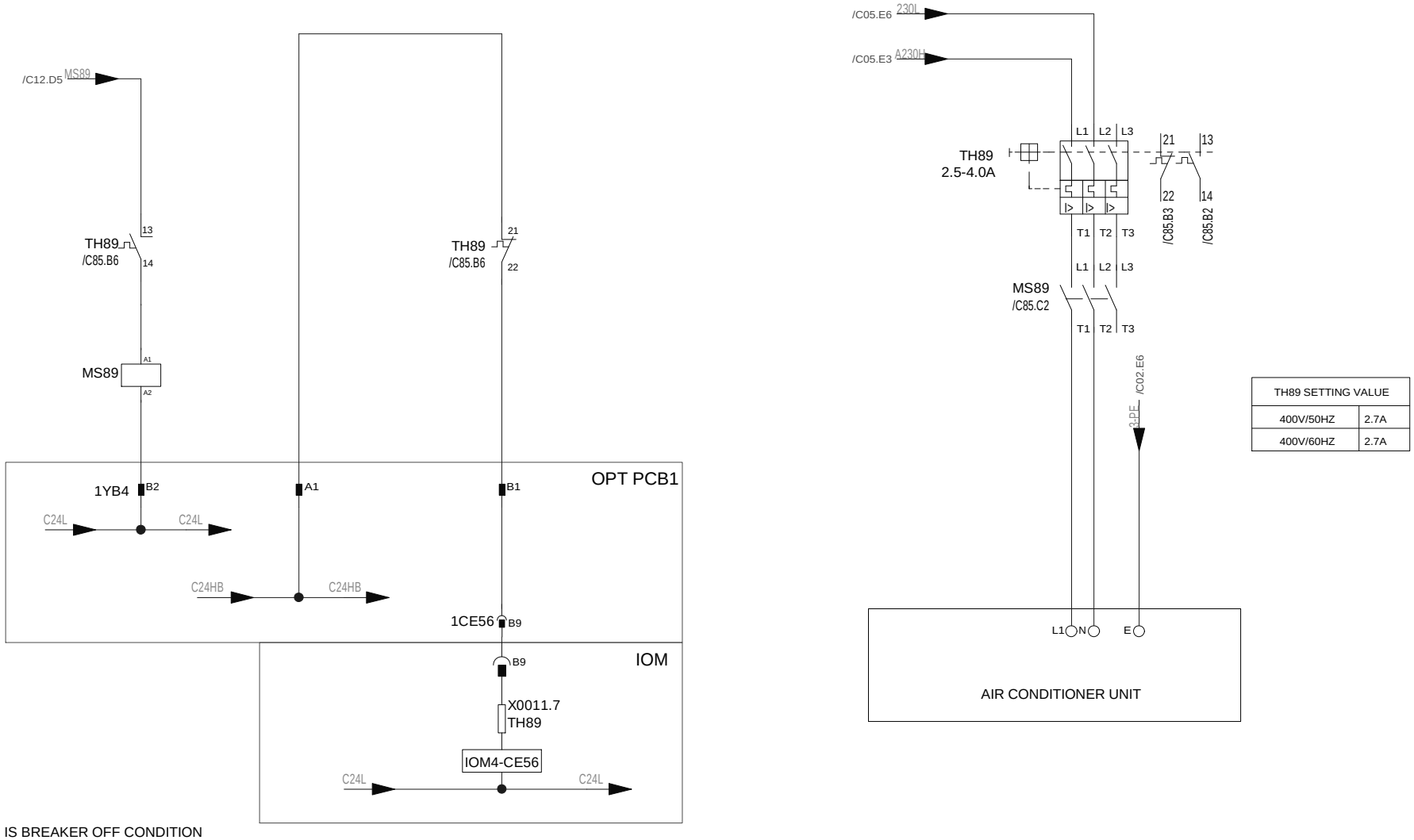
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| AUTO GREASE | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C83 |




| REVISIONS | | | | |
|---|---|---|---|---|
| ④ | | | DATE | NAME |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| MTC FAN | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C84 |




| TH89 SETTING VALUE | |
|---|---|
| 400V/50HZ | 2.7A |
| 400V/60HZ | 2.7A |

*NOTE:
TH STATUS SHOWN IS BREAKER OFF CONDITION

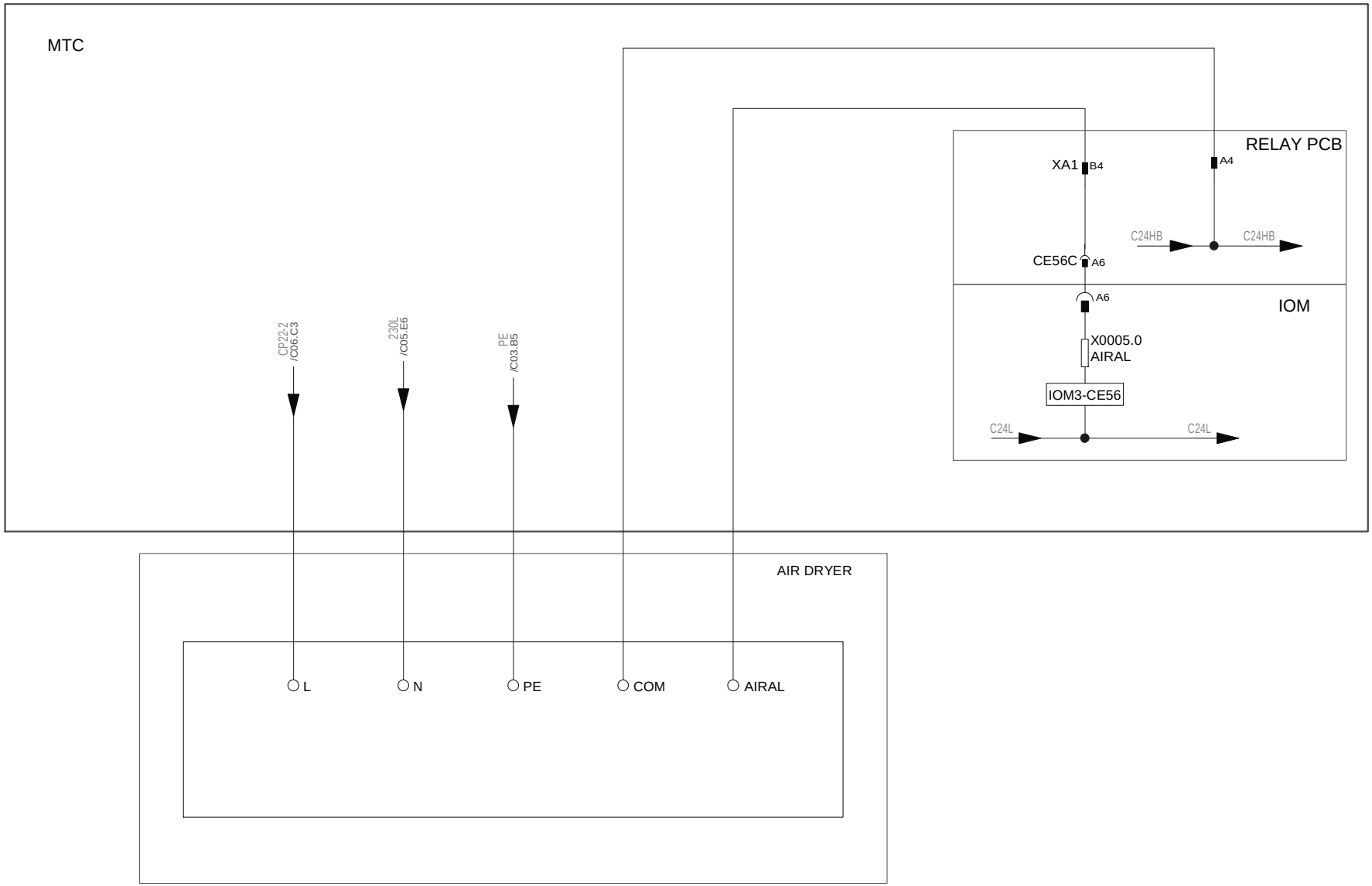
AIR CON CONTROL

AIR CON THERMAL TRIP ALARM

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

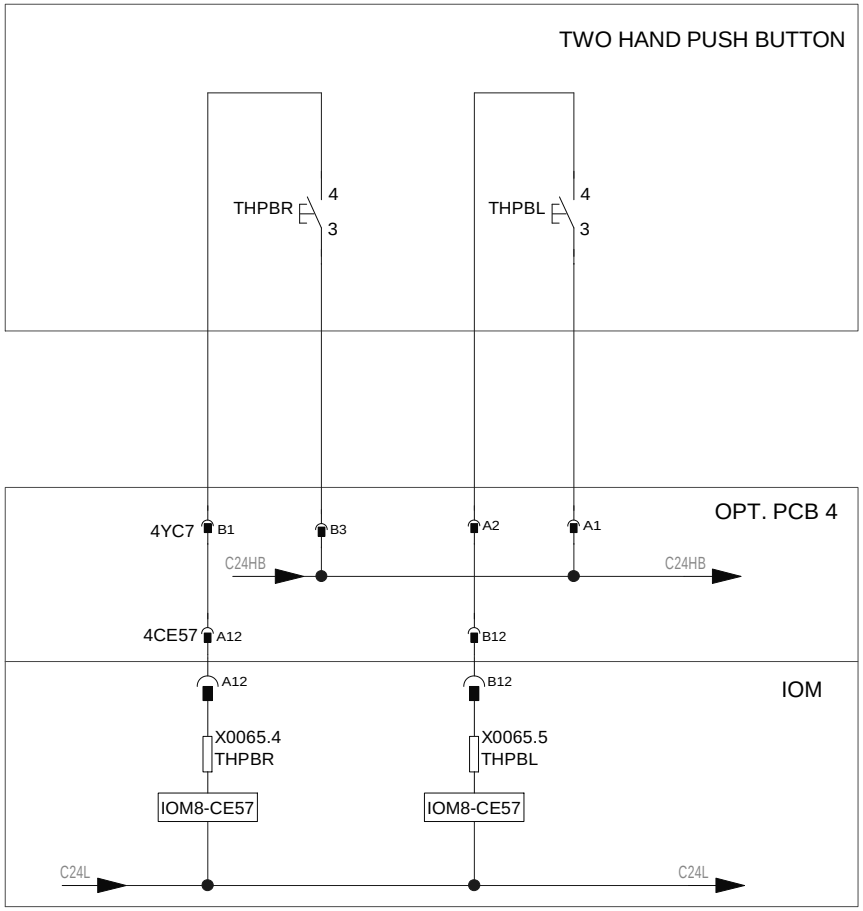
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| MTC AIR CONDITIONER (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | C85 |




| REVISIONS | | | | |
|---|---|---|---|---|
| ④ | | | DATE | NAME |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| AIR DRYER (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C86 |

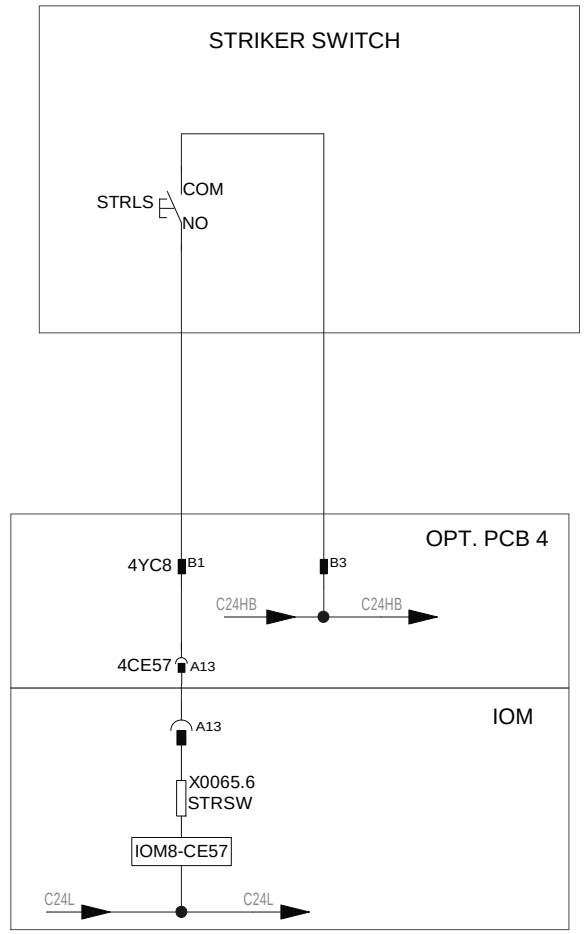



TWO HAND PUSHBOTTON RIGHT | TWO HAND PUSHBOTTON LEFT

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| TWO HAND PUSH BUTTON (OPT) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C87 |

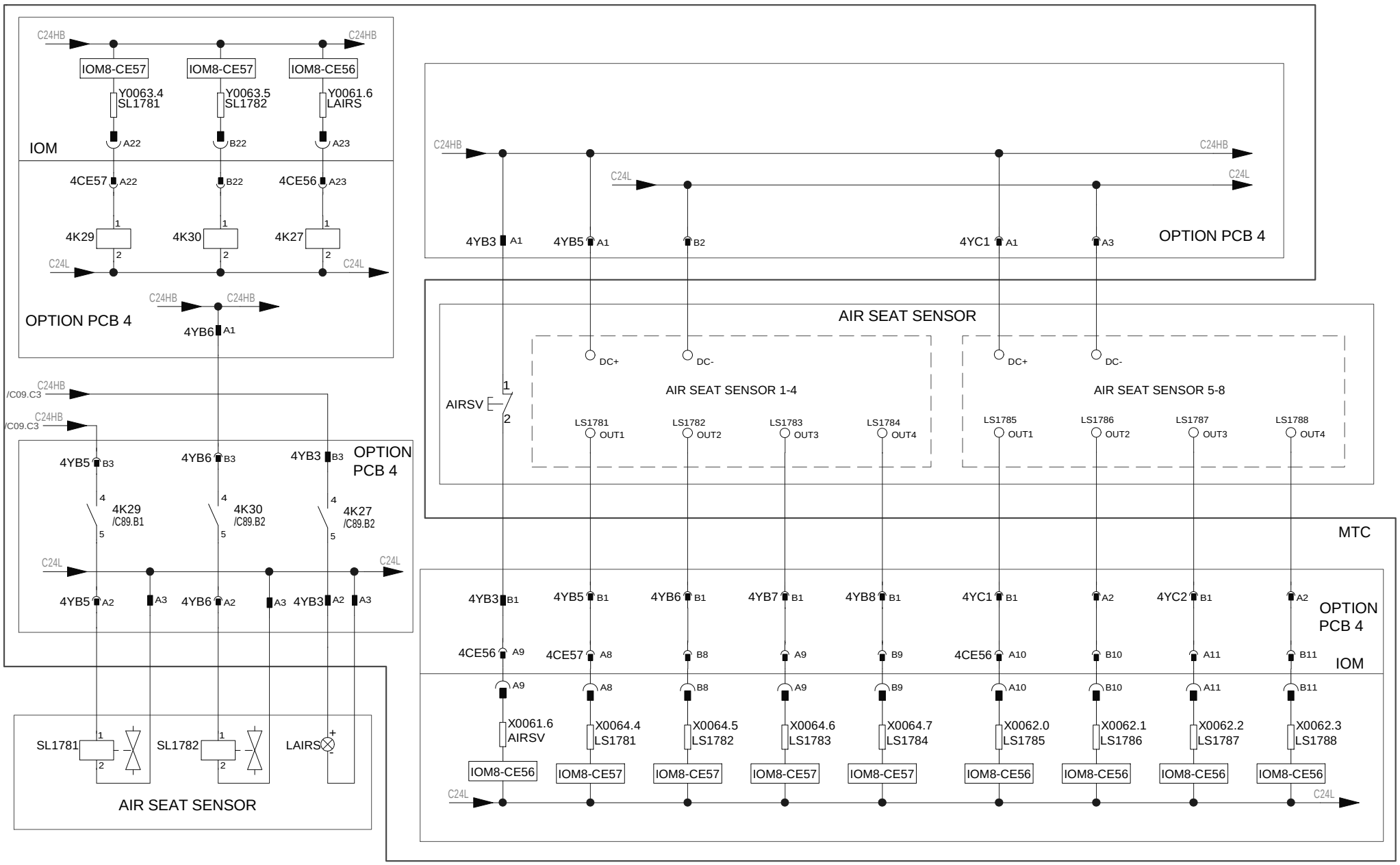



STRIKER SWITCH

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| STRIKER SWITCH (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C88 |

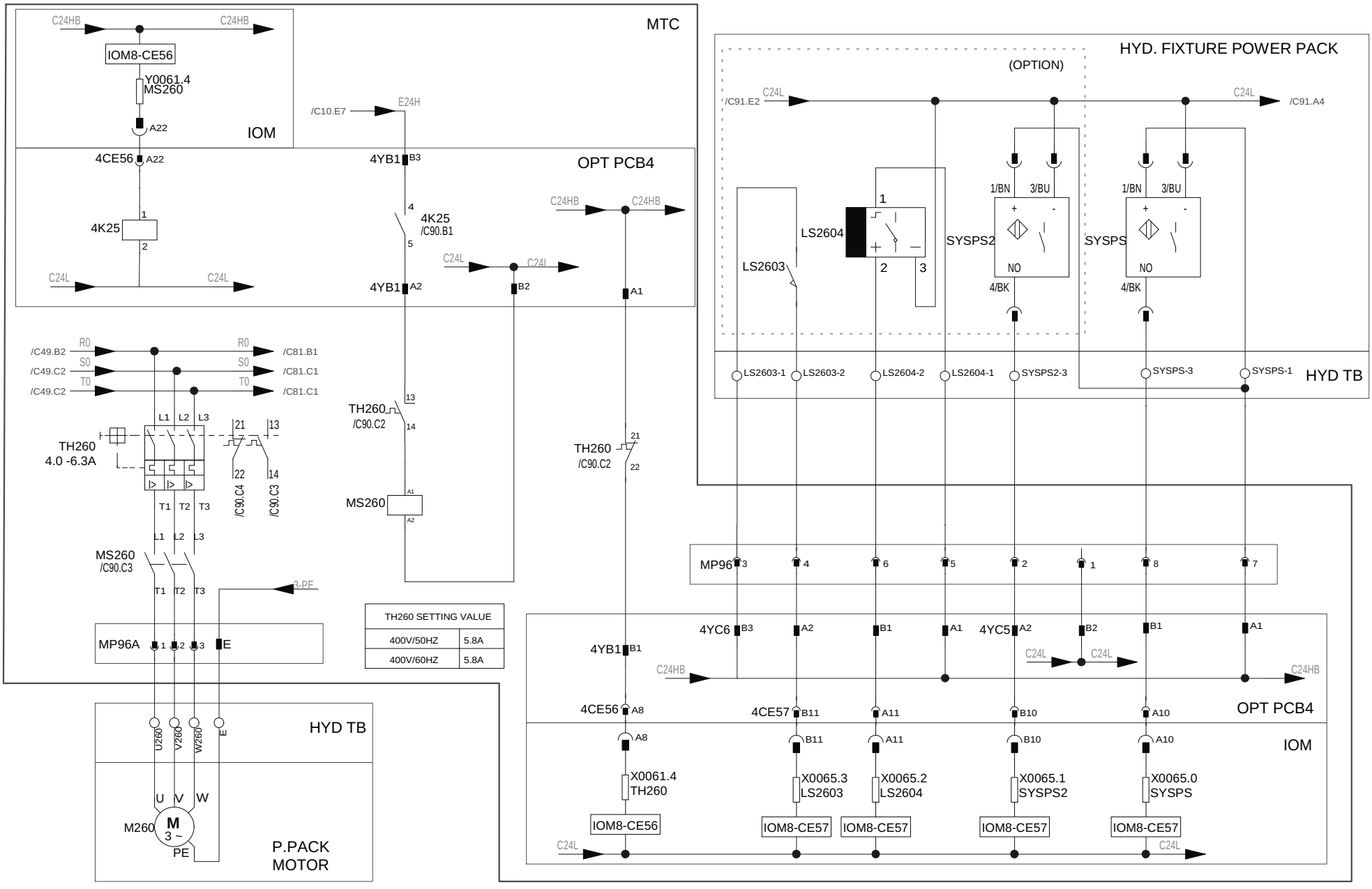



| AIR SEAT SOLENOID 1 | AIR SEAT SOLENOID 2 | AIR SEAT CHECK LAMP | | AIR SEAT CHECK SW | AIR SEAT SENSOR 1 | AIR SEAT SENSOR 2 | AIR SEAT SENSOR 3 | AIR SEAT SENSOR 4 | AIR SEAT SENSOR 5 | AIR SEAT SENSOR 6 | AIR SEAT SENSOR 7 | AIR SEAT SENSOR 8 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| AIR SEAT CHECK (1-8 SENSOR) (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C89 |



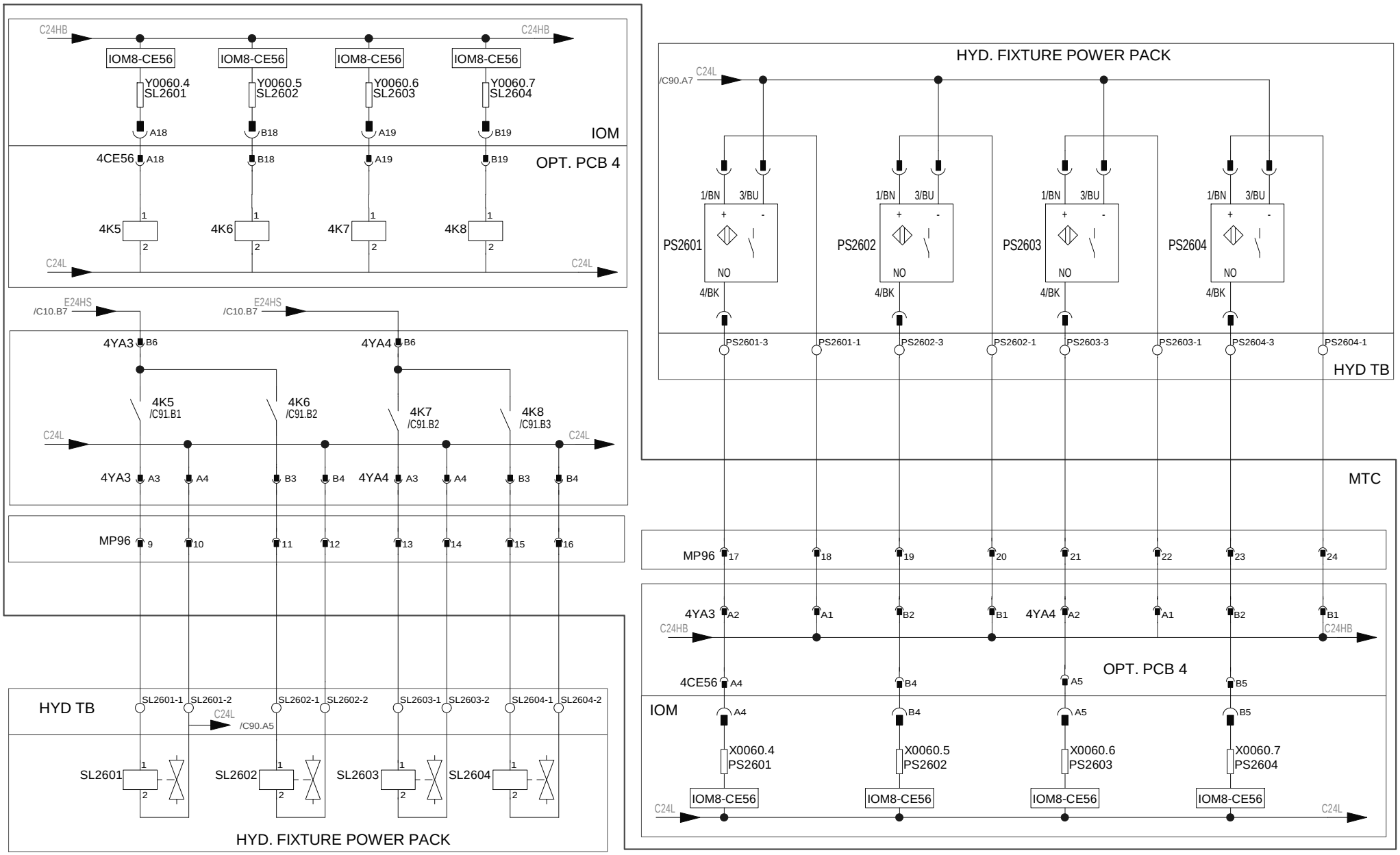

HYDRAULIC PP ON

| HYDRAULIC PP THERMAL | HYDRAULIC OIL LEVEL FLOAT | HYDRAULIC OIL TEMP SENS | HYDRAULIC PP SYTEM PRESSURE SW (LOW) | HYDRAULIC PP SYTEM PRESSURE SW (HIGH) |
|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| HYDRAULIC FIXTURE (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C90 |

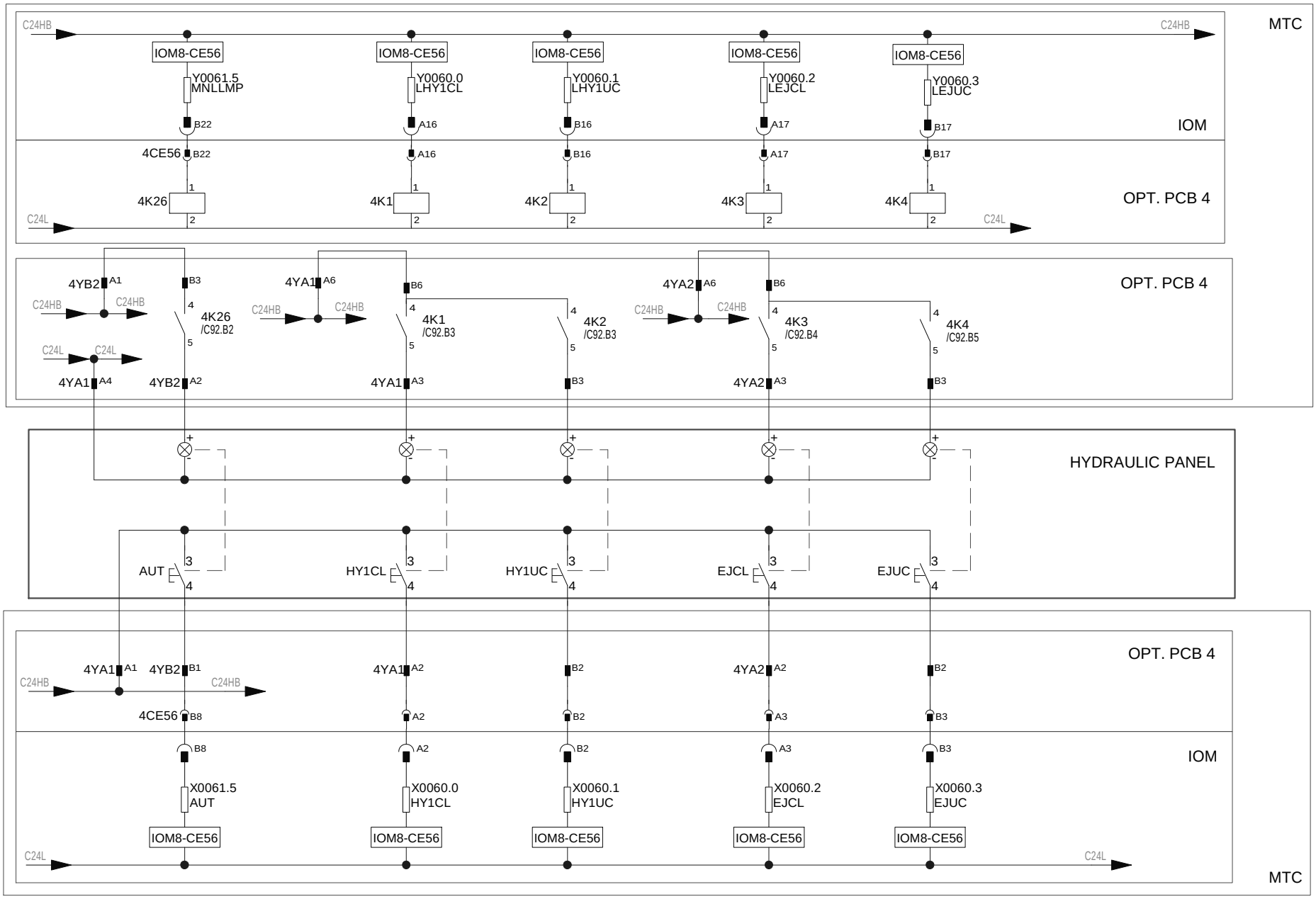



| FIXTURE SOL CLAMP 1 | FIXTURE SOL UNCLAMP 1 | FIXTURE SOL CLAMP 2 | FIXTURE SOL UNCLAMP 2 | PRE SWITCH CLAMP 1 | PRE SWITCH UNCLAMP 1 | PRE SWITCH CLAMP 2 | PRE SWITCH UNCLAMP 2 |
|---|---|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| HYDRAULIC FIXTURE SOL/PS (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C91 |

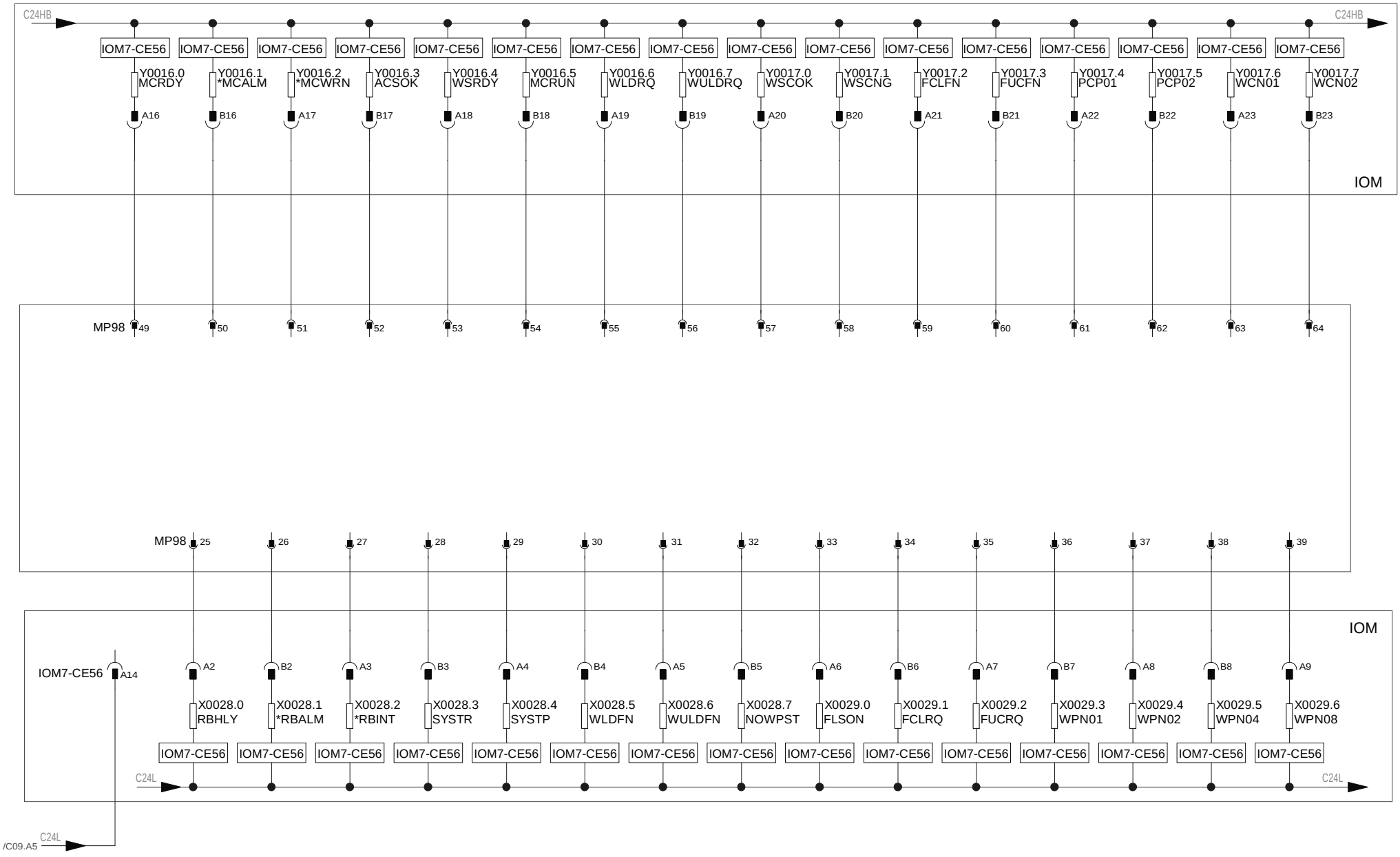



| AUTO MODE SELECTION | HYD CLAMP PB/LAMP | HYD UNCLAMP PB/LAMP | EJEC CLAMP PB/LAMP | EJEC UNCLAMP PB/LAMP |
|---|---|---|---|---|

| REVISIONS | | DATE | NAME |
|---|---|---|---|
| ④ | | | |
| ③ | APPROVED | 13/06/2019 | AVINASH |
| ② | CHECKED | 13/06/2019 | AVINASH |
| ① | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

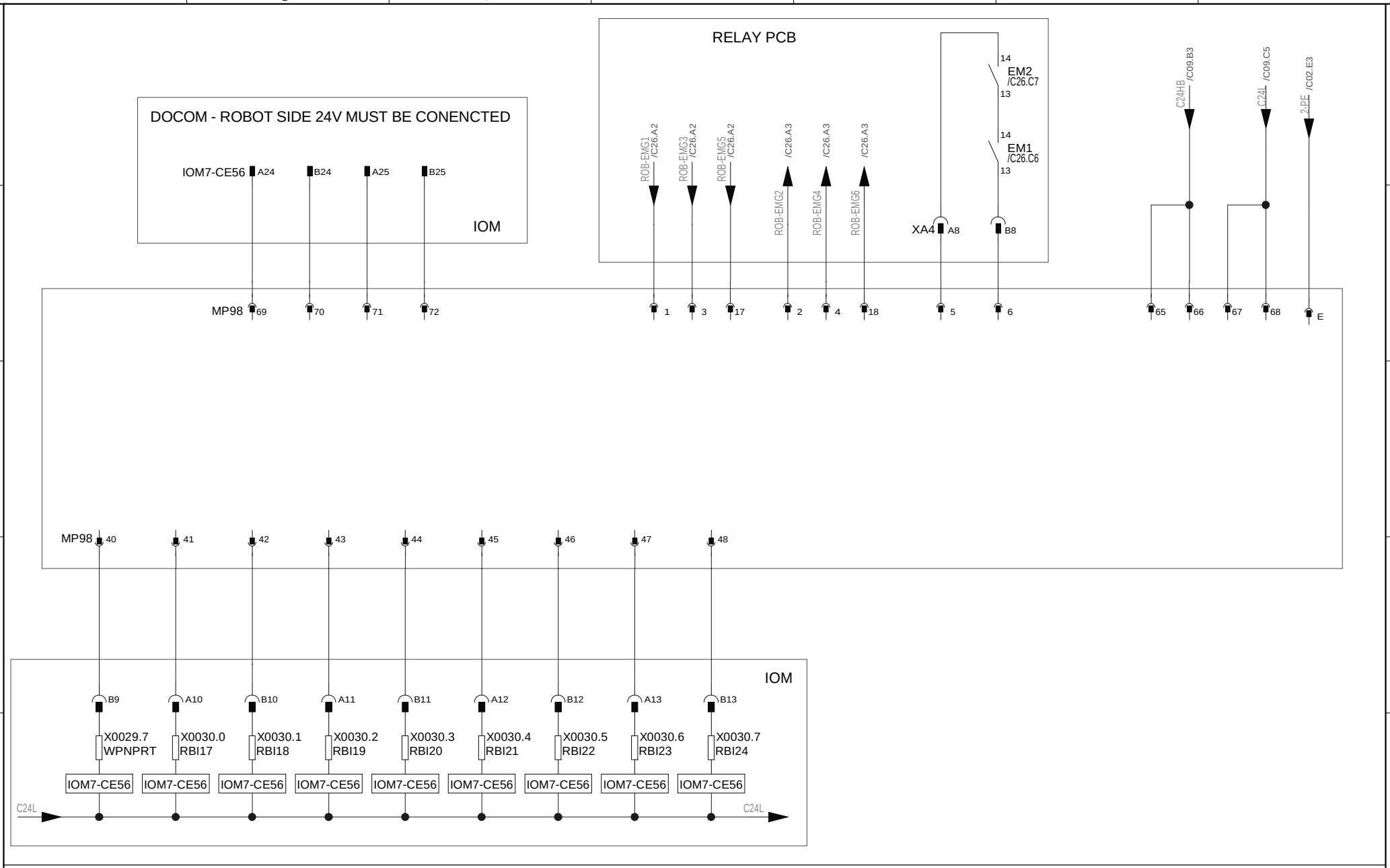
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| HYDRAULIC PANEL (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | C92 |




| REVISIONS | | | | |
|---|---|---|---|---|
| ④ | | | DATE | NAME |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

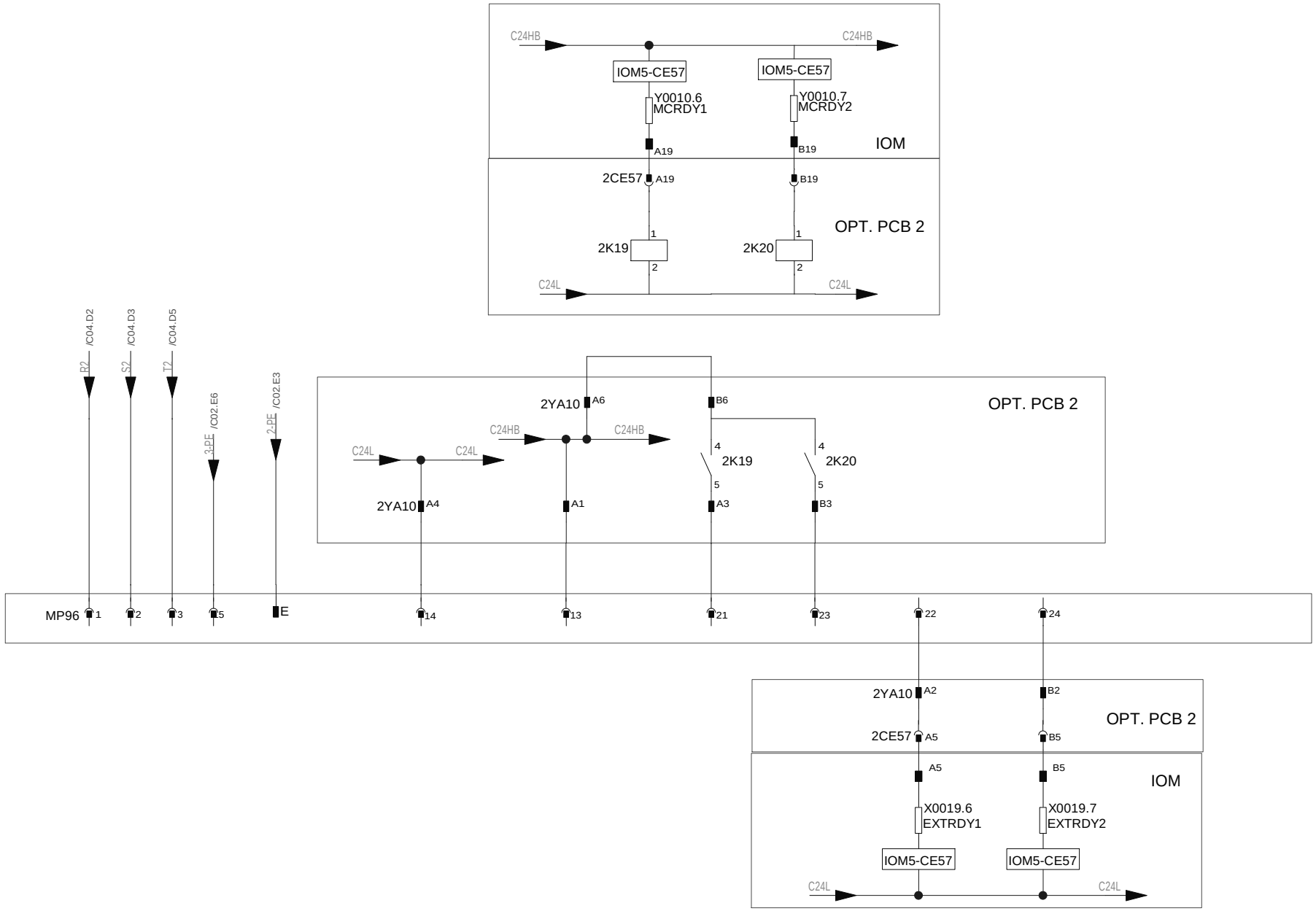
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| ROBOT INTERFACE (OPT) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C93 |




| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

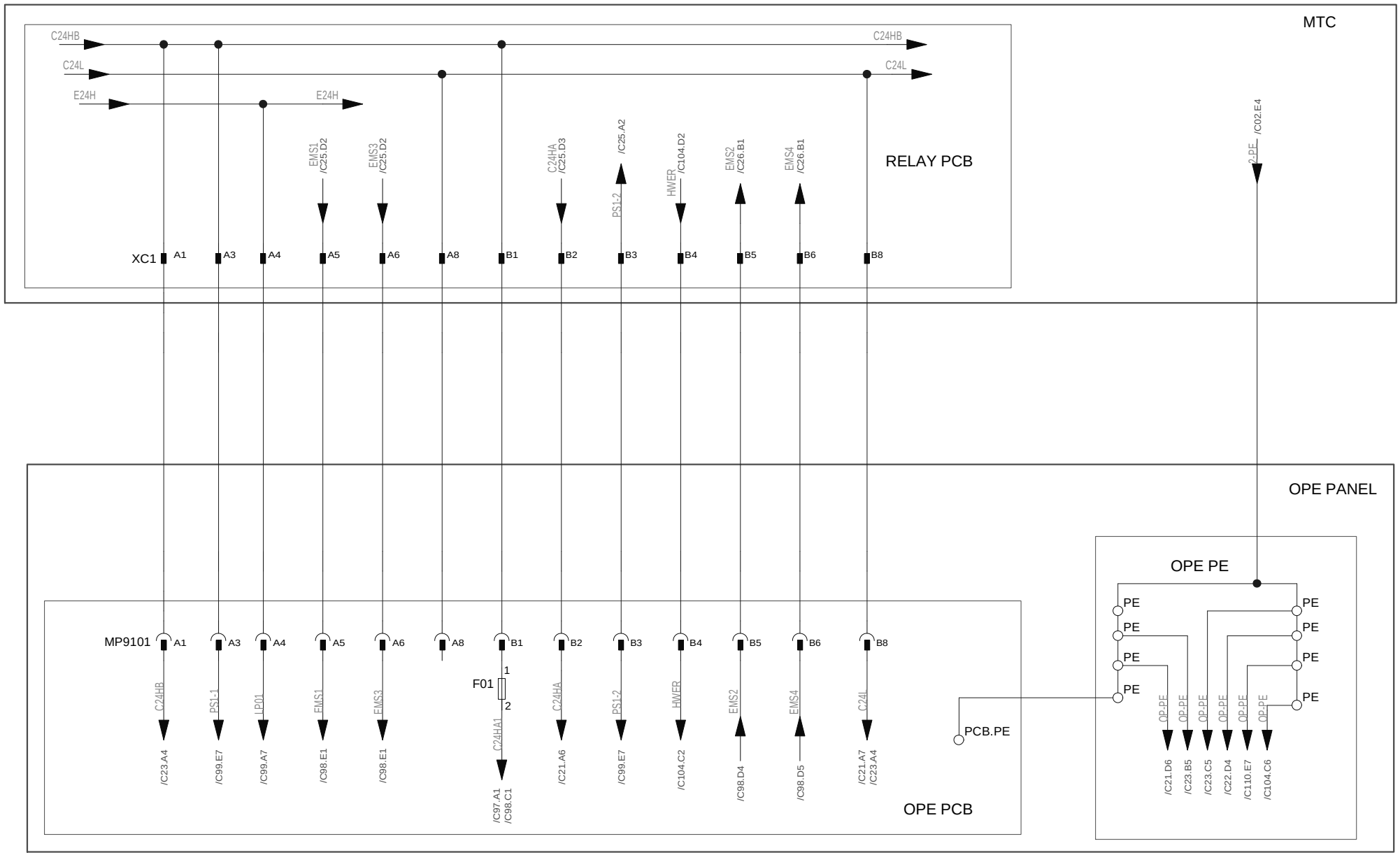
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| ROBOT INTERFACE (OPT) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C94 |




| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| EXTERNAL HYDRAULIC PUMP I/F(OPT) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C95 |




| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

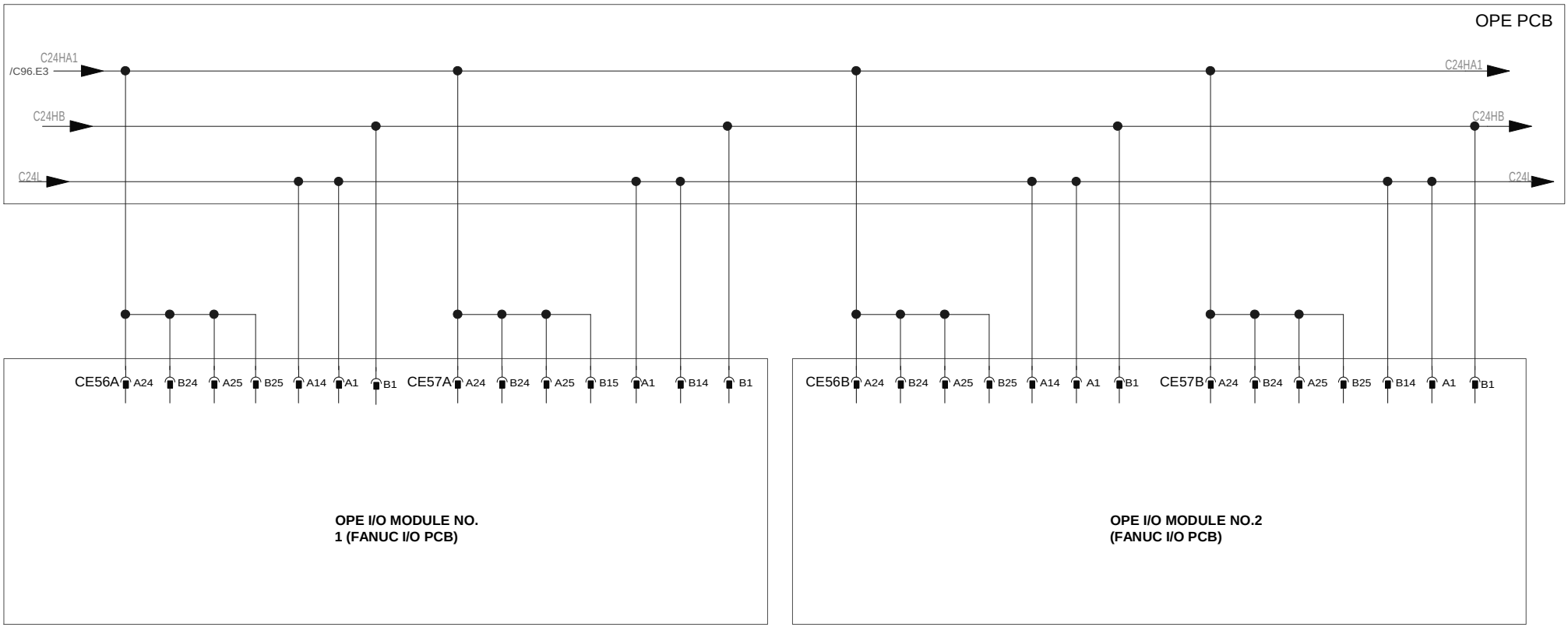
MAKINO
MAKINO ASIA PTE LTD

**TITLE**

MAIN RELAY PCB - OPERATOR PANEL PCB INTERFACE

| MACHINE TYPE | PS65/PS105 |
|---|---|
| NC TYPE | 0i-MF |

| DRAWING No. | GROUP |
|---|---|
| V210I-C1E-XG | = |
| | SHEET No. |
| | C96 |

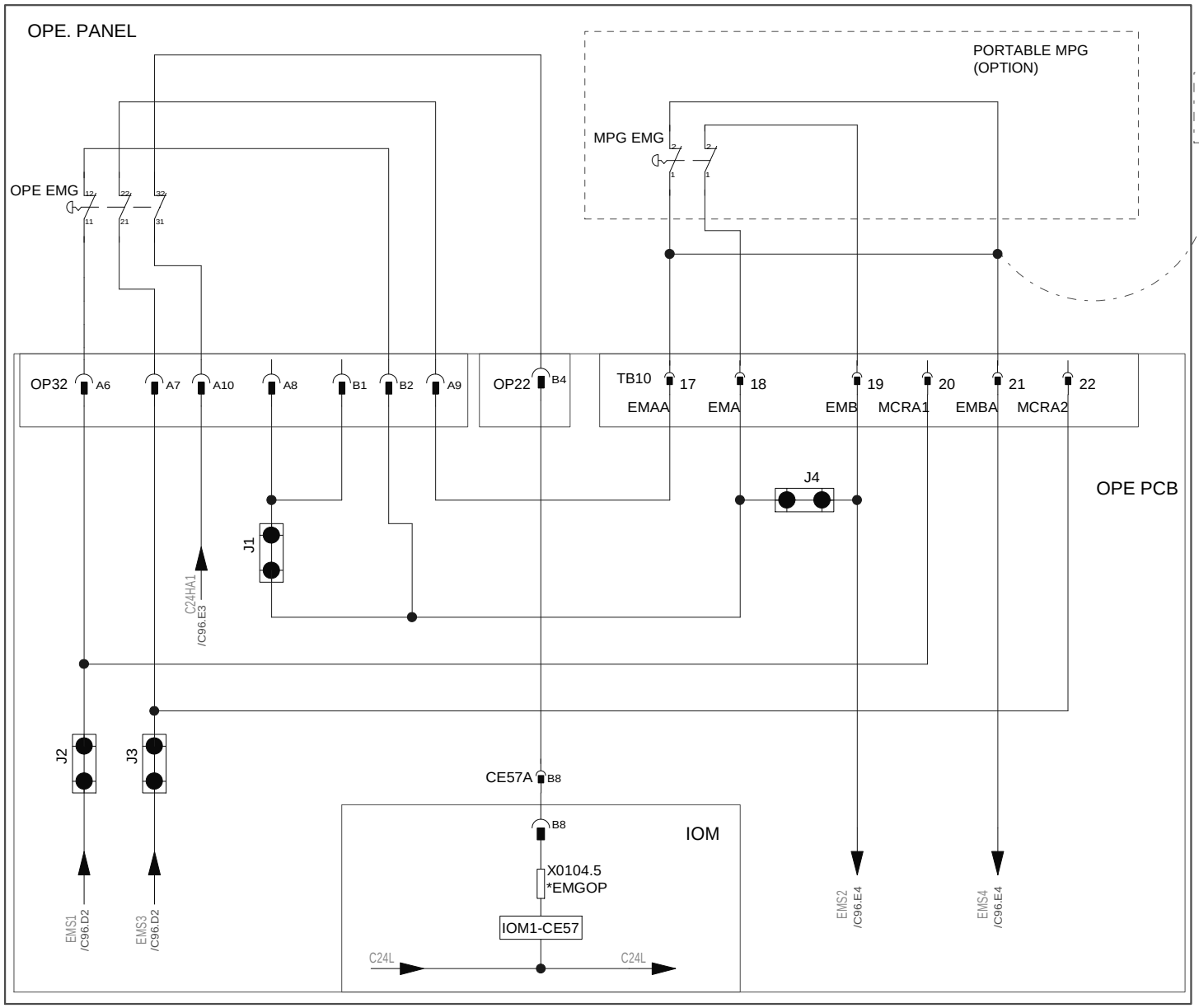



OPEARATOR I/O MODULE POWER

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| 24V FOR OPE PCB & I/O MODULE | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C97 |




NOTE:
FOR M/C WITH PORTALE MPG,
REMOVE JUMPER TB10-17 ~ TB10-21

FOR STANDARD
J1 = IGNORE
J2 = SHORT
J3 = SHORT
J4 = SHORT
J5 = OPEN

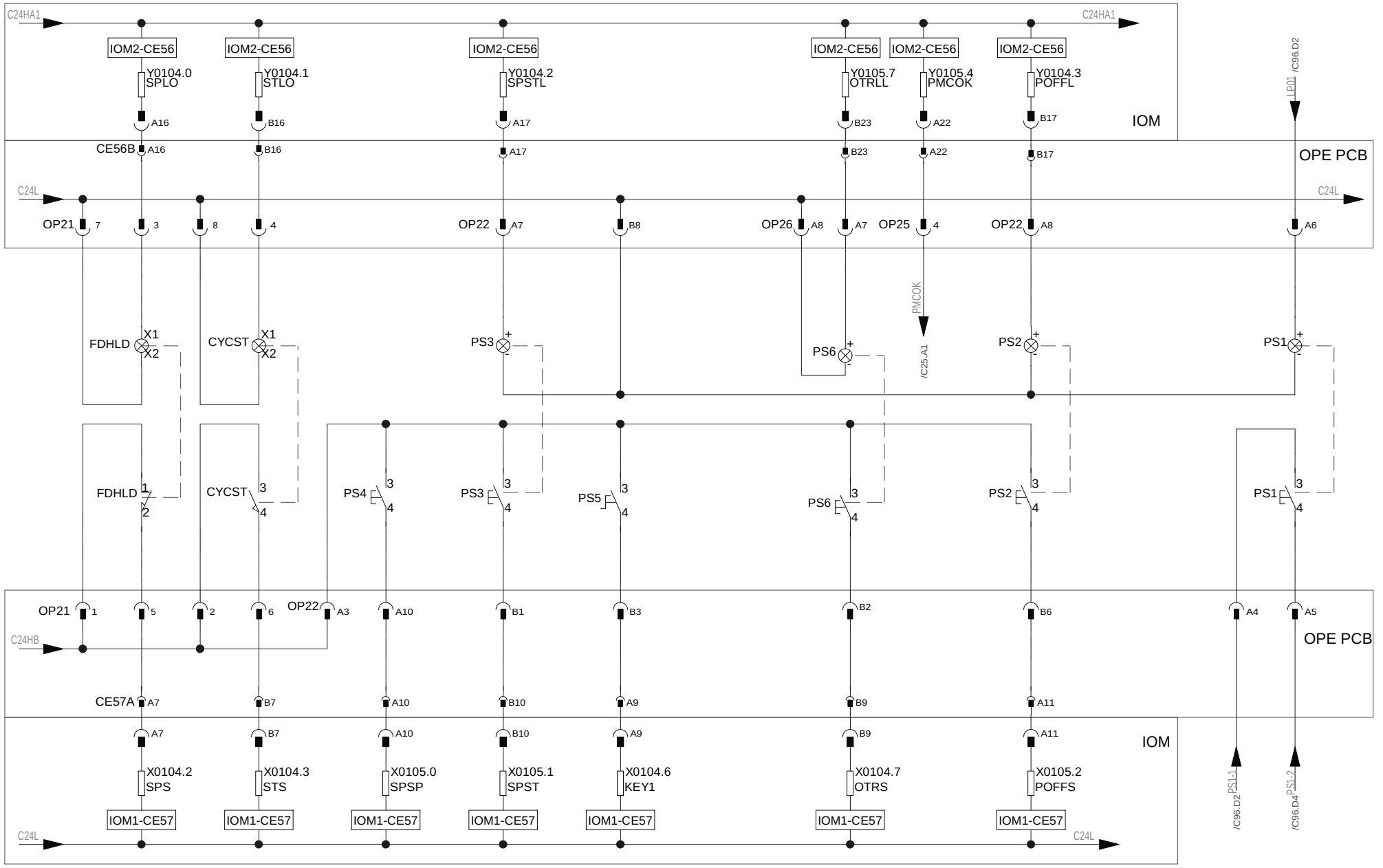
FOR PROTABLE MPG
J1 = IGNORE
J2 = SHORT
J3 = SHORT
J4 = OPEN
J5 = OPEN

OPE PANEL EMG STOP

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| OPERATOR PANEL EMG CIRCUIT | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | C98 |

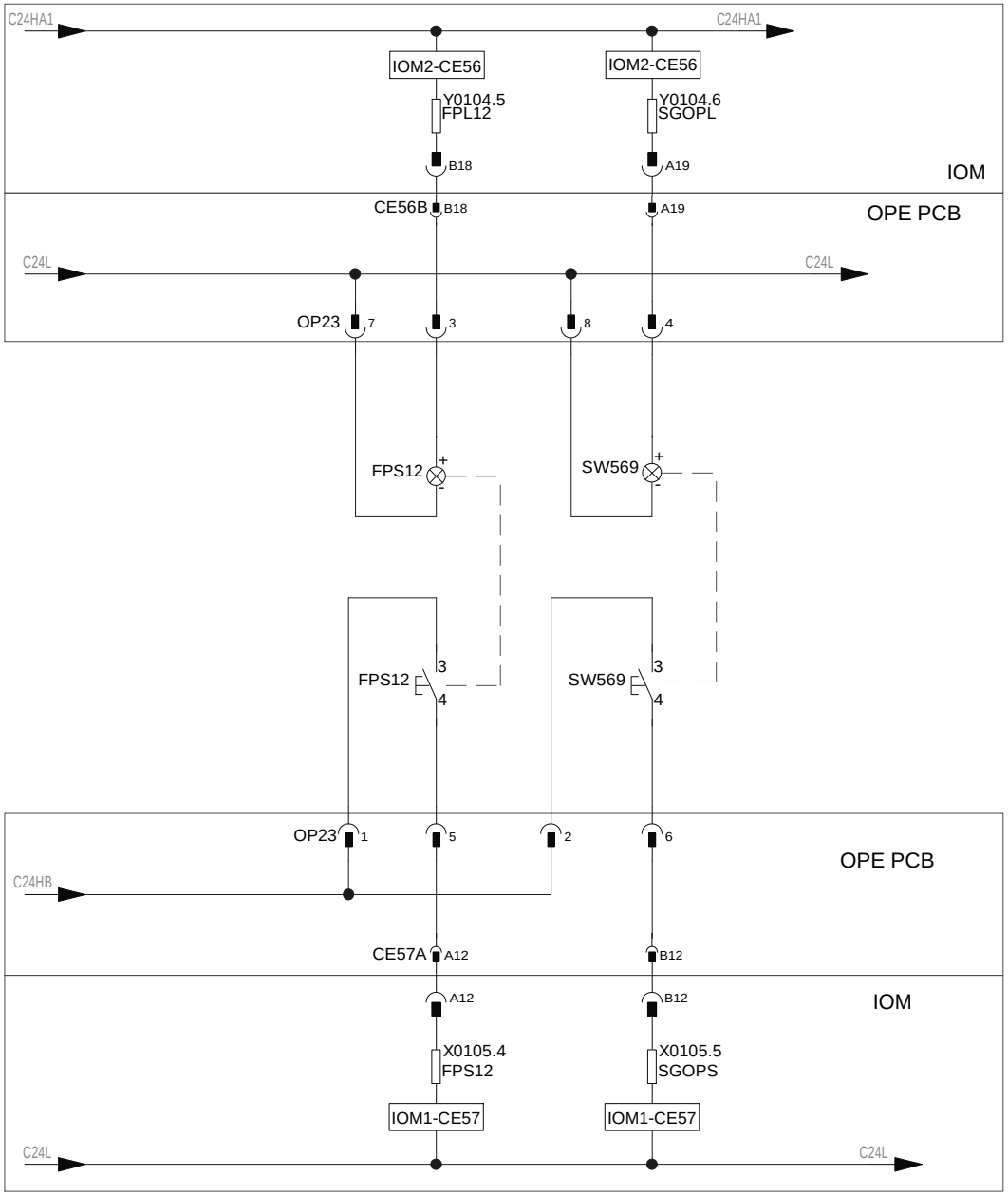



| FEED HOLD | CYCLE START | SPINDLE OFF | SPINDLE ON | MEMORY PROTECT KEY | | OT RELEASE | PMC RUNNING | POWER OFF | | POWER ON |
|---|---|---|---|---|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| OPERATOR PANEL SWITCHES | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C99 |

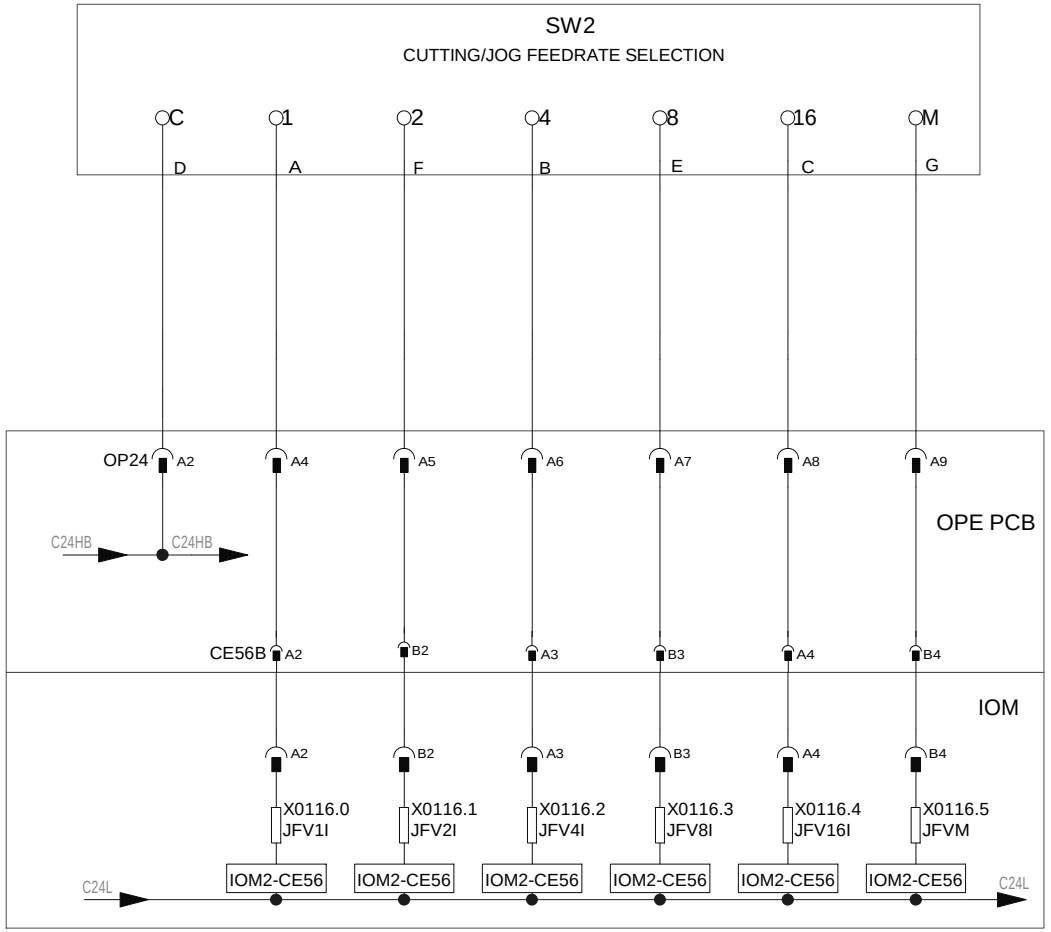



| CONVEYOR ON/OFF | SG DOOR UNLOCK |
|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| OPERATOR PANEL SWITCHES | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C100 |



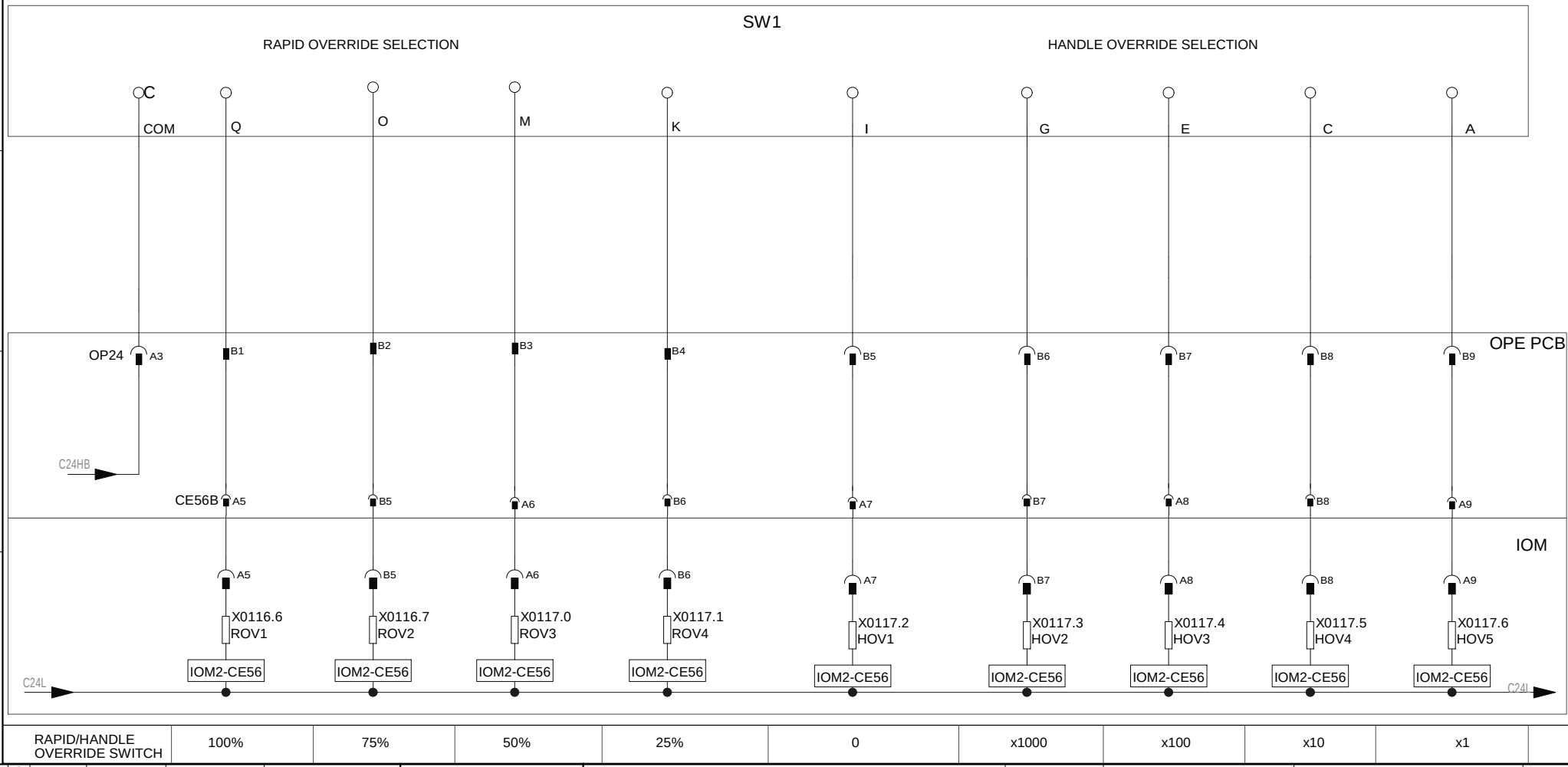

| CUTTING/JOG FEED OVERRIDE SWITCH | 1 | 2 | 4 | 8 | 16 | INHIBIT |
|---|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| CUTTING/JOG OVERRIDE CONTROL | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C101 |

*NOTE:
APPLICABLE WITH STANDARD MPG



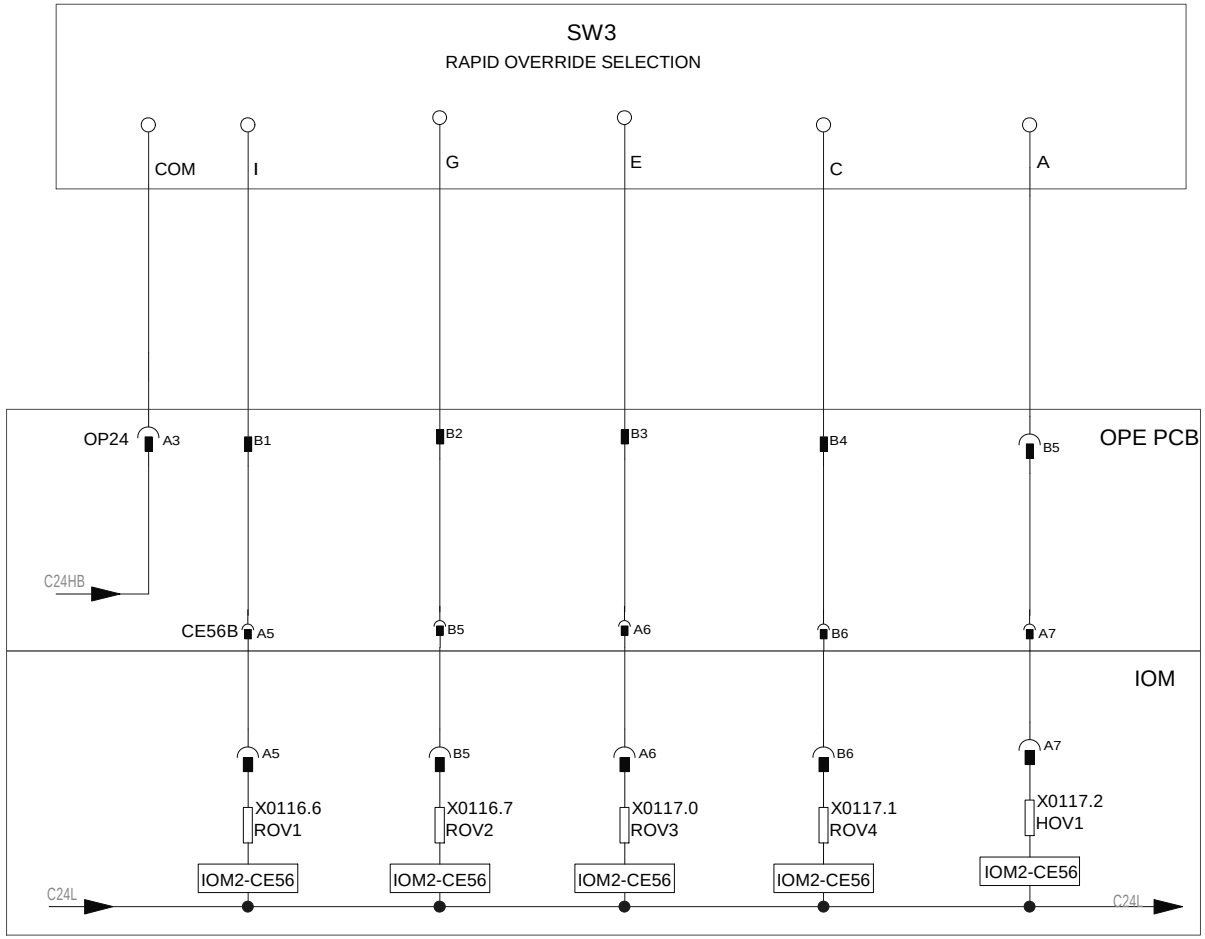

| RAPID/HANDLE OVERRIDE SWITCH | 100% | 75% | 50% | 25% | 0 | x1000 | x100 | x10 | x1 |
|---|---|---|---|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| RAPID/HANDLE OVERRIDE CONTROL | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C102 |

*NOTE:
APPLICABLE WITH PORTABLE MPG

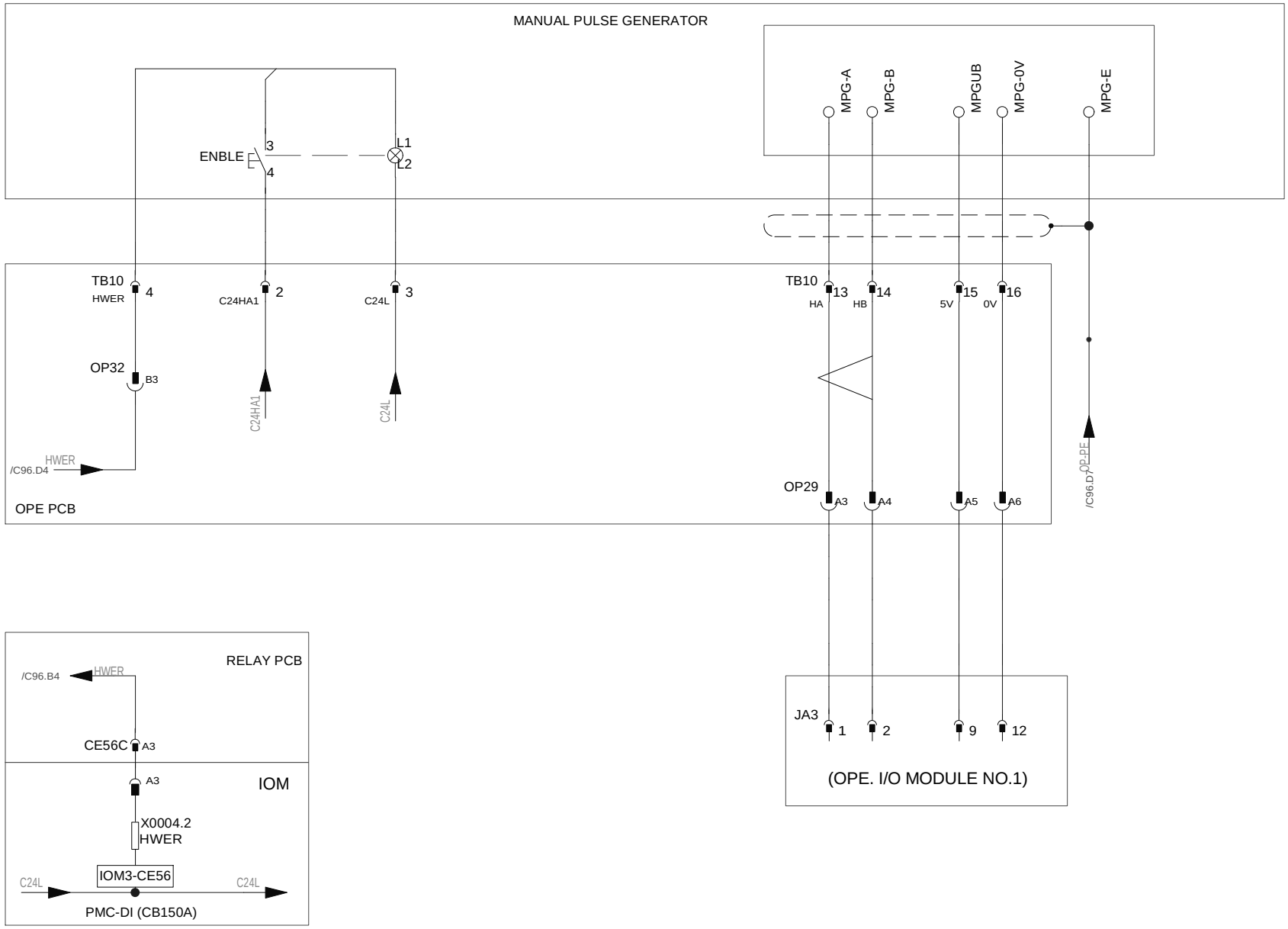



| RAPID OVERRIDE SWITCH | 100% | 75% | 50% | 25% | 0 |
|---|---|---|---|---|---|

| REVISIONS | | | | |
|---|---|---|---|---|
| ④ | | | DATE | NAME |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| RAPID OVERRIDE CONTROL | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C103 |

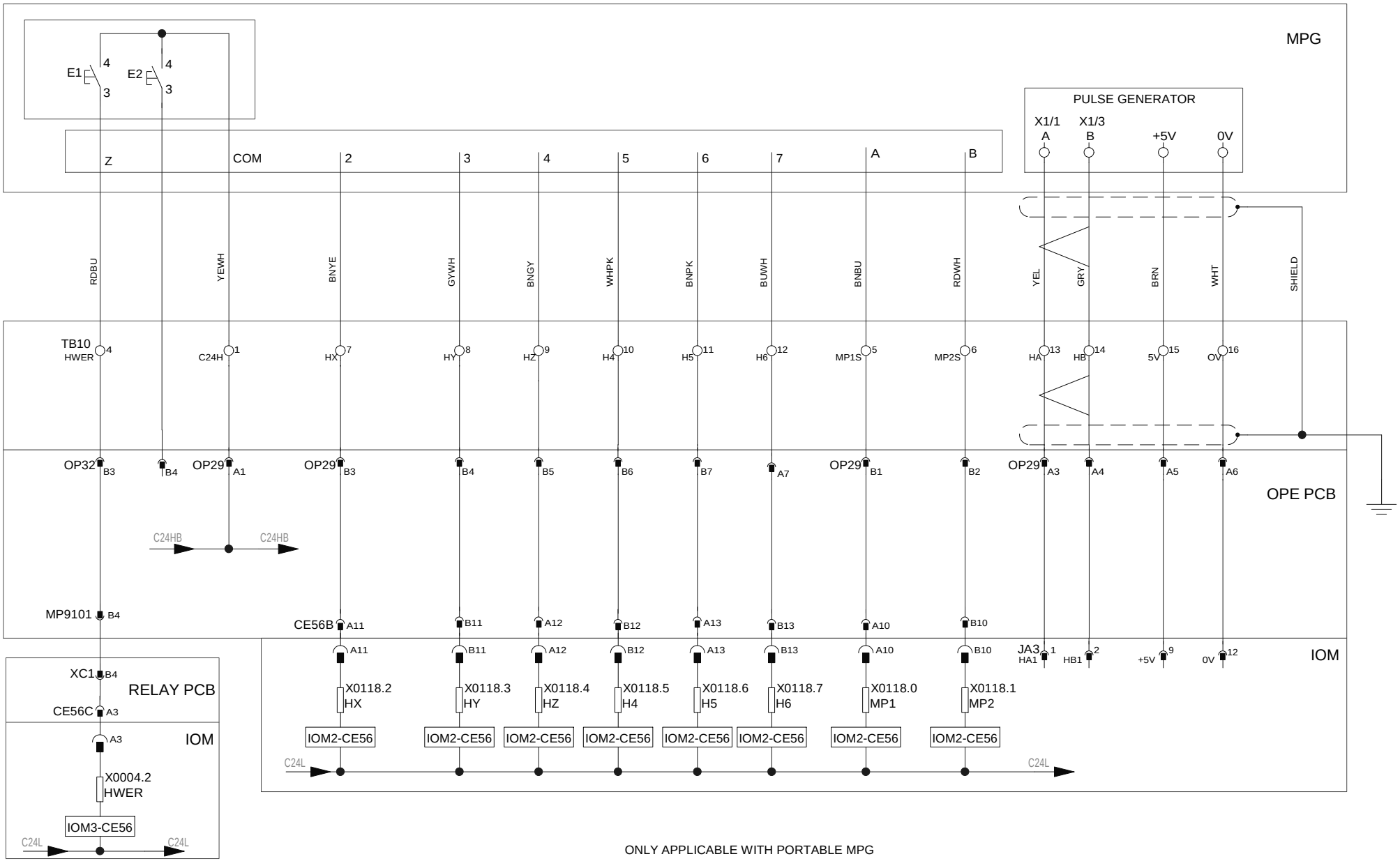



| | | |
|---|---|---|
| HWER ENABLE | ENABLE SWITCH | MANUAL PULSE GENERATOR |

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| MANUAL PULSE GENERATOR | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C104 |

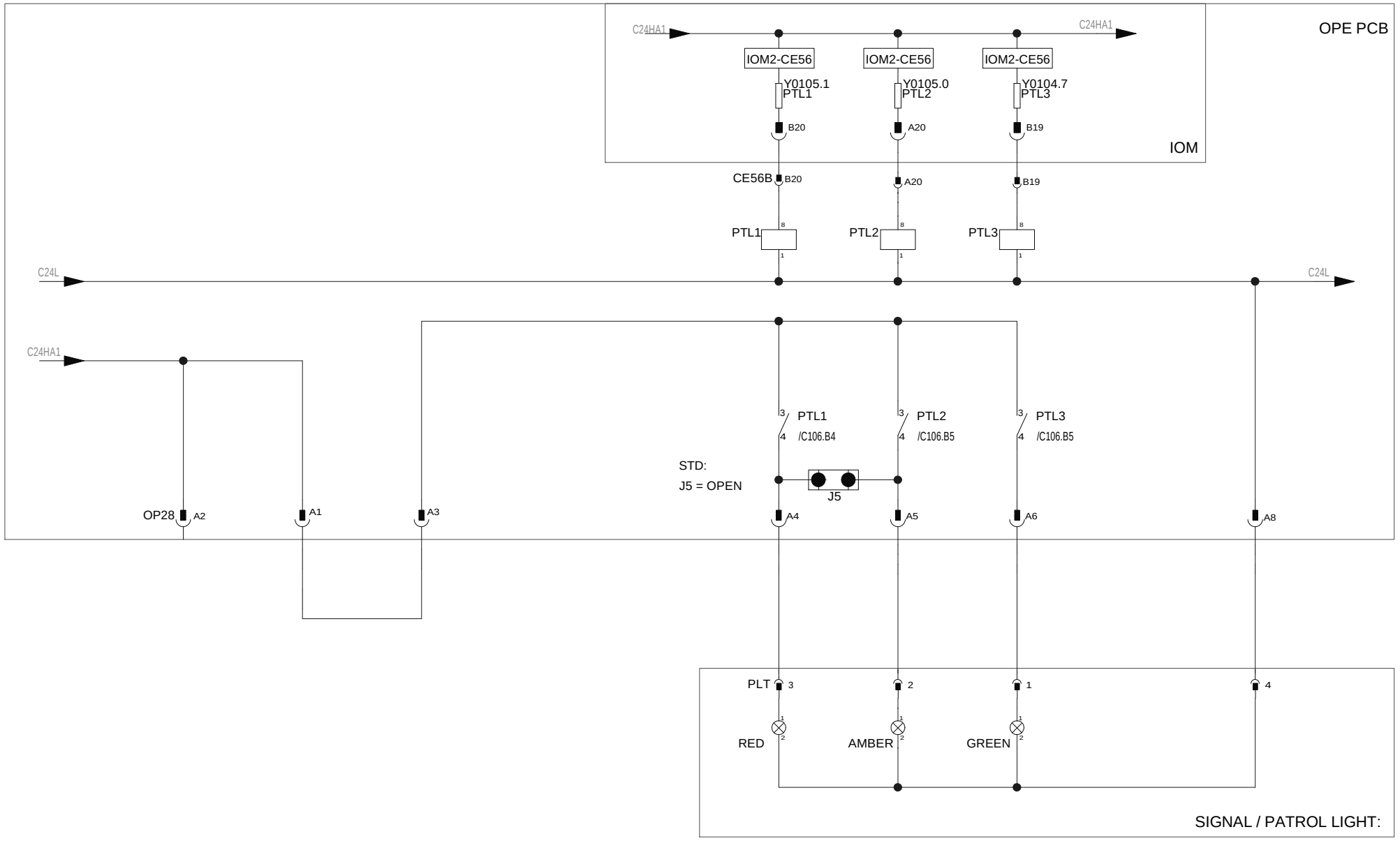



| HWER | | X-AXIS SELECT | Y-AXIS SELECT | Z-AXIS SELECT | 4 AXIS SELECT | 5 AXIS SELECT | 6 AXIS SELECT | MULTIPLY X10 | MULTIPLY X100 | 5V PULSE GENERATOR |
|---|---|---|---|---|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| MANUAL PULSE GENERATOR (PORTABLE) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C105 |

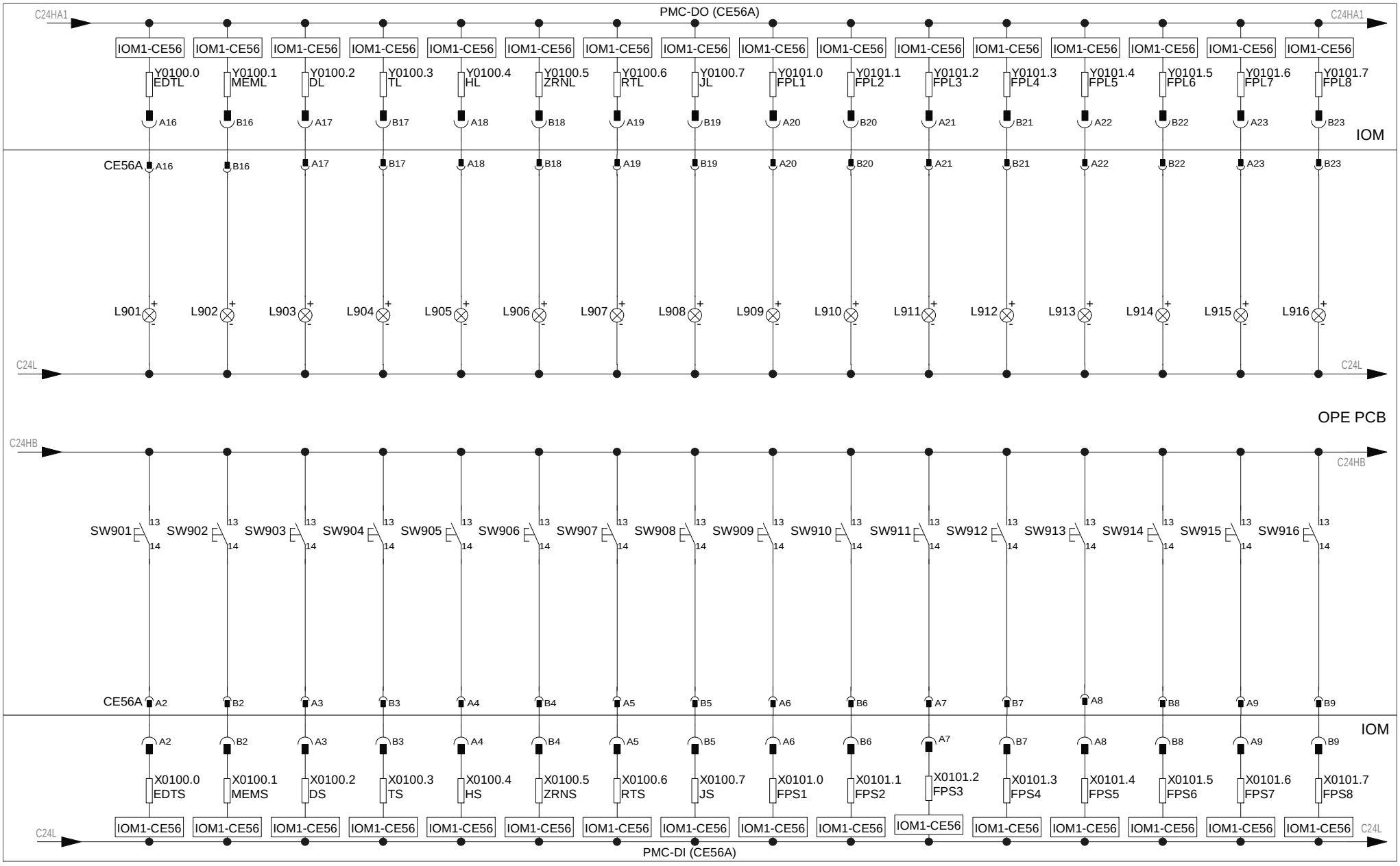



| SIGNAL LIGHT 1 | SIGNAL LIGHT 2 | SIGNAL LIGHT 3 |
|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| SIGNAL / PATROL LIGHT | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. C106 |

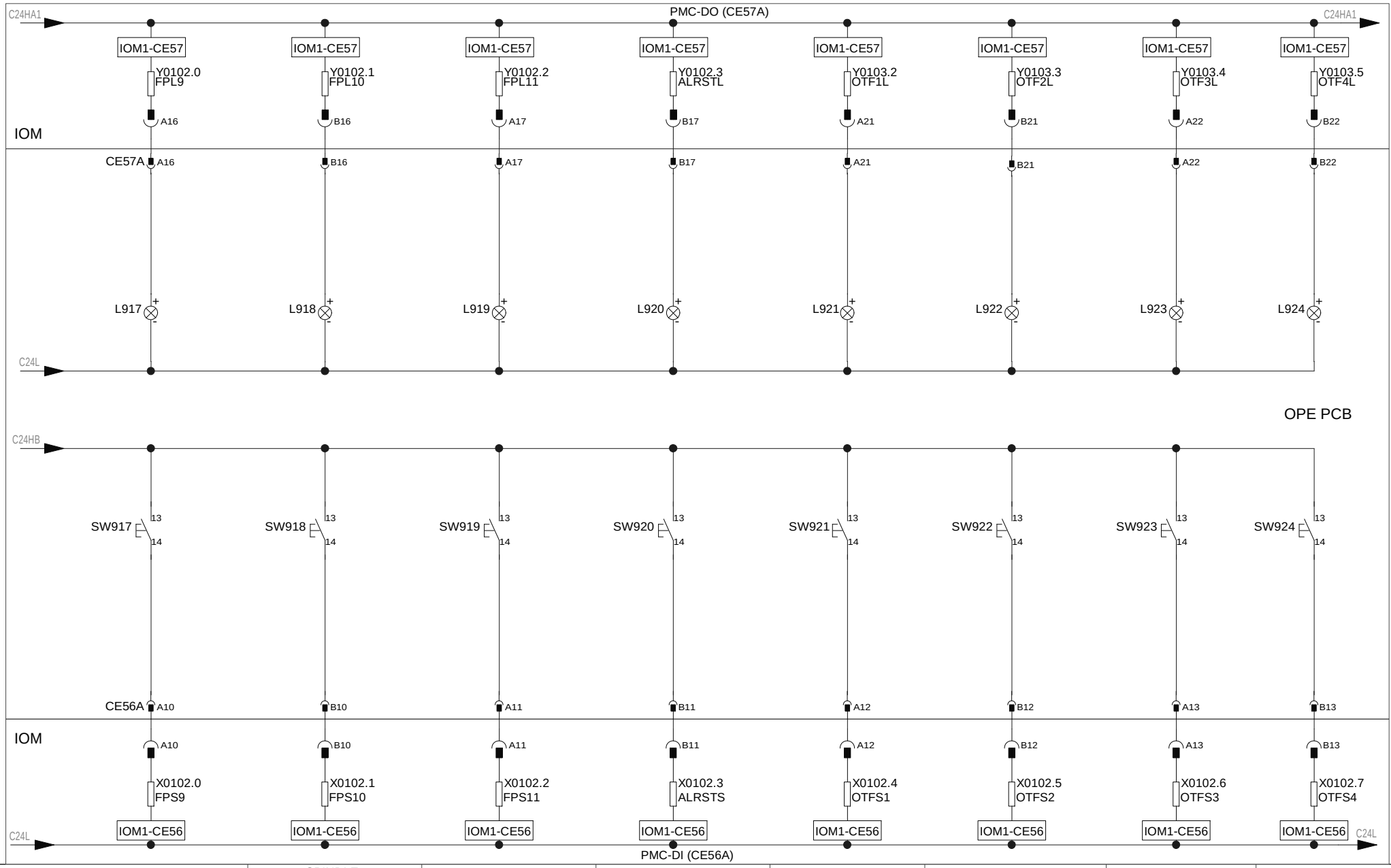



| EDIT | MEMORY | MDI | TAPE | HANDLE | REF | RAPID | JOG | SINGLE BLK | BLK SKIP | OPTIONAL STOP | DRY RUN | COOLANT ON/OFF | NOZZLE COOLANT | AIR BLOW | TS COOLANT |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| OPERATOR PANEL SOFT KEYS 1 | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C107 |

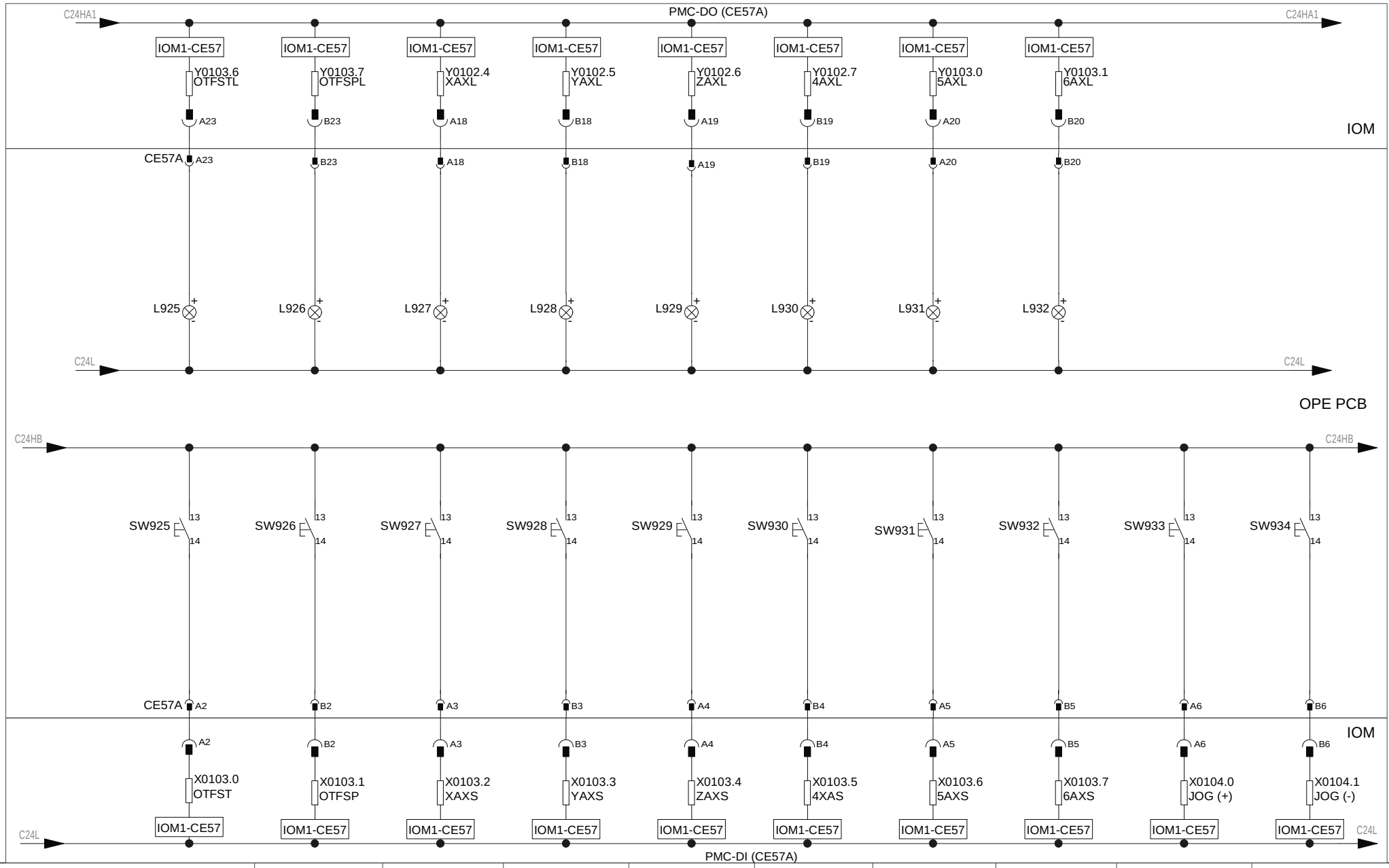



| SPINDLE SPEED (-) | SPINDLE SPEED (+) | SPINDLE ORIENTATION | ALARM RESET | AUTO ZERO | TOOL CHANGE | SETUP POSITION | CUSTOM |
|---|---|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| OPERATOR PANEL SOFT KEYS 2 | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C108 |

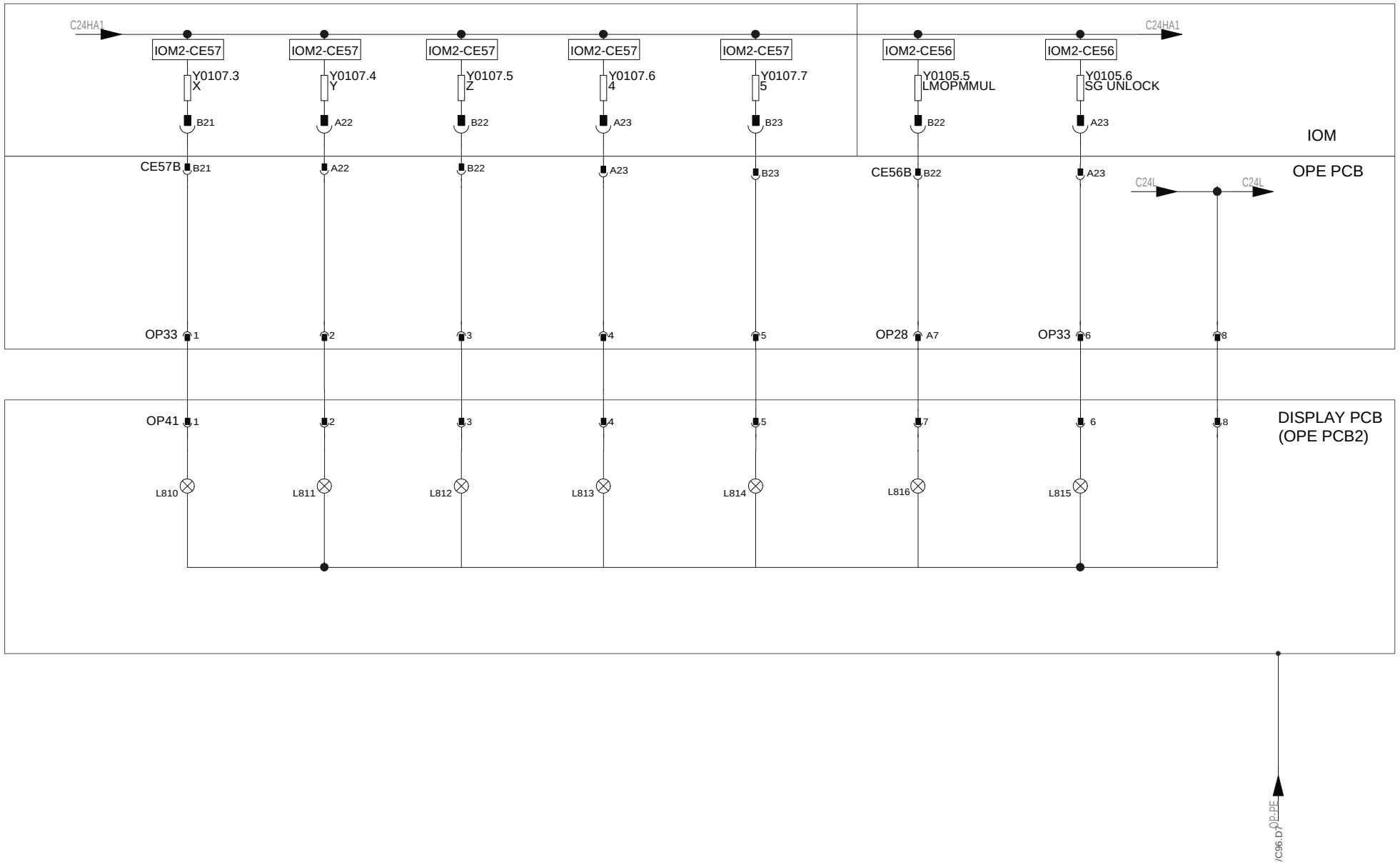



| FUNCTION START | FUNCTION STOP | AXIS SELECT X | AXIS SELECT Y | AXIS SELECT Z | AXIS SELECT 4th | AXIS SELECT 5th | AXIS SELECT Cs | JOG (+) SWITCH | JOG (-) SWITCH |
|---|---|---|---|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| AXIS SELECTION | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. C109 |

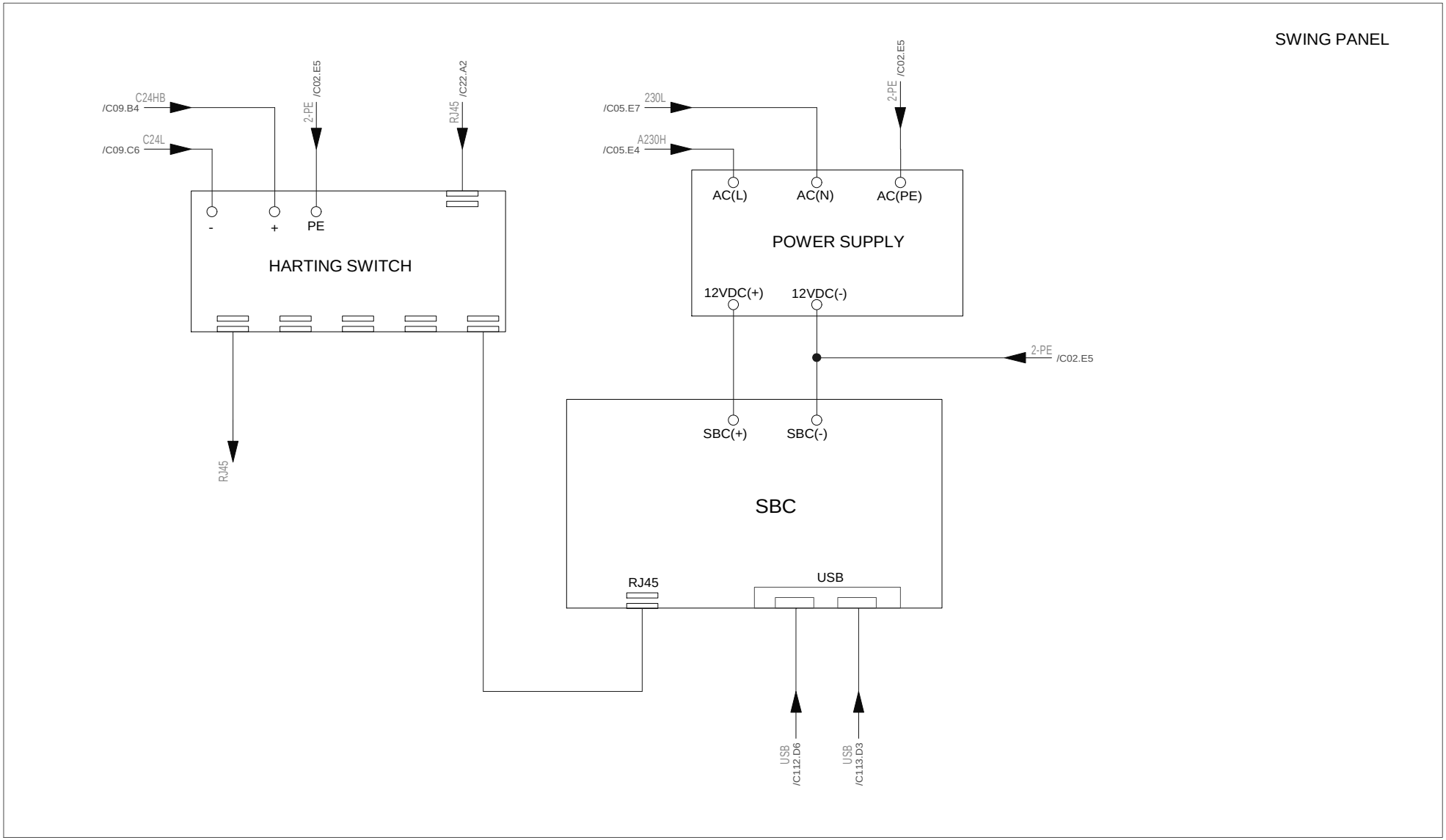



| | X AXIS 1ST REF | Y AXIS 1ST REF | Z AXIS 1ST REF | 4TH AXIS 1ST REF | 5TH AXIS 1ST REF | MODE OP. MAN<br>MODE UNLCK LAMP | S/G REL LAMP | |
|---|---|---|---|---|---|---|---|---|

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

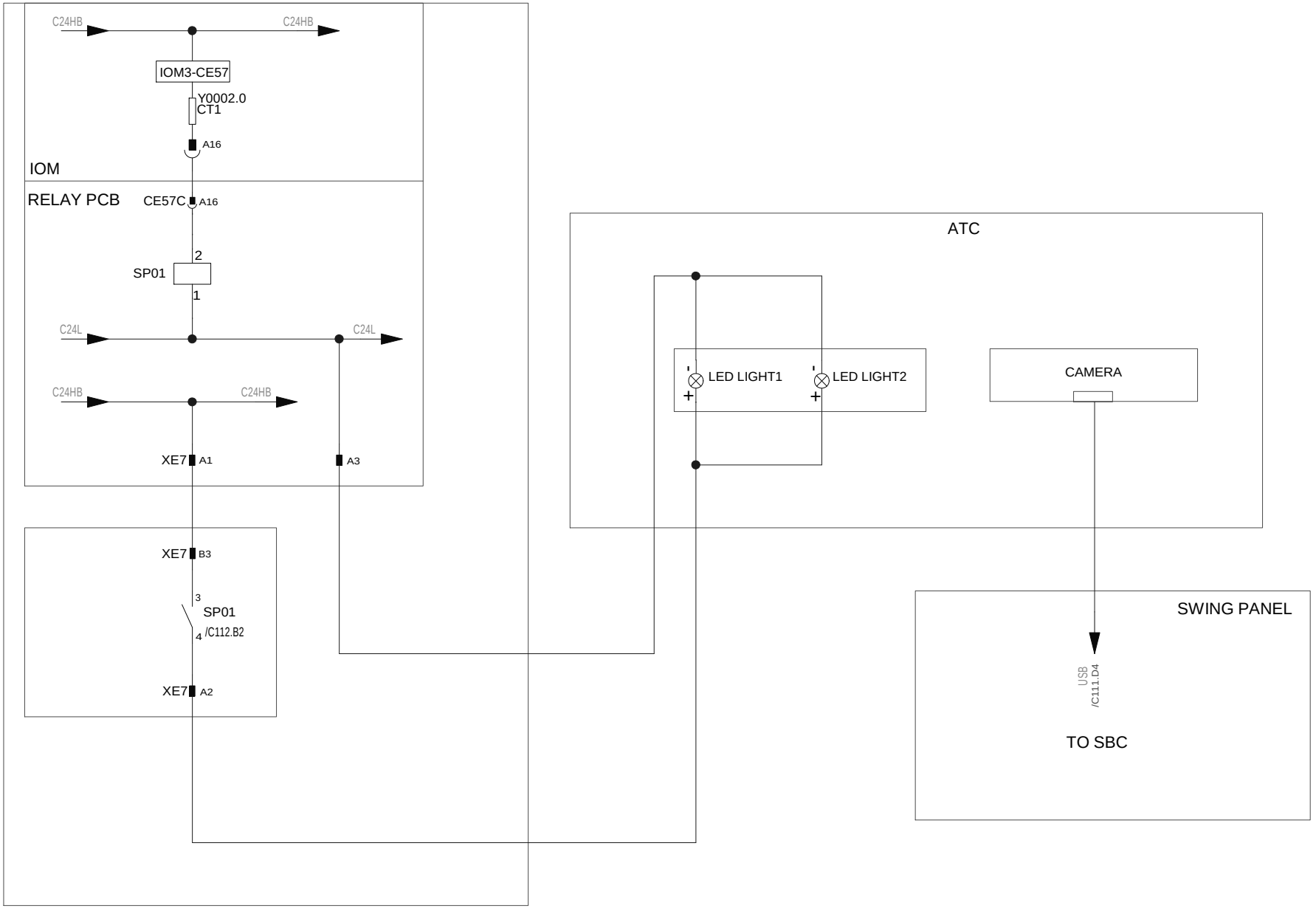
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| INDICATION LED | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C110 |




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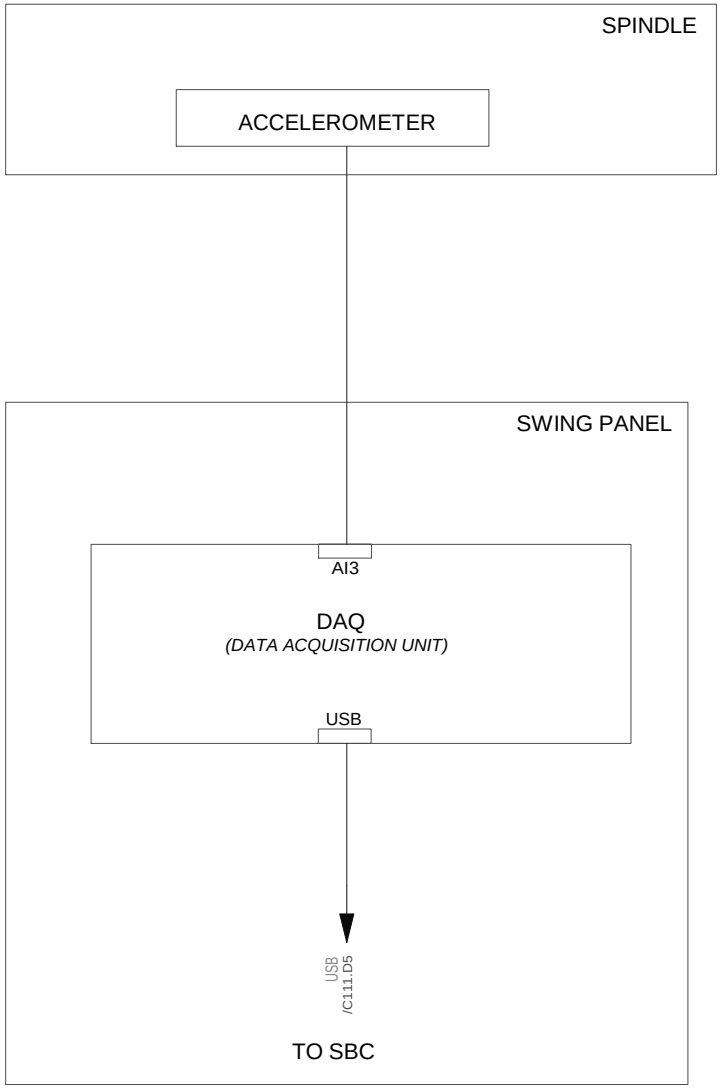
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| AUXILIARY COMPUTER SYSTEM (SBC) (OPT) | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C111 |




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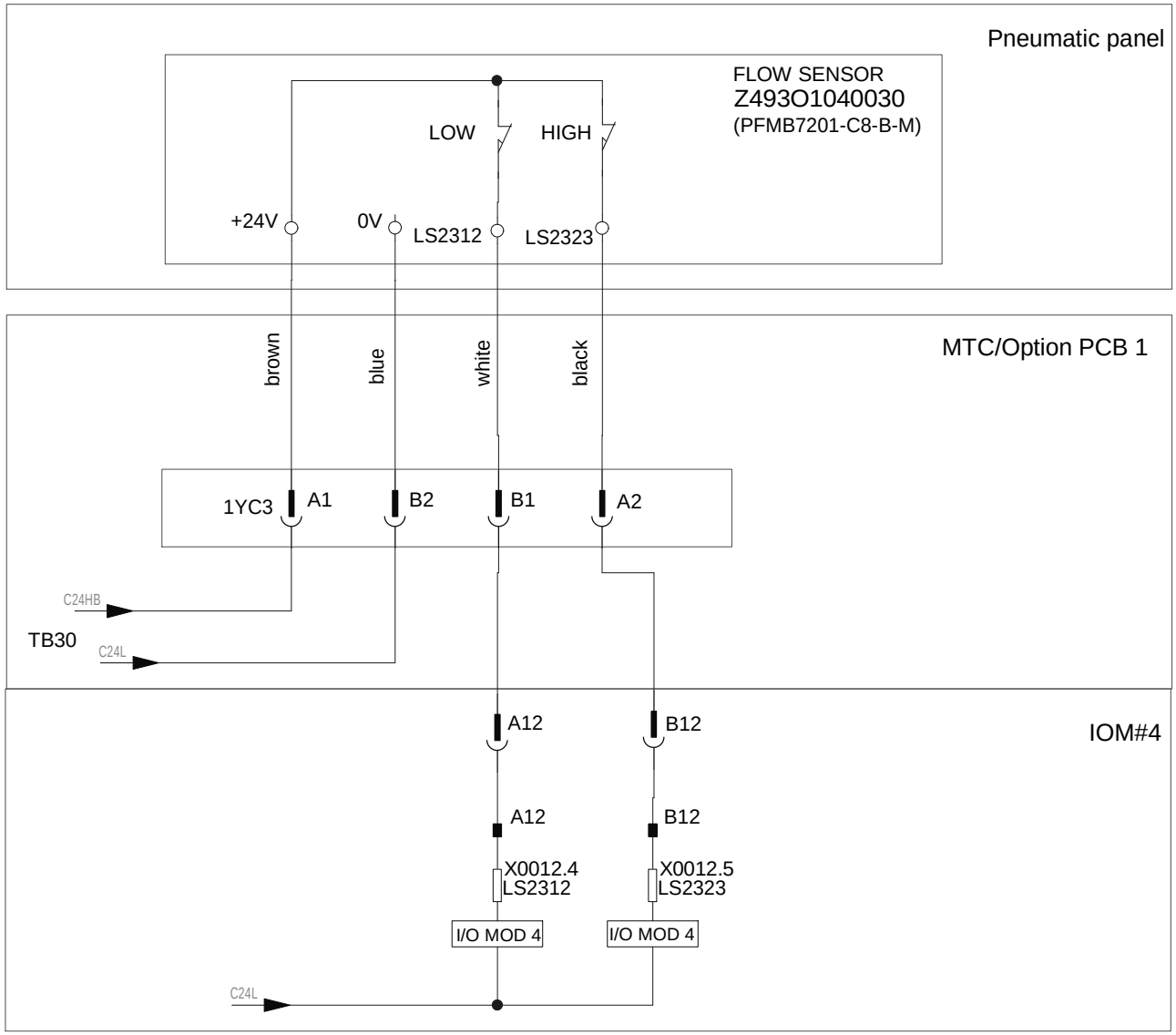
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| VISION BTS (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C112 |




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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
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| SPINDLE ACTIVE CARE (OPT) | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | C113 |

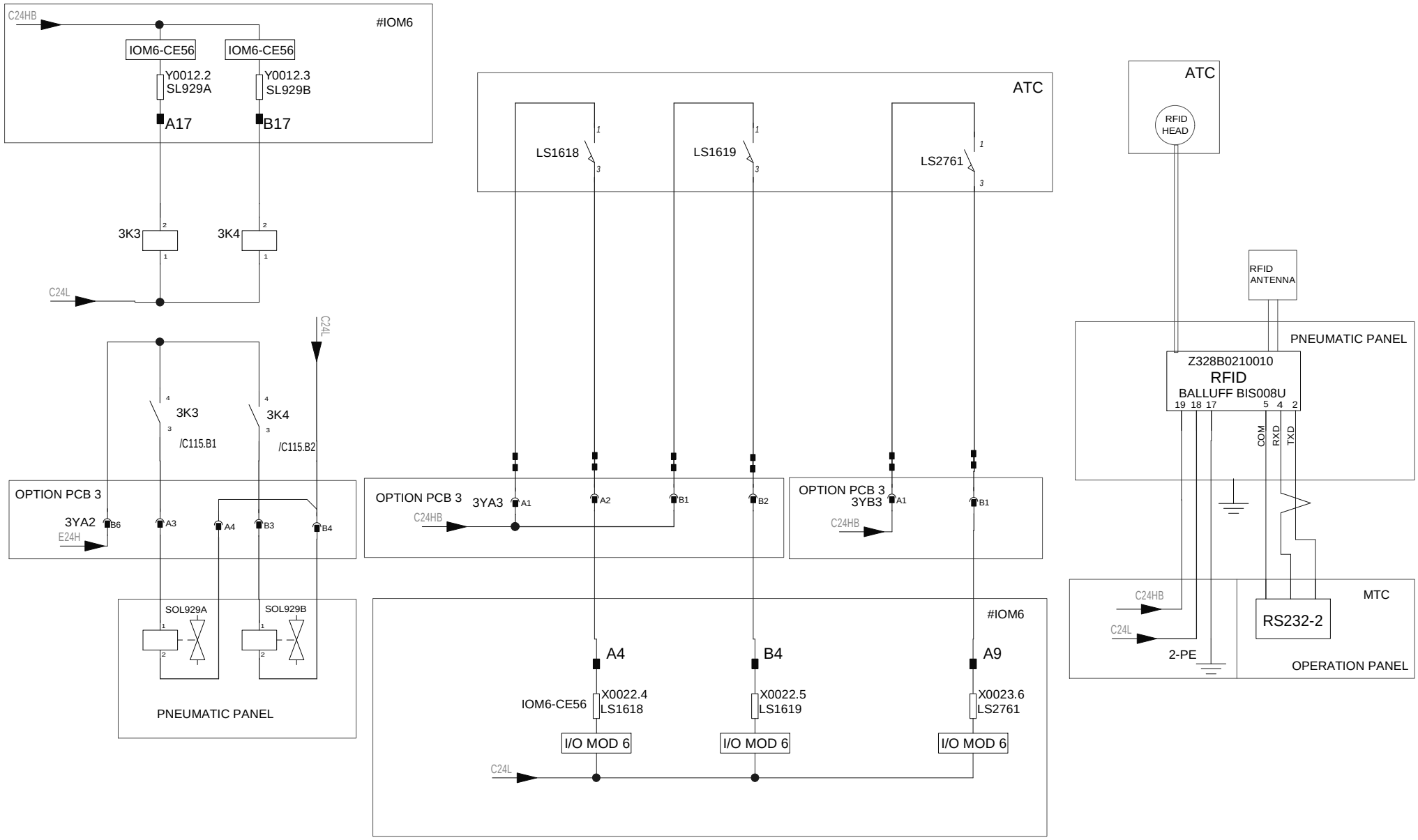



FLOW LOW LEVEL | FLOW HIGH LEVEL

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|---|---|---|---|---|
| OIL AIR FLOW SENSOR | NC TYPE | 0iMF | V210I-C1E-XG | = |
| | | | | SHEET No. C114 |




| TOOL ID ARM ADVANCE SOLENOID | TOOL ID ARM RETRACT SOLENOID | TOOL ID ARM RETRACT LS | TOOL ID ARM ADVANCE LS | TOOL ID POSITION TOOL COMFIRM |
|---|---|---|---|---|

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|---|---|---|---|---|
| RFID (OPTION) | NC TYPE | 0iMF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | C115 |

# IOM1-CE56

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 1 | | PAGE |
|---|---|---|---|---|---|
| X0100.7 | JS | JOG FEED MODE | B05 | COMMON SIGNALS | /C107.E4 |
| X0100.6 | RTS | RAPID TRAVERSE MODE | A05 | | /C107.E4 |
| X0100.5 | ZRNS | ZERO RETURN MODE | B04 | +24V (B01) | /C107.E3 |
| X0100.4 | HS | HANDLE MODE | A04 | | /C107.E3 |
| X0100.3 | TS | EXTERNAL INPUT | B03 | C24L (A01) | /C107.E3 |
| X0100.2 | DS | MDI MODE | A03 | | /C107.E2 |
| X0100.1 | MEMS | MEMORY MODE | B02 | | /C107.E2 |
| X0100.0 | EDTS | EDIT MODE | A02 | DICOM (A14) | /C107.E2 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 1 | | PAGE |
|---|---|---|---|---|---|
| X0101.7 | FPS8 | TSC COOLANT | B09 | COMMON SIGNALS | /C107.E7 |
| X0101.6 | FPS7 | AIR BLOW | A09 | | /C107.E7 |
| X0101.5 | FPS6 | NOZZLE COOLANT | B08 | +24V (B01) | /C107.E6 |
| X0101.4 | FPS5 | COOLANT ON/OFF | A08 | | /C107.E6 |
| X0101.3 | FPS4 | DRY RUN | B07 | C24L (A01) | /C107.E6 |
| X0101.2 | FPS3 | OPTIONAL STOP | A07 | | /C107.E5 |
| X0101.1 | FPS2 | BLOCK SKIP | B06 | | /C107.E5 |
| X0101.0 | FPS1 | SINGLE BLOCK | A06 | DICOM (A14) | /C107.E5 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 1 | | PAGE |
|---|---|---|---|---|---|
| X0102.7 | OTFS4 | CUSTOM | B13 | COMMON SIGNALS | /C108.E7 |
| X0102.6 | OTFS3 | SETUP POSITION | A13 | | /C108.E7 |
| X0102.5 | OTFS2 | TOOL CHANGE | B12 | +24V (B01) | /C108.E6 |
| X0102.4 | OTFS1 | AUTO ZERO | A12 | | /C108.E5 |
| X0102.3 | ALRSTS | ALARM RESET | B11 | C24L (A01) | /C108.E4 |
| X0102.2 | FPS11 | SP ORIENTATION | A11 | | /C108.E3 |
| X0102.1 | FPS10 | SP SPEED (+) | B10 | | /C108.E2 |
| X0102.0 | FPS9 | SP SPEED (-) | A10 | DICOM (A14) | /C108.E2 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 1 | | PAGE |
|---|---|---|---|---|---|
| Y0100.7 | JL | JOG MODE SWITCH LAMP | B19 | COMMON SIGNALS | /C107.A4 |
| Y0100.6 | RTL | RAPID MODE SWITCH LAMP | A19 | | /C107.A4 |
| Y0100.5 | ZRNL | REF MODE SWITCH LAMP | B18 | C24H (A24) (B24) (A25) (B25) | /C107.A3 |
| Y0100.4 | HL | HANDLE MODE SWITCH LAMP | A18 | | /C107.A3 |
| Y0100.3 | TL | TAPE MODE SWITCH LAMP | B17 | | /C107.A3 |
| Y0100.2 | DL | MDI MODE SWITCH LAMP | A17 | | /C107.A2 |
| Y0100.1 | MEML | MEMORY MODE SWITCH LAMP | B16 | | /C107.A2 |
| Y0100.0 | EDTL | EDIT MODE SWITCH LAMP | A16 | C24L (A01) | /C107.A2 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 1 | | PAGE |
|---|---|---|---|---|---|
| Y0101.7 | FPL8 | TSC COOLANT LAMP | B23 | COMMON SIGNALS | /C107.A7 |
| Y0101.6 | FPL7 | AIR BLOW LAMP | A23 | | /C107.A7 |
| Y0101.5 | FPL6 | NOZZLE COOLANT LAMP | B22 | C24H (A24) (B24) (A25) (B25) | /C107.A6 |
| Y0101.4 | FPL5 | COOLANT ON/OFF LAMP | A22 | | /C107.A6 |
| Y0101.3 | FPL4 | DRY RUN LAMP | B21 | | /C107.A6 |
| Y0101.2 | FPL3 | OPTIONAL STOP LAMP | A21 | | /C107.A5 |
| Y0101.1 | FPL2 | BLOCK SKIP LAMP | B20 | | /C107.A5 |
| Y0101.0 | FPL1 | SINGLE BLOCK LAMP | A20 | C24L (A01) | /C107.A5 |

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| | | | | SHEET No. |
| | | | | E01 |

# IOM1-CE57

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 1 | | PAGE |
|---|---|---|---|---|---|
| X0103.7 | 6AXS | AXIS SELECT CS | B05 | COMMON SIGNALS | /C109.E6 |
| X0103.6 | 5AXS | AXIS SELECT 5TH | A05 | | /C109.E5 |
| X0103.5 | 4XAS | AXIS SELECT 4TH | B04 | +24V (B01) | /C109.E5 |
| X0103.4 | ZAXS | AXIS SELECT Z | A04 | | /C109.E4 |
| X0103.3 | YAXS | AXIS SELECT Y | B03 | C24L (A01) | /C109.E4 |
| X0103.2 | XAXS | AXIS SELECT X | A03 | | /C109.E3 |
| X0103.1 | OTFSP | ONE TOUCH FUNCTION STOP | B02 | | /C109.E2 |
| X0103.0 | OTFST | ONE TOUCH FUNCTION START | A02 | DICOM (A14) | /C109.E2 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 1 | | PAGE |
|---|---|---|---|---|---|
| X0104.7 | OTRS | OT RELEASE SWITCH | B09 | COMMON SIGNALS | /C99.E5 |
| X0104.6 | KEY1 | MEMORY PROTECT KEY SWITCH | A09 | | /C99.E4 |
| X0104.5 | *EMGOP | EMERGENCY STOP OP PANEL | B08 | +24V (B01) | /C98.E3 |
| X0104.4 | | | A08 | | |
| X0104.3 | STS | CYCLE START SWITCH | B07 | C24L (A01) | /C99.E2 |
| X0104.2 | SPS | FEED HOLD SWITCH | A07 | | /C99.E2 |
| X0104.1 | JOG (-) | JOG - SWITCH | B06 | | /C109.E7 |
| X0104.0 | JOG (+) | JOG + SWITCH | A06 | DICOM (A14) | /C109.E7 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 1 | | PAGE |
|---|---|---|---|---|---|
| X0105.7 | | | B13 | COMMON SIGNALS | |
| X0105.6 | STUCS | TOOL UNCL SWITCH | A13 | | /C55.E3 |
| X0105.5 | SGOPS | SG DOOR UNLCK | B12 | +24V (B01) | /C100.E3 |
| X0105.4 | FPS12 | CONVEYOR ON | A12 | | /C100.E2 |
| X0105.3 | | | B11 | C24L (A01) | |
| X0105.2 | POFFS | POWER OFF | A11 | | /C99.E6 |
| X0105.1 | SPST | SPINDLE START SWITCH | B10 | | /C99.E3 |
| X0105.0 | SPSP | SPINDLE STOP SWITCH | A10 | DICOM (A14) | /C99.E3 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 1 | | PAGE |
|---|---|---|---|---|---|
| Y0102.7 | 4AXL | 4TH AXIS SELECT LAMP | B19 | COMMON SIGNALS | /C109.A5 |
| Y0102.6 | ZAXL | Z AXIS SELECT LAMP | A19 | | /C109.A4 |
| Y0102.5 | YAXL | Y AXIS SELECT LAMP | B18 | C24H (A24) (B24) (A25) (B25) | /C109.A4 |
| Y0102.4 | XAXL | X AXIS SELECT LAMP | A18 | | /C109.A3 |
| Y0102.3 | ALRSTL | ALARM RESET LAMP | B17 | | /C108.A4 |
| Y0102.2 | FPL11 | SPINDLE ORIENTATION | A17 | | /C108.A3 |
| Y0102.1 | FPL10 | SPINDLE SPEED (+) | B16 | | /C108.A2 |
| Y0102.0 | FPL9 | SPINDLE SPEED (-) | A16 | C24L (A01) | /C108.A2 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 1 | | PAGE |
|---|---|---|---|---|---|
| Y0103.7 | OTFSPL | ONE TOUCH FUNCTION STOP LAMP | B23 | COMMON SIGNALS | /C109.A2 |
| Y0103.6 | OTFSTL | ONE TOUCH FUNCTION START LAMP | A23 | | /C109.A2 |
| Y0103.5 | OTF4L | CUSTOM LAMP | B22 | C24H (A24) (B24) (A25) (B25) | /C108.A7 |
| Y0103.4 | OTF3L | SETUP POSITION LAMP | A22 | | /C108.A7 |
| Y0103.3 | OTF2L | TOOL CHANGE LAMP | B21 | | /C108.A6 |
| Y0103.2 | OTF1L | AUTO ZERO LAMP | A21 | | /C108.A5 |
| Y0103.1 | 6AXL | CS AXIS SELECT LAMP | B20 | | /C109.A6 |
| Y0103.0 | 5AXL | 5TH AXIS SELECT LAMP | A20 | C24L (A01) | /C109.A5 |

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| DI/DO ADDRESS TABLE - IO MODULE 1 | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | E02 |

# IOM2-CE56

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 2 | | PAGE |
|---|---|---|---|---|---|
| X0116.7 | ROV2 | RAPID OVERRIDE SWITCH (2) | B05 | COMMON SIGNALS | /C102.E2 |
| X0116.6 | ROV1 | RAPID OVERRIDE SWITCH (1) | A05 | | /C102.E2 |
| X0116.5 | JFVM | CUTTING/JOG FEED OVER SW (INHIBIT) | B04 | +24V (B01) | /C101.E5 |
| X0116.4 | JFV16I | CUTTING/JOG FEED OVERRIDE SW (16) | A04 | | /C101.E5 |
| X0116.3 | JFV8I | CUTTING/JOG FEED OVERRIDE SW (8) | B03 | C24L (A01) | /C101.E4 |
| X0116.2 | JFV4I | CUTTING/JOG FEED OVERRIDE SW (4) | A03 | | /C101.E4 |
| X0116.1 | JFV2I | CUTTING/JOG FEED OVERRIDE SW (2) | B02 | | /C101.E3 |
| X0116.0 | JFV1I | CUTTING/JOG FEED OVERRIDE SW (1) | A02 | DICOM (A14) | /C101.E3 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 2 | | PAGE |
|---|---|---|---|---|---|
| X0117.7 | | | B09 | COMMON SIGNALS | |
| X0117.6 | HOV5 | HANDLE OVERRIDE SWITCH (5) | A09 | | /C102.E7 |
| X0117.5 | HOV4 | HANDLE OVERRIDE SWITCH (4) | B08 | +24V (B01) | /C102.E6 |
| X0117.4 | HOV3 | HANDLE OVERRIDE SWITCH (3) | A08 | | /C102.E6 |
| X0117.3 | HOV2 | HANDLE OVERRIDE SWITCH (2) | B07 | C24L (A01) | /C102.E5 |
| X0117.2 | HOV1 | HANDLE OVERRIDE SWITCH (1) | A07 | | /C102.E4 |
| X0117.1 | ROV4 | RAPID OVERRIDE SWITCH (4) | B06 | | /C102.E4 |
| X0117.0 | ROV3 | RAPID OVERRIDE SWITCH (3) | A06 | DICOM (A14) | /C102.E3 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 2 | | PAGE |
|---|---|---|---|---|---|
| X0118.7 | H6 | HANDLE SELECT (CS) | B13 | COMMON SIGNALS | /C105.E5 |
| X0118.6 | H5 | HANDLE SELECT (5TH) | A13 | | /C105.E4 |
| X0118.5 | H4 | HANDLE SELECT (4TH) | B12 | +24V (B01) | /C105.E4 |
| X0118.4 | HZ | HANDLE SELECT (Z) | A12 | | /C105.E3 |
| X0118.3 | HY | HANDLE SELECT (Y) | B11 | C24L (A01) | /C105.E3 |
| X0118.2 | HX | HANDLE SELECT (X) | A11 | | /C105.E3 |
| X0118.1 | MP2 | HANDLE MULTIPLY *100 | B10 | | /C105.E5 |
| X0118.0 | MP1 | HANDLE MULTIPLY *10 | A10 | DICOM (A14) | /C105.E5 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 2 | | PAGE |
|---|---|---|---|---|---|
| Y0104.7 | PTL3 | SIGNAL LIGHT 3 LAMP | B19 | COMMON SIGNALS | /C106.A6 |
| Y0104.6 | SGOPL | SG DOOR RELEASE LAMP | A19 | | /C100.A3 |
| Y0104.5 | FPL12 | CONVEYOR ON LAMP | B18 | C24H (A24) (B24) (A25) (B25) | /C100.A2 |
| Y0104.4 | | | A18 | | |
| Y0104.3 | POFFL | POWER OFF LAMP | B17 | | /C99.A6 |
| Y0104.2 | SPSTL | SP START LAMP | A17 | | /C99.A3 |
| Y0104.1 | STLO | CYCLE START LAMP | B16 | | /C99.A2 |
| Y0104.0 | SPLO | FEED HOLD LAMP | A16 | C24L (A01) | /C99.A2 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 2 | | PAGE |
|---|---|---|---|---|---|
| Y0105.7 | OTRLL | OT RELEASE LAMP | B23 | COMMON SIGNALS | /C99.A5 |
| Y0105.6 | SG UNLOCK | SG UNLOCK LAMP | A23 | | /C110.A6 |
| Y0105.5 | LMOPMMUL | OPER MANUAL MODE UNLOCK LAMP | B22 | C24H (A24) (B24) (A25) (B25) | /C110.A5 |
| Y0105.4 | PMCOK | PMC RUNNING | A22 | | /C99.A5 |
| Y0105.3 | STUCL | SP INDLETOOL UNCLAMP SWITCH LAMP | B21 | | /C55.A1 |
| Y0105.2 | MLT | MC LIGHT IN SPLASH GUARD | A21 | | /C82.A2 |
| Y0105.1 | PTL1 | SIGNAL LIGHT 1 | B20 | | /C106.A5 |
| Y0105.0 | PTL2 | SIGNAL LIGHT 2 | A20 | C24L (A01) | /C106.A5 |

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| DI/DO ADDRESS TABLE - IO MODULE 2 | NC TYPE | 0i-MF | V210I-C1E-XG | = |
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| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 2 | | PAGE |
|---|---|---|---|---|---|
| X0119.7 | BTSOT | BTS OVERTRAVEL | B05 | COMMON SIGNALS | /C61.E7 |
| X0119.6 | ROTB | ROTARY AXIS OVERTRAVEL B | A05 | | /C27.E6 |
| X0119.5 | ROTA | ROTARY AXIS OVERTRAVEL A | B04 | +24V (B01) | /C27.E5 |
| X0119.4 | *EMFX | EMERGENCY STOP FIXTURE | A04 | | /C26.E6 |
| X0119.3 | *EMLUC | EMERGENCY STOP LUCC | B03 | C24L (A01) | /C26.E5 |
| X0119.2 | *EMEXT | EMERGENCY STOP APC | A03 | | /C26.E4 |
| X0119.1 | *EMRB | EMERGENCY STOP Robot | B02 | DICOM (A14) | /C26.E3 |
| X0119.0 | *EMATA | EMERGENCY STOP ATC | A02 | | /C26.E2 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 2 | | PAGE |
|---|---|---|---|---|---|
| X0120.7 | | | B09 | COMMON SIGNALS | |
| X0120.6 | | | A09 | | |
| X0120.5 | | | B08 | +24V (B01) | |
| X0120.4 | | | A08 | | |
| X0120.3 | | | B07 | C24L (A01) | |
| X0120.2 | | | A07 | | |
| X0120.1 | | | B06 | DICOM (A14) | |
| X0120.0 | | | A06 | | |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 2 | | PAGE |
|---|---|---|---|---|---|
| X0121.7 | | | B13 | COMMON SIGNALS | |
| X0121.6 | | | A13 | | |
| X0121.5 | | | B12 | +24V (B01) | |
| X0121.4 | | | A12 | | |
| X0121.3 | | | B11 | C24L (A01) | |
| X0121.2 | ADCLS | AUTO DOOR CLOSE | A11 | | /C75.E7 |
| X0121.1 | ADOPN | AUTO DOOR OPEN | B10 | DICOM (A14) | /C75.E6 |
| X0121.0 | | | A10 | | |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 2 | | PAGE |
|---|---|---|---|---|---|
| Y0106.7 | | | B19 | COMMON SIGNALS | |
| Y0106.6 | | | A19 | | |
| Y0106.5 | | | B18 | C24H (A24) (B24) (A25) (B25) | |
| Y0106.4 | | | A18 | | |
| Y0106.3 | | | B17 | | |
| Y0106.2 | LADCLS | AUTO DOOR CLOSE LAMP | A17 | | /C75.A7 |
| Y0106.1 | LADOPN | AUTO DOOR OPEN LAMP | B16 | | /C75.A6 |
| Y0106.0 | | | A16 | C24L (A01) | |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 2 | | PAGE |
|---|---|---|---|---|---|
| Y0107.7 | 5 | 5 REF AXIS LAMP | B23 | COMMON SIGNALS | /C110.A5 |
| Y0107.6 | 4 | 4 REF AXIS LAMP | A23 | | /C110.A4 |
| Y0107.5 | Z | Z REF AXIS LAMP | B22 | C24H (A24) (B24) (A25) (B25) | /C110.A3 |
| Y0107.4 | Y | Y REF AXIS LAMP | A22 | | /C110.A3 |
| Y0107.3 | X | X REF AXIS LAMP | B21 | | /C110.A2 |
| Y0107.2 | | | A21 | | |
| Y0107.1 | | | B20 | C24L (A01) | |
| Y0107.0 | | | A20 | | |

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| DI/DO ADDRESS TABLE - IO MODULE 2 | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | E04 |

# IOM3-CE56

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 3 | | PAGE |
|---|---|---|---|---|---|
| X0004.7 | SKIP | SKIP SIGNAL | B05 | COMMON SIGNALS | /C61.E7 |
| X0004.6 | MSALB | MEAS SYSTEM BATTERY ERROR | A05 | | /C61.E5 |
| X0004.5 | MSALA | MEAS SYSTEM RECEPTION ALM | B04 | +24V (B01) | /C61.E5 |
| X0004.4 | STA | STATIC | A04 | | /C61.E6 |
| X0004.3 | LSROK | LASER OK | B03 | C24L (A01) | /C61.E6 |
| X0004.2 | HWER | HANDLE ENABLE | A03 | | /C104.E2 |
| X0004.1 | *LS170 | SPINDLE LUB OIL FLOW SW | B02 | | /C53.E5 |
| X0004.0 | *OMAL1 | SPINDLE LUB OIL UNIT ALM | A02 | DICOM (A14) | /C53.E4 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 3 | | PAGE |
|---|---|---|---|---|---|
| X0005.7 | LS2308 | OIL AIR UNIT AIR PRESS DETECT | B09 | COMMON SIGNALS | /C50.E7 |
| X0005.6 | *LS2307 | OIL AIR UNIT LOW LEVEL | A09 | | /C50.E6 |
| X0005.5 | TH154 | OIL AIR UNIT THERMAL ALARM | B08 | +24V (B01) | /C50.E5 |
| X0005.4 | *LS173 | SP LUB OIL FIL PRESS LS | A08 | | /C53.E5 |
| X0005.3 | | | B07 | C24L (A01) | |
| X0005.2 | *OMAL2 | COOLANT CHILLER UNIT ALARM | A07 | | /C51.E4 |
| X0005.1 | *LS1319 | COOLANT TANK EMPTY DETECT | B06 | | /C45.E6 |
| X0005.0 | AIRAL | AIR DRYER ALARM | A06 | DICOM (A14) | /C86.B6 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 3 | | PAGE |
|---|---|---|---|---|---|
| X0006.7 | GRAY7 | ATC ARM BIT 7 | B13 | COMMON SIGNALS | /C57.E6 |
| X0006.6 | GRAY6 | ATC ARM BIT 6 | A13 | | /C57.E5 |
| X0006.5 | GRAY5 | ATC ARM BIT 5 | B12 | +24V (B01) | /C57.E5 |
| X0006.4 | GRAY4 | ATC ARM BIT 4 | A12 | | /C57.E4 |
| X0006.3 | GRAY3 | ATC ARM BIT 3 | B11 | C24L (A01) | /C57.E4 |
| X0006.2 | GRAY2 | ATC ARM BIT 2 | A11 | | /C57.E3 |
| X0006.1 | GRAY1 | ATC ARM BIT 1 | B10 | | /C57.E3 |
| X0006.0 | GRAY0 | ATC ARM BIT 0 | A10 | DICOM (A14) | /C57.E2 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 3 | | PAGE |
|---|---|---|---|---|---|
| Y0000.7 | POUT | AUTO POWER OFF | B19 | COMMON SIGNALS | /C12.A2 |
| Y0000.6 | | | A19 | | |
| Y0000.5 | | | B18 | C24H (A24) (B24) (A25) (B25) | |
| Y0000.4 | | | A18 | | |
| Y0000.3 | | | B17 | | |
| Y0000.2 | | | A17 | | |
| Y0000.1 | OTR | OVER TRAVEL RELEASE | B16 | | /C27.A7 |
| Y0000.0 | SA | SERVO READY | A16 | C24L (A01) | /C20.A2 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 3 | | PAGE |
|---|---|---|---|---|---|
| Y0001.7 | RCHP | SPINDLE STATOR SWITCH | B23 | COMMON SIGNALS | /C52.A2 |
| Y0001.6 | CLOMON | COOLANT CHILLER ON | A23 | | /C51.A1 |
| Y0001.5 | OMON | SPINDLE CHILLER ON | B22 | C24H (A24) (B24) (A25) (B25) | /C53.A1 |
| Y0001.4 | SL2300 | SPINDLE LUBRICATION AIR (OILAIR UNIT) | A22 | | /C50.A2 |
| Y0001.3 | SL18 | SPINDLE AIR SEAL | B21 | | /C54.A2 |
| Y0001.2 | SL192 | THROUGH SPINDLE AIR | A21 | | /C47.A3 |
| Y0001.1 | SL09 | TAPER CLEANING AIR 1 | B20 | | /C54.A1 |
| Y0001.0 | SL181 | AIR BLOW | A20 | C24L (A01) | /C47.A2 |

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|---|---|---|---|---|
| DI/DO ADDRESS TABLE - IO MODULE 3 | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | E05 |

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| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 3 | | PAGE |
|---|---|---|---|---|---|
| X0007.7 | | | B05 | COMMON SIGNALS | |
| X0007.6 | | | A05 | | |
| X0007.5 | LS942 | ATC MAGAZINE DOOR CLOSE LS | B04 | +24V (B01) | /C74.E4 |
| X0007.4 | LS912 | OPERATOR DOOR INTERLOCK LS | A04 | | /C72.E4 |
| X0007.3 | RCH2 | SPINDLE HIGH SPEED | B03 | C24L (A01) | /C52.E7 |
| X0007.2 | RCH1 | SPINDLE LOW SPEED | A03 | | /C52.E6 |
| X0007.1 | LS2466 | AREA SENSOR | B02 | | /C76.E5 |
| X0007.0 | GRAY8 | ATC ARM BIT 8 | A02 | DICOM (A14) | /C57.E6 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 3 | | PAGE |
|---|---|---|---|---|---|
| X0008.7 | OT2Z | OVER TRAVEL Z | B09 | COMMON SIGNALS | /C27.E4 |
| X0008.6 | OT2Y | OVER TRAVEL Y | A09 | | /C27.E3 |
| X0008.5 | OT2X | OVER TRAVEL X | B08 | +24V (B01) | /C27.E2 |
| X0008.4 | *ESP | EMERGENCY STOP SIGNAL | A08 | | /C27.E7 |
| X0008.3 | *EMI | EMERGENCY STOP SW | B07 | C24L (A01) | /C10.E3 |
| X0008.2 | FPTP | FUSE BLOWN | A07 | | /C26.E7 |
| X0008.1 | CPTP | CP TRIPPED | B06 | | /C07.E3 |
| X0008.0 | | | A06 | DICOM (A14) | |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 3 | | PAGE |
|---|---|---|---|---|---|
| X0009.7 | 24VOK | POWER SUPPLY OK | B13 | COMMON SIGNALS | /C08.E4 |
| X0009.6 | SPARE3 | SPARE3 | A13 | | |
| X0009.5 | SPARE2 | SPARE2 | B12 | +24V (B01) | |
| X0009.4 | SPARE1 | SPARE1 | A12 | | |
| X0009.3 | *DECM | MAG AXS REF RTN DECEL | B11 | C24L (A01) | /C27.E4 |
| X0009.2 | *DECZ | 3RD AXS REF RTN DECEL | A11 | | /C27.E3 |
| X0009.1 | *DECY | 2ND AXS REF RTN DECEL | B10 | | /C27.E2 |
| X0009.0 | *DECX | 1ST AXS REF RTN DECEL | A10 | DICOM (A14) | /C27.E1 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 3 | | PAGE |
|---|---|---|---|---|---|
| Y0002.7 | | | B19 | COMMON SIGNALS | |
| Y0002.6 | | | A19 | | |
| Y0002.5 | SL913 | ATC DOOR UNLOCK SOLENOID | B18 | C24H (A24) (B24) (A25) (B25) | /C74.A2 |
| Y0002.4 | SL912 | FRONT DOOR UNLOCK SOLENOID | A18 | | /C72.A2 |
| Y0002.3 | MS154 | OIL AIR UNIT PUMP ON | B17 | | /C50.A1 |
| Y0002.2 | SPARE3 | SPARE3 | A17 | | |
| Y0002.1 | SPARE2 | SPARE2 | B16 | | |
| Y0002.0 | CT1 | VISION BTS | A16 | C24L (A01) | /C112.A2 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 3 | | PAGE |
|---|---|---|---|---|---|
| Y0003.7 | PRBSEL | PROBE SELECT | B23 | COMMON SIGNALS | /C61.A2 |
| Y0003.6 | OPPON | MEASURING PROBE POWER ON | A23 | | /C61.A2 |
| Y0003.5 | SL302 | TOOL CLEANING AIR | B22 | C24H (A24) (B24) (A25) (B25) | /C61.A3 |
| Y0003.4 | SL303 | PURGE AIR | A22 | | /C61.A3 |
| Y0003.3 | SL304 | LASER SHUTTER OPEN | B21 | | /C61.A4 |
| Y0003.2 | LSRCL | LASER CLEARANCE (ENABLE 0) | A21 | | /C61.A3 |
| Y0003.1 | ENBL2 | ENABLE 2 | B20 | | /C61.A5 |
| Y0003.0 | ENBL1 | ENABLE 1 | A20 | C24L (A01) | /C61.A4 |

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|---|---|---|---|---|
| DI/DO ADDRESS TABLE - IO MODULE 3 | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | E06 |

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| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 4 | | PAGE |
|---|---|---|---|---|---|
| X0010.7 | LS138L | AUTO FRONT DOOR OPEN LEFT | B05 | COMMON SIGNALS | /C75.E4 |
| X0010.6 | LS138R | AUTO FRONT DOOR OPEN RIGHT | A05 | | /C75.E4 |
| X0010.5 | LS2064 | AUTO GREASE SPLY UNIT LEVEL SW | B04 | +24V (B01) | /C83.E7 |
| X0010.4 | LS1698 | AUTO GREASE INJECTOR PRESS UP | A04 | | /C83.E6 |
| X0010.3 | TH16 | BASE COOLANT TH ALM | B03 | C24L (A01) | /C30.E5 |
| X0010.2 | TH2 | NOZZLE COOLANT MOTOR TH ALARM | A03 | | /C29.E5 |
| X0010.1 | LS01 | TOOL UNCLAMP | B02 | | /C55.E7 |
| X0010.0 | LS00 | TOOL CLAMP | A02 | DICOM (A14) | /C55.E6 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 4 | | PAGE |
|---|---|---|---|---|---|
| X0011.7 | TH89 | MTC AIR CON THERMAL ALARM | B09 | COMMON SIGNALS | /C85.E4 |
| X0011.6 | | | A09 | | |
| X0011.5 | *LS905 | AIR PRESS SWITCH | B08 | +24V (B01) | /C48.E5 |
| X0011.4 | TH4 | HYDRAULIC UNIT TH ALARM | A08 | | /C49.E4 |
| X0011.3 | LS748 | ATC SHUTTER OPEN | B07 | C24L (A01) | /C59.E6 |
| X0011.2 | LS747 | ATC SHUTTER CLOSE | A07 | | /C59.E5 |
| X0011.1 | TH7A | SPIRAL CHIP CONV MTR TH ALARM A | B06 | | /C28.E7 |
| X0011.0 | TH7 | SPIRAL CHIP CONV MTR TH ALARM | A06 | DICOM (A14) | /C28.E6 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 4 | | PAGE |
|---|---|---|---|---|---|
| X0012.7 | LS3320 | TSC PRESSURE SWITCH | B13 | COMMON SIGNALS | /C38.E4 |
| X0012.6 | *LS184 | SEC TANK EMPTY DETECT | A13 | | /C46.D5 |
| X0012.5 | LS2323 | OIL AIR FLOW HIGH LEVEL | B12 | +24V (B01) | /C114.E5 |
| X0012.4 | LS2312 | OIL AIR FLOW LOW LEVEL | A12 | | /C114.E4 |
| X0012.3 | LS602 | SPINDLE TOOL CONFIRM | B11 | C24L (A01) | /C55.E5 |
| X0012.2 | LS601 | SPINDLE TOOL SEATING | A11 | | /C55.E6 |
| X0012.1 | LS968 | SPINDLE HYD UNIT PRESS SW (HIGH) | B10 | | /C49.E7 |
| X0012.0 | LS969 | SPINDLE HYD UNIT PRESS SW (LOW) | A10 | DICOM (A14) | /C49.E6 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 4 | | PAGE |
|---|---|---|---|---|---|
| Y0004.7 | SL132A-1 | AUTO FRONT DOOR CLOSE | B19 | COMMON SIGNALS | /C75.A2 |
| Y0004.6 | SL132B-1 | AUTO FRONT DOOR OPEN | A19 | | /C75.A1 |
| Y0004.5 | ALSXYF | AUTO GREASE INJECTOR CW | B18 | C24H (A24) (B24) (A25) (B25) | /C83.A2 |
| Y0004.4 | | | A18 | | |
| Y0004.3 | MS16 | BASE COOLANT | B17 | | /C30.A1 |
| Y0004.2 | MS2 | NOZZLE COOLANT | A17 | | /C29.A2 |
| Y0004.1 | SL01B | SPINDLE TOOL UNCLAMP SOLENOID | B16 | | /C54.A4 |
| Y0004.0 | SL01A | SPINDLE TOOL CLAMP SOLENOID | A16 | C24L (A01) | /C54.A3 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 4 | | PAGE |
|---|---|---|---|---|---|
| Y0005.7 | | | B23 | COMMON SIGNALS | |
| Y0005.6 | | | A23 | | |
| Y0005.5 | SL905 | MASTER SOLENOID VALVE | B22 | C24H (A24) (B24) (A25) (B25) | /C48.A2 |
| Y0005.4 | MS4 | HYDRAULIC UNIT ON | A22 | | /C49.A1 |
| Y0005.3 | SL711B | ATC SHUTTER OPEN | B21 | | /C59.A3 |
| Y0005.2 | SL711A | ATC SHUTTER CLOSE | A21 | | /C59.A2 |
| Y0005.1 | MS7R | SPIRAL CONVEYOR REV | B20 | | /C28.A2 |
| Y0005.0 | MS7F | SPIRAL CONVEYOR FWD | A20 | C24L (A01) | /C28.A1 |

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|---|---|---|---|---|
| DI/DO ADDRESS TABLE - IO MODULE 4 | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | E07 |

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| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 4 | | PAGE |
|---|---|---|---|---|---|
| X0013.7 | | | B05 | COMMON SIGNALS | |
| X0013.6 | | | A05 | | |
| X0013.5 | MGCCW | ATC MAGAZINE CCW SW | B04 | +24V (B01) | /C60.E6 |
| X0013.4 | MGCW | ATC MAGAZINE CW SW | A04 | | /C60.E5 |
| X0013.3 | ATCUNC | ATC DOOR UNLOCK SW | B03 | C24L (A01) | /C60.E3 |
| X0013.2 | ATCMAS | ATC MANUAL INTERRUPT | A03 | | /C60.E2 |
| X0013.1 | LS22 | POT VERTICAL | B02 | | /C58.E6 |
| X0013.0 | LS21 | POT HORIZONTAL | A02 | DICOM (A14) | /C58.E5 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 4 | | PAGE |
|---|---|---|---|---|---|
| X0014.7 | | | B09 | COMMON SIGNALS | |
| X0014.6 | | | A09 | | |
| X0014.5 | TH18 | OIL SKIMMER MTR TH ALM | B08 | +24V (B01) | /C43.E5 |
| X0014.4 | LS137R | AUTO DOOR CLOSE SLOW DOWN | A08 | | /C75.E5 |
| X0014.3 | THARM | INVERTER ALARM | B07 | C24L (A01) | /C56.E5 |
| X0014.2 | TMATC | ATC ARM MTR OVERHEAT | A07 | | /C56.E4 |
| X0014.1 | | | B06 | | |
| X0014.0 | | | A06 | DICOM (A14) | |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 4 | | PAGE |
|---|---|---|---|---|---|
| X0015.7 | | | B13 | COMMON SIGNALS | |
| X0015.6 | | | A13 | | |
| X0015.5 | | | B12 | +24V (B01) | |
| X0015.4 | | | A12 | | |
| X0015.3 | | | B11 | C24L (A01) | |
| X0015.2 | | | A11 | | |
| X0015.1 | LS192 | SEC FIL TANK LOWER FLOAT | B10 | | |
| X0015.0 | LS191 | SEC FIL TANK UPPER FLOAT | A10 | DICOM (A14) | |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 4 | | PAGE |
|---|---|---|---|---|---|
| Y0006.7 | PIVRM | INVERTER SPEED 2 (MID SPEED) | B19 | COMMON SIGNALS | /C56.A4 |
| Y0006.6 | PIVRH | INVERTER SPEED 1 (HIGH SPEED) | A19 | | /C56.A3 |
| Y0006.5 | LMCW | ATC MAGAZINE CW LAMP | B18 | C24H (A24) (B24) (A25) (B25) | /C60.A5 |
| Y0006.4 | LMCCW | ATC MAGAZINE CCW LAMP | A18 | | /C60.A6 |
| Y0006.3 | LMGSB | ATC DOOR UNLOCK LAMP | B17 | | /C60.A3 |
| Y0006.2 | ATCMAL | ATC MANUAL INTERRUPT LAMP | A17 | | /C60.A2 |
| Y0006.1 | SL20 | POT HORIZONTAL | B16 | | /C58.A3 |
| Y0006.0 | SL21 | POT VERTICAL | A16 | C24L (A01) | /C58.A2 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 4 | | PAGE |
|---|---|---|---|---|---|
| Y0007.7 | | | B23 | COMMON SIGNALS | |
| Y0007.6 | NZSLVLV | NOZZLE SOL VALVE | A23 | | /C29.A2 |
| Y0007.5 | MS18 | OIL SKIMMER | B22 | C24H (A24) (B24) (A25) (B25) | /C43.A2 |
| Y0007.4 | SL132A-2 | AUTO DOOR CLOSE SLOW DOWN | A22 | | /C75.A3 |
| Y0007.3 | MS78R | ATC MOTOR REVERSE | B21 | | /C56.A3 |
| Y0007.2 | MS78F | ATC MOTOR FORWARD | A21 | | /C56.A2 |
| Y0007.1 | MS78B | ATC MOTOR BRAKE | B20 | | /C56.A1 |
| Y0007.0 | PIVRL | INVERTER SPEED 3 (LOW SPEED) | A20 | C24L (A01) | /C56.A2 |

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|---|---|---|---|---|
| DI/DO ADDRESS TABLE - IO MODULE 4 | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | E08 |

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| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 5 | | PAGE |
|---|---|---|---|---|---|
| X0016.7 | LS1338 | TSC PRESSURE CHECK | B05 | COMMON SIGNALS | /C38.E5 |
| X0016.6 | LS193 | TSC FLOW SWITCH | A05 | | /C35.E4 |
| X0016.5 | LS1311 | TSC LEAKAGE DETECT | B04 | +24V (B01) | /C33.E4 |
| X0016.4 | | | A04 | | |
| X0016.3 | LS155 | 5TH AXIS UNCLAMP SWITCH | B03 | C24L (A01) | /C79.E5 |
| X0016.2 | LS154 | 5TH AXIS CLAMP SWITCH | A03 | | /C79.E5 |
| X0016.1 | LS152 | 4TH AXIS UNCLAMP SWITCH | B02 | | /C77.E5 |
| X0016.0 | LS151 | 4TH AXIS CLAMP SWITCH | A02 | DICOM (A14) | /C77.E5 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 5 | | PAGE |
|---|---|---|---|---|---|
| X0017.7 | | | B09 | COMMON SIGNALS | |
| X0017.6 | TH19 | MIST COLLECTOR MOTOR TH ALM | A09 | | /C44.E5 |
| X0017.5 | TH10 | OVERHEAD SHOWER COOLANT TH ALM | B08 | +24V (B01) | /C31.E5 |
| X0017.4 | TH8 | TSC PUMP MOTOR TH ALM | A08 | | /C32.E7 |
| X0017.3 | LCMAS | LUCC MANUAL INTERRUPT SW | B07 | C24L (A01) | /C42.E4 |
| X0017.2 | TH6 | LUCC MOTOR TH ALM | A07 | | /C41.E7 |
| X0017.1 | LCREV | LUCC REVERSE SW | B06 | | /C42.E5 |
| X0017.0 | LCFWD | LUCC FORWARD SW | A06 | DICOM (A14) | /C42.E5 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 5 | | PAGE |
|---|---|---|---|---|---|
| X0018.7 | ROTA / +L4 | 4TH AXIS OVERTRAVEL | B13 | COMMON SIGNALS | /C77.E4 |
| X0018.6 | 4NG | 4TH AXS NEGLECT | A13 | | /C77.E3 |
| X0018.5 | ROTB / +L5 | 5TH AXIS OVERTRAVEL | B12 | +24V (B01) | /C79.E4 |
| X0018.4 | 5NG | 5TH AXS NEGLECT | A12 | | /C79.E3 |
| X0018.3 | -L5 | 5TH AXS MINUS SIDE STROKE END | B11 | C24L (A01) | |
| X0018.2 | -L4 | 4TH AXS MINUS SIDE STROKE END | A11 | | |
| X0018.1 | | | B10 | | |
| X0018.0 | | | A10 | DICOM (A14) | |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 5 | | PAGE |
|---|---|---|---|---|---|
| Y0008.7 | SL1310 | TS SUCTION SOLENOID | B19 | COMMON SIGNALS | /C33.A7 |
| Y0008.6 | | | A19 | | |
| Y0008.5 | SL1309 | TS UNION DRAIN | B18 | C24H (A24) (B24) (A25) (B25) | /C33.A5 |
| Y0008.4 | SL1308 | TS UNION BACKUP AIR | A18 | | /C33.A6 |
| Y0008.3 | SL151B | 5TH AXIS BRAKE/CLAMP | B17 | | /C80.A2 |
| Y0008.2 | SL151 | 5TH AXIS UNCLAMP | A17 | | /C79.A1 |
| Y0008.1 | SL150B | 4TH AXIS BRAKE/CLAMP | B16 | | /C78.A2 |
| Y0008.0 | SL150 | 4TH AXIS UNCLAMP | A16 | C24L (A01) | /C77.A1 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 5 | | PAGE |
|---|---|---|---|---|---|
| Y0009.7 | | | B23 | COMMON SIGNALS | |
| Y0009.6 | MS19 | MIST COLLECTOR ON | A23 | | /C44.A2 |
| Y0009.5 | MS10 | OVERHEAD SHOWER COOLANT ON | B22 | C24H (A24) (B24) (A25) (B25) | /C31.A1 |
| Y0009.4 | MS8 | THROUGH SPINDLE COOLANT ON | A22 | | /C32.A4 |
| Y0009.3 | MS6R | LUCC REVERSE ON | B21 | | /C41.A2 |
| Y0009.2 | MS6F | LUCC FORWARD ON | A21 | | /C41.A1 |
| Y0009.1 | LCMAL | LUCC MANUAL INTERRUPT LAMP | B20 | | /C42.A2 |
| Y0009.0 | | | A20 | C24L (A01) | |

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| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 5 | | PAGE |
|---|---|---|---|---|---|
| X0019.7 | EXTRDY2 | EXTERNAL READY SIGNAL (AWC) | B05 | COMMON SIGNALS | /C95.E6 |
| X0019.6 | EXTRDY1 | EXTERNAL READY SIGNAL (WORK) | A05 | | /C95.E5 |
| X0019.5 | | | B04 | +24V (B01) | |
| X0019.4 | | | A04 | | |
| X0019.3 | | | B03 | C24L (A01) | |
| X0019.2 | | | A03 | | |
| X0019.1 | | | B02 | DICOM (A14) | |
| X0019.0 | | | A02 | | |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 5 | | PAGE |
|---|---|---|---|---|---|
| X0020.7 | TH260-2 | 2nd HYD PP MTR THERMAL ALARM | B09 | COMMON SIGNALS | /C81.E4 |
| X0020.6 | TH71 | SEC FILTER TANK PUMP TH ALM | A09 | | /C39.E5 |
| X0020.5 | TH6A | DRUM PUMP TH ALARM | B08 | +24V (B01) | /C40.E5 |
| X0020.4 | LS1306 | LUCC SLIP DETECT | A08 | | /C42.E7 |
| X0020.3 | | | B07 | C24L (A01) | |
| X0020.2 | | | A07 | | |
| X0020.1 | | | B06 | DICOM (A14) | |
| X0020.0 | | | A06 | | |

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|---|---|---|---|---|---|
| X0021.7 | | | B13 | COMMON SIGNALS | |
| X0021.6 | | | A13 | | |
| X0021.5 | TSCCG2 | 75% CLOG, WARNING | B12 | +24V (B01) | /C34.E4 |
| X0021.4 | TSCCG1 | 100% CLOG, ALARM | A12 | | /C34.E3 |
| X0021.3 | SYSPS-2-2 | 2nd HYD PP SYSTEM PRESS (Low) | B11 | C24L (A01) | /C81.E5 |
| X0021.2 | SYSPS-2 | 2nd HYD PP SYSTEM PRESS (High) | A11 | | /C81.E6 |
| X0021.1 | *DEC5 | 5TH AXS REF RETURN DECEL | B10 | DICOM (A14) | /C79.E4 |
| X0021.0 | *DEC4 | 4TH AXS REF RETURN DECEL | A10 | | /C77.E4 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 5 | | PAGE |
|---|---|---|---|---|---|
| Y0010.7 | MCRDY2 | MACHINE READY (DOOR/ATC) | B19 | COMMON SIGNALS | /C95.A5 |
| Y0010.6 | MCRDY1 | MACHINE READY (Stop) | A19 | | /C95.A4 |
| Y0010.5 | | | B18 | C24H (A24) (B24) (A25) (B25) | |
| Y0010.4 | | | A18 | | |
| Y0010.3 | | | B17 | | |
| Y0010.2 | | | A17 | | |
| Y0010.1 | | | B16 | C24L (A01) | |
| Y0010.0 | | | A16 | | |

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|---|---|---|---|---|---|
| Y0011.7 | M260-2 | 2nd HDY POWER PACK | B23 | COMMON SIGNALS | /C81.A2 |
| Y0011.6 | MS71 | SEC FILTER TANK PUMP ON | A23 | | /C39.A1 |
| Y0011.5 | MS6A | DRUM FILTER CLEANING PUMP ON | B22 | C24H (A24) (B24) (A25) (B25) | /C40.A1 |
| Y0011.4 | | | A22 | | |
| Y0011.3 | | | B21 | | |
| Y0011.2 | | | A21 | | |
| Y0011.1 | | | B20 | C24L (A01) | |
| Y0011.0 | | | A20 | | |

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|---|---|---|---|---|---|
| X0022.7 | | | B05 | COMMON SIGNALS | |
| X0022.6 | | | A05 | | |
| X0022.5 | LS1619 | TOOL ID ARM ADVANCE LS | B04 | +24V (B01) | /C115.E4 |
| X0022.4 | LS1618 | TOOL ID ARM RETRACT LS | A04 | | /C115.E4 |
| X0022.3 | | | B03 | C24L (A01) | |
| X0022.2 | | | A03 | | |
| X0022.1 | | | B02 | DICOM (A14) | |
| X0022.0 | | | A02 | | |

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|---|---|---|---|---|---|
| X0023.7 | | | B09 | COMMON SIGNALS | |
| X0023.6 | LS2761 | TOOL ID POSITION TOOL COMFIRM | A09 | | /C115.E5 |
| X0023.5 | | | B08 | +24V (B01) | |
| X0023.4 | | | A08 | | |
| X0023.3 | | | B07 | C24L (A01) | |
| X0023.2 | | | A07 | | |
| X0023.1 | | | B06 | DICOM (A14) | |
| X0023.0 | | | A06 | | |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 6 | | PAGE |
|---|---|---|---|---|---|
| X0024.7 | | | B13 | COMMON SIGNALS | |
| X0024.6 | | | A13 | | |
| X0024.5 | | | B12 | +24V (B01) | |
| X0024.4 | | | A12 | | |
| X0024.3 | | | B11 | C24L (A01) | |
| X0024.2 | | | A11 | | |
| X0024.1 | | | B10 | DICOM (A14) | |
| X0024.0 | | | A10 | | |

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|---|---|---|---|---|---|
| Y0012.7 | | | B19 | COMMON SIGNALS | |
| Y0012.6 | | | A19 | | |
| Y0012.5 | | | B18 | C24H (A24) (B24) (A25) (B25) | |
| Y0012.4 | | | A18 | | |
| Y0012.3 | SL929B | TOOL ID ARM RETRACT SOLENOID | B17 | | /C115.A2 |
| Y0012.2 | SL929A | TOOL ID ARM ADVANCE SOLENOID | A17 | | /C115.A2 |
| Y0012.1 | | | B16 | C24L (A01) | |
| Y0012.0 | | | A16 | | |

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|---|---|---|---|---|---|
| Y0013.7 | | | B23 | COMMON SIGNALS | |
| Y0013.6 | | | A23 | | |
| Y0013.5 | | | B22 | C24H (A24) (B24) (A25) (B25) | |
| Y0013.4 | | | A22 | | |
| Y0013.3 | | | B21 | | |
| Y0013.2 | | | A21 | | |
| Y0013.1 | | | B20 | C24L (A01) | |
| Y0013.0 | | | A20 | | |

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|---|---|---|---|---|---|
| X0025.7 | USDI4 | USER DI 4 | B05 | COMMON SIGNALS | /C71.E4 |
| X0025.6 | USDI3 | USER DI 3 | A05 | | /C71.E4 |
| X0025.5 | USDI2 | USER DI 2 | B04 | +24V (B01) | /C71.E3 |
| X0025.4 | USDI1 | USER DI 1 | A04 | | /C71.E2 |
| X0025.3 | | | B03 | C24L (A01) | |
| X0025.2 | | | A03 | | |
| X0025.1 | | | B02 | | |
| X0025.0 | | | A02 | DICOM (A14) | |

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|---|---|---|---|---|---|
| X0026.7 | | | B09 | COMMON SIGNALS | |
| X0026.6 | | | A09 | | |
| X0026.5 | | | B08 | +24V (B01) | |
| X0026.4 | | | A08 | | |
| X0026.3 | USDI8 | USER DI 8 | B07 | C24L (A01) | /C71.E7 |
| X0026.2 | USDI7 | USER DI 7 | A07 | | /C71.E6 |
| X0026.1 | USDI6 | USER DI 6 | B06 | | /C71.E6 |
| X0026.0 | USDI5 | USER DI 5 | A06 | DICOM (A14) | /C71.E5 |

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|---|---|---|---|---|---|
| X0027.7 | | | B13 | COMMON SIGNALS | |
| X0027.6 | | | A13 | | |
| X0027.5 | | | B12 | +24V (B01) | |
| X0027.4 | | | A12 | | |
| X0027.3 | | | B11 | C24L (A01) | |
| X0027.2 | | | A11 | | |
| X0027.1 | | | B10 | | |
| X0027.0 | | | A10 | DICOM (A14) | |

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|---|---|---|---|---|---|
| Y0014.7 | USDO4 | USER DO4 | B19 | COMMON SIGNALS | /C71.A4 |
| Y0014.6 | USDO3 | USER DO3 | A19 | | /C71.A3 |
| Y0014.5 | USDO2 | USER DO2 | B18 | C24H (A24) (B24) (A25) (B25) | /C71.A3 |
| Y0014.4 | USDO1 | USER DO1 | A18 | | /C71.A2 |
| Y0014.3 | | | B17 | | |
| Y0014.2 | | | A17 | | |
| Y0014.1 | | | B16 | | |
| Y0014.0 | | | A16 | C24L (A01) | |

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|---|---|---|---|---|---|
| Y0015.7 | | | B23 | COMMON SIGNALS | |
| Y0015.6 | | | A23 | | |
| Y0015.5 | | | B22 | C24H (A24) (B24) (A25) (B25) | |
| Y0015.4 | | | A22 | | |
| Y0015.3 | USDO8 | USER DO8 | B21 | | /C71.A7 |
| Y0015.2 | USDO7 | USER DO7 | A21 | | /C71.A6 |
| Y0015.1 | USDO6 | USER DO6 | B20 | | /C71.A5 |
| Y0015.0 | USDO5 | USER DO5 | A20 | C24L (A01) | /C71.A5 |

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|---|---|---|---|---|---|
| X0028.7 | NOWPST | AUTO START WITHOUT WORK | B05 | COMMON SIGNALS | /C93.E4 |
| X0028.6 | WULDFN | WORK UNLOAD FINISH | A05 | | /C93.E4 |
| X0028.5 | WLDFN | WORK LOAD FINISH | B04 | +24V (B01) | /C93.E4 |
| X0028.4 | SYSTP | AUTO STOP INSTRUCTION | A04 | | /C93.E3 |
| X0028.3 | SYSTR | AUTO START INSTRUCTION | B03 | C24L (A01) | /C93.E3 |
| X0028.2 | *RBINT | ROBOT IN INTERFERENCE ZONE | A03 | | /C93.E3 |
| X0028.1 | *RBALM | ROBOT ALARM | B02 | | /C93.E2 |
| X0028.0 | RBHLY | ROBOT HEALTHY | A02 | DICOM (A14) | /C93.E2 |

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|---|---|---|---|---|---|
| X0029.7 | WPNPRT | WORK NUMBER PARITY | B09 | COMMON SIGNALS | /C94.E1 |
| X0029.6 | WPN08 | WORK NUMBER 08 | A09 | | /C93.E7 |
| X0029.5 | WPN04 | WORK NUMBER 04 | B08 | +24V (B01) | /C93.E7 |
| X0029.4 | WPN02 | WORK NUMBER 02 | A08 | | /C93.E6 |
| X0029.3 | WPN01 | WORK NUMBER 01 | B07 | C24L (A01) | /C93.E6 |
| X0029.2 | FUCRQ | FIXTURE UNCLAMP REQUEST | A07 | | /C93.E6 |
| X0029.1 | FCLRQ | FIXTURE CLAMP REQUEST | B06 | | /C93.E5 |
| X0029.0 | FLSON | FIXTURE WASH INSTRUCTION | A06 | DICOM (A14) | /C93.E5 |

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|---|---|---|---|---|---|
| X0030.7 | RBI24 | ROBOT INTERFACE INPUT 24 | B13 | COMMON SIGNALS | /C94.E4 |
| X0030.6 | RBI23 | ROBOT INTERFACE INPUT 23 | A13 | | /C94.E4 |
| X0030.5 | RBI22 | ROBOT INTERFACE INPUT 22 | B12 | +24V (B01) | /C94.E4 |
| X0030.4 | RBI21 | ROBOT INTERFACE INPUT 21 | A12 | | /C94.E3 |
| X0030.3 | RBI20 | ROBOT INTERFACE INPUT 20 | B11 | C24L (A01) | /C94.E3 |
| X0030.2 | RBI19 | ROBOT INTERFACE INPUT 19 | A11 | | /C94.E2 |
| X0030.1 | RBI18 | ROBOT INTERFACE INPUT 18 | B10 | | /C94.E2 |
| X0030.0 | RBI17 | ROBOT INTERFACE INPUT 17 | A10 | DICOM (A14) | /C94.E2 |

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|---|---|---|---|---|---|
| Y0016.7 | WULDRQ | WORK UNLOAD REQUEST | B19 | COMMON SIGNALS | /C93.A4 |
| Y0016.6 | WLDRQ | WORK LOAD REQUEST | A19 | | /C93.A4 |
| Y0016.5 | MCRUN | MACHINE RUNNING | B18 | C24H (A24) (B24) (A25) (B25) | /C93.A3 |
| Y0016.4 | WSRDY | START RESERVATION CONDITION | A18 | | /C93.A3 |
| Y0016.3 | ACSOK | INTERVENTION PREMISSION | B17 | | /C93.A3 |
| Y0016.2 | *MCWRN | MACHINE WARNING | A17 | | /C93.A2 |
| Y0016.1 | *MCALM | MACHINE ALARM | B16 | | /C93.A2 |
| Y0016.0 | MCRDY | MACHINE READY | A16 | C24L (A01) | /C93.A1 |

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|---|---|---|---|---|---|
| Y0017.7 | WCN02 | WORK CONDITION NO 2 | B23 | COMMON SIGNALS | /C93.A7 |
| Y0017.6 | WCN01 | WORK CONDITION NO 1 | A23 | | /C93.A7 |
| Y0017.5 | PCP02 | STOCKER PALLET NO 2 | B22 | C24H (A24) (B24) (A25) (B25) | /C93.A6 |
| Y0017.4 | PCP01 | STOCKER PALLET NO 1 | A22 | | /C93.A6 |
| Y0017.3 | FUCFN | FIXTURE UNCLAMP FINISH | B21 | | /C93.A6 |
| Y0017.2 | FCLFN | FIXTURE CLAMP FINISH | A21 | | /C93.A5 |
| Y0017.1 | WSCNG | WORK SEATING NG | B20 | | /C93.A5 |
| Y0017.0 | WSCOK | WORK SEATING OK | A20 | C24L (A01) | /C93.A4 |

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|---|---|---|---|---|---|
| X0031.7 | | | B05 | COMMON SIGNALS | |
| X0031.6 | | | A05 | | |
| X0031.5 | | | B04 | +24V (B01) | |
| X0031.4 | | | A04 | | |
| X0031.3 | | | B03 | C24L (A01) | |
| X0031.2 | | | A03 | | |
| X0031.1 | | | B02 | DICOM (A14) | |
| X0031.0 | | | A02 | | |

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|---|---|---|---|---|---|
| X0032.7 | | | B09 | COMMON SIGNALS | |
| X0032.6 | | | A09 | | |
| X0032.5 | | | B08 | +24V (B01) | |
| X0032.4 | | | A08 | | |
| X0032.3 | | | B07 | C24L (A01) | |
| X0032.2 | | | A07 | | |
| X0032.1 | | | B06 | DICOM (A14) | |
| X0032.0 | | | A06 | | |

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| X0033.7 | | | B13 | COMMON SIGNALS | |
| X0033.6 | | | A13 | | |
| X0033.5 | | | B12 | +24V (B01) | |
| X0033.4 | | | A12 | | |
| X0033.3 | | | B11 | C24L (A01) | |
| X0033.2 | | | A11 | | |
| X0033.1 | | | B10 | DICOM (A14) | |
| X0033.0 | | | A10 | | |

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| Y0018.7 | | | B19 | COMMON SIGNALS | |
| Y0018.6 | | | A19 | | |
| Y0018.5 | | | B18 | C24H (A24) (B24) (A25) (B25) | |
| Y0018.4 | | | A18 | | |
| Y0018.3 | | | B17 | | |
| Y0018.2 | | | A17 | | |
| Y0018.1 | | | B16 | C24L (A01) | |
| Y0018.0 | | | A16 | | |

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| Y0019.7 | | | B23 | COMMON SIGNALS | |
| Y0019.6 | | | A23 | | |
| Y0019.5 | | | B22 | C24H (A24) (B24) (A25) (B25) | |
| Y0019.4 | | | A22 | | |
| Y0019.3 | | | B21 | | |
| Y0019.2 | | | A21 | | |
| Y0019.1 | | | B20 | C24L (A01) | |
| Y0019.0 | | | A20 | | |

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|---|---|---|---|---|---|
| X0060.7 | PS2604 | HYDRAULIC PRESSURE SW 4 | B05 | COMMON SIGNALS | /C91.E7 |
| X0060.6 | PS2603 | HYDRAULIC PRESSURE SW 3 | A05 | | /C91.E6 |
| X0060.5 | PS2602 | HYDRAULIC PRESSURE SW 2 | B04 | +24V (B01) | /C91.E5 |
| X0060.4 | PS2601 | HYDRAULIC PRESSURE SW 1 | A04 | | /C91.E4 |
| X0060.3 | EJUC | STOCKER SIDE EJECT UNCLAMP PB | B03 | C24L (A01) | /C92.E5 |
| X0060.2 | EJCL | STOCKER SIDE EJECT CLAMP PB | A03 | | /C92.E4 |
| X0060.1 | HY1UC | HYDRAULIC UNCLAMP PB | B02 | | /C92.E4 |
| X0060.0 | HY1CL | HYDRAULIC CLAMP PB | A02 | DICOM (A14) | /C92.E3 |

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|---|---|---|---|---|---|
| X0061.7 | | | B09 | COMMON SIGNALS | |
| X0061.6 | AIRSV | AIR SEAT CHECK SW | A09 | | /C89.E3 |
| X0061.5 | AUT | AUTO MODE SELECTION SW | B08 | +24V (B01) | /C92.E2 |
| X0061.4 | TH260 | HYDRAULIC MOTOR OVERLOAD | A08 | | /C90.E4 |
| X0061.3 | | | B07 | C24L (A01) | |
| X0061.2 | | | A07 | | |
| X0061.1 | | | B06 | | |
| X0061.0 | | | A06 | DICOM (A14) | |

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| X0062.7 | | | B13 | COMMON SIGNALS | |
| X0062.6 | | | A13 | | |
| X0062.5 | | | B12 | +24V (B01) | |
| X0062.4 | | | A12 | | |
| X0062.3 | LS1788 | AIR SEAT SENSOR 8 | B11 | C24L (A01) | /C89.E7 |
| X0062.2 | LS1787 | AIR SEAT SENSOR 7 | A11 | | /C89.E7 |
| X0062.1 | LS1786 | AIR SEAT SENSOR 6 | B10 | | /C89.E6 |
| X0062.0 | LS1785 | AIR SEAT SENSOR 5 | A10 | DICOM (A14) | /C89.E6 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 8 | | PAGE |
|---|---|---|---|---|---|
| Y0060.7 | SL2604 | HYDRAULIC SOLENOID 4 | B19 | COMMON SIGNALS | /C91.A3 |
| Y0060.6 | SL2603 | HYDRAULIC SOLENOID 3 | A19 | | /C91.A3 |
| Y0060.5 | SL2602 | HYDRAULIC SOLENOID 2 | B18 | C24H (A24) (B24) (A25) (B25) | /C91.A2 |
| Y0060.4 | SL2601 | HYDRAULIC SOLENOID 1 | A18 | | /C91.A1 |
| Y0060.3 | LEJUC | EJECTOR UNCLAMP LAMP | B17 | | /C92.A5 |
| Y0060.2 | LEJCL | EJECTOR CLAMP LAMP | A17 | | /C92.A4 |
| Y0060.1 | LHY1UC | FIXTURE UNCLAMP LAMP | B16 | | /C92.A4 |
| Y0060.0 | LHY1CL | FIXTURE CLAMP LAMP | A16 | C24L (A01) | /C92.A3 |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 8 | | PAGE |
|---|---|---|---|---|---|
| Y0061.7 | | | B23 | COMMON SIGNALS | |
| Y0061.6 | LAIRS | AIR SEAT CHECK VALID LAMP | A23 | | /C89.A2 |
| Y0061.5 | MNLLMP | FIXTURE MANUAL LAMP | B22 | C24H (A24) (B24) (A25) (B25) | /C92.A2 |
| Y0061.4 | MS260 | HYDRAULIC MOTOR ON | A22 | | /C90.A2 |
| Y0061.3 | | | B21 | | |
| Y0061.2 | | | A21 | | |
| Y0061.1 | | | B20 | | |
| Y0061.0 | | | A20 | C24L (A01) | |

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| | ④ | | | |
| | ③ | APPROVED | 13/06/2019 | AVINASH |
| | ② | CHECKED | 13/06/2019 | AVINASH |
| | ① | DRAWN | 13/06/2019 | JAVAHAR |

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| DI/DO ADDRESS TABLE - IO MODULE 8 | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. E15 |

# IOM8-CE57

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 8 | | PAGE |
|---|---|---|---|---|---|
| X0063.7 | | | B05 | COMMON SIGNALS | |
| X0063.6 | | | A05 | | |
| X0063.5 | | | B04 | +24V (B01) | |
| X0063.4 | | | A04 | | |
| X0063.3 | | | B03 | C24L (A01) | |
| X0063.2 | | | A03 | | |
| X0063.1 | | | B02 | DICOM (A14) | |
| X0063.0 | | | A02 | | |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 8 | | PAGE |
|---|---|---|---|---|---|
| X0064.7 | LS1784 | AIR SEAT SENSOR 4 | B09 | COMMON SIGNALS | /C89.E5 |
| X0064.6 | LS1783 | AIR SEAT SENSOR 3 | A09 | | /C89.E5 |
| X0064.5 | LS1782 | AIR SEAT SENSOR 2 | B08 | +24V (B01) | /C89.E4 |
| X0064.4 | LS1781 | AIR SEAT SENSOR 1 | A08 | | /C89.E4 |
| X0064.3 | | | B07 | C24L (A01) | |
| X0064.2 | | | A07 | | |
| X0064.1 | | | B06 | DICOM (A14) | |
| X0064.0 | | | A06 | | |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 8 | | PAGE |
|---|---|---|---|---|---|
| X0065.7 | | | B13 | COMMON SIGNALS | |
| X0065.6 | STRSW | STRIKER SW | A13 | | /C88.E4 |
| X0065.5 | THPBL | TWO HAND PB LEFT | B12 | +24V (B01) | /C87.E4 |
| X0065.4 | THPBR | TWO HAND PB RIGHT | A12 | | /C87.E4 |
| X0065.3 | LS2603 | HYDRAULIC OIL LEVEL FLOAT SW | B11 | C24L (A01) | /C90.E5 |
| X0065.2 | LS2604 | HYDRAULIC OIL TEMP SENSOR | A11 | | /C90.E5 |
| X0065.1 | SYSPS2 | HYDRAULIC PP SYSTEM PRESS (LOW) | B10 | DICOM (A14) | /C90.E6 |
| X0065.0 | SYSPS | HYDRAULIC PP SYSTEM PRESS (HIGH) | A10 | | /C90.E6 |

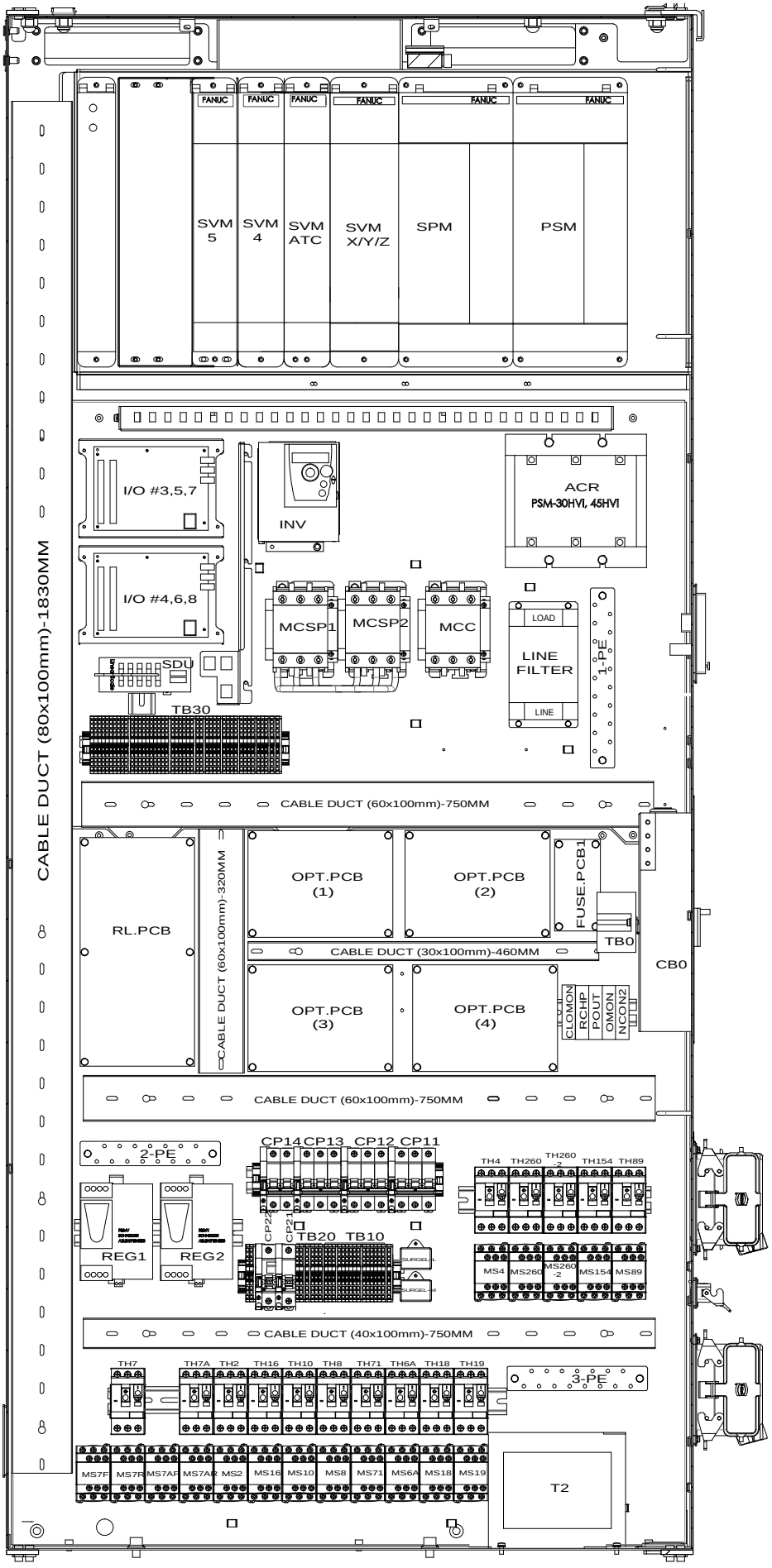
| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 8 | | PAGE |
|---|---|---|---|---|---|
| Y0062.7 | | | B19 | COMMON SIGNALS | |
| Y0062.6 | | | A19 | | |
| Y0062.5 | | | B18 | C24H (A24) (B24) (A25) (B25) | |
| Y0062.4 | | | A18 | | |
| Y0062.3 | | | B17 | | |
| Y0062.2 | | | A17 | | |
| Y0062.1 | | | B16 | C24L (A01) | |
| Y0062.0 | | | A16 | | |

| DGN No. | SYMBOL | SIGNAL NAME | CONNECTOR NO. I/O MOD 8 | | PAGE |
|---|---|---|---|---|---|
| Y0063.7 | | | B23 | COMMON SIGNALS | |
| Y0063.6 | | | A23 | | |
| Y0063.5 | SL1782 | AIR SEAT CHECK SOLENOID 2 | B22 | C24H (A24) (B24) (A25) (B25) | /C89.A2 |
| Y0063.4 | SL1781 | AIR SEAT CHECK SOLENOID 1 | A22 | | /C89.A1 |
| Y0063.3 | | | B21 | | |
| Y0063.2 | | | A21 | | |
| Y0063.1 | | | B20 | C24L (A01) | |
| Y0063.0 | | | A20 | | |

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| DI/DO ADDRESS TABLE - IO MODULE 8 | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | E16 |

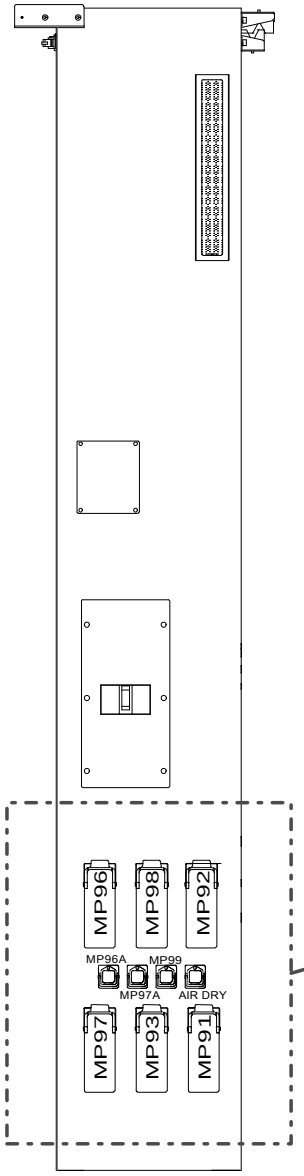



| | | | |
|---|---|---|---|
| GROUP | = | | |
| SHEET No. | F01 | | |
| DRAWING No. | V210I-C1E-XG | | |
| MACHINE TYPE | PS65/PS105 | | |
| NC TYPE | 0i-MF | | |
| TITLE | MTC LAYOUT DETAILS | | |

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OUTSIDE (RIGHT WALL) MTC VIEW

| CONNECTOR No. | OPTION NAME | REMARKS |
|---|---|---|
| MP91 | COOLANT POWER | 24 WAY |
| MP92 | COOLANT SIGNAL | 46 WAY |
| MP93 | LUCC | 24 WAY |
| MP94 | ATC PANEL | |
| MP95 | FLUID OPTIONS | MTC INTERNAL CONNECTOR |
| MP96 | FIXTURE SOL/PS | 24 WAY |
| MP96A | FIXTURE PP1 | 4 WAY |
| MP97 | FIXTURE SOL/PS | 24 WAY |
| MP97A | FIXTURE PP2 | 4 WAY |
| MP98 | ROBOT INTERFACE | 72 WAY |
| MP99 | MIST COLLECTOR | 4 WAY |

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| MTC LAYOUT DETAILS | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | F02 |

| DEVICE DESIGNATION | PART DESCRIPTION | MANUFACTURER | MFR PART NO. | MAKINO PART NO. |
|---|---|---|---|---|
| PSM | αiPS 45HV-B | FANUC | A06B-6252-H030 | Z513A0302510 |
| SPM | αiSP 45HV-B | FANUC | A06B-6272-H045#H610 | Z513A0302520 |
| SVM X/Y/Z | αiSV 40/40/40HV-B | FANUC | A06B-6290-H328 | Z513A0302360 |
| SVM ATC | αiSV 10HV-B | FANUC | A06B-6290-H122 | Z513A0302370 |
| SVM 4 / 5 | αiSV 10HV-B | FANUC | A06B-6290-H122 | Z513A0302370 |
| | αiSV 20HV-B | FANUC | A06B-6290-H123 | Z513A0302380 |
| | αiSV 30HV-B | FANUC | A06B-6290-H124 | Z513A0302390 |
| | αiSV 40HV-B | FANUC | A06B-6290-H125 | Z513A0302400 |
| ACR | AC REACTOR FOR αiPS 30/45H | FANUC | A81L-0001-0191 | A81L-0001-0191 |
| I/O # 3 ~ 8 | I/O MODULE 48/32 | FANUC | A03B-0824-K203 | A03B-0824-K203 |
| SDU | SDU UNIT FOR FEED BACK SCALE | FANUC | A02B-0323-C205 | A02B-0323-C205 |
| LINE FILTER | LINE FILTER 100A | SOSHIN | NF3100C-YXZ | Z721A0010180 |
| INV | INVERTER 1.5KW 3P 415VAC | SCHNEIDER | ATV312HU15N4 | L17B102 |
| MCC | CONTACTOR 95A | SCHNEIDER | LC1D95P7 | L16A100 |
| | SUPPRESSOR BLOCK | SCHNEIDER | LA4DA2U | L16A102 |
| | CONTACT BLOCK | SCHNEIDER | LAD8N11 | L16A103 |
| MCSP1, MCSP2 | CONTACTOR 80A | SCHNEIDER | LC1D80P7 | L16A101 |
| | SUPPRESSOR BLOCK | SCHNEIDER | LA4DA2U | L16A102 |
| | CONTACT BLOCK | SCHNEIDER | LAD8N11 | L16A103 |
| TB10, TB20, TB30 | TB GREY 2.5mm 4W | WAGO | 2002-1401 | Z697I0140003 |
| | TB Y/G 2.5mm 4W | WAGO | 2002-1407 | Z697I0140016 |
| | TB JUMPER 10W | WAGO | 2002-410 | Z697I0140023 |
| | TB JUMPER 2W | WAGO | 2002-402 | Z697I0130004 |
| | TB END PLATE 4 CONN | WAGO | 2002-1492 | L15C232 |
| | TB END STOPPER 6MM | WAGO | 249-116 | L15C218 |
| RELAY PCB | RELAY PCB | ST CIRCUITS | E21VS690A100-XX | E21VS690A100-XX |
| OPTION PCB 1 ~ 4 | OPTION PCB | ST CIRCUITS | E21VS690A101-XX | E21VS690A101-XX |
| FUSE PCB1 | FUSE PCB | ST CIRCUITS | E21VS690A102-XX | E21VS690A102-XX |
| TB0 | M.I.E TERMINAL BLOCK 1P 90A | SCL | IN60BK-C | L15C252 |
| CB0 | MCCB 75A | FUJI | BW125JAG-3P075F=AC240V | L14A160 |
| | MCCB TERM COVER | FUJI | BW9BTCA-L3 L | L14A060 |
| | MCCB 75A UL (FOR MM ONLY) | FUJI | BW125JAGU-3P075F=AC240V | L14A163 |
| CB0 ROTARY | MCCB HANDLE (STD FOR CE AND GB) | FUJI | BW9NOCA | L14A143 |
| RCHP, POUT, OMON, CLOMON, NCON2 | RELAY | PANASONIC | AHN22324 | Z643L1300201 |
| | RELAY SOCKET | PANASONIC | AHNA21 | Z643L1310201 |
| REG1 | DC REGULATOR 3P 415VAC 20A | SCHNEIDER | ABL8WPS24200 | L35A020 |
| REG2 | DC REGULATOR | SCHNEIDER | ABL8RPS24100 | L35A014 |

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| ① | DRAWN | 13/06/2019 | JAVAHAR |

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| MTC COMPONENT DETAILS | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | F03 |

| DEVICE DESIGNATION | PART DESCRIPTION | MANUFACTURER | MFR PART NO. | MAKINO PART NO. |
|---|---|---|---|---|
| CP11,CP13 | CB IC60N 3P 10A C | SCHNEIDER | A9F74310, | L14A112 |
| | AUX SWITCH IC60N | SCHNEIDER | A9A26924 | L14A097 |
| CP12 | CB IC60N 3P 16A C | SCHNEIDER | A9F74316, | L14A107 |
| | AUX SWITCH IC60N | SCHNEIDER | A9A26924 | L14A097 |
| CP14 | CB ACTI9 D60H 2P 6A | SCHNEIDER | A9F85206 | L14A164 |
| | AUX SWITCH IC60N | SCHNEIDER | A9A26924 | L14A097 |
| CP21 | CB IC60N 1P 6A C | SCHNEIDER | A9F74106 | L14A098 |
| | AUX SWITCH IC60N | SCHNEIDER | A9A26924 | L14A097 |
| CP22 | CB IC60N 1P 2A C | SCHNEIDER | A9F74102 | L14A176 |
| | AUX SWITCH IC60N | SCHNEIDER | A9A26924 | L14A097 |
| MS2/TH2 | THERMAL CONTACTOR 2.5A~4A | SCHNEIDER | GV2~ME08 | L16A134 |
| | STARTER CONT.3 POLE 24VDC | SCHNEIDER | LC1D09BD | L16A107 |
| | COMBINATION BLOCK | SCHNEIDER | GV2-AF3 | Z643Y2130020 |
| | AUX CONTACT SW -1NO 1NC | SCHNEIDER | GV AE11 | L16A092 |
| MS4/TH4 | THERMAL CONTACTOR 1A~1.6A | SCHNEIDER | GV2~ME06 | L16A133 |
| | STARTER CONT.3 POLE 24VDC | SCHNEIDER | LC1D09BD | L16A107 |
| | COMBINATION BLOCK | SCHNEIDER | GV2-AF3 | Z643Y2130020 |
| | AUX CONTACT SW -1NO 1NC | SCHNEIDER | GV AE11 | L16A092 |
| MS7F/MS7R/TH7, MS7AF/MS7AR/TH7A | MOTOR STARTER, THERMAL CONTACTOR 0.63~1A | SCHNEIDER | GV2DM205BD | L16A144 |
| MS6A/TH6A, MS10/TH10, MS16/TH16, MS71/TH71, MS89/TH89 | MOTOR STARTER, THERMAL CONTACTOR 2.5-4A | SCHNEIDER | GV2DM108BD | L16A136 |
| | AUX CONTACT SW -1NO 1NC | SCHNEIDER | GV AE11 | L16A092 |
| MS8/TH8, MS260/TH260 | MOTOR STARTER, THERMAL CONTACTOR 4~6.3A | SCHNEIDER | GV2DM110BD | L16A139 |
| | AUX CONTACT SW -1NO 1NC | SCHNEIDER | GV AE11 | L16A092 |
| MS18/TH18 | 0.16-0.25A | SCHNEIDER | GV2DM102BD | L16A140 |
| | AUX CONTACT SW -1NO 1NC | SCHNEIDER | GV AE11 | L16A092 |
| MS19/TH19, MS154/TH154 | MOTOR STARTER, THERMAL CONTACTOR 1.6~2.5A | SCHNEIDER | GV2DM107BD | L16A119 |
| | AUX CONTACT SW -1NO 1NC | SCHNEIDER | GV AE11 | L16A092 |
| SURGE P.L~L | SURGE P.L~L | OKAYA | RAV-152BYZ-2A | L13C014 |
| SURGE P.L~G | SURGE P.L~G | OKAYA | RAV-801BXZ-4 | L13C015 |
| T2 | TRANSFORMER 2KVA(CE,UL) 380/400/415V-200/230V | SHIN NIPPON | L29A201 | L29A201 |
| SW296 | KEY SWITCH | DECA | ADA16K6-AA6-DF-03 | L11B198 |
| LS1001 | DOOR LIMIT SWITCH | AZBIL | SL1-PK | Z621A0180110 |
| FAN | FAN 230VAC 150D | SUNON | A2175-HBT.T.GN + FG-17B | L19A022 |
| AIR CON | AIR CON | RITTAL | SK3303500 | L19A020 |

| REVISIONS | | DATE | NAME |
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| ③ | APPROVED | 13/06/2019 | AVINASH |
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| ① | DRAWN | 13/06/2019 | JAVAHAR |

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| MTC COMPONENT DETAILS | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | F04 |

| MP81 | | | |
|---|---|---|---|
| A1 | *LS1319 | B1 | OM2-RE1 |
| A2 | *OMAL2 | B2 | OM2-RE2 |
| A3 | *LS1343 | B3 | |
| A4 | *LS184 | B4 | |
| A5 | LS3320 | B5 | LS193 |
| A6 | LS181 | B6 | LS1338 |
| A7 | LS191 | B7 | LS1339 |
| A8 | LS192 | B8 | SPARE1 |
| A9 | LS1325 | B9 | SPARE2 |
| A10 | LS1337 | B10 | SLATCF |

| MP82 | | | |
|---|---|---|---|
| A1 | SL750 | B1 | *OMAL3 |
| A2 | SPARE3 | B2 | OM3-RE1 |
| A3 | OM1-RE1 | B3 | OM3-RE2 |
| A4 | OM1-RE2 | B4 | *LS2340 |
| A5 | LS170 | B5 | |
| A6 | LS173 | B6 | |
| A7 | *OMAL1 | B7 | TSCCG1 |
| A8 | C24HB | B8 | TSCCG2 |
| A9 | C24L | B9 | |
| A10 | | B10 | |

| MP84 | | | |
|---|---|---|---|
| A1 | SL1310 | B1 | SL8022 |
| A2 | SL1306 | B2 | SL22 |
| A3 | SL1308 | B3 | SL181 |
| A4 | SL302 | B4 | SL09 |
| A5 | SL303 | B5 | SL192 |
| A6 | SL304 | B6 | SL18 |
| A7 | MQLON | B7 | |
| A8 | SP1 | B8 | |
| A9 | SP2 | B9 | |
| A10 | SP3 | B10 | |

| MP85 | | | |
|---|---|---|---|
| A1 | LS905 | B1 | SL132B |
| A2 | SL905 | B2 | SL132A |
| A3 | SL2300 | B3 | SL711A |
| A4 | *LS2307 | B4 | SL711B |
| A5 | LS2308 | B5 | |
| A6 | LS1698 | B6 | |
| A7 | LS2064 | B7 | |
| A8 | ALSXYF | B8 | |
| A9 | SP4 | B9 | |
| A10 | SL132A-2 | B10 | |

| MP94 | | | |
|---|---|---|---|
| A1 | C24HB | B1 | ATCMA-(4) |
| A2 | ATCUNC-(4) | B2 | ATCMA-(+) |
| A3 | ATCUNC-(+) | B3 | C24L |
| A4 | MGCW-(4) | B4 | MGCCW-4 |
| A5 | MGCCW-(+) | B5 | MGCW-(+) |
| A6 | | B6 | |
| A7 | | B7 | |
| A8 | EMG-11 | B8 | EMG-12 |
| A9 | EMG-21 | B9 | EMG-22 |
| A10 | EMG-31 | B10 | EMG-32 |

| MP95 | | | |
|---|---|---|---|
| A1 | SL8022 | B1 | SL1310 |
| A2 | SL22 | B2 | SL1306 |
| A3 | SP4 | B3 | SL1308 |
| A4 | | B4 | SL302 |
| A5 | | B5 | SL303 |
| A6 | | B6 | SL304 |
| A7 | | B7 | MQLON |
| A8 | | B8 | SP1 |
| A9 | LS1311 | B9 | SP2 |
| A10 | SL1309 | B10 | SP3 |

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| ① | | DRAWN | 13/06/2019 | JAVAHAR |

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| CONNECTOR PIN DETAILS | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | F05 |

| MP91 | | | |
|---|---|---|---|
| 1 | U2 | 13 | U10 |
| 2 | V2 | 14 | V10 |
| 3 | W2 | 15 | W10 |
| 4 | U16 | 16 | U8 |
| 5 | V16 | 17 | V8 |
| 6 | W16 | 18 | W8 |
| 7 | U6A | 19 | U18 |
| 8 | V6A | 20 | V18 |
| 9 | W6A | 21 | W18 |
| 10 | U71 | 22 | R2 |
| 11 | V71 | 23 | S2 |
| 12 | W71 | 24 | T2 |

| MP92 | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1 | *LS1319 | 14 | TSCCG2 | 24 | C24L | 34 | OM3-RE2 |
| 2 | *OMAL2 | 15 | LS193 | 25 | OM1-RE1 | 35 | *LS2340 |
| 3 | *LS1343 | 16 | LS1338 | 26 | OM1-RE2 | 36 | |
| 4 | *LS184 | 17 | LS1339 | 27 | LS170 | 37 | |
| 5 | LS3320 | 18 | SPARE1 | 28 | LS173 | 38 | |
| 6 | LS181 | 19 | SPARE2 | 29 | *OMAL1 | 39 | |
| 7 | LS191 | 20 | SLATCF | 30 | C24HB | 40 | |
| 8 | LS192 | 21 | SL750 | 31 | C24L | 41 | |
| 9 | LS1325 | 22 | SPARE3 | 32 | *OMAL3 | 42 | |
| 10 | LS1337 | 23 | C24HB | 33 | OM3-RE1 | 43 | |
| 11 | OM2-RE1 | | | | | 44 | |
| 12 | OM2-RE2 | | | | | 45 | |
| 13 | TSCCG1 | | | | | 46 | |

| MP93 | | | |
|---|---|---|---|
| 1 | TH6-L1 | 13 | |
| 2 | TH6-L2 | 14 | |
| 3 | TH6-L3 | 15 | |
| 4 | TH6-14 | 16 | C24L |
| 5 | M6R-22 | 17 | MAS-(+) |
| 6 | M6F-22 | 18 | EMG31 |
| 7 | TH6-22 | 19 | MAS-(4) |
| 8 | TH6-21 | 20 | FWD-(4) |
| 9 | EMG-12 | 21 | REV-(4) |
| 10 | EMG-11 | 22 | C24HB |
| 11 | EMG-22 | 23 | |
| 12 | EMG-21 | 24 | |

| MP96 | | | |
|---|---|---|---|
| 1 | C24L | 13 | SL2603-1 |
| 2 | SYSPS2-3 | 14 | SL2603-2 |
| 3 | LS2603-1 | 15 | SL2604-1 |
| 4 | LS2603-2 | 16 | SL2604-2 |
| 5 | LS2604-1 | 17 | PS2601-3 |
| 6 | LS2604-2 | 18 | PS260-1 |
| 7 | SYSPS-1 | 19 | PS2602-3 |
| 8 | SYSPS-3 | 20 | PS2602-1 |
| 9 | SL2601-1 | 21 | PS2603-3 |
| 10 | SL2601-2 | 22 | PS2603-1 |
| 11 | SL2602-1 | 23 | PS2604-3 |
| 12 | SL2602-2 | 24 | PS2604-1 |

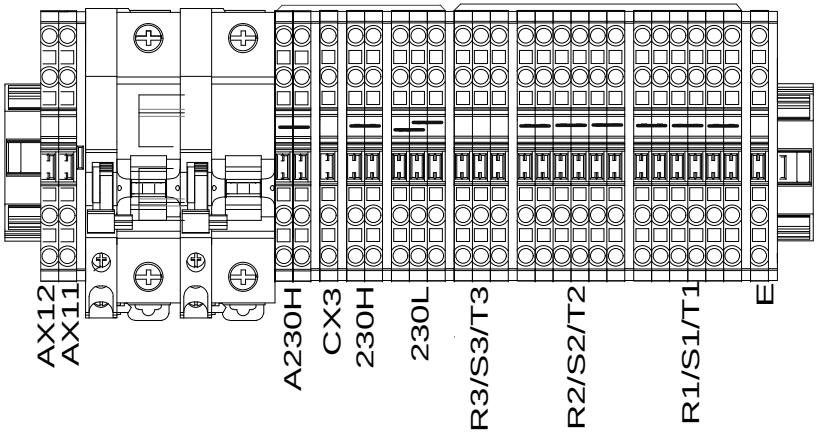
| MP98 | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 1 | ROB-EMG1 | 13 | | 25 | RBHLY | 37 | WPN02 | 49 | MCRDY | 61 | PCP01 |
| 2 | ROB-EMG2 | 14 | | 26 | *RBALM | 38 | WPN04 | 50 | *MCALM | 62 | PCP02 |
| 3 | ROB-EMG3 | 15 | | 27 | *RBINT | 39 | WPN08 | 51 | *MCWRN | 63 | WCN01 |
| 4 | ROB-EMG4 | 16 | | 28 | SYSTR | 40 | WPNPRT | 52 | ACSOK | 64 | WCN02 |
| 5 | EM1-1 | 17 | ROB-EMG5 | 29 | SYSTP | 41 | RBI17 | 53 | WSRDY | 65 | C24HB |
| 6 | EM1-2 | 18 | ROB-EMG6 | 30 | WLDFN | 42 | RBI18 | 54 | MCRUN | 66 | C24HB |
| 7 | EM2-1 | 19 | | 31 | WULDFN | 43 | RBI19 | 55 | WLDRQ | 67 | C24L |
| 8 | EM2-2 | 20 | | 32 | NOWPST | 44 | RBI20 | 56 | WULDRQ | 68 | C24L |
| 9 | | 21 | | 33 | FLSON | 45 | RBI21 | 57 | WSCOK | 69 | DOCOM |
| 10 | | 22 | | 34 | FCLRQ | 46 | RBI22 | 58 | WSCNG | 70 | DOCOM |
| 11 | | 23 | | 35 | FUCRQ | 47 | RBI23 | 59 | FCLFN | 71 | DOCOM |
| 12 | | 24 | | 36 | WPN01 | 48 | RBI24 | 60 | FUCFN | 72 | DOCOM |

| REVISIONS | | | | |
|---|---|---|---|---|
| ④ | | | DATE | NAME |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

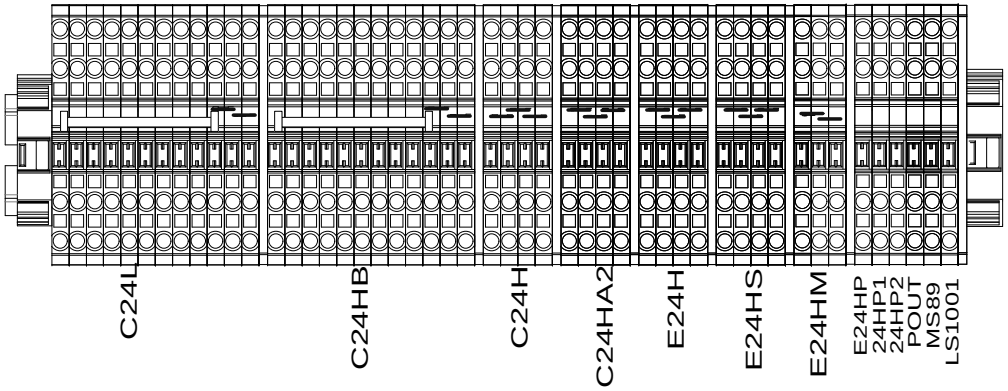
MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| CONNECTOR PIN DETAILS | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | F06 |

## TB20 TB10



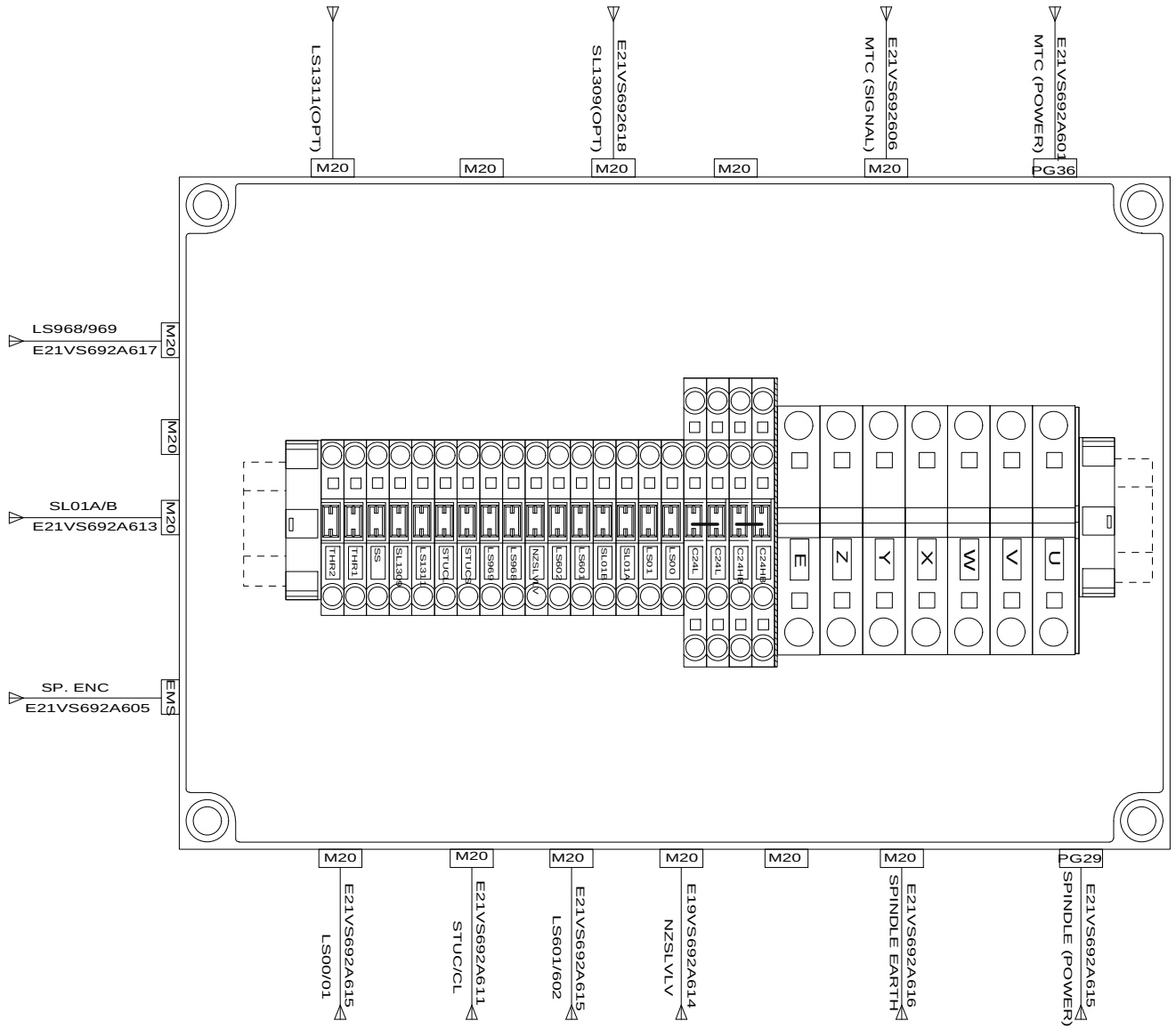

## TB30




| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

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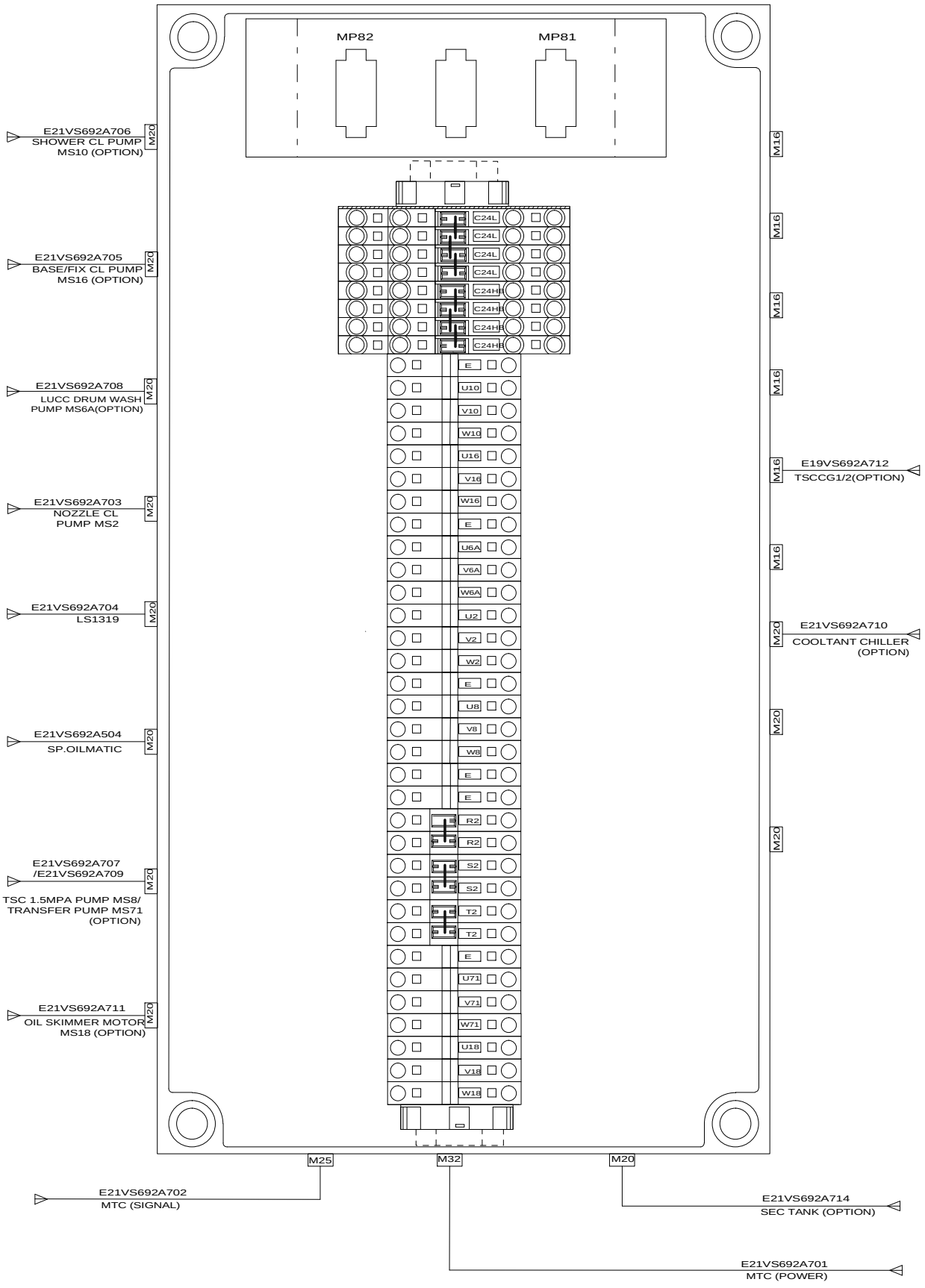
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| TB10 / TB20 / TB30 LAYOUT | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | F07 |




| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

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MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| TB40 SPINDLE TERMINAL BOX | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | F08 |




| | | | |
|---|---|---|---|
| GROUP | = | SHEET No. | F09 |
| DRAWING No. | | V210I-C1E-XG | |
| MACHINE TYPE | PS65/PS105 | NC TYPE | 0i-MF |

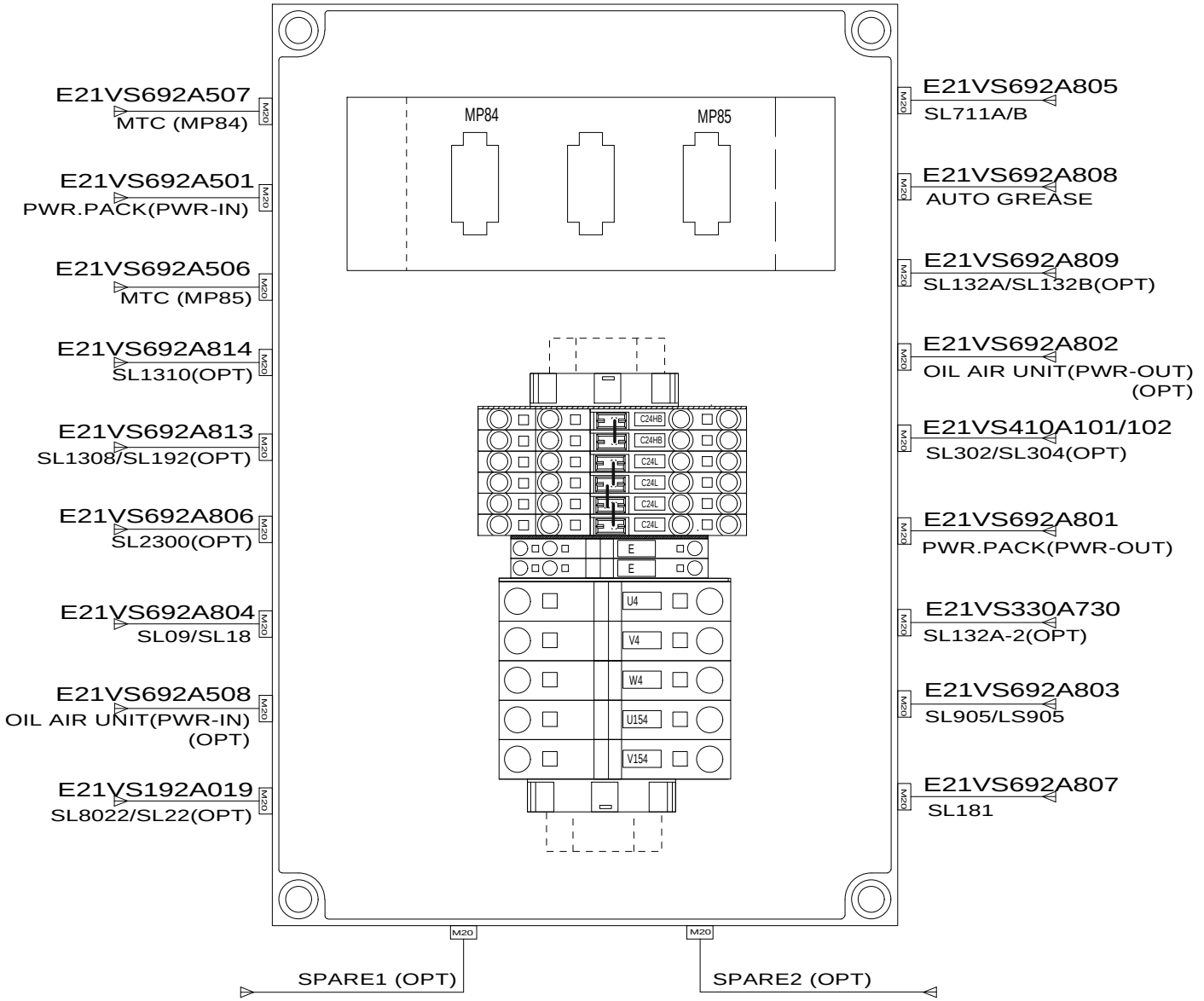
**TITLE**

# TB50 COOLANT TERMINAL BOX

**MAKINO**
MAKINO ASIA PTE LTD

| | | DATE | NAME |
|---|---|---|---|
| | APPROVED | 13/06/2019 | AVINASH |
| | CHECKED | 13/06/2019 | AVINASH |
| | DRAWN | 13/06/2019 | JAVAHAR |

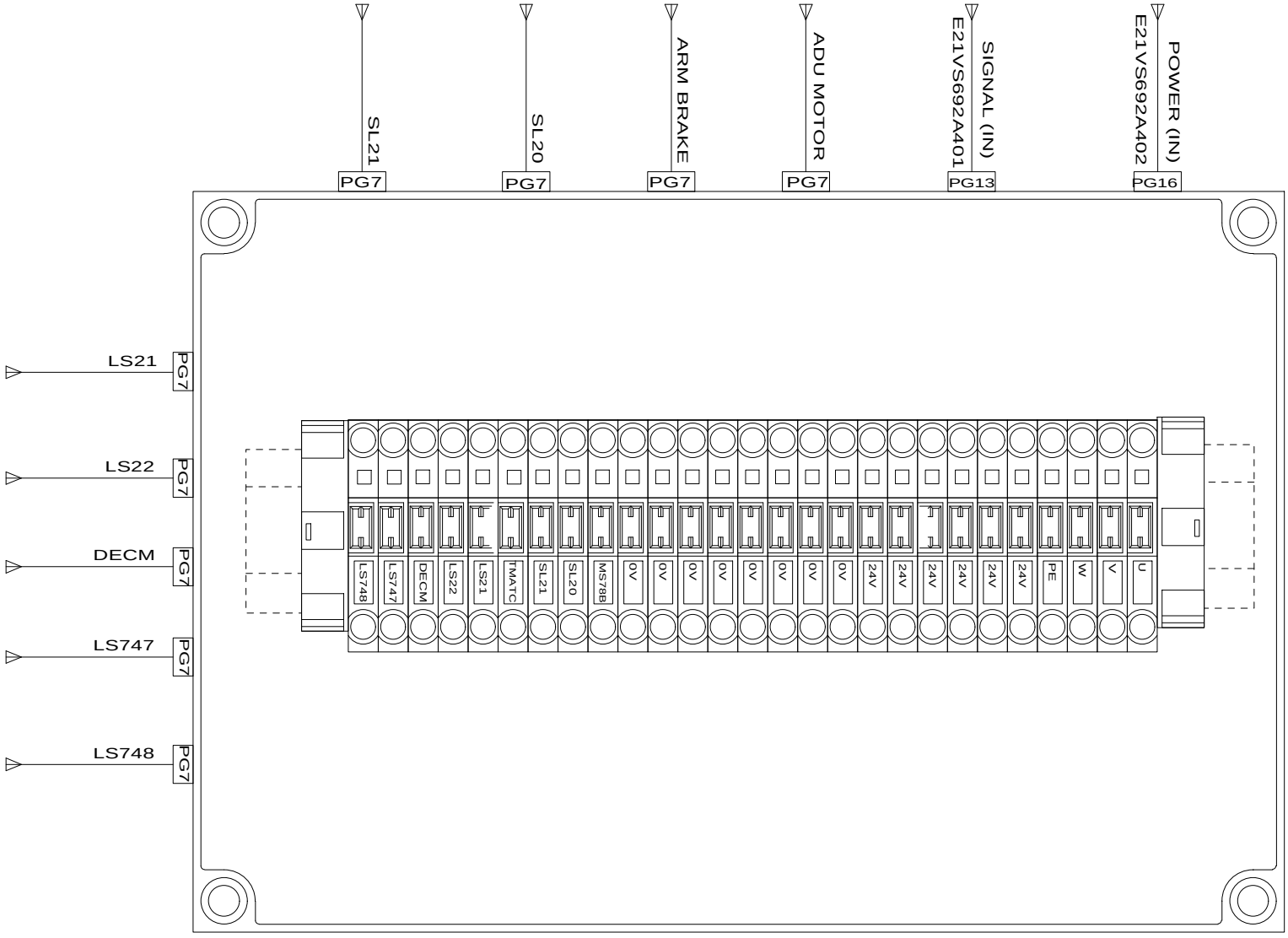
| REVISIONS |
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| ④ |
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| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

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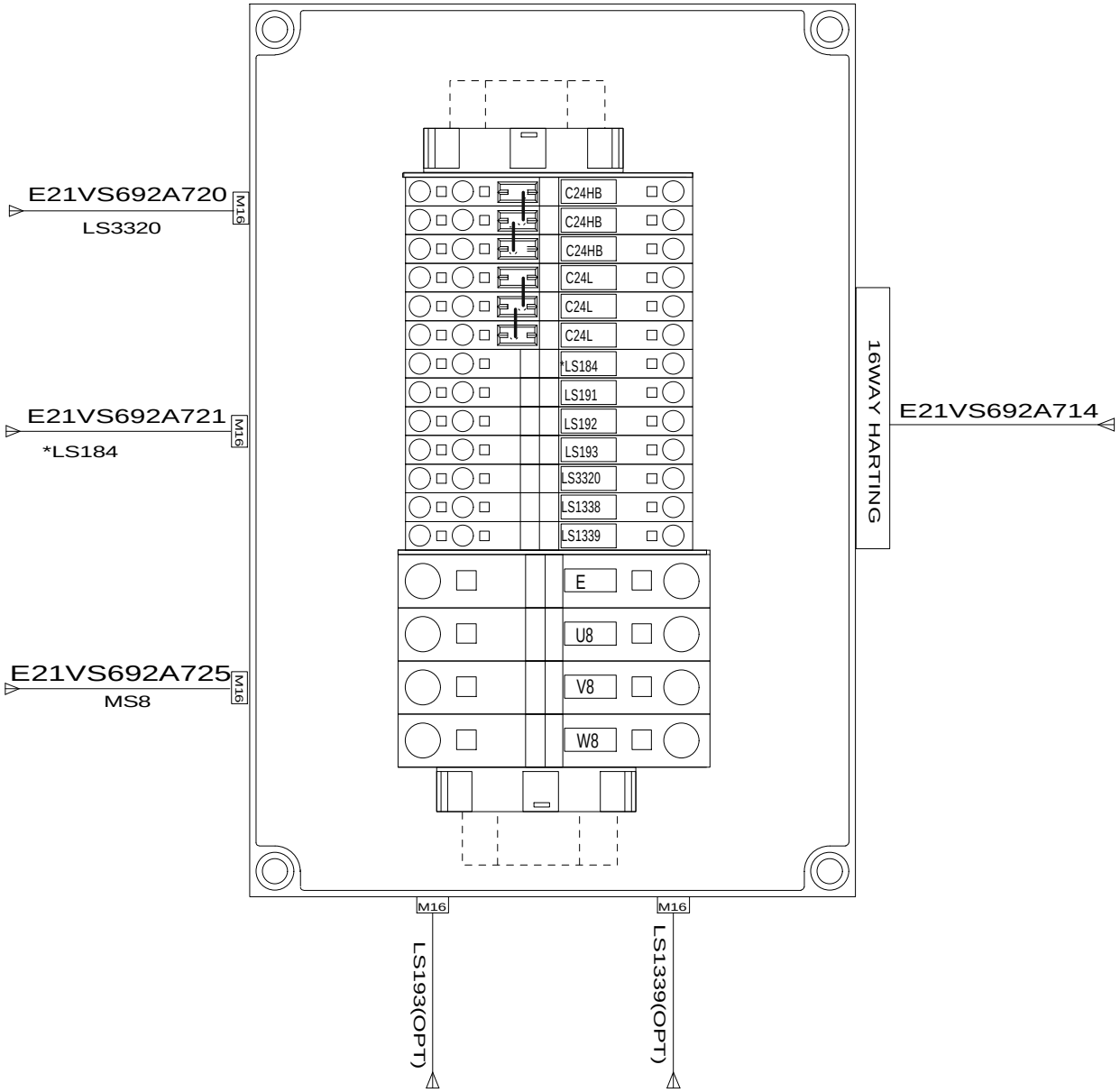
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| TB60 FLUID PANEL TERMINAL BOX | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | F10 |




| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

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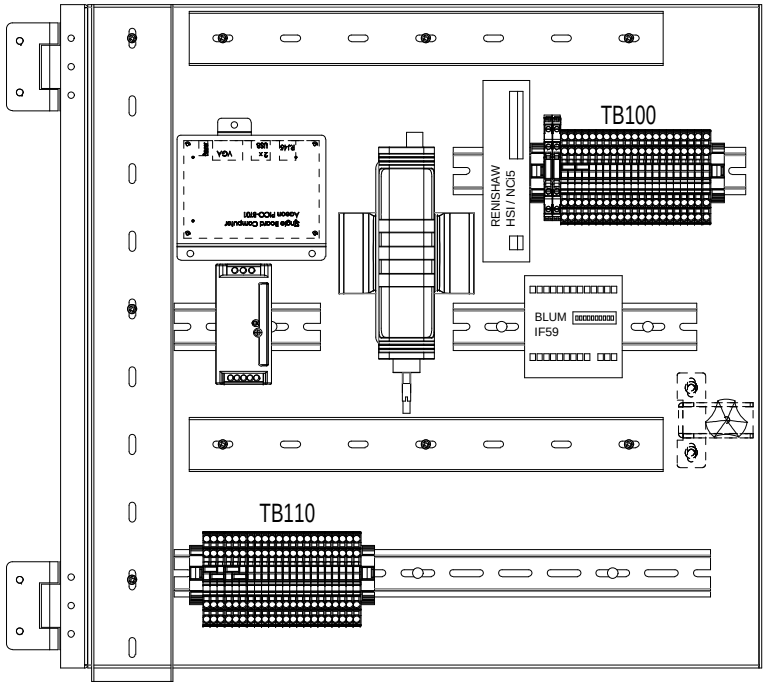
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| TB70 ATC TERMINAL BOX | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. F11 |

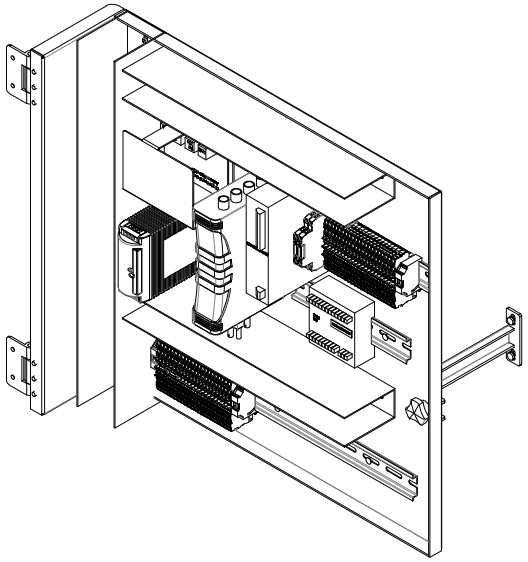



| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

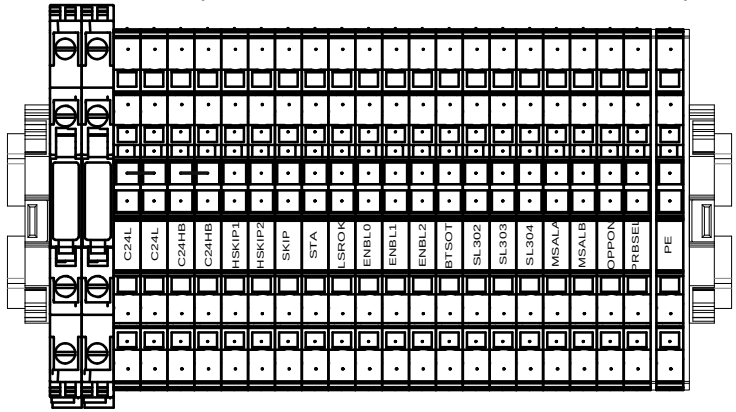
MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| TB90 SEC TANK TERMINAL BOX | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | F12 |

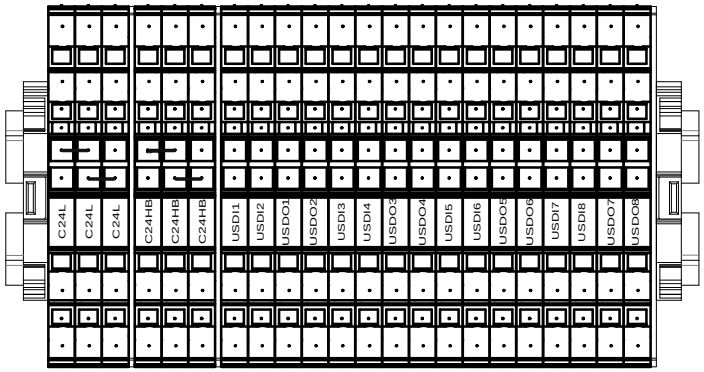


TB100 (MEASUREMENT SYSTEM INTERFACE)



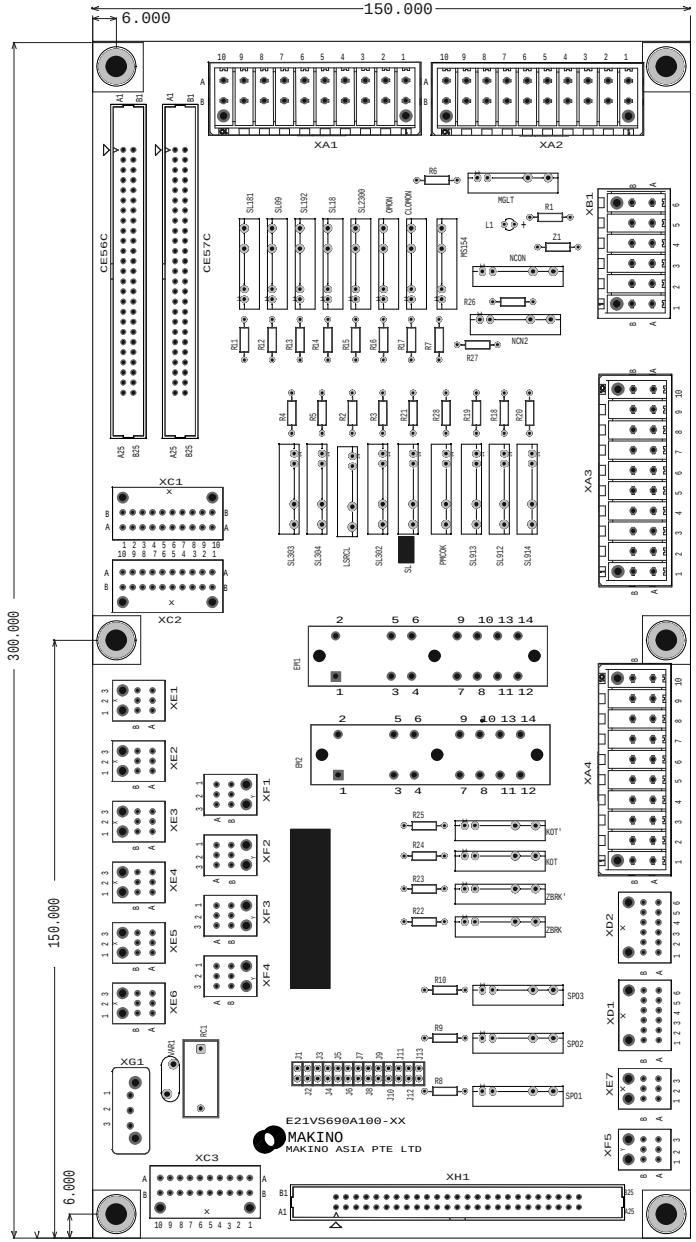

TB110 (USER M-CODE INTERFACE)




| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| SWING PANEL -TB100/TB110 LAYOUT | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | F13 |



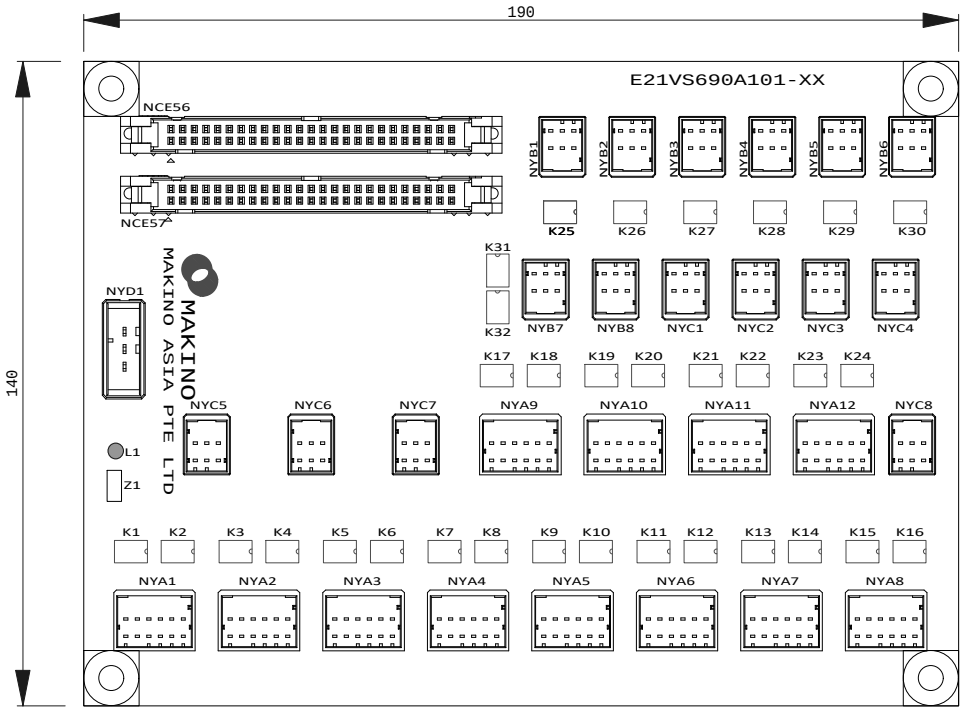

| S.NO | Description | Specification | Designation | Qty |
|---|---|---|---|---|
| 1 | PCB | 300 x 150mm | | 1 |
| 2 | Panasonic failsafe Relay + Base | SFS-5-L-24V + SFS6-PS | EM1, EM2 | 2 |
| 3 | Panasonic Photomos | AQZ102 | | 27 |
| 4 | Resistor - Reputed Brand | 2.2KOhms , 1/4W | R1 - R28 | 28 |
| 5 | Varistor Panasonic | ERZV10D820 | VAR | 1 |
| 6 | Spark Killer OKAYA | XEB0471 / XE0471 | RC | 1 |
| 7 | 20 WAY AMP D3100 Series | 316517-5 | XA1, XA2, XA3, XA4 | 4 |
| 8 | 50 WAY HIROSE / AUK | HIF3CB-50PA-2-54D5A LD20 (AUK) | XH1, CE56C, CE57C | 3 |
| 9 | 20 WAY D2100 Series | 1318127-1 | XC1, XC2, XC3 | 3 |
| 10 | 12 WAY D3100 Series | 316516-5 | XB1 | 1 |
| 11 | 12 WAY D2100 Series | 1318126-1 | XD1, XD2 | 2 |
| 12 | 6 WAY X-KEY D2100 Series | 1318124-1 | XE1, XE2, XE3, XE4, XE5, XE6, XE7 | 7 |
| 13 | 6 WAY Y-KEY D2100 Series | 1318124-2 | XF1, XF2, XF3, XF4, XF5 | 5 |
| 14 | 3 WAY D3100 Series | 1-178313-5 | XG1 | 1 |
| 15 | 2 WAY JUMPERS | REPUTED BRAND | J1 - J13 | 13 |
| 16 | Zener Overvoltage Transient Suppressors | ST - 1.5KE33A | Z1 | 1 |

E21VS690A100-XX

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| | ④ | | | |
| | ③ | APPROVED | 13/06/2019 | AVINASH |
| | ② | CHECKED | 13/06/2019 | AVINASH |
| | ① | DRAWN | 13/06/2019 | JAVAHAR |

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| RELAY PCB LAYOUT & PARTS DETAILS | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. F14 |



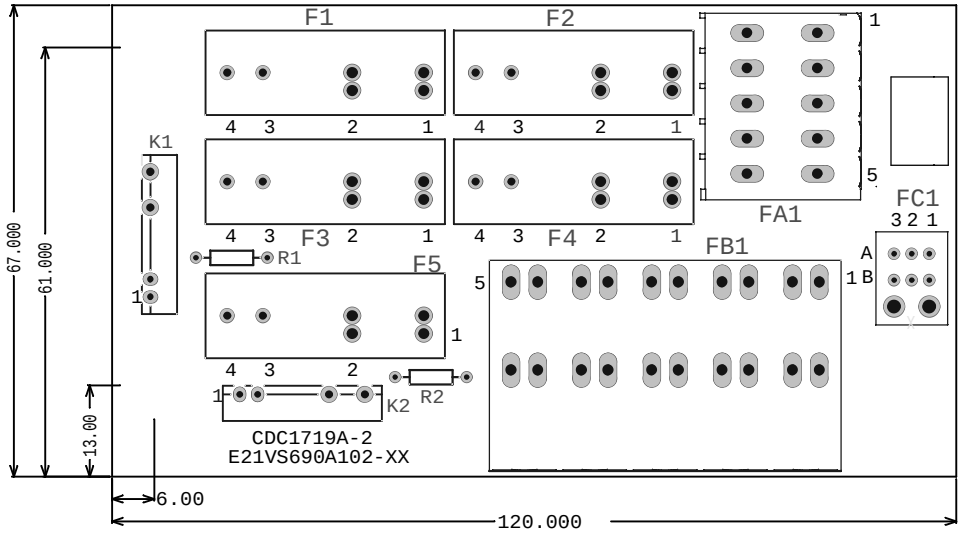

| S.NO | Description | Specification | Designation | Qty |
|---|---|---|---|---|
| 1 | PCB | 190 x 140mm | | 1 |
| 2 | LED | DC 2.1V,Green 3mm | L1 | 1 |
| 3 | Panasonic Photomos | AQV252G | K1 - K32 | 32 |
| 4 | Resistor | 3KOhms , 1/4W, +/-5% | R1 - R33 | 33 |
| 5 | 50 WAY FLAT CABLE CONNECTOR | HIF3BB-50PA-2.54DSA (HIROSE) LD50 (AUK) | NCE56, NCE57 | 2 |
| 6 | 12 WAY D2100 Series | 1318126-1 | NYA1 - NYA2 | 12 |
| 7 | 6 WAY X-KEY D2100 Series | 1318124-1 | NYB1 - NYB8 | 8 |
| 8 | 6 WAY Y-KEY D2100 Series | 1318124-2 | NYC1-NYC8 | 8 |
| 9 | 3 WAY D3100 Series | 1-178313-5 | NYD1 | 1 |
| 10 | Zener Overvoltage Transient Suppressors | ST - 1.5KE33A | Z1 | 1 |

E21VS690A101-XX

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| | ④ | | | |
| | ③ | APPROVED | 13/06/2019 | AVINASH |
| | ② | CHECKED | 13/06/2019 | AVINASH |
| | ① | DRAWN | 13/06/2019 | JAVAHAR |

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| OPTION PCB LAYOUT & PARTS DETAILS | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | F15 |



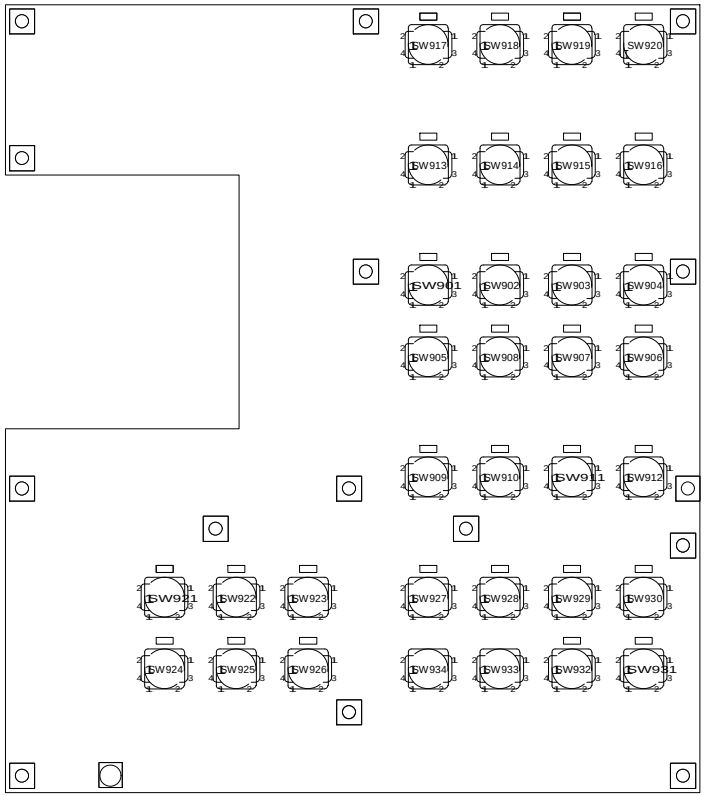

| S.NO | Description | Specification | Designation | Qty |
|---|---|---|---|---|
| 1 | PCB | 67 x 120mm | | 1 |
| 2 | 6 WAY AMP SERIES | 1318124-1 | FC1 | 1 |
| 3 | PHOTOMOS PANASONIC | AQZ404 | K1, K2 | 2 |
| 4 | RESISTOR | 3KOhms , 1/4W, +/-5% | R1, R2 | 2 |
| 5 | FUSE HOLDER DAITO | GPH-4V | F1-F5 | 5 |
| 6 | 5 WAY WAGO TERMINAL BLOCK | 745-105 | FA1 | 1 |
| 7 | 5 WAY WAGO TERMINAL BLOCK | 2716-105 | FB1 | 1 |
| 8 | FUSE DAITO 5A | GP50 | F1, F3, F4, F5 | 4 |
| 9 | FUSE DAITO 15A | GP150 | F2 | 1 |

E21VS690A102-XX

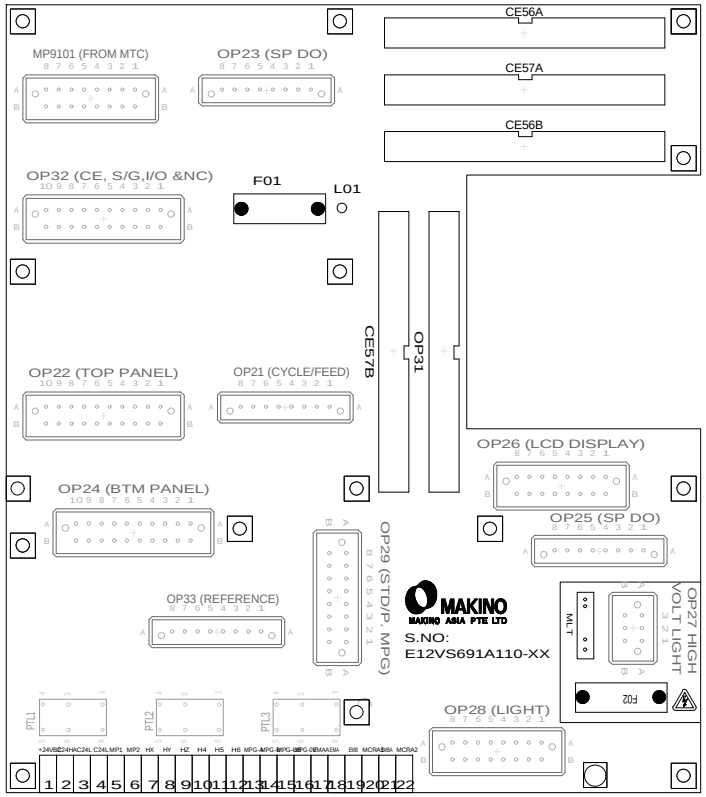
| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| FUSE PCB LAYOUT & PARTS DETAILS | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | F16 |




FRONT VIEW FOR PANEL MOUNTING




BACK VIEW FOR CABLE CONNECTIONS

E12VS691A110-XX

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

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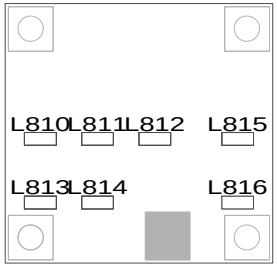
| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| | | | | = |
| OPERATOR PCB LAYOUT DETAILS | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | F17 |

| S. NO | DESCRIPTION | SPECIFICATION | DESIGNATION | QTY |
|---|---|---|---|---|
| 1 | 16 WAY, AMP | 178327-5 | MP9101, OP26, OP28, OP29 | 4 |
| 2 | 8 WAY, AMP | 1-178317-2 | OP21, OP23, OP25, OP33 | 4 |
| 3 | 20 WAY, AMP | 178328-5 | OP22, OP24, OP32 | 3 |
| 4 | CONN. TAP HRS 50 WAY, HIROSE | HIF3CB-50PA-2.54DSA | OP31, CE56A, CE56B, CE57A, CE57B | 5 |
| 5 | SW CAP DAILYWELL, SW TACTILE FB DAILYWELL | KTSC-21K, DTSP-24R-V | SW901~SW934 | 34 |
| 6 | DIODE 1N4007 | REPUTED BRAND | D10~D13 | 4 |
| 7 | RESISTOR, 1/2W, 2.2K | REPUTED BRAND | R901~R932, R13A,R13B, R101,R10~R12 | 38 |
| 8 | LED 2MM x 5MM GREEN, AVAGO/FARNEL | HLMP-S501/1003238 | L901~L919, L921~L932 | 31 |
| 9 | SHORTING LINK | REPUTED BRAND | J1~J5 | 5 |
| 10 | OMRON G6B SERIES | G6B-2214P-US DC24 | PTL1, PTL2, PLT3 | 3 |
| 11 | FUSE (5A) | MCHTC-100M, MCF05G-5A | F01 | 1 |
| 12 | M4 EARTH STUB | REPUTED BRAND | EARTH STUB | 1 |
| 13 | GREEN LED (3MM) | REPUTED BRAND | L01, L10~L13 | 5 |
| 14 | PCB TERMMINAL, 2WAY, PHOENIX | 5430409 | TB10 | 11 |
| 15 | PHOTO MOS (PA RELAY),MATSUSHITA | PA1A-24V | MLT | 1 |
| 16 | FUSE (2A) | MCHTC-100M, MCF05G-2A | F02 | 1 |
| 17 | CONN AMP 6 WAY (XY TYPED) | 3-178140-5 | OP27 | 1 |
| 18 | LED 2MM x 5MM RED, AVAGO/FARNEL | HLMP-S201/1003236 | L920 | 1 |

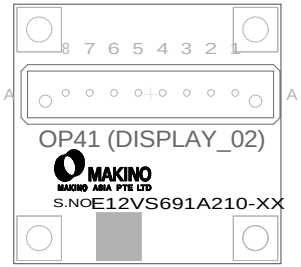
| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| OPERATOR PCB PARTS DETAILS | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | F18 |




FRONT VIEW FOR PANEL MOUNTING




BACK VIEW FOR CABLE CONNECTIONS

| S. NO | DESCRIPTION | SPECIFICATION | DESIGNATION | QTY |
|---|---|---|---|---|
| 1 | 8 WAY, AMP | 1-0178317-2 | OP41 | 1 |
| 2 | LED 2MM x 5MM GREEN, AVAGO/FARNEL | HLMP-S501/1003238 | L810~L816 | 7 |
| 3 | M4 EARTH STUB | REPUTED BRAND | EARTH STUB | 1 |
| 4 | RESISTOR, 1/2W, 2.2K | REPUTED BRAND | R810~R816 | 7 |

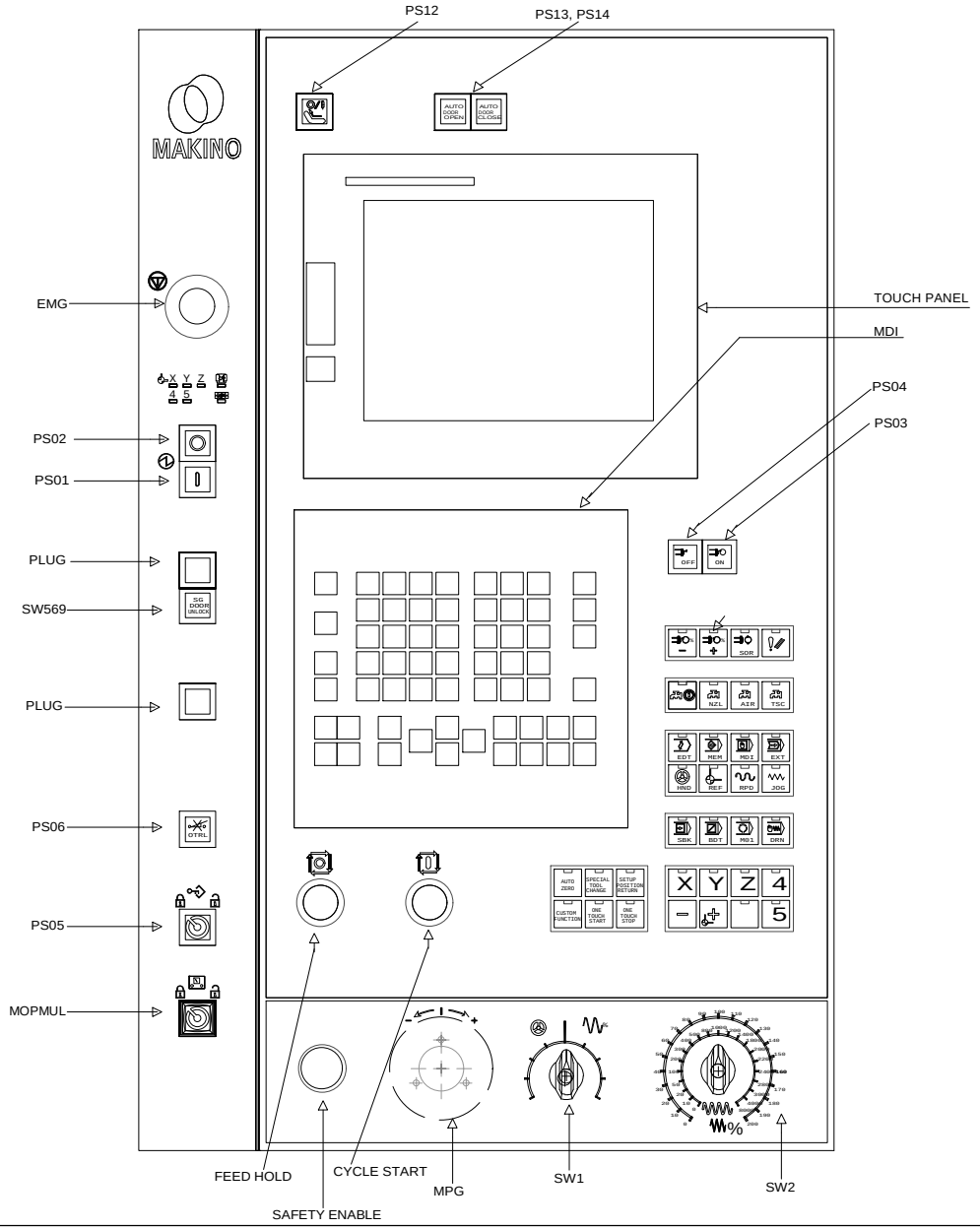
DISPLAY PCB (OPE PCB2)

E12VS691A210-XX

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| DISPLAY PCB LAYOUT & PARTS DETAILS | | | | = |
| | NC TYPE | 0i-MF | V210I-C1E-XG | SHEET No. |
| | | | | F19 |

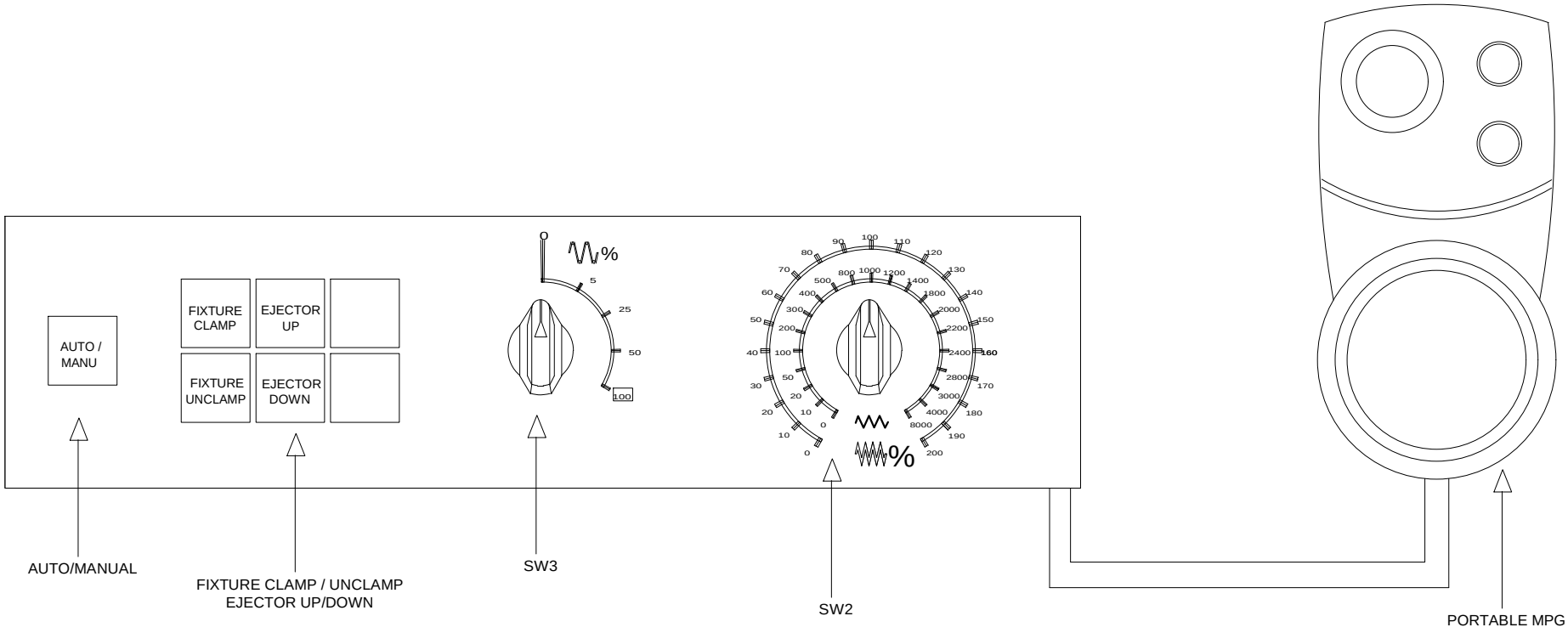



| NO. | DESIGNATION | DESCRIPTION | MAKE | MFR PART NO. | MAKINO PART NO. |
|---|---|---|---|---|---|
| 1 | EMG | PB EMERGENCY | DECA | A20B-V4E03R,3NC,C/W | L11B227 |
| 2 | PS02 | POWER OFF | DECA | ADA16S6-MA6-B10R<br>MG-6UR-24VAC/DC | L11B189<br>L11B193 |
| 3 | PS01 | POWER ON | DECA | ADA16S6~MA6-B10G<br>MG-2PG-24VAC/DC | L11B188<br>L11B192 |
| 4 | PS03 | SPINDLE ON | DECA | ADA16S6-MA6-B10C+PC<br>MG-6UY-24VAC/DC | L11B191<br>L11B194 |
| 5 | PS04 | SPINDLE OFF | DECA | ADA16S6-MA6-B10C | L11B190 |
| 6 | PS05 | MEMORY KEY | DECA | ADA16K6-AS3-DF | L11B238 |
| 7 | MOPMUL | (RESERVED FOR OPTION) | DECA | SS0-013-05 | L11B121 |
| 8 | PS07<br>PS12<br>PS06<br>PS13, PS14 | S/G DOOR UNLOCK PB<br>SCREW CONVEYOR<br>OT RELEASE<br>AUTO DOOR | DECA | ADA16S6-MA6-B10C<br>MG-6UY-24VAC/DC | L11B190<br>L11B194 |
| 9 | PLUG | (RESERVED FOR OPTION) | DECA | S50-013-03 | L11B199 |
| 10 | FEED HOLD | FEED HOLD | DECA | A204L-MF2E11Q0R<br>LEBA9S-6UR-24VAC/DC | L11B222<br>L11B224 |
| 11 | CYCLE START | CYCLE START | DECA | A204L-MF2E11Q0G<br>LEBA9S-2PG-24VAC/DC | L11B221<br>L11B225 |
| 12 | MPG | MANUAL PULSE GENERATOR | EUCHNER | HKD100S100G05C2136<br>075977 | L11C053<br>L11C054 |
| 13 | SW1 | ROTARY SWITCH, KNOB | TOSOKU | RS400P-1-1-9BP 30 16R<br>BS60 | L11C029, L11C076 |
| 14 | SW2 | ROTARY SWITCH, KNOB | TOSOKU | DPP01020J16R<br>BS60 | L11C029, L11C040 |
| 15 | MDI | MANUAL DATA INPUT | FANUC | A02B-0323-C125#M | A02B-0323-C125#M |
| 16 | TOUCH PANEL | TOUCH PANEL CONTROL | FANUC | A02B-0338-H142 | Z513A0121760 |
| 17 | OPE PCB<br>OPE PCB2 | PCB-1, OPERATOR PANEL<br>DISPLAY PCB / PCB-2, OPER PANEL | ST<br>ST | E12VS691A110-XX<br>E12VS691A210-XX | E12VS691A110-XX<br>E12VS691A210-XX |
| 18 | ENABLE SW | SAFETY ENABLE | FUJI | AR22G4L-10E3W | L11B056 |

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

MAKINO
MAKINO ASIA PTE LTD

| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| OPERATORS PANEL LAYOUT, PARTS | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | F20 |




| DEVICE DESIGNATION | MANUFACTURER | MFR PART NO. | MAKINO PART NO. |
|---|---|---|---|
| PORTABLE MPG | EUCHNER | P-MPG HBA-119782 | Z318R0001300 |
| SW2 | TOSOKU | DPP01020J16R<br>BS60 | L11C029<br>L11C040 |
| SW3 | TOSOKU | RS400P-1-1-9BP 30 16R<br>BS60 | L11C029<br>L11C076 |
| AUTO/MANUAL | DECA | ADA16S6~MA6~B10C+PC<br>MG~6UY~24VAC/DC | L11B191<br>L11B194 |
| FIXTURE CLAMP / UNCLAMP, EJECTOR UP/DOWN | DECA | ADA16S6~MA6~B10C<br>MG~6UY~24VAC/DC | L11B190<br>L11B194 |

| REVISIONS | | | DATE | NAME |
|---|---|---|---|---|
| ④ | | | | |
| ③ | | APPROVED | 13/06/2019 | AVINASH |
| ② | | CHECKED | 13/06/2019 | AVINASH |
| ① | | DRAWN | 13/06/2019 | JAVAHAR |

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| TITLE | MACHINE TYPE | PS65/PS105 | DRAWING No. | GROUP |
|---|---|---|---|---|
| PORTABLE / FIXTURE MPG PLATE | NC TYPE | 0i-MF | V210I-C1E-XG | = |
| | | | | SHEET No. |
| | | | | F21 |